

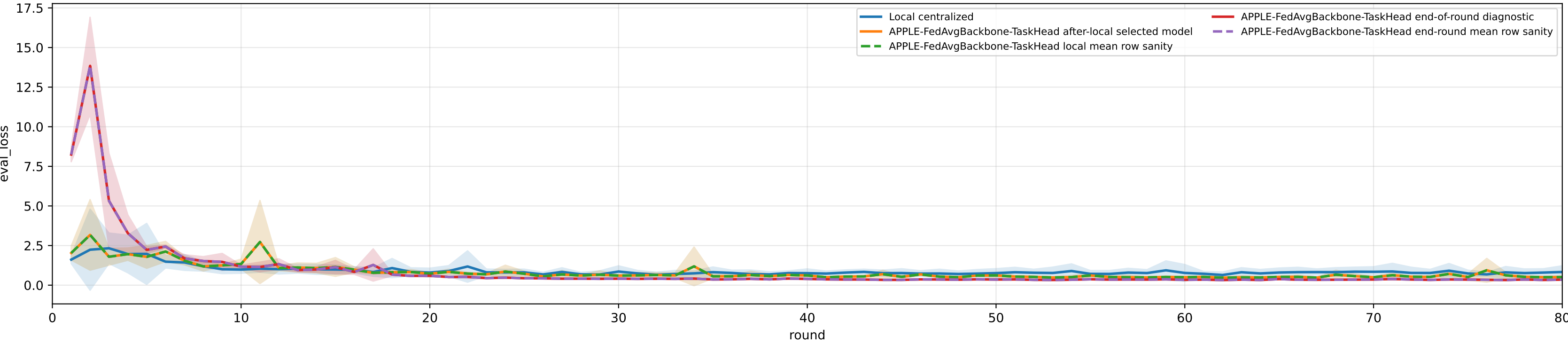
Controlled q-label heterogeneity: iid | **APPLE-FedAvgBackbone-TaskHead diagnostics** |
aggregated / pooled over 5 clients |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

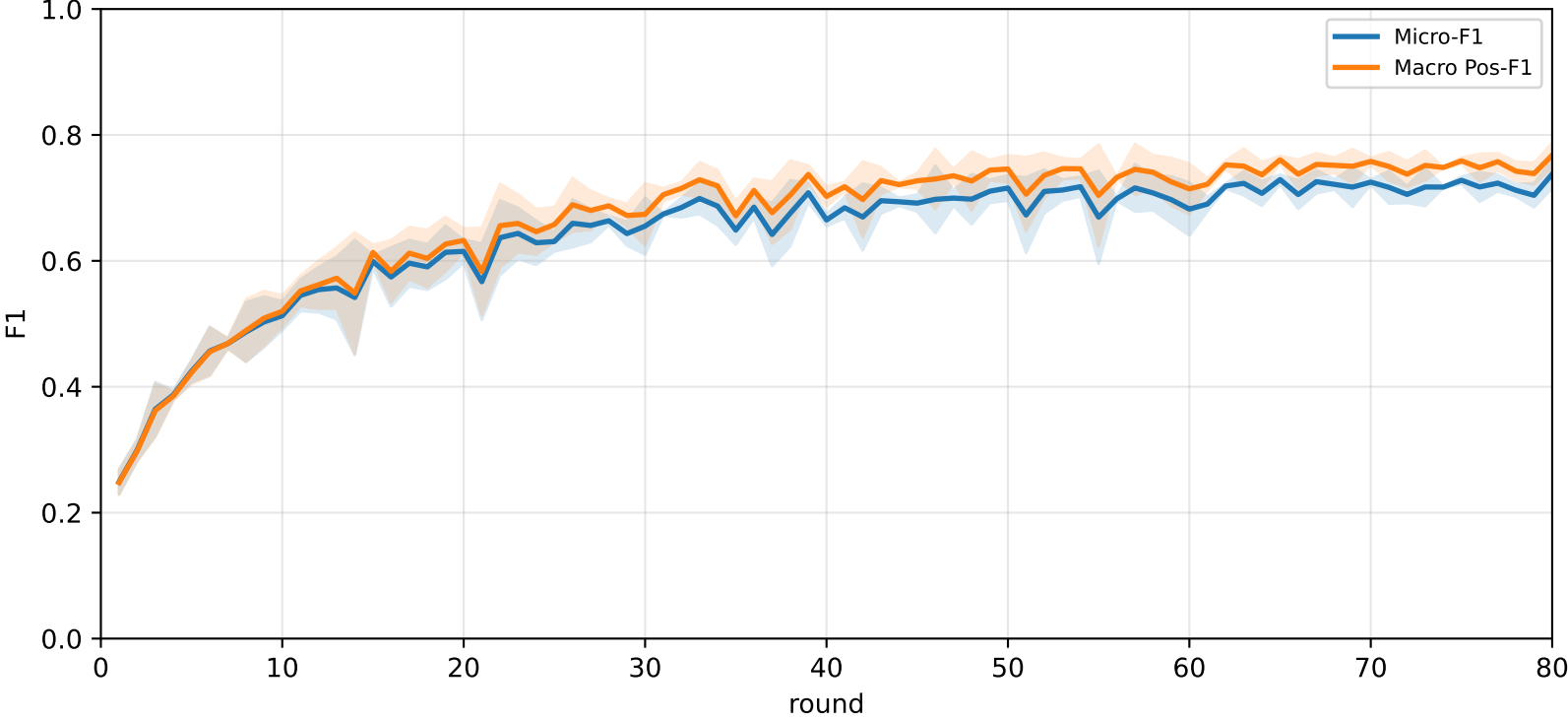
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	64.2% ± 0.6%	67.8% ± 0.9%	79.6% ± 4.0%	78.3% ± 5.1%	66.8% ± 1.9%	61.9% ± 2.0%	48.9% ± 0.2%
APPLE-FedAvgBackbone-TaskHead	72.4% ± 1.2%	75.0% ± 0.7%	85.2% ± 3.6%	87.6% ± 0.9%	71.0% ± 2.4%	71.2% ± 3.2%	59.2% ± 3.5%

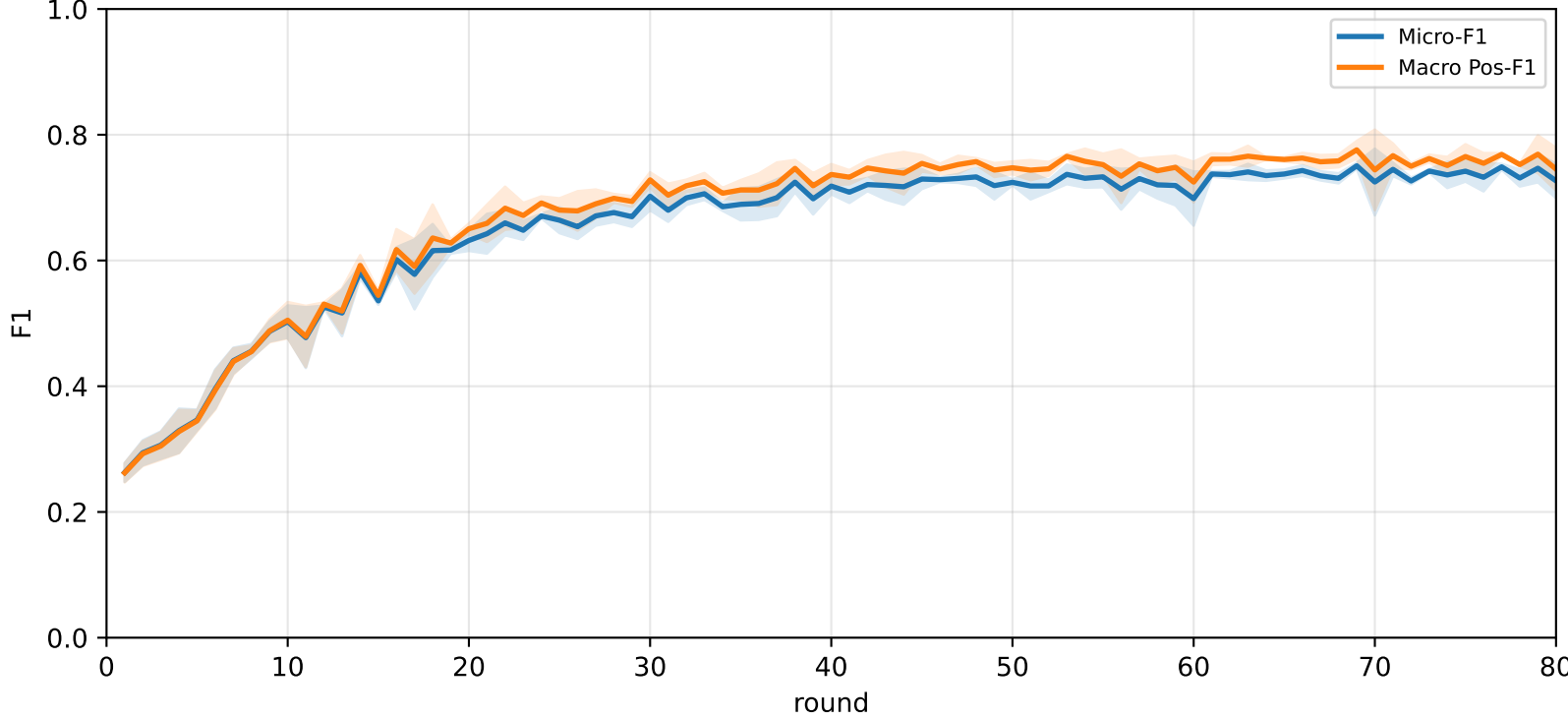
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



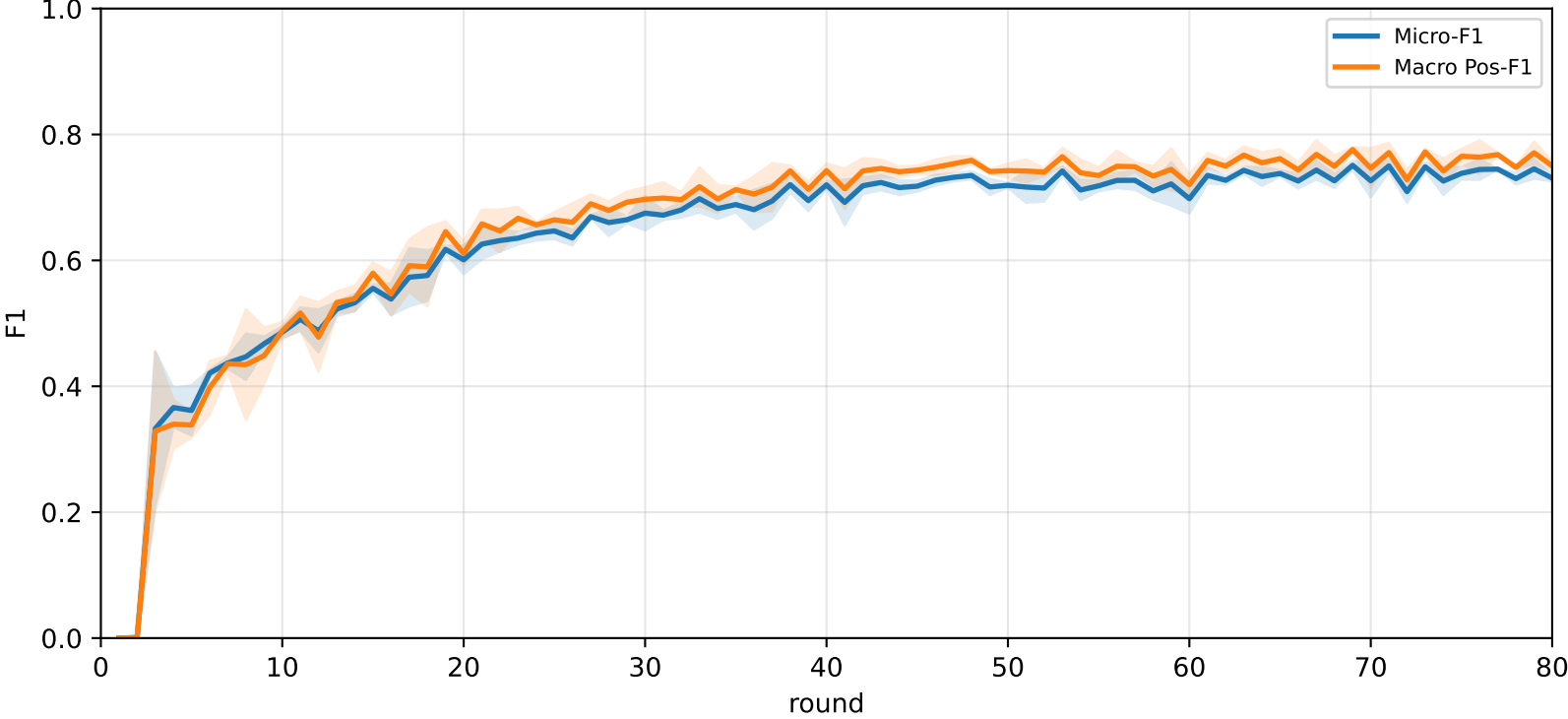
Pooled Local baseline micro / macro F1



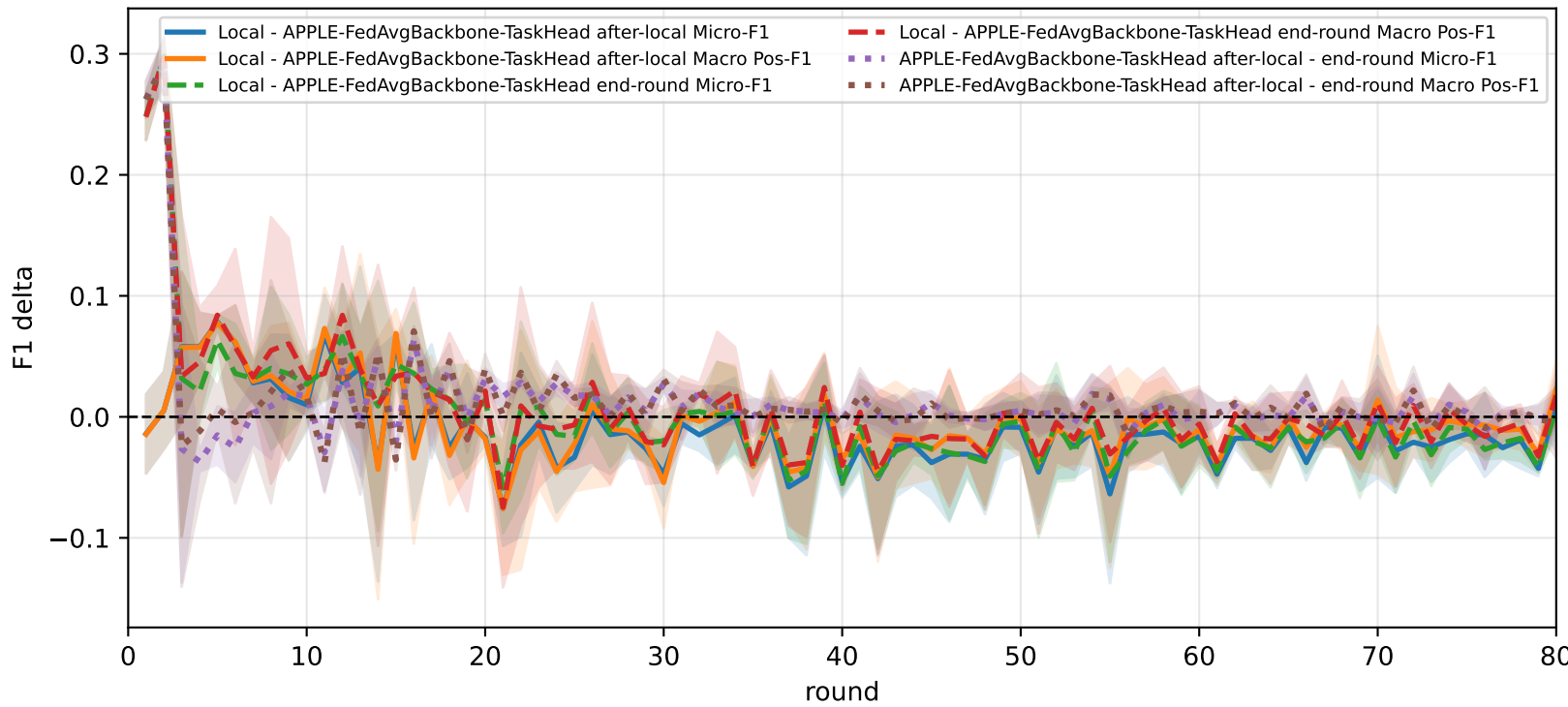
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



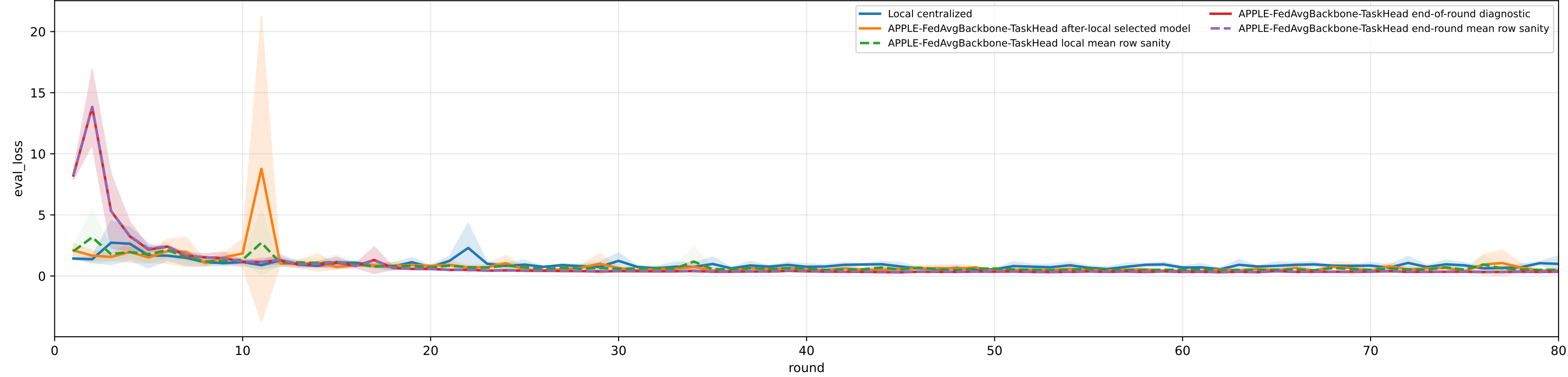
Controlled q-label heterogeneity: iid | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q task_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

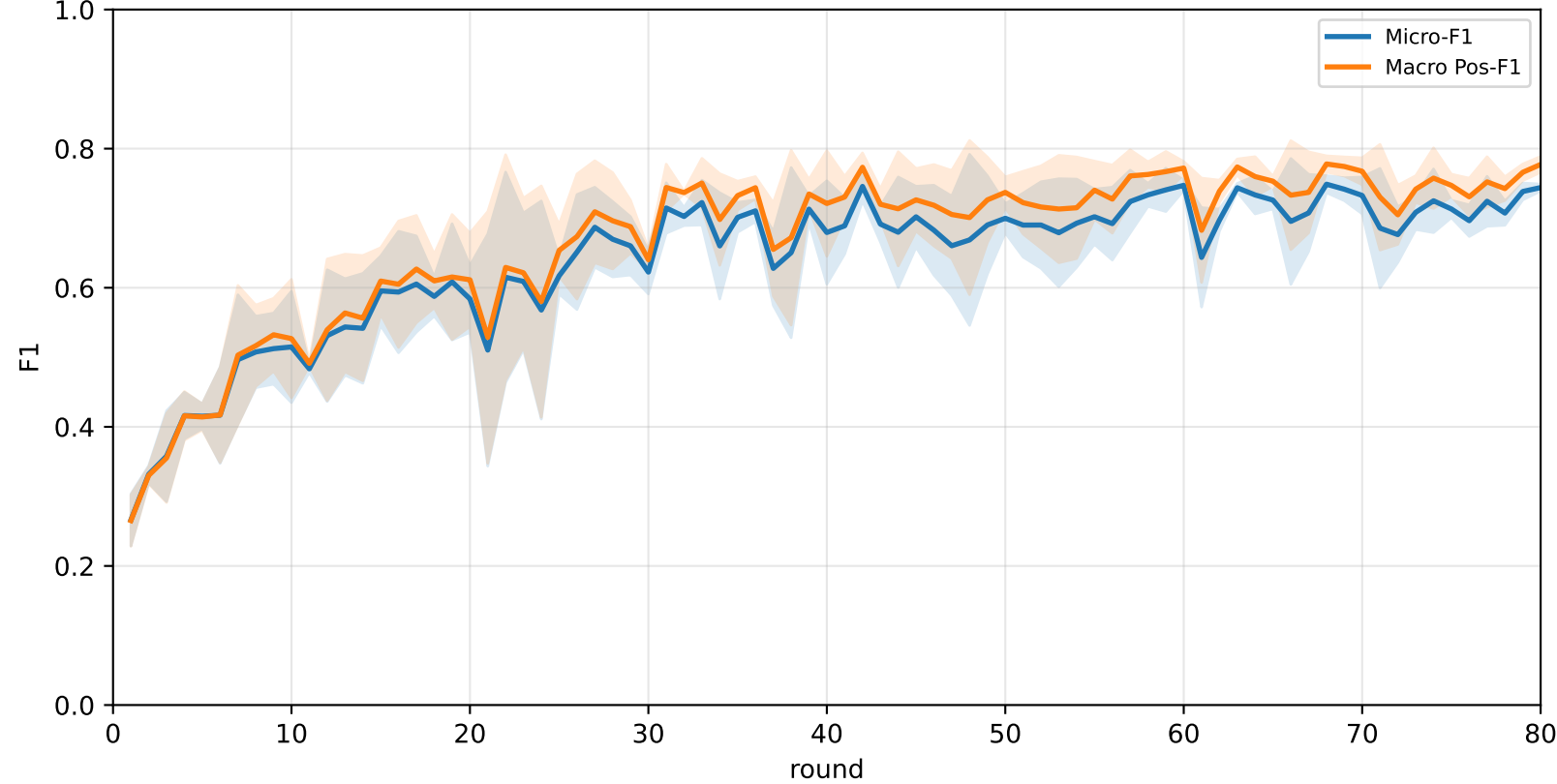
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	61.7% ± 2.9%	65.8% ± 2.5%	83.4% ± 5.2%	77.4% ± 2.3%	64.8% ± 2.9%	56.1% ± 5.8%	47.3% ± 2.8%
APPLE-FedAvgBackbone-TaskHead	73.9% ± 0.5%	76.5% ± 1.2%	85.6% ± 9.0%	89.5% ± 1.8%	74.3% ± 4.7%	71.7% ± 3.8%	61.4% ± 5.4%

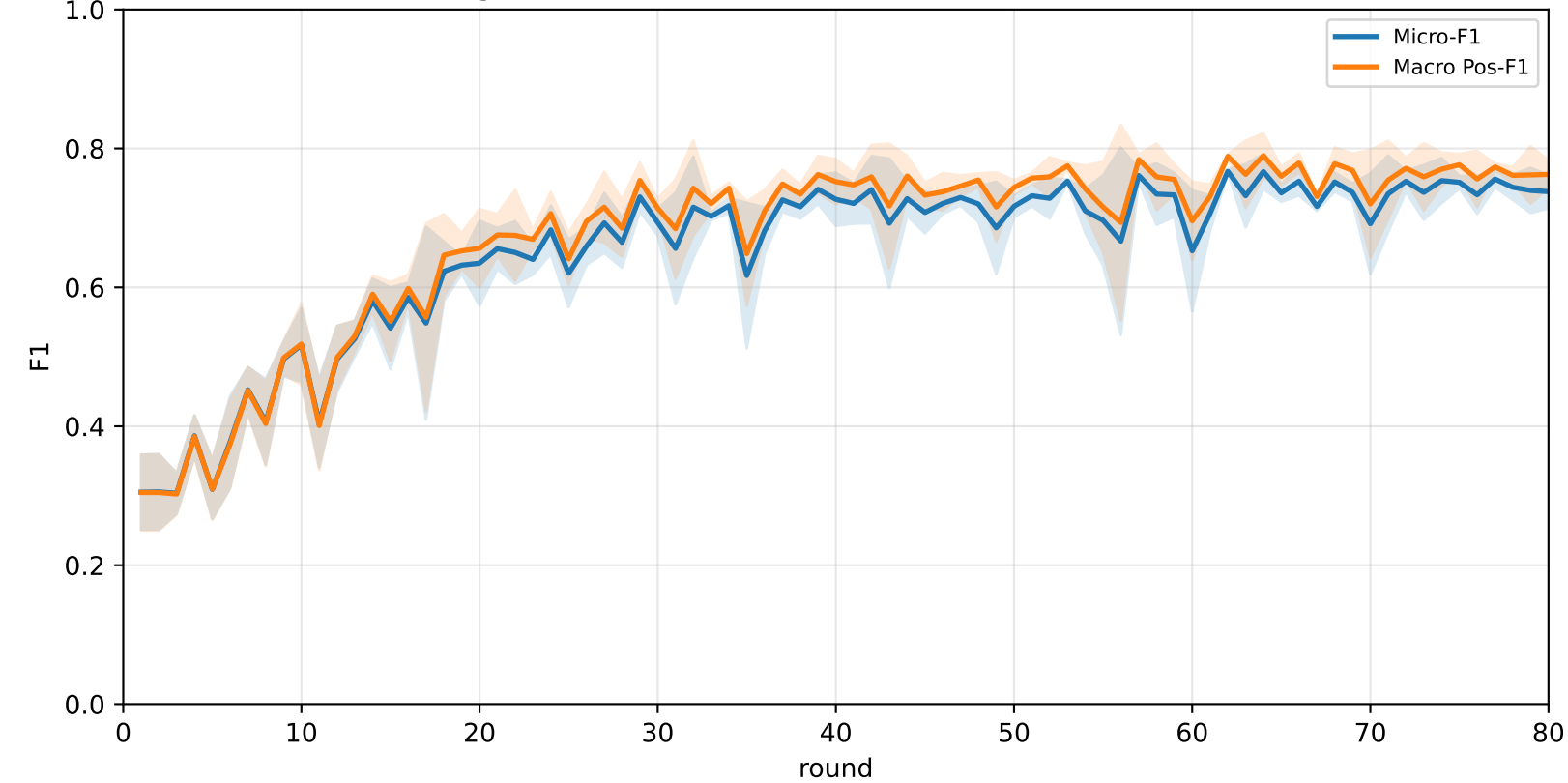
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



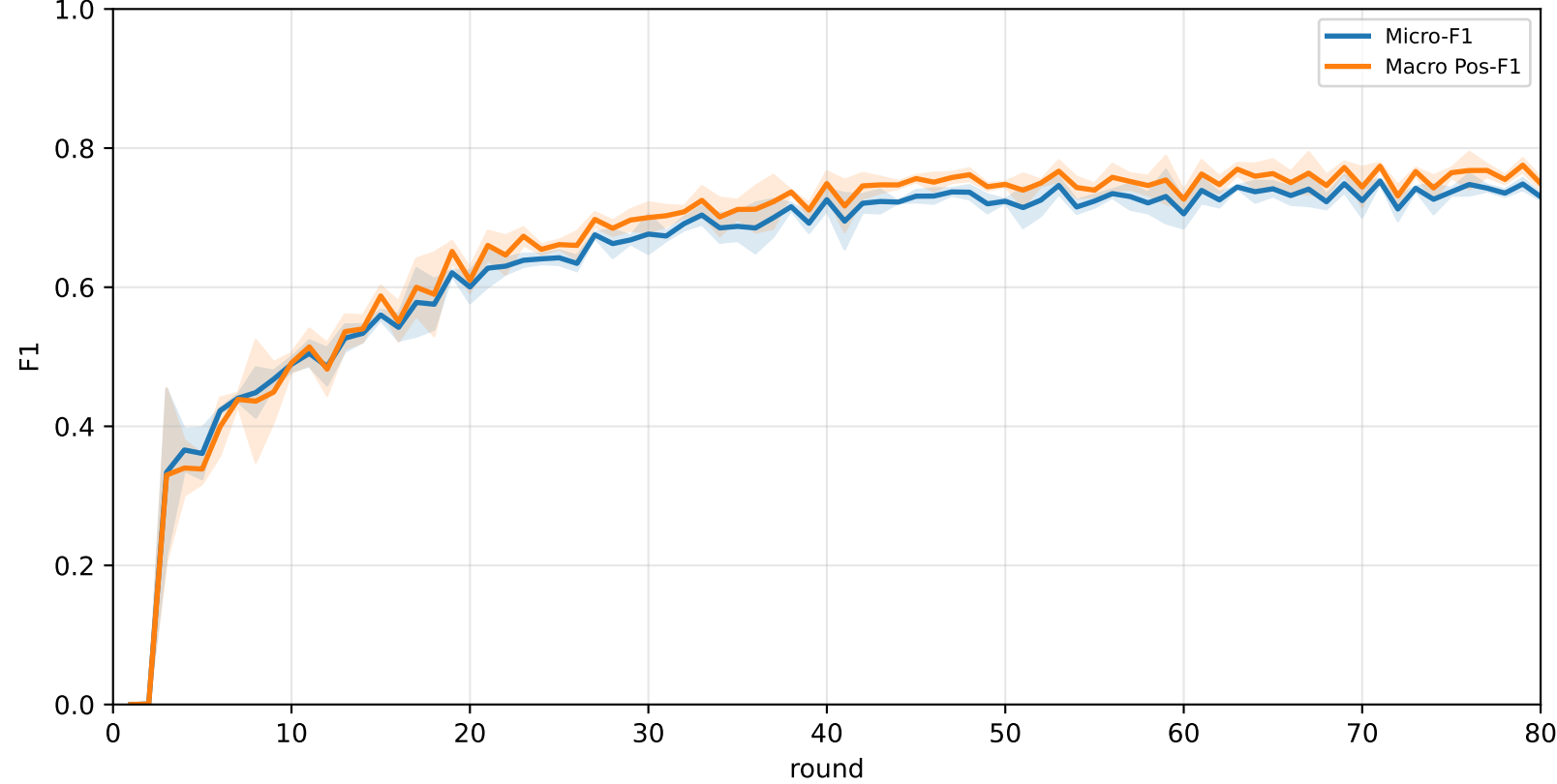
Pooled Local baseline micro / macro F1



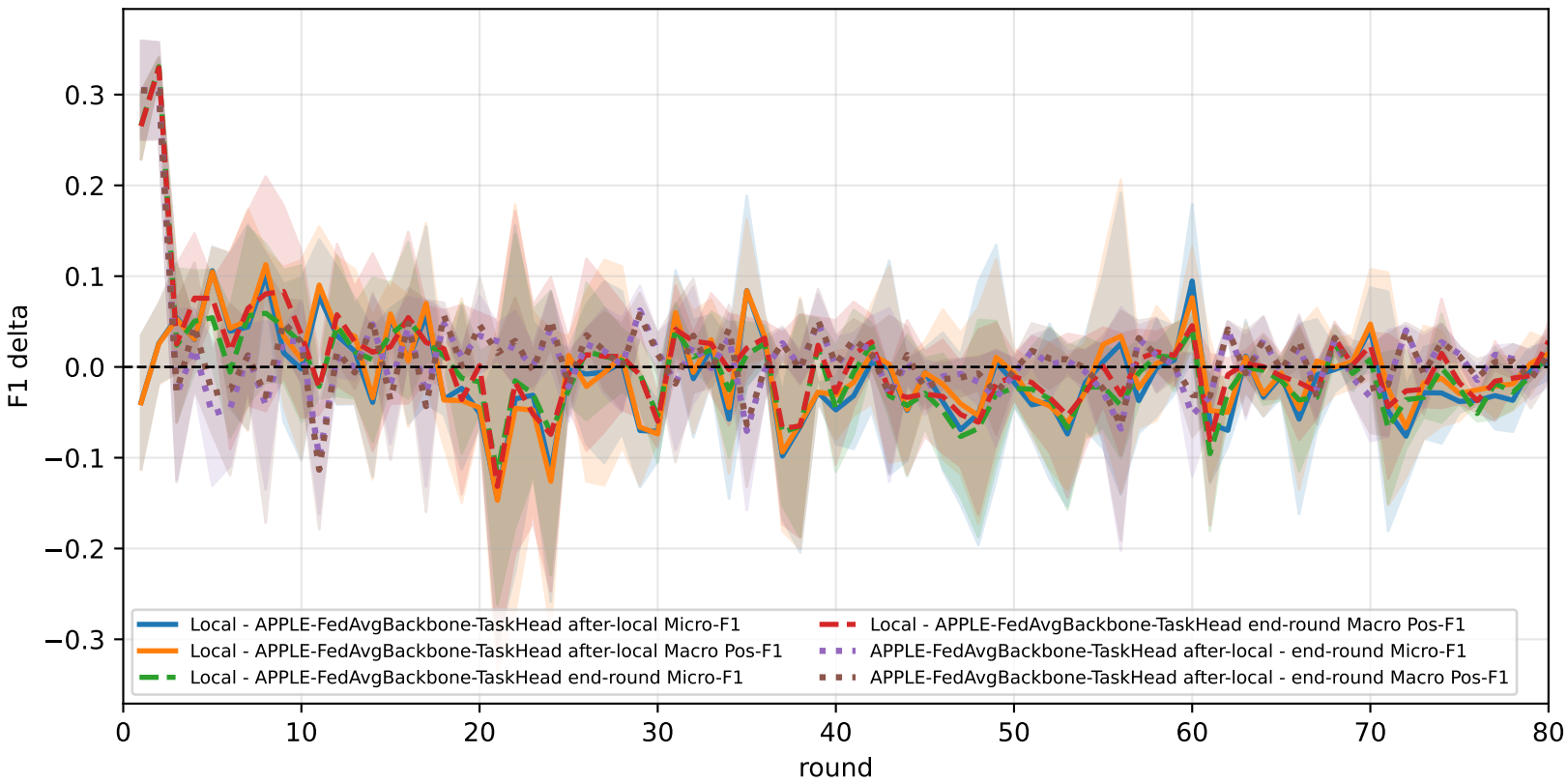
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



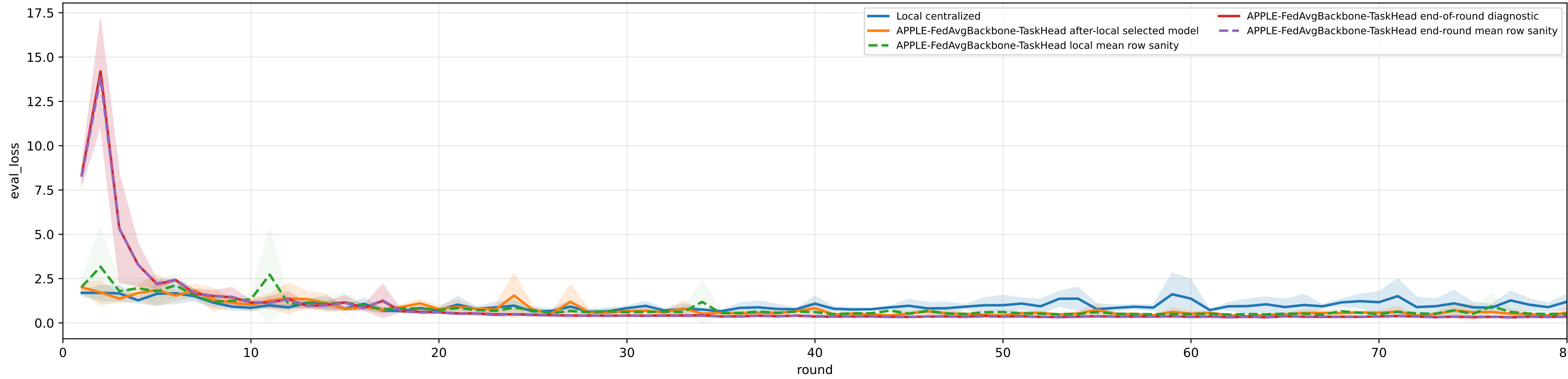
Controlled q-label heterogeneity: iid | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q_task_cycle3 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

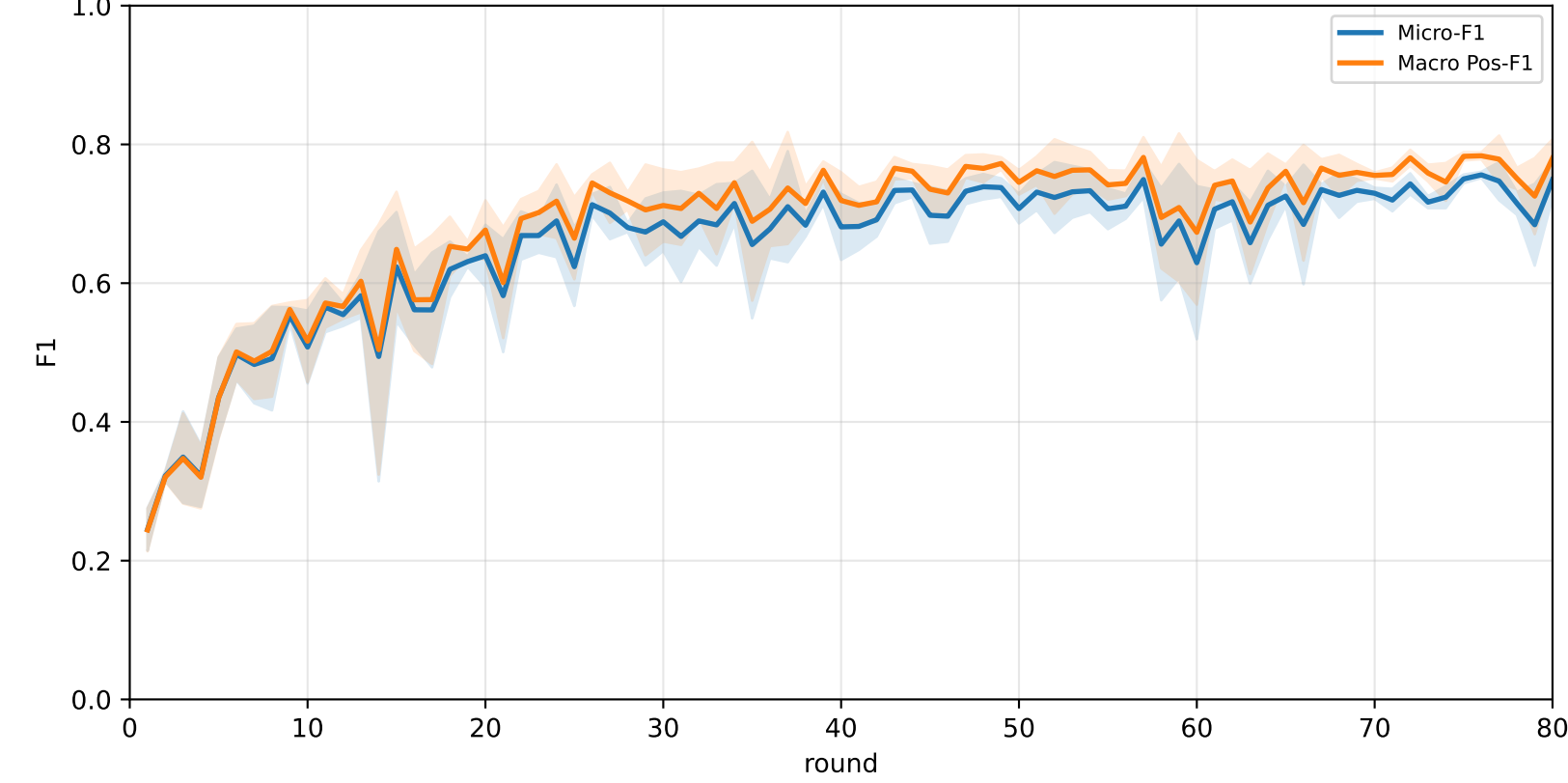
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	59.3% ± 1.8%	62.3% ± 1.2%	70.6% ± 12.2%	70.7% ± 7.8%	63.8% ± 2.1%	59.4% ± 6.7%	46.7% ± 4.1%
APPLE-FedAvgBackbone-TaskHead	73.0% ± 2.3%	76.0% ± 2.7%	88.2% ± 6.2%	89.9% ± 1.2%	68.4% ± 9.4%	73.2% ± 2.3%	60.4% ± 1.0%

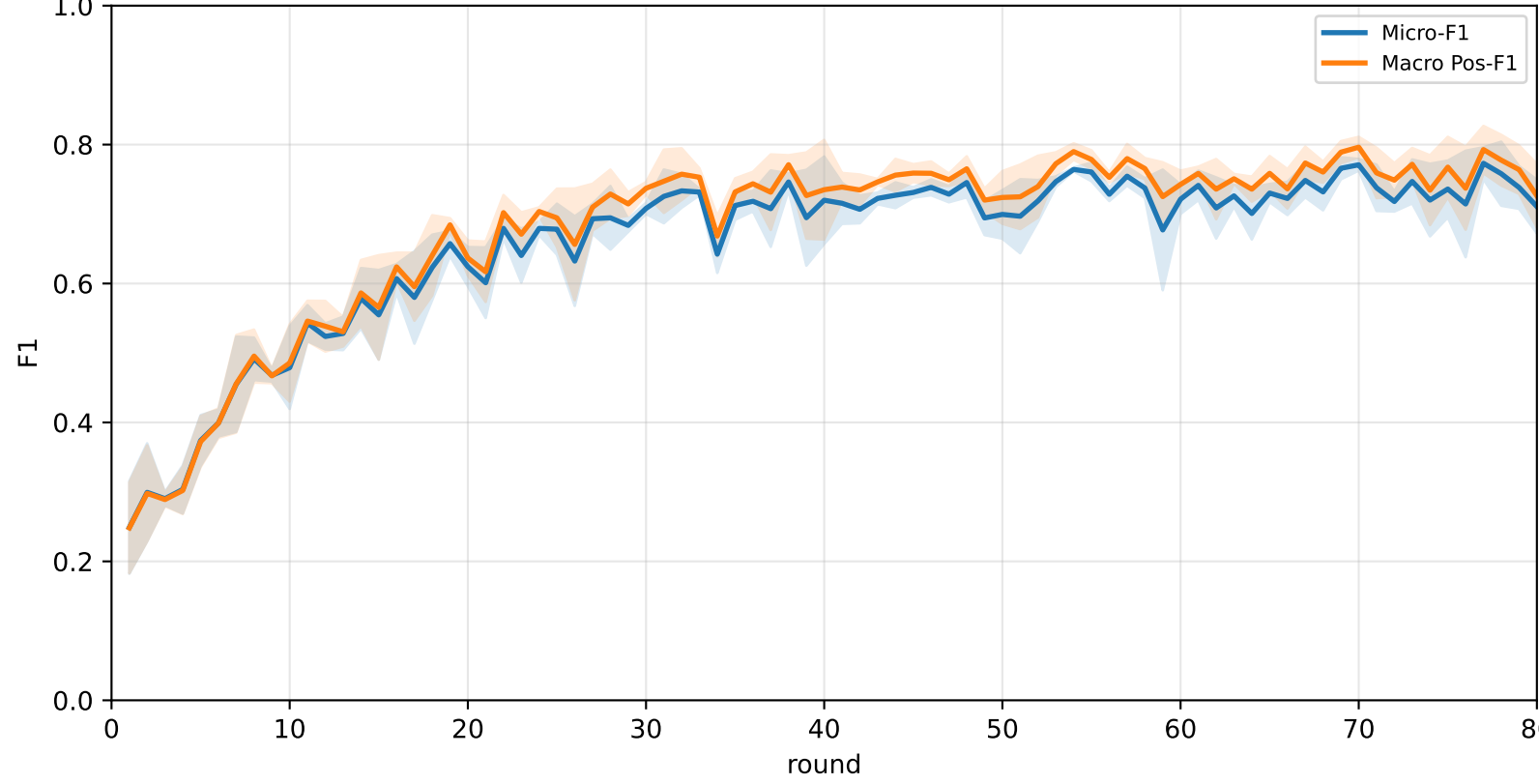
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



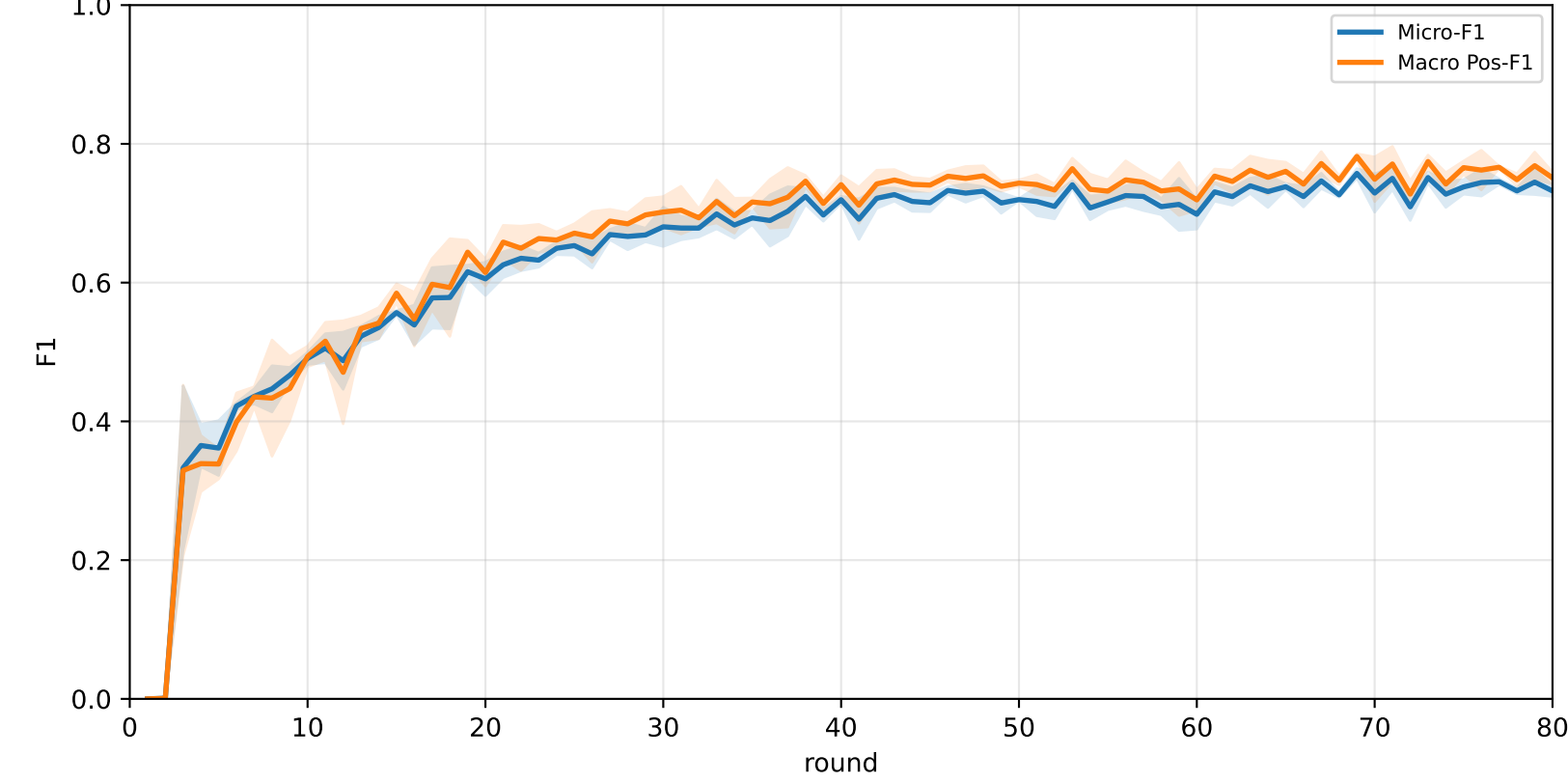
Pooled Local baseline micro / macro F1



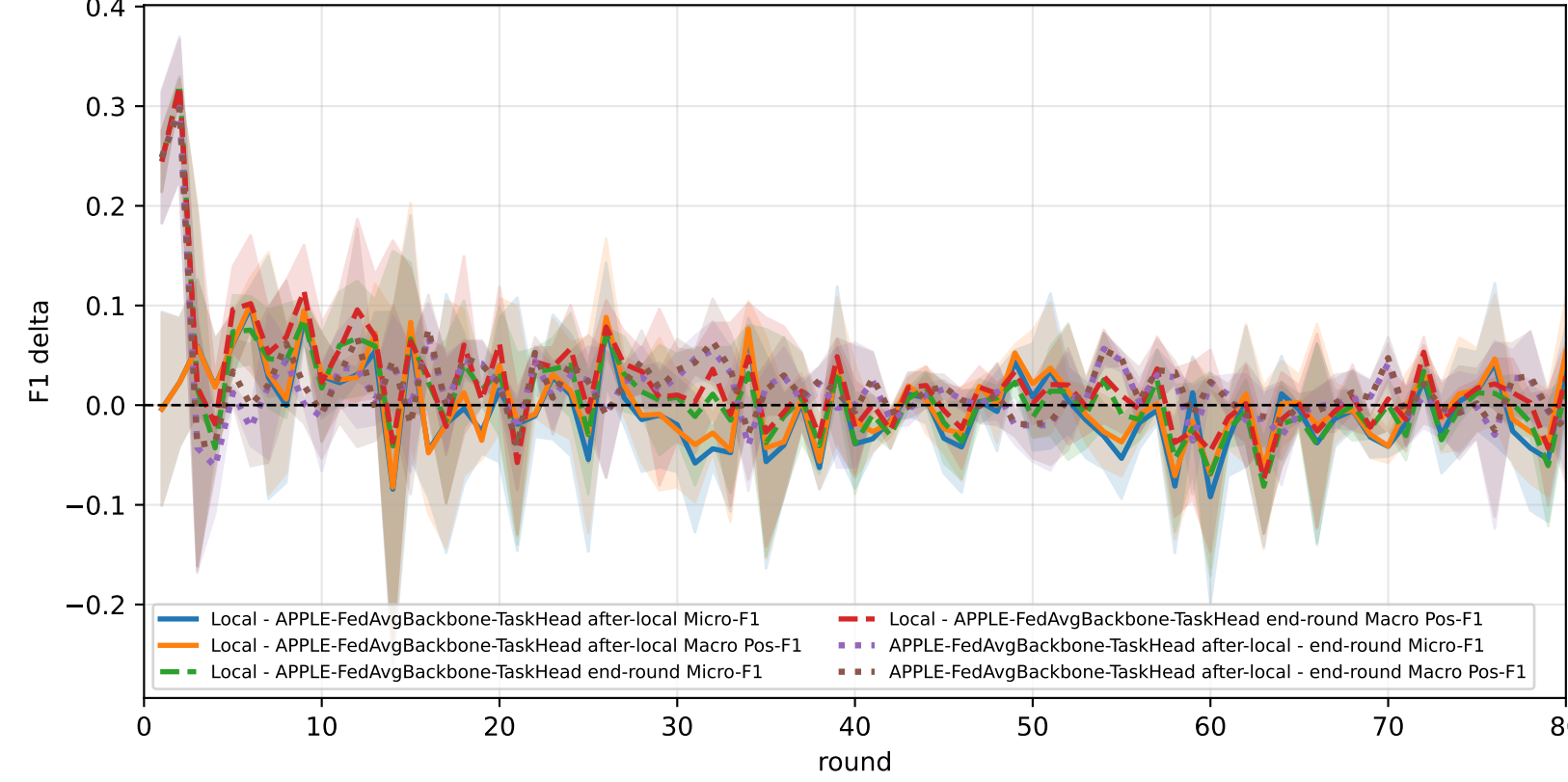
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



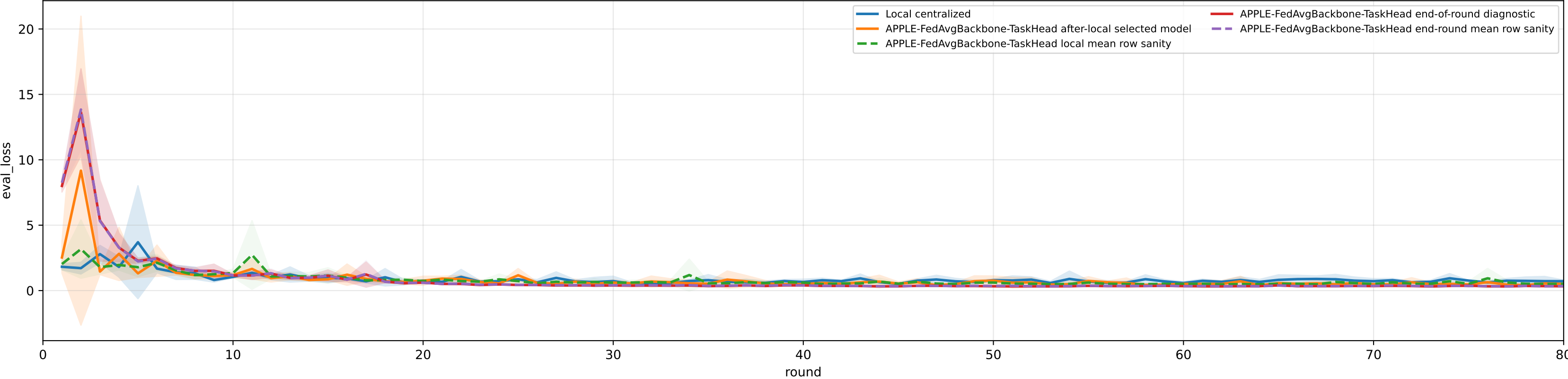
Controlled q-label heterogeneity: iid | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q task_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

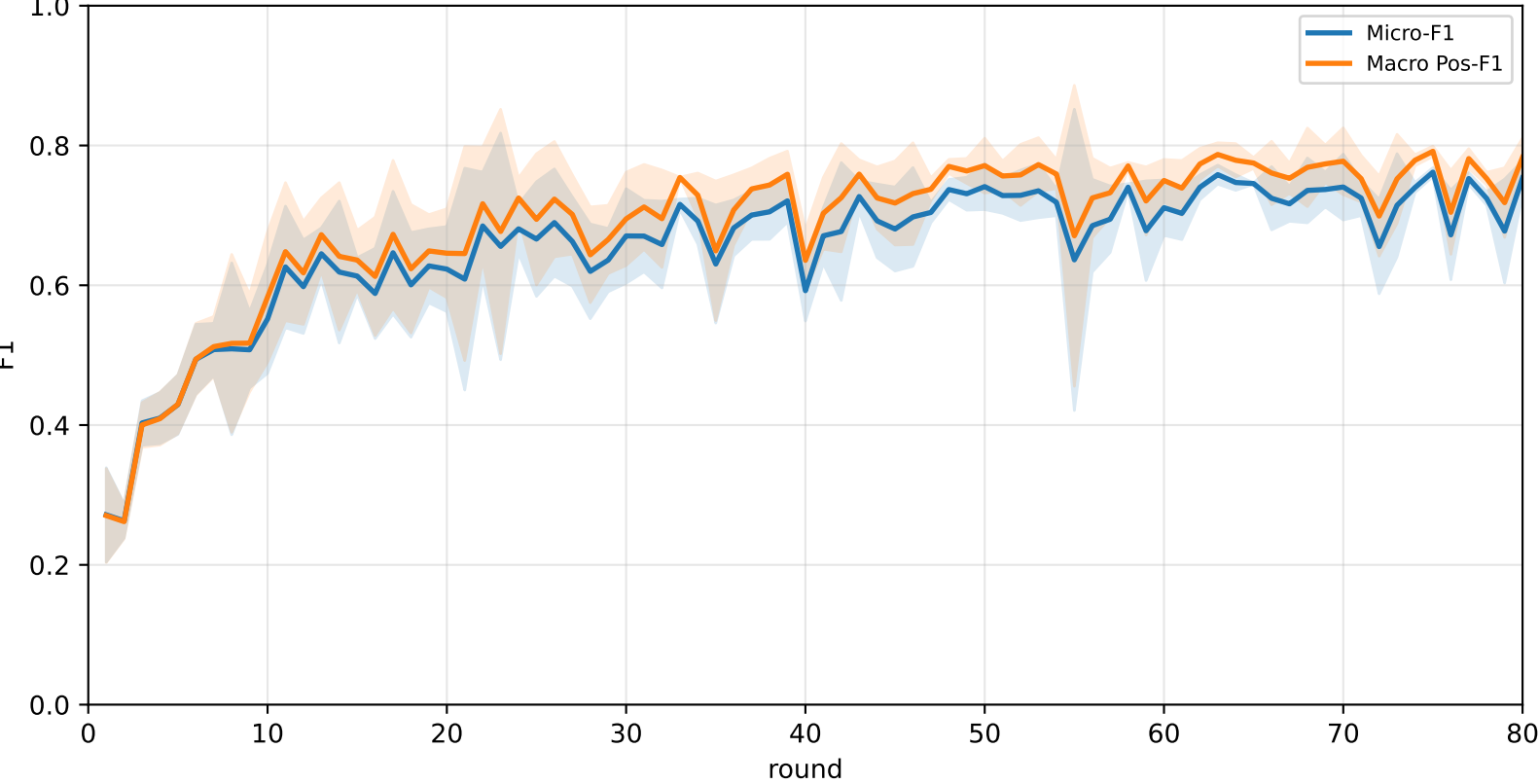
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	65.1% ± 6.6%	68.5% ± 6.7%	78.1% ± 11.0%	82.9% ± 5.3%	69.7% ± 5.5%	62.2% ± 8.5%	49.8% ± 5.3%
APPLE-FedAvgBackbone-TaskHead	71.5% ± 2.5%	74.1% ± 1.9%	81.7% ± 5.1%	87.7% ± 1.8%	74.1% ± 1.5%	69.4% ± 5.2%	57.6% ± 3.8%

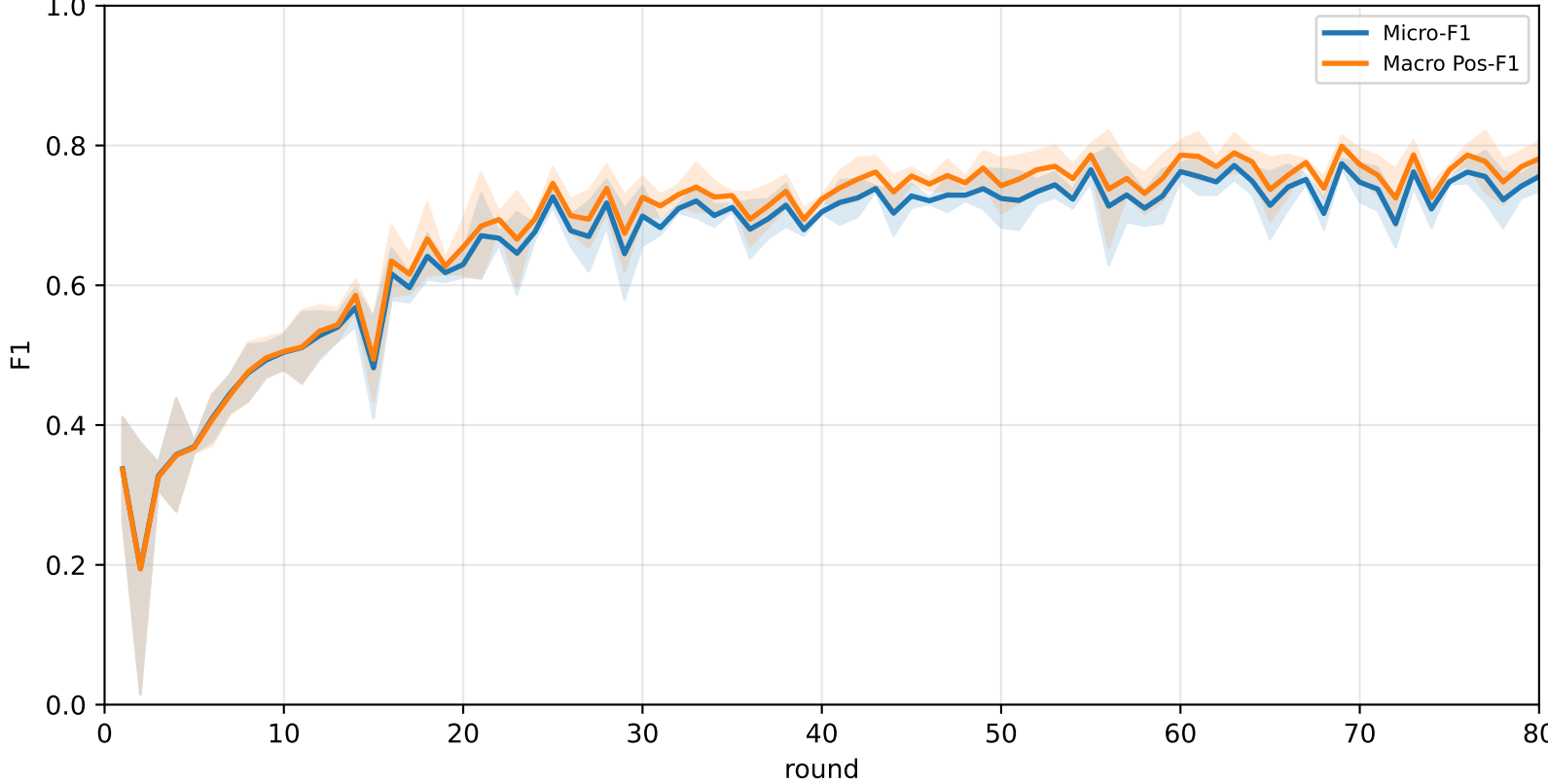
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



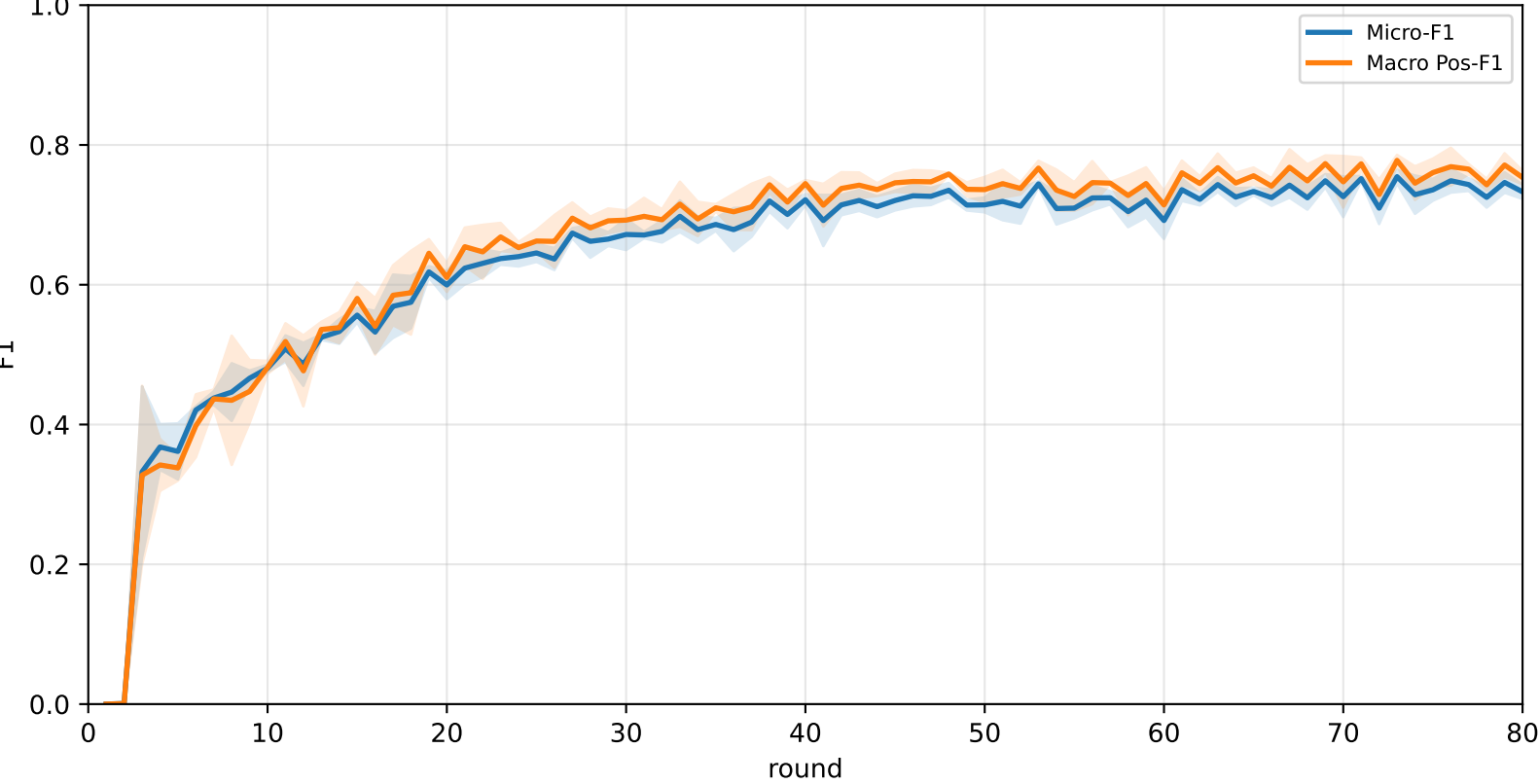
Pooled Local baseline micro / macro F1



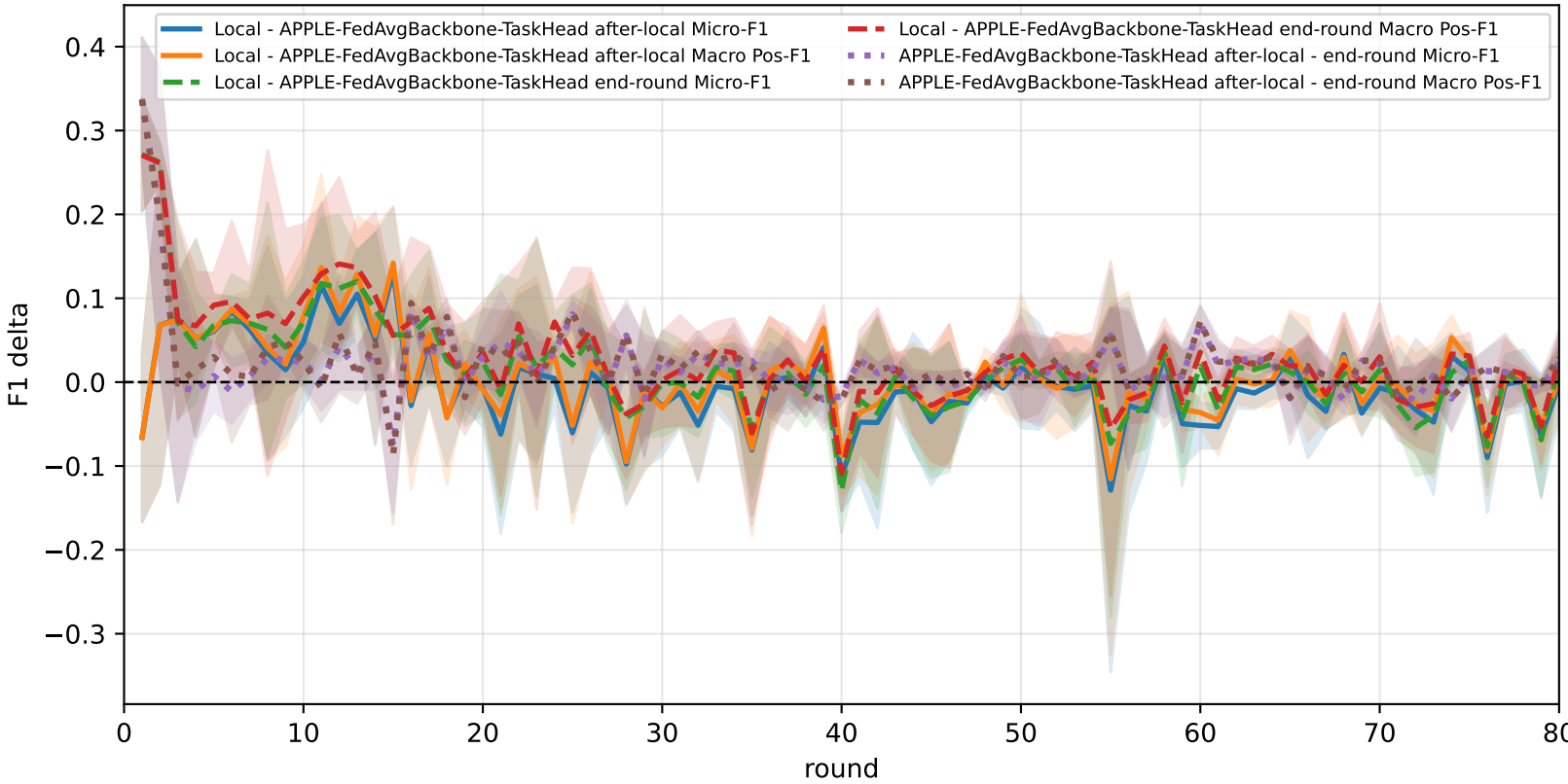
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



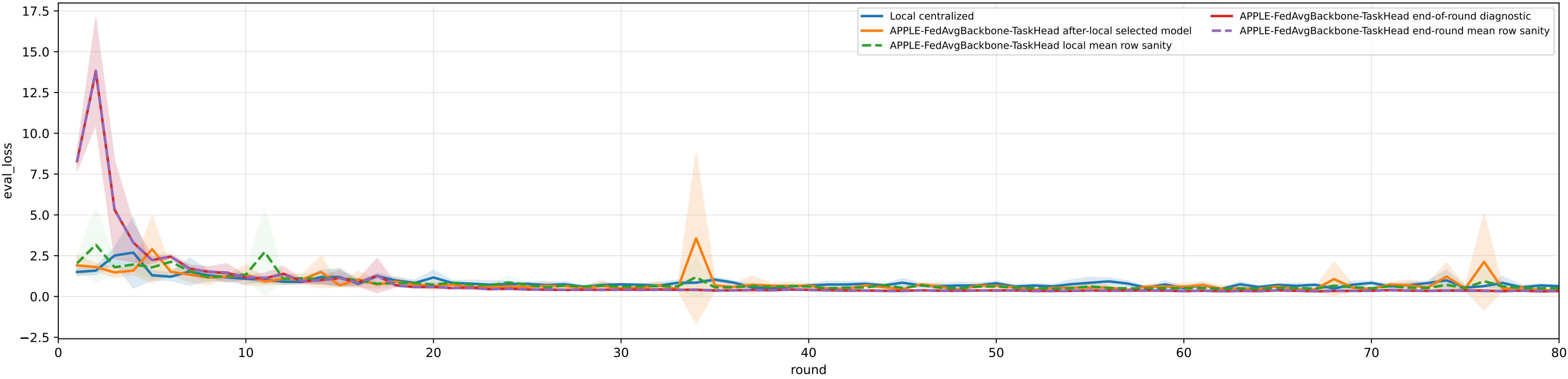
Controlled q-label heterogeneity: iid | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q_task_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

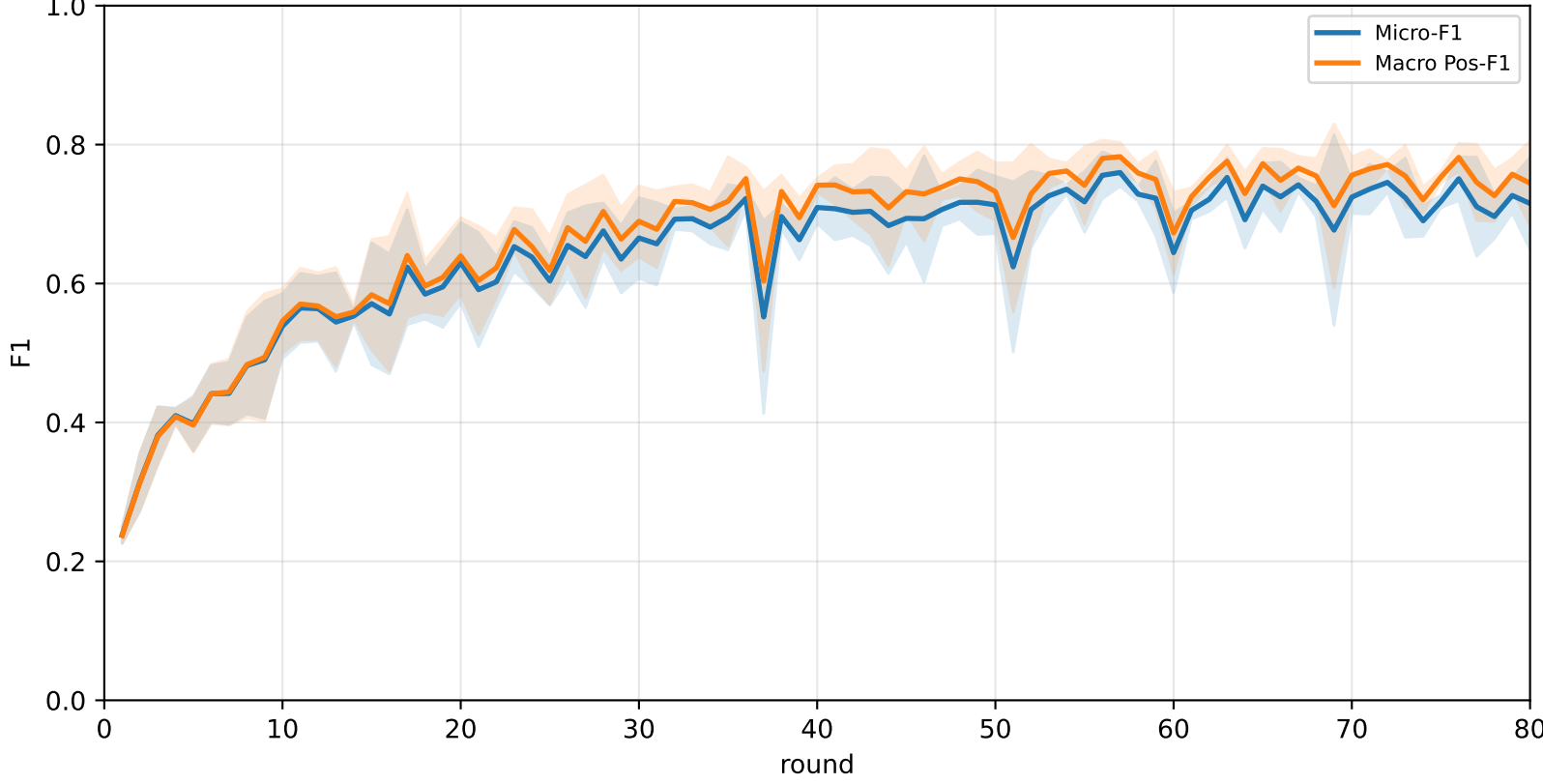
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	66.7% ± 3.2%	69.8% ± 3.5%	83.1% ± 3.5%	76.9% ± 8.5%	67.2% ± 1.6%	69.7% ± 4.1%	52.1% ± 4.2%
APPLE-FedAvgBackbone-TaskHead	72.1% ± 3.7%	74.4% ± 3.3%	85.4% ± 3.4%	85.5% ± 3.8%	70.9% ± 3.7%	70.5% ± 1.4%	59.7% ± 5.3%

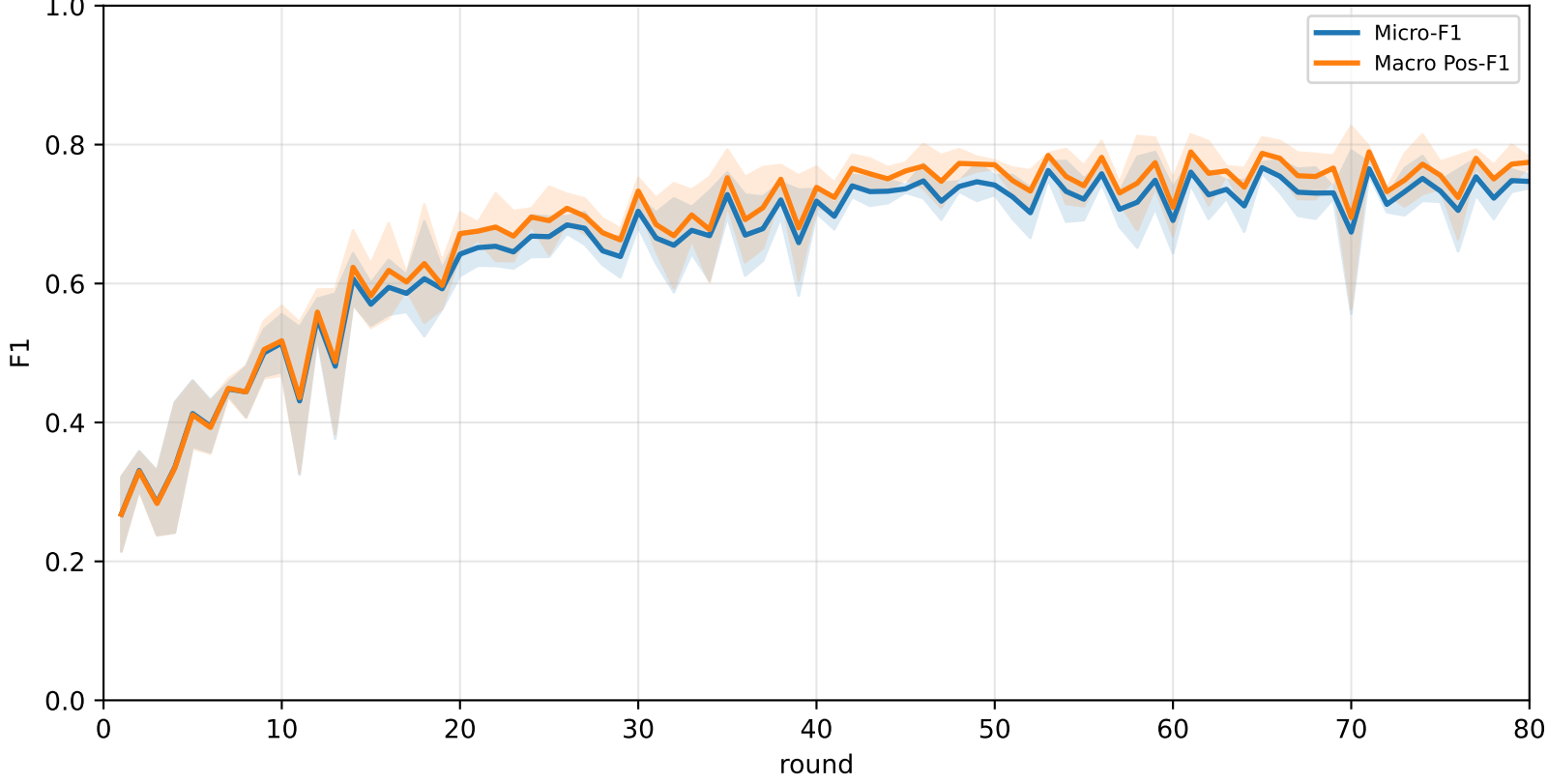
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



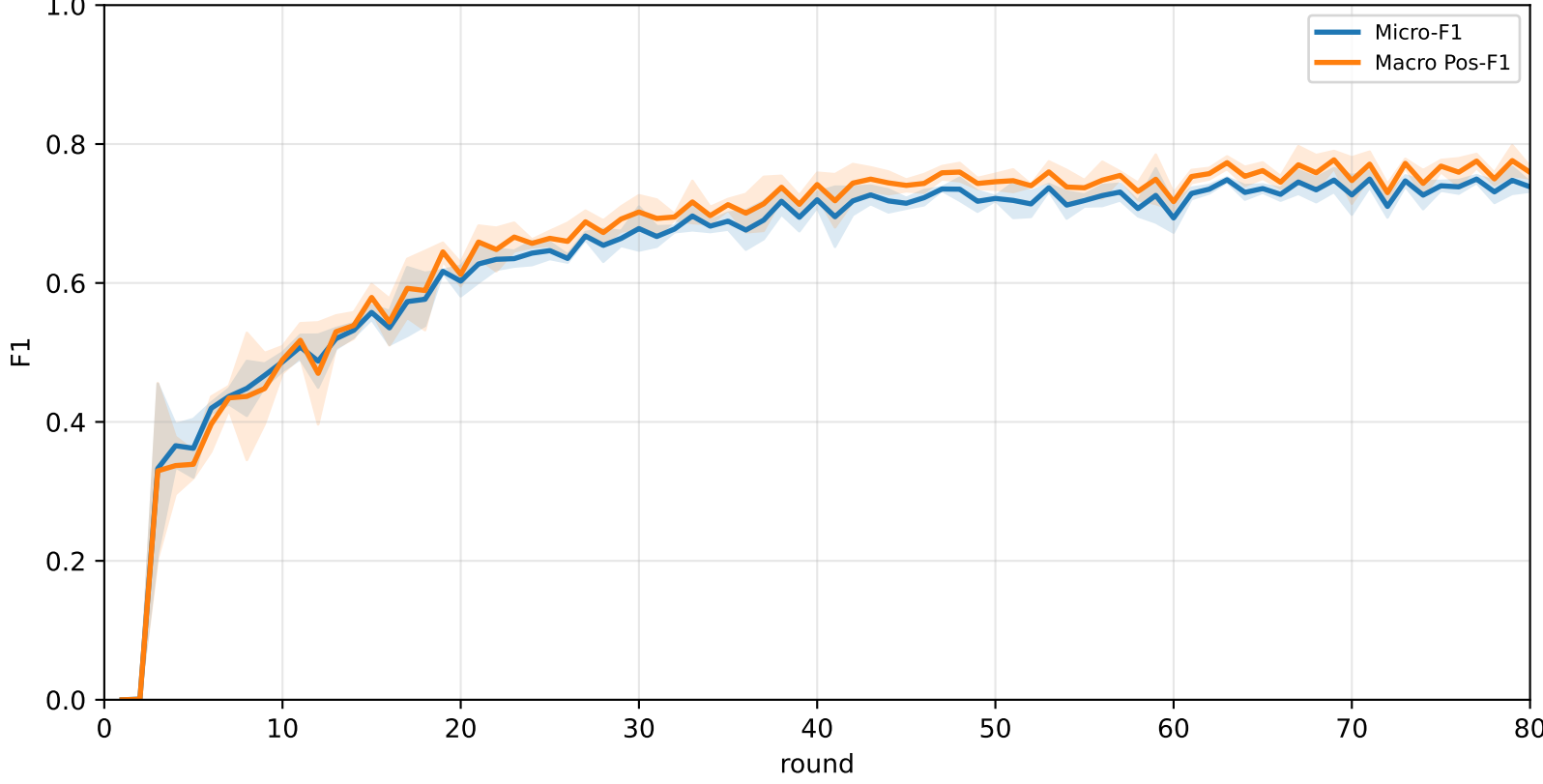
Pooled Local baseline micro / macro F1



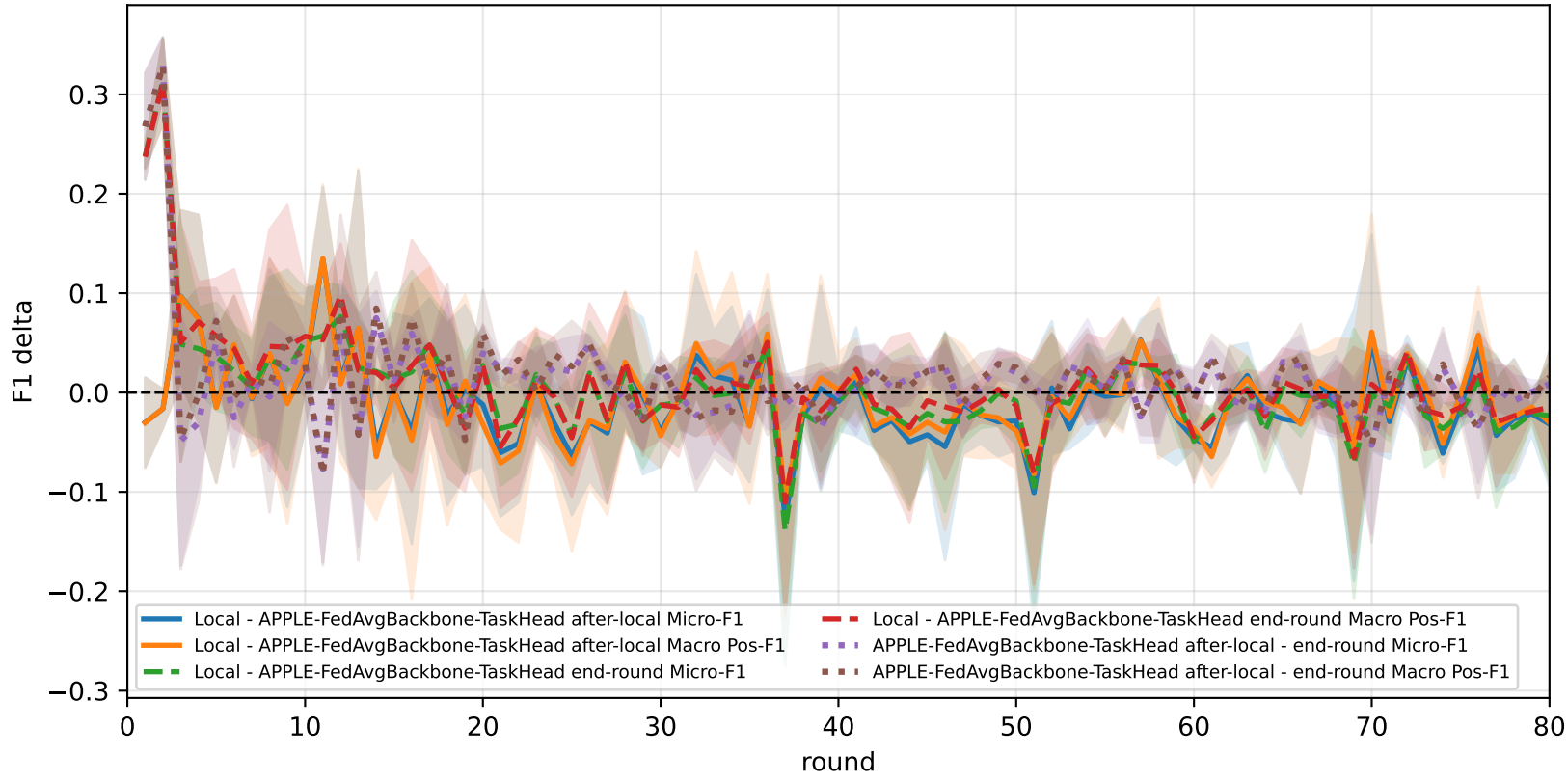
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



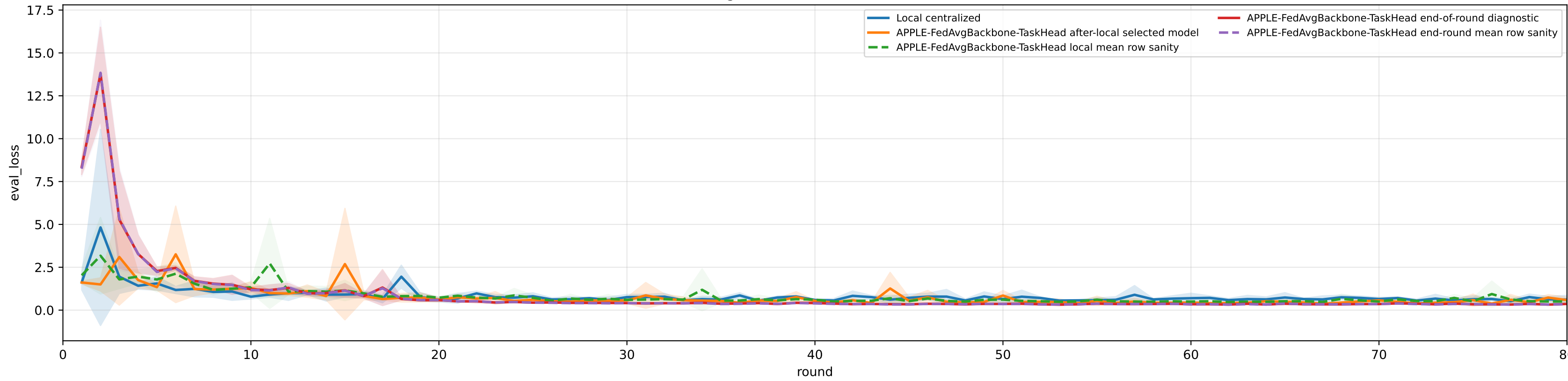
Controlled q-label heterogeneity: iid | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q_task_cycle6 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

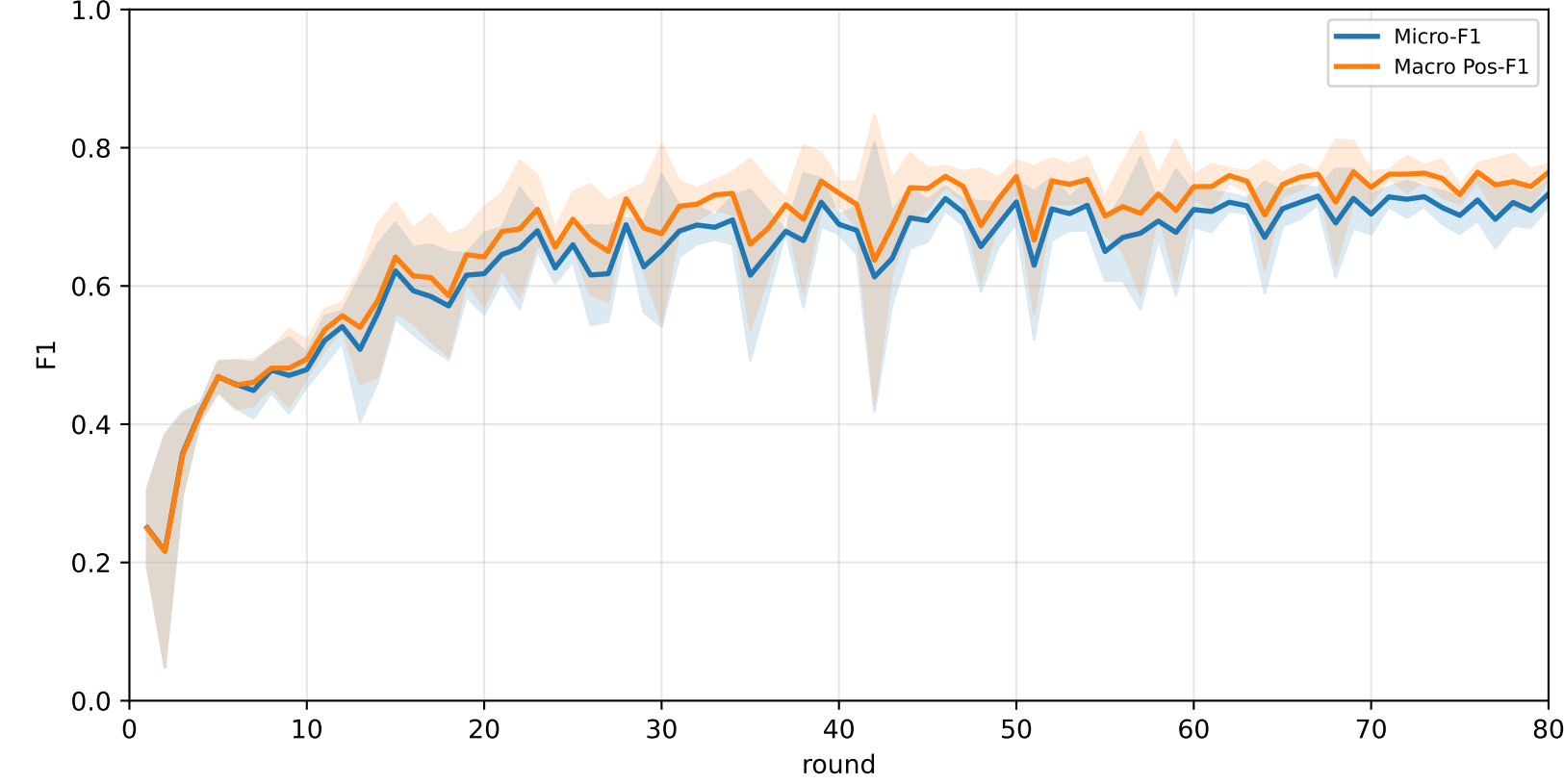
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	68.1% ± 1.5%	72.6% ± 1.2%	89.4% ± 1.7%	86.4% ± 0.7%	69.7% ± 3.6%	67.1% ± 2.2%	50.4% ± 1.9%
APPLE-FedAvgBackbone-TaskHead	71.4% ± 2.5%	74.2% ± 1.9%	86.4% ± 1.9%	85.8% ± 2.3%	69.3% ± 1.5%	71.8% ± 3.9%	57.7% ± 4.5%

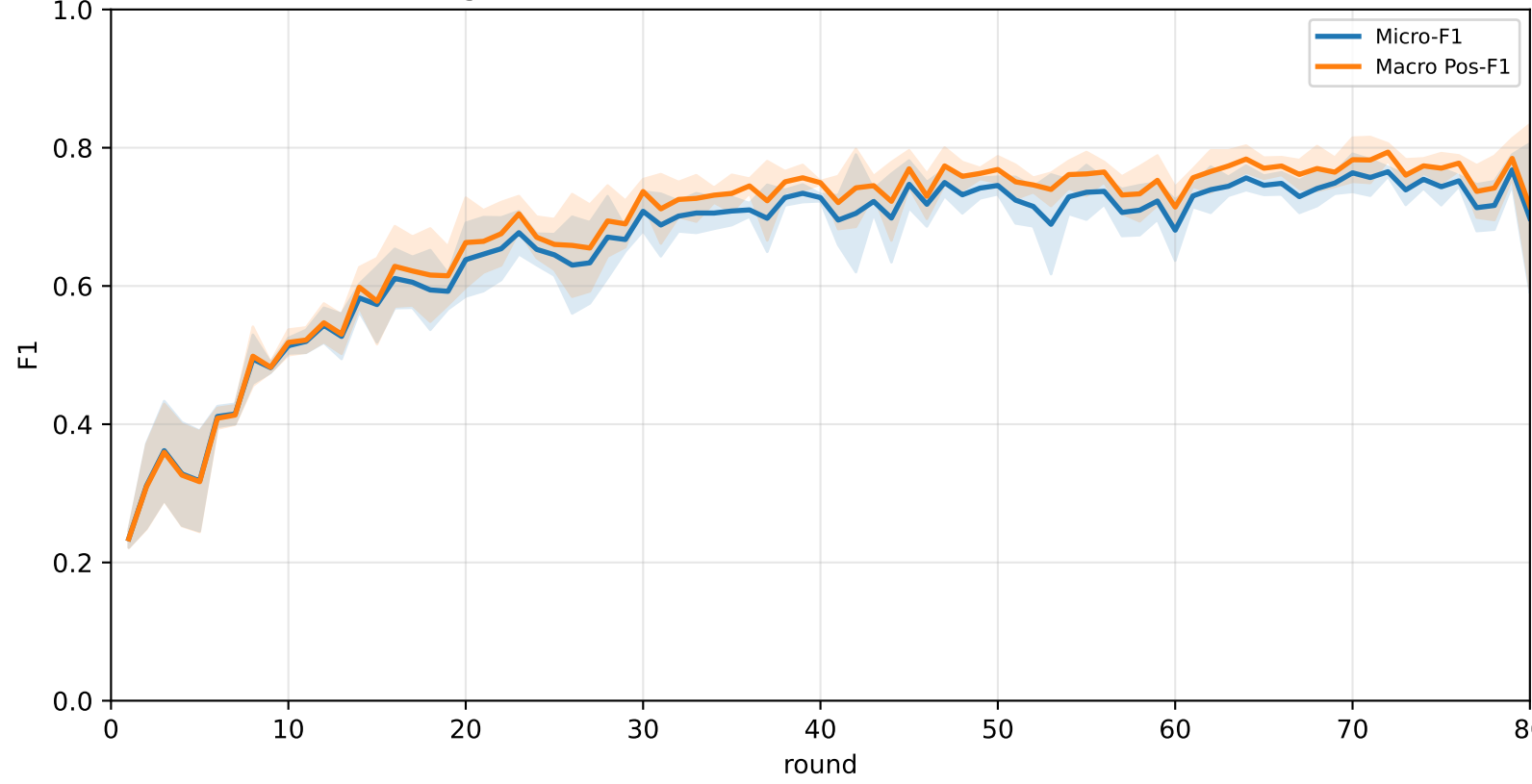
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



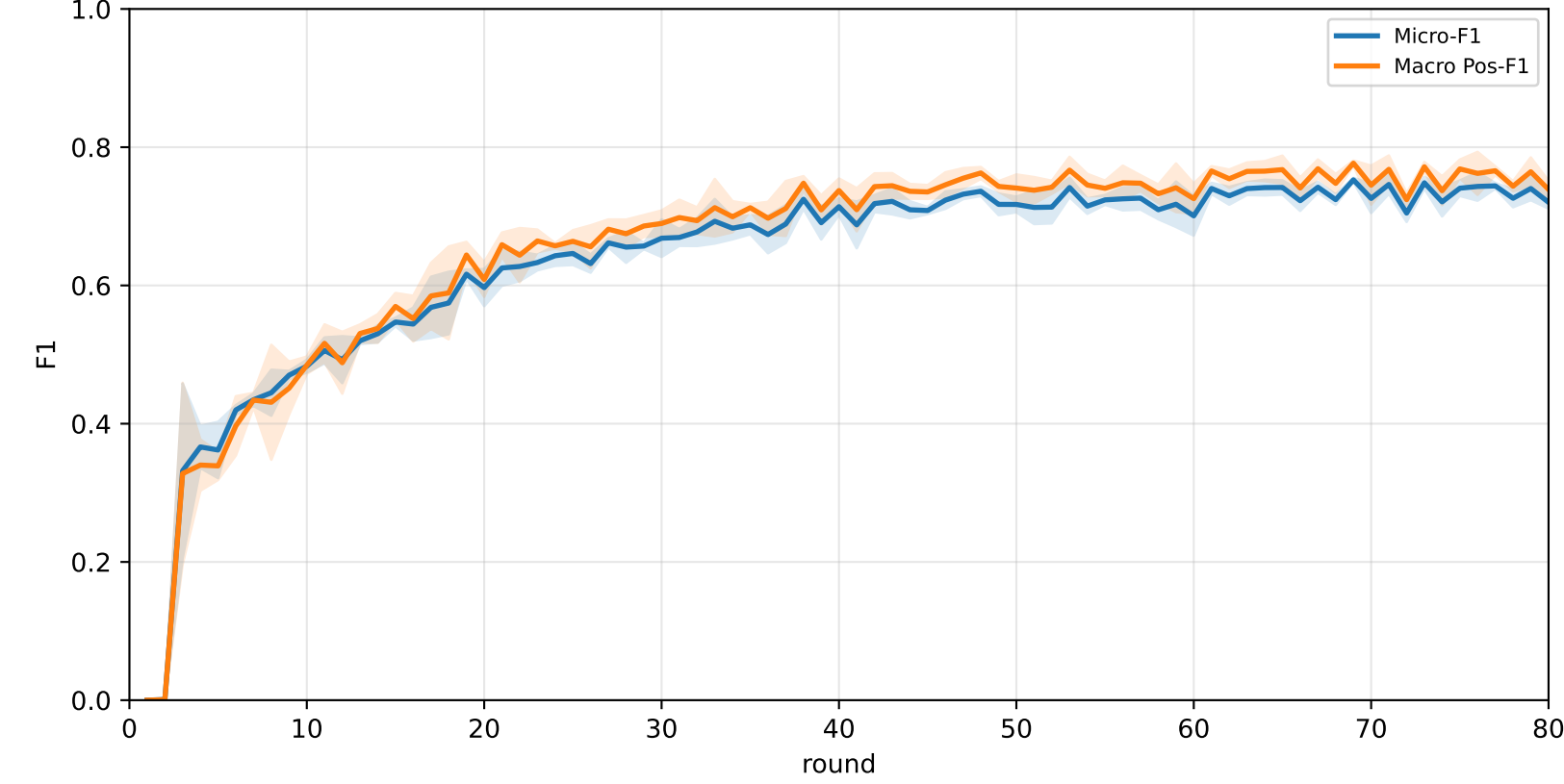
Pooled Local baseline micro / macro F1



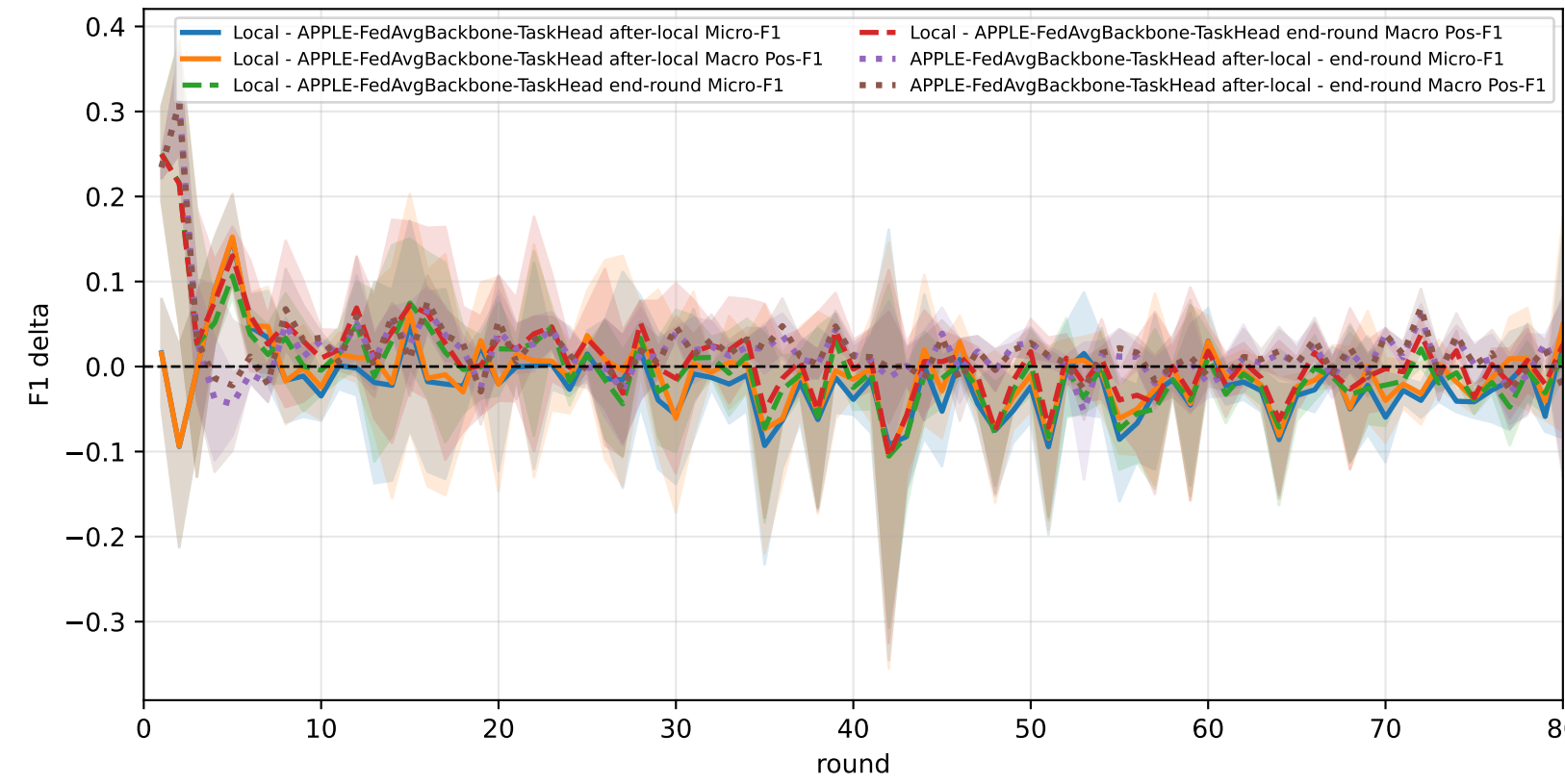
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



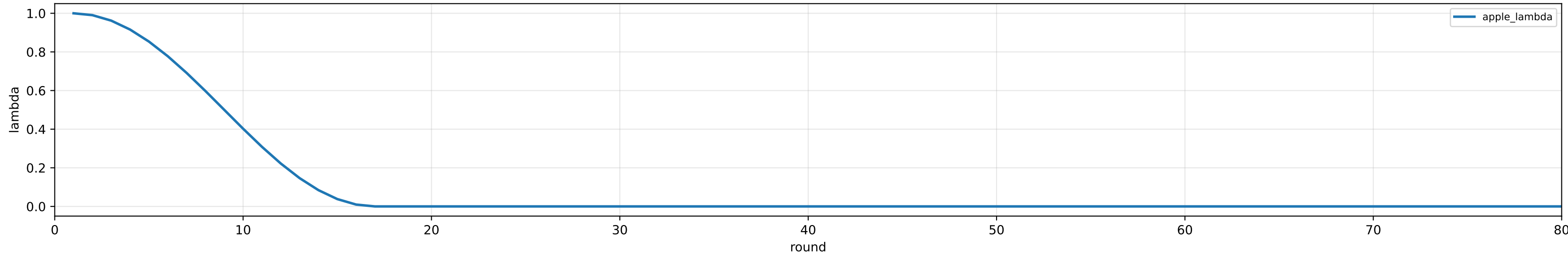
Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



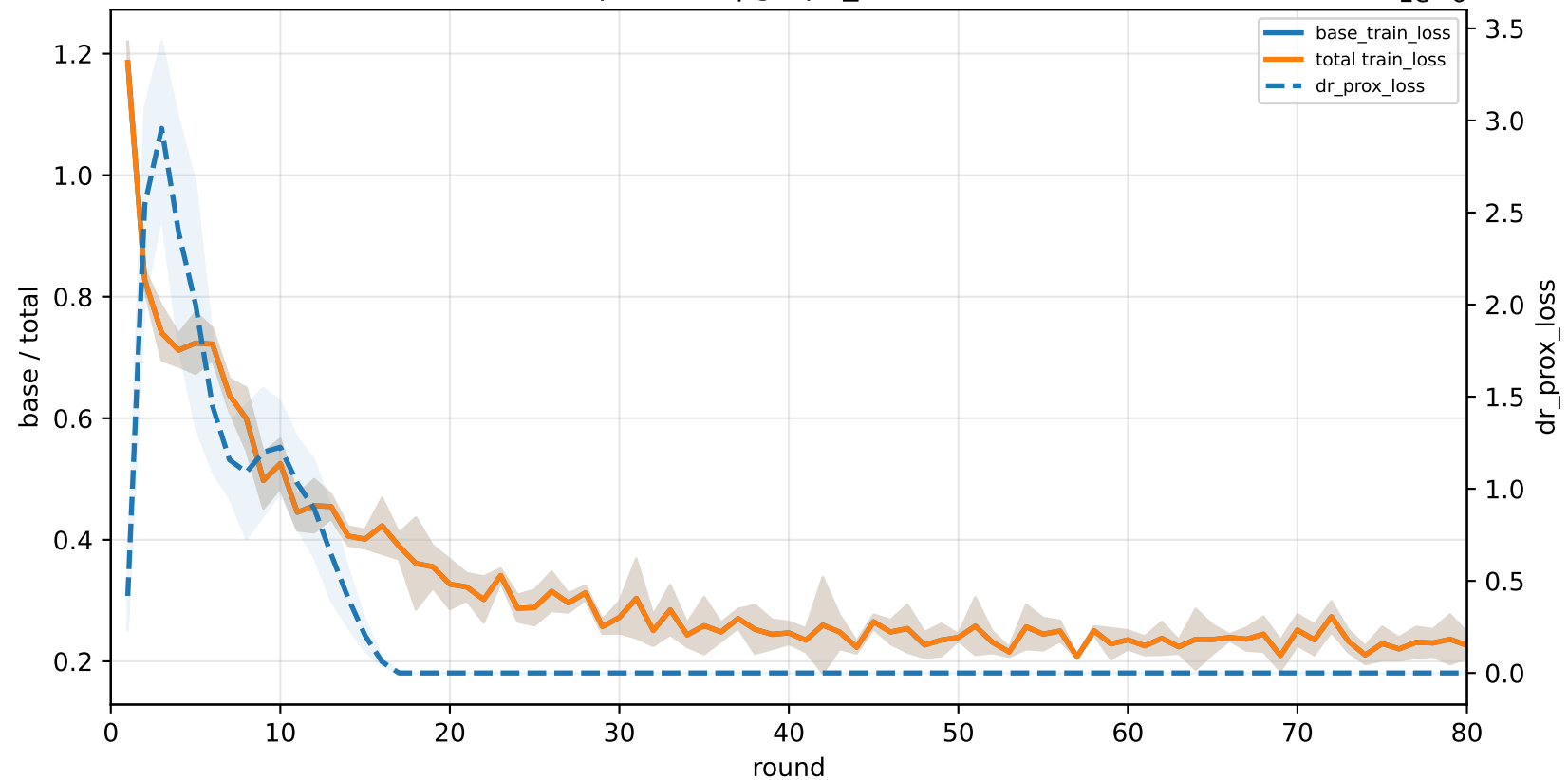
APPLE-FedAvgBackbone-TaskHead training-loss decomposition | iid heterogeneity

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

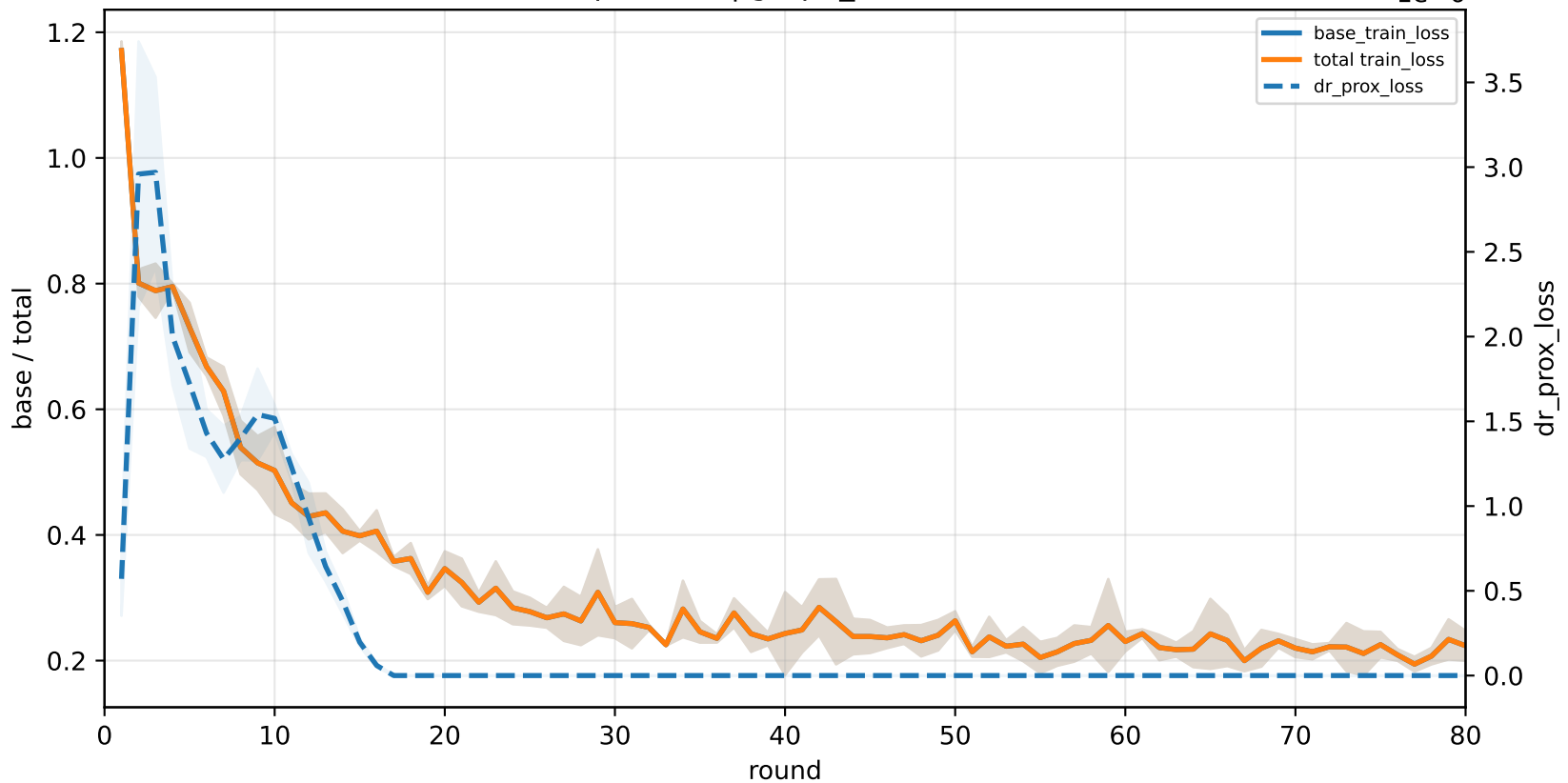
APPLE lambda scheduler | apple_mu=0.001



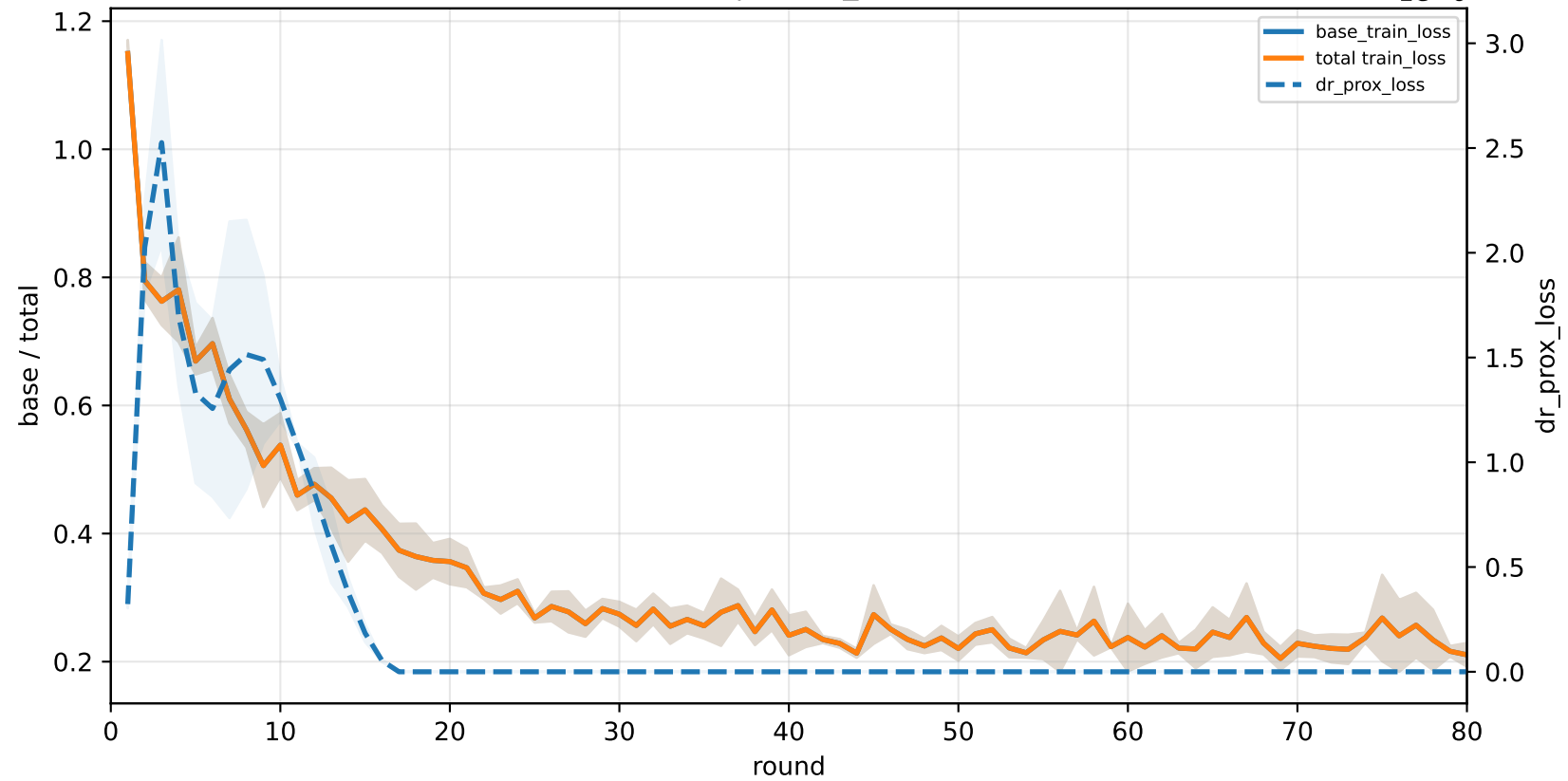
C2-specialist | graph_id=3000000



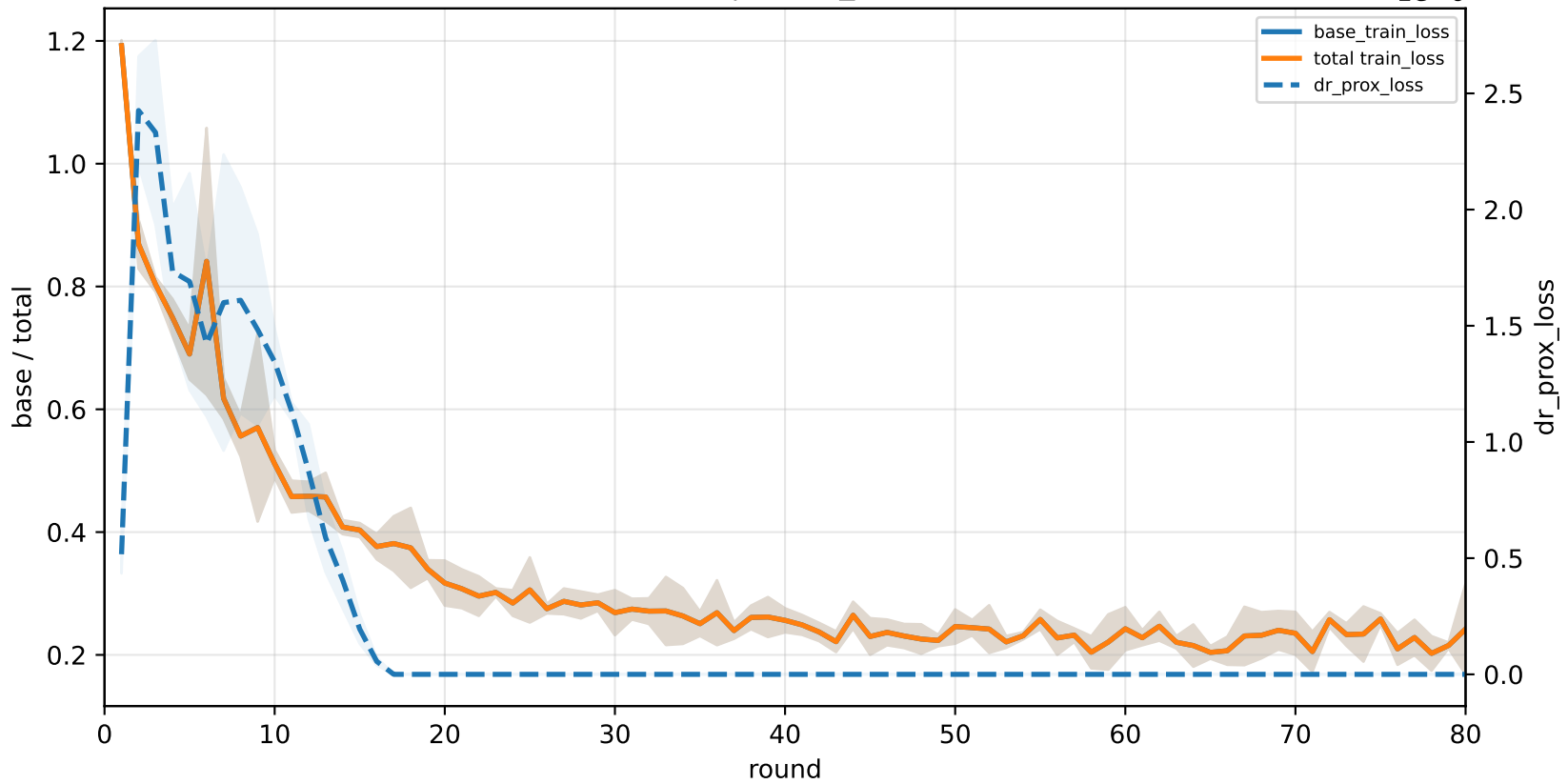
C3-specialist | graph_id=3000001



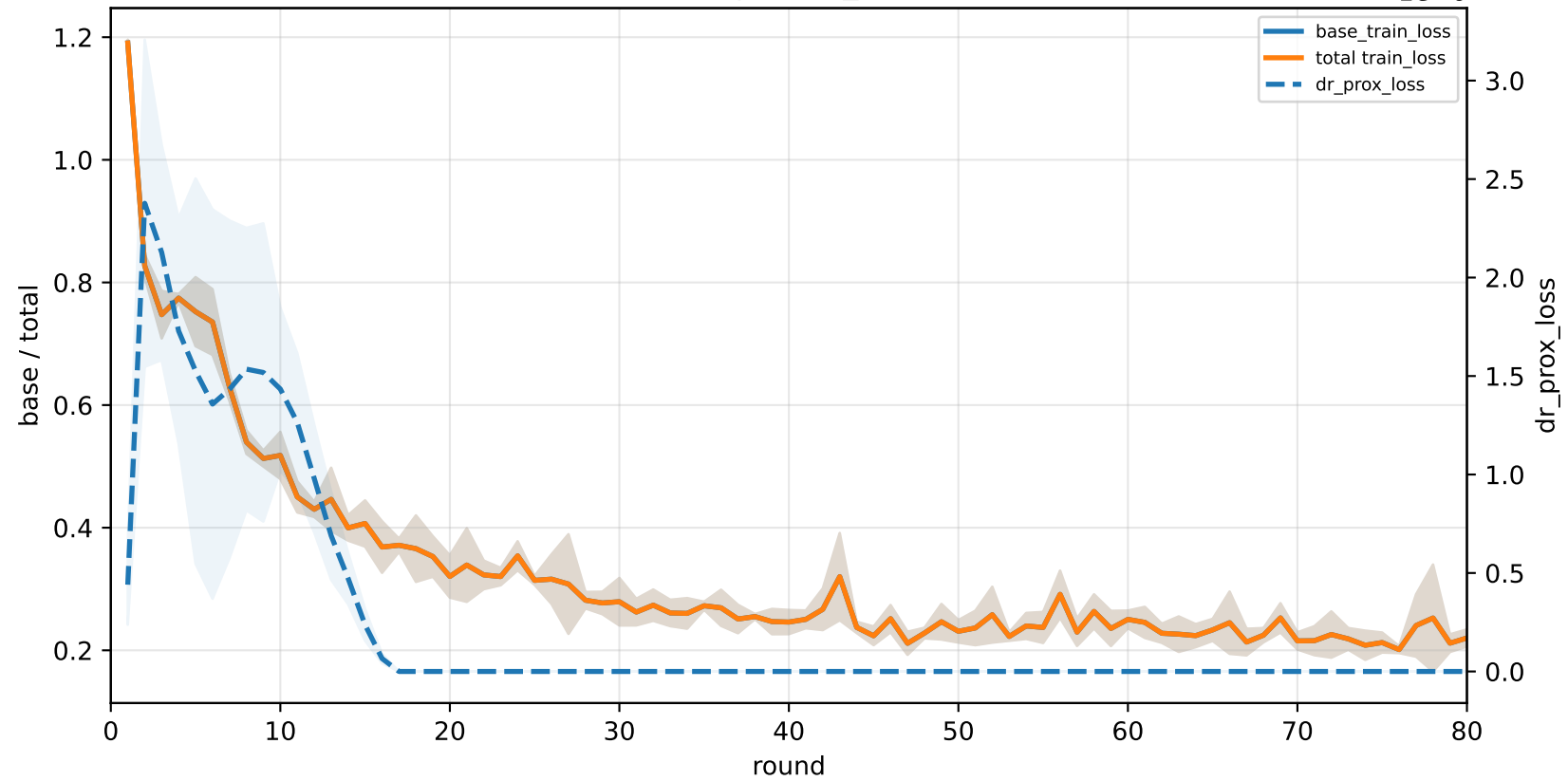
C4-specialist | graph_id=3000002



C5-specialist | graph_id=3000003

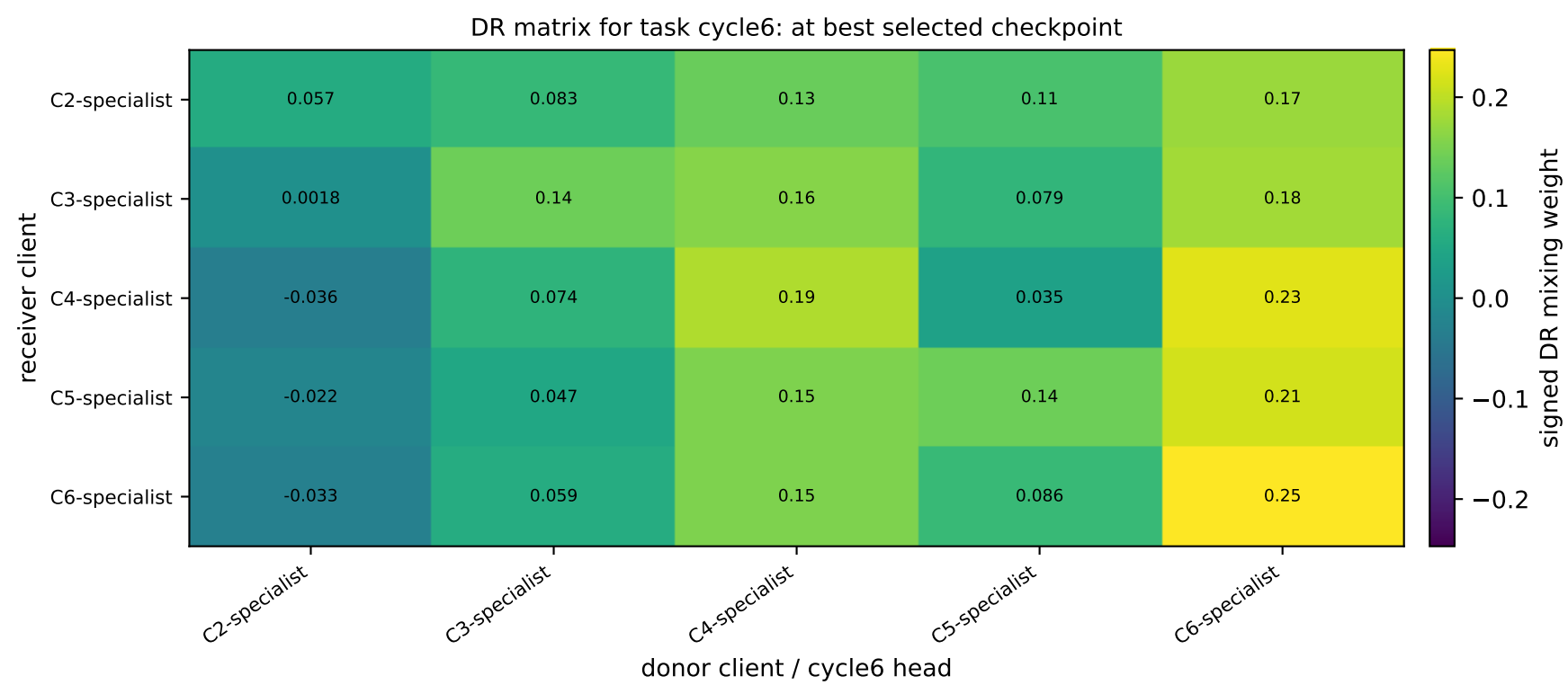
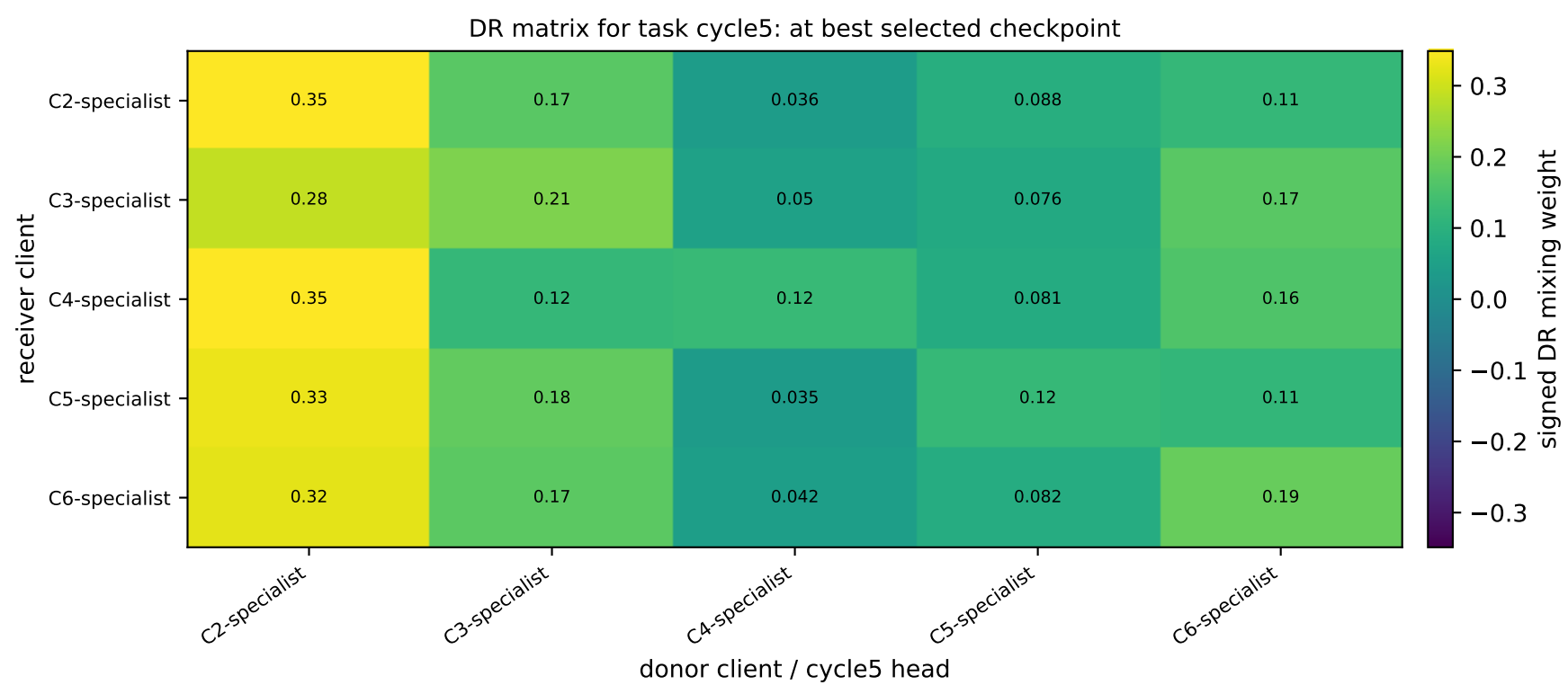
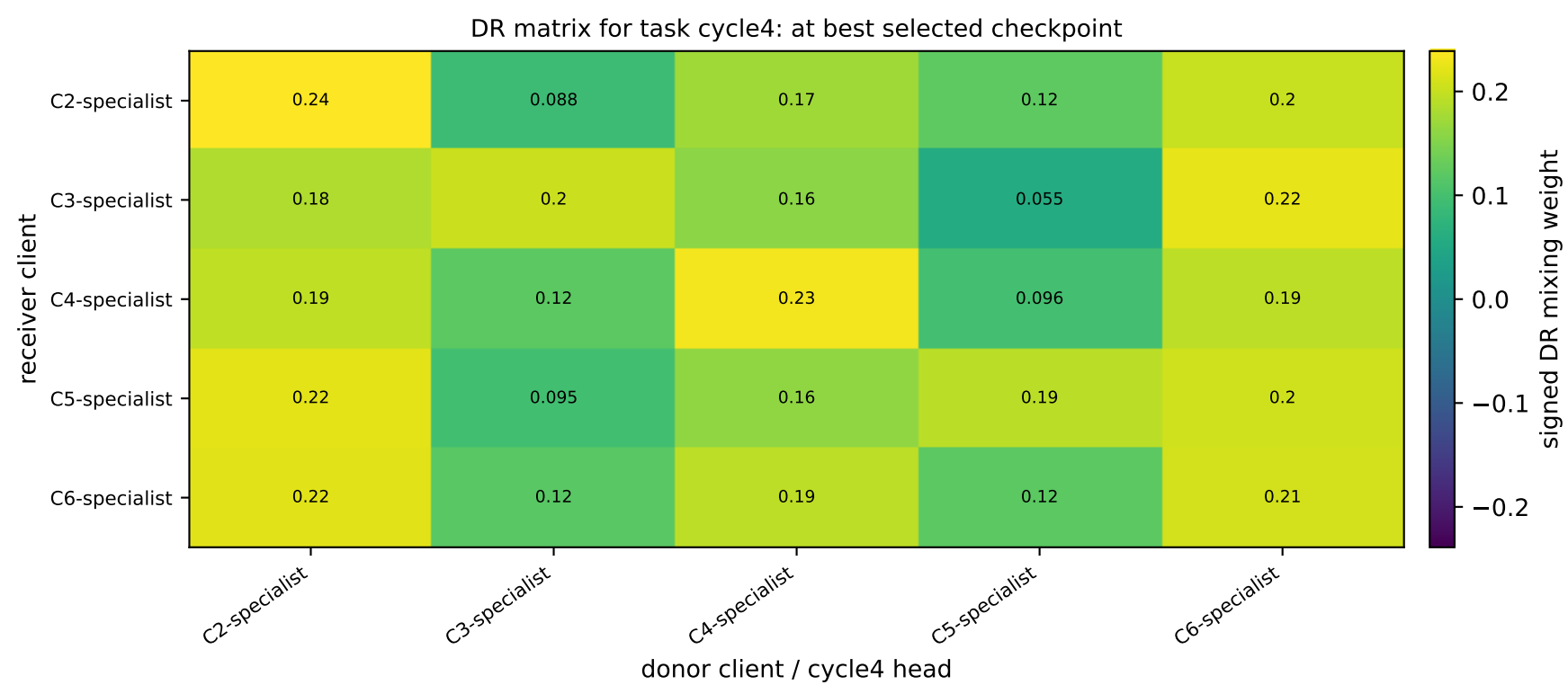
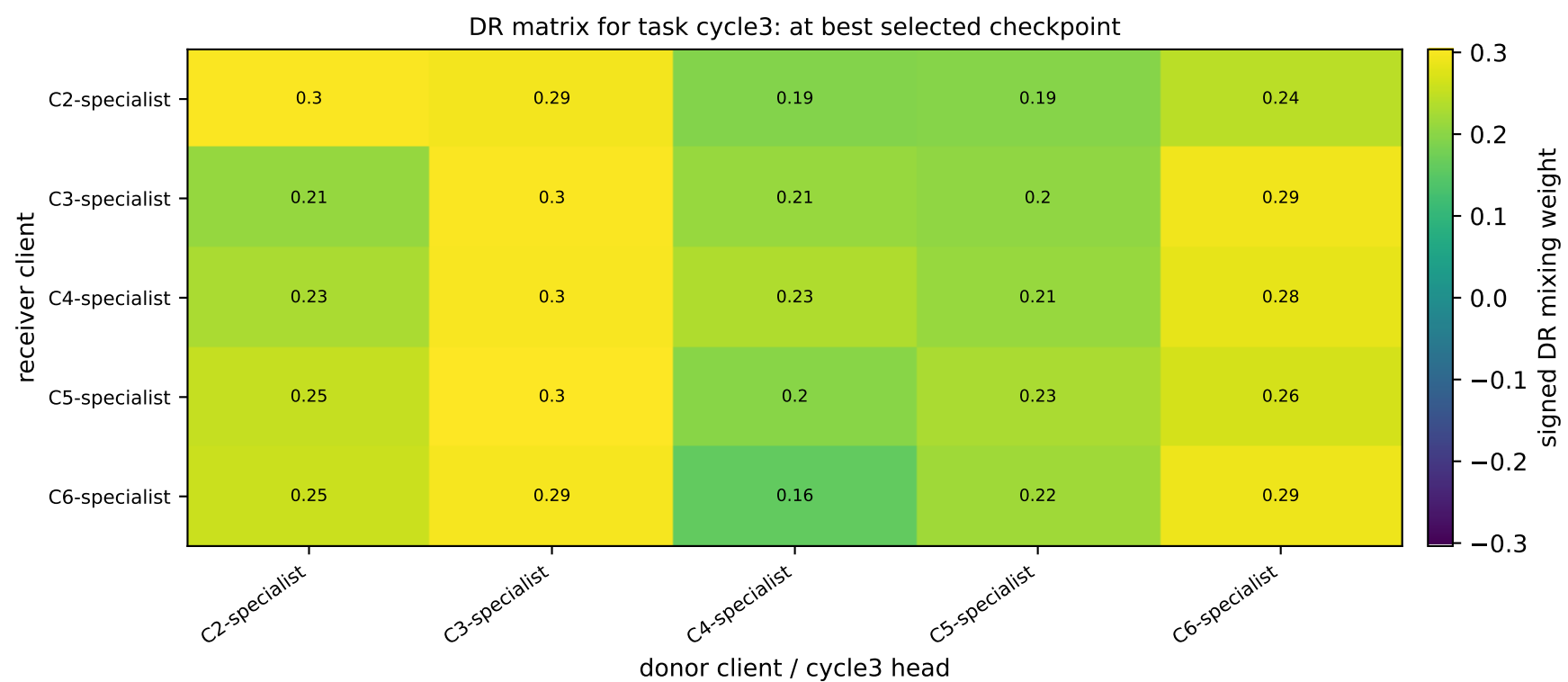
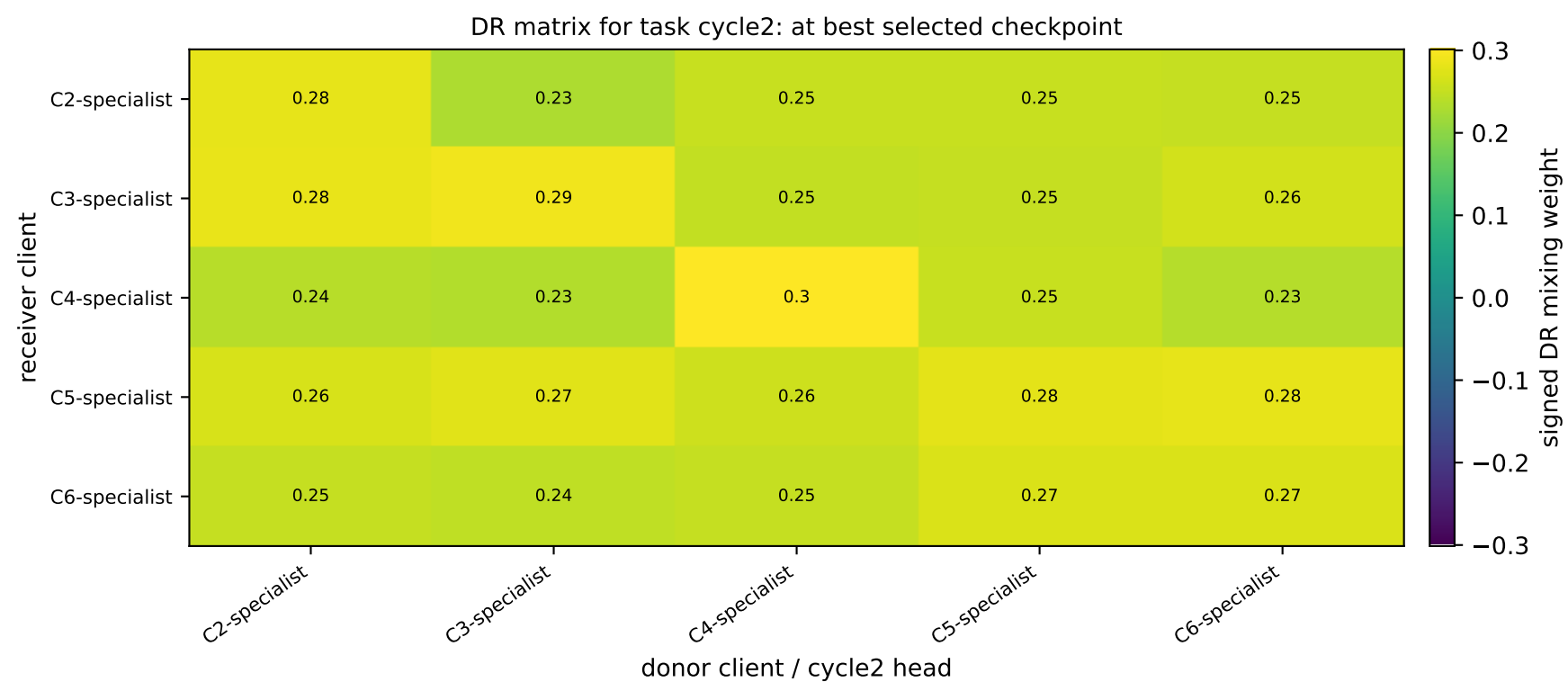


C6-specialist | graph_id=3000004



APPLE-FedAvgBackbone-TaskHead task-wise DR matrices: receiver clients × donor task-heads | iid

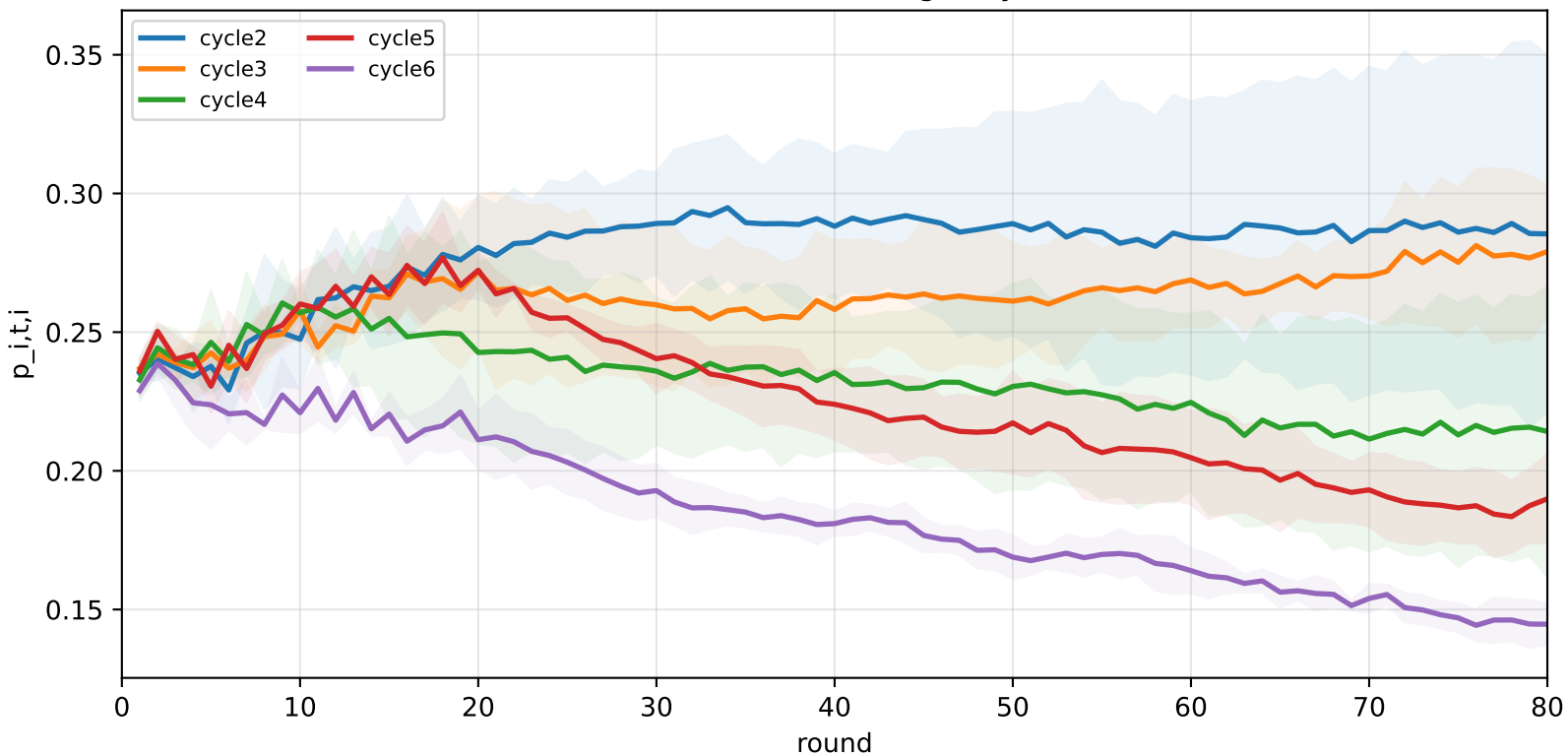
family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000



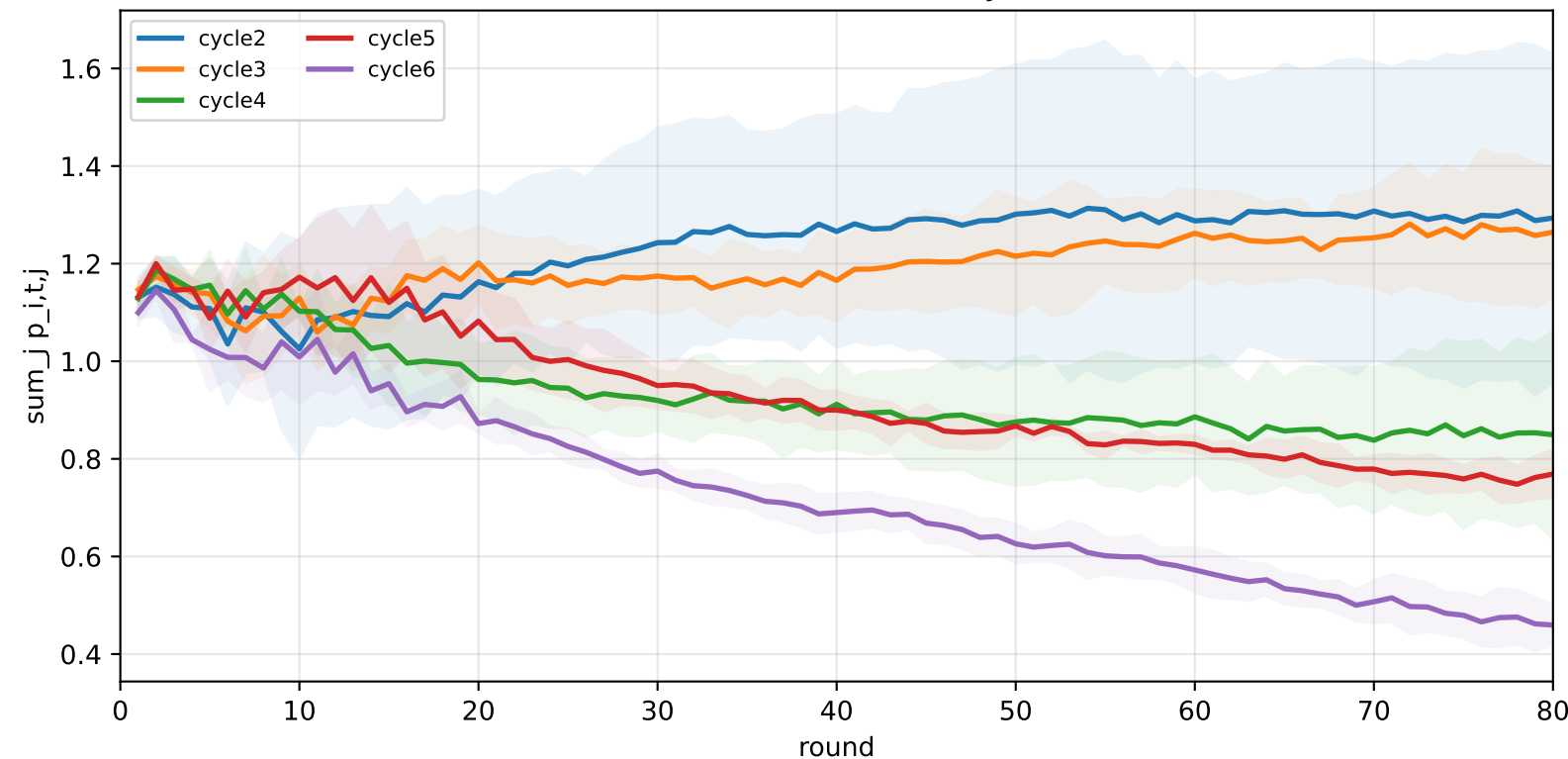
APPLE-FedAvgBackbone-TaskHead task-wise DR dynamics | iid

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
 cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
 family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
 cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
 family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
 cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
 family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
 cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
 family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
 cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
 heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

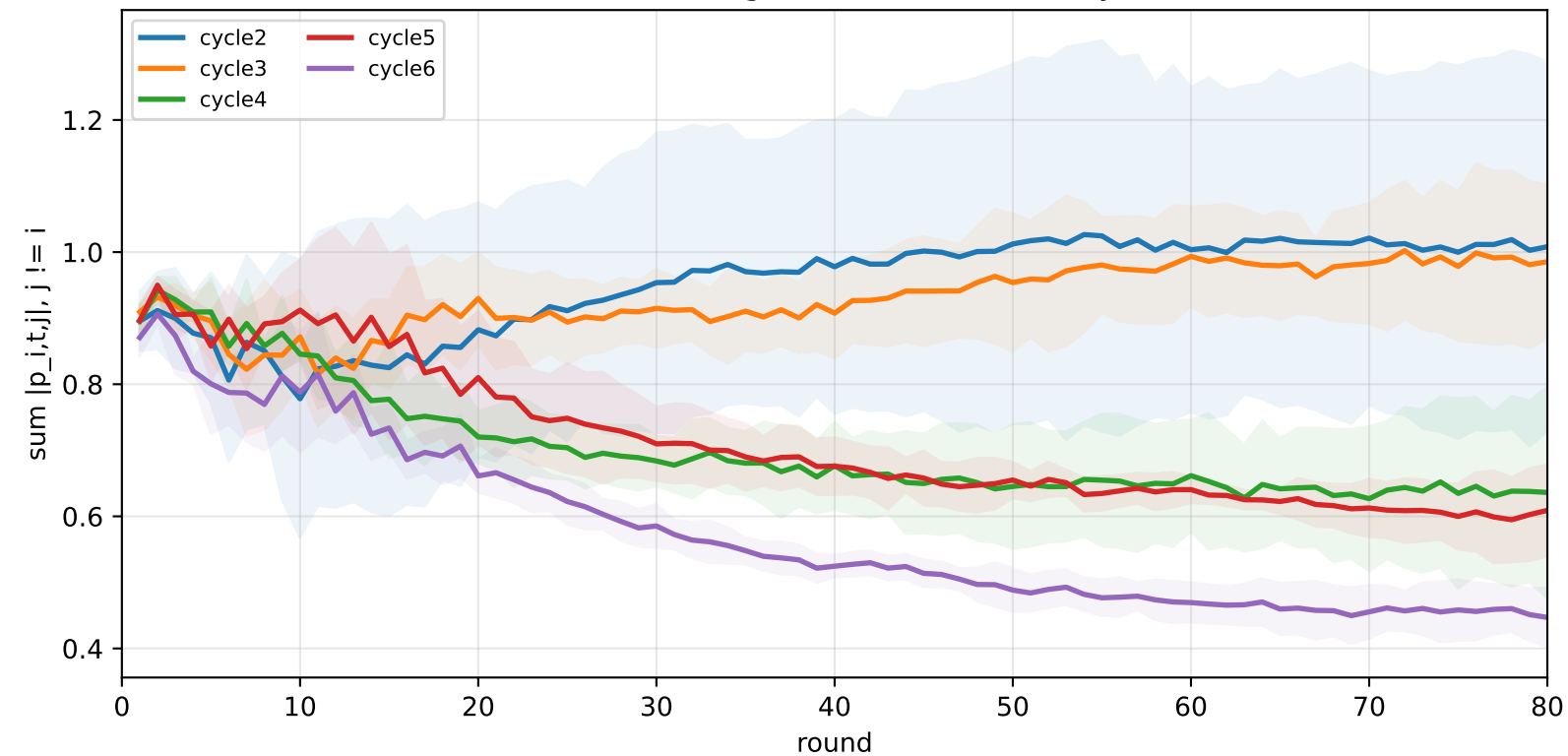
Task-head DR self-weight by task



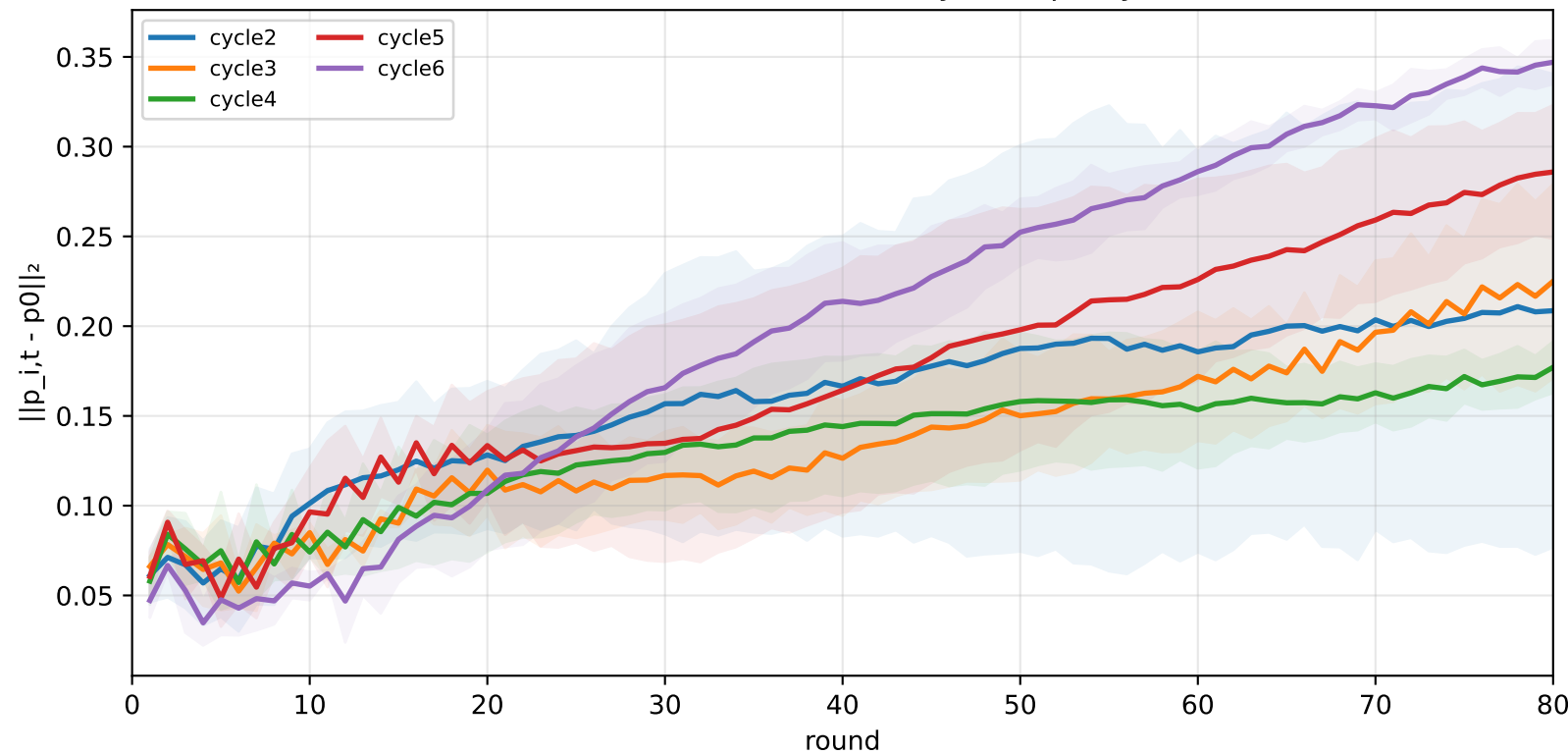
Task-head DR row sum by task



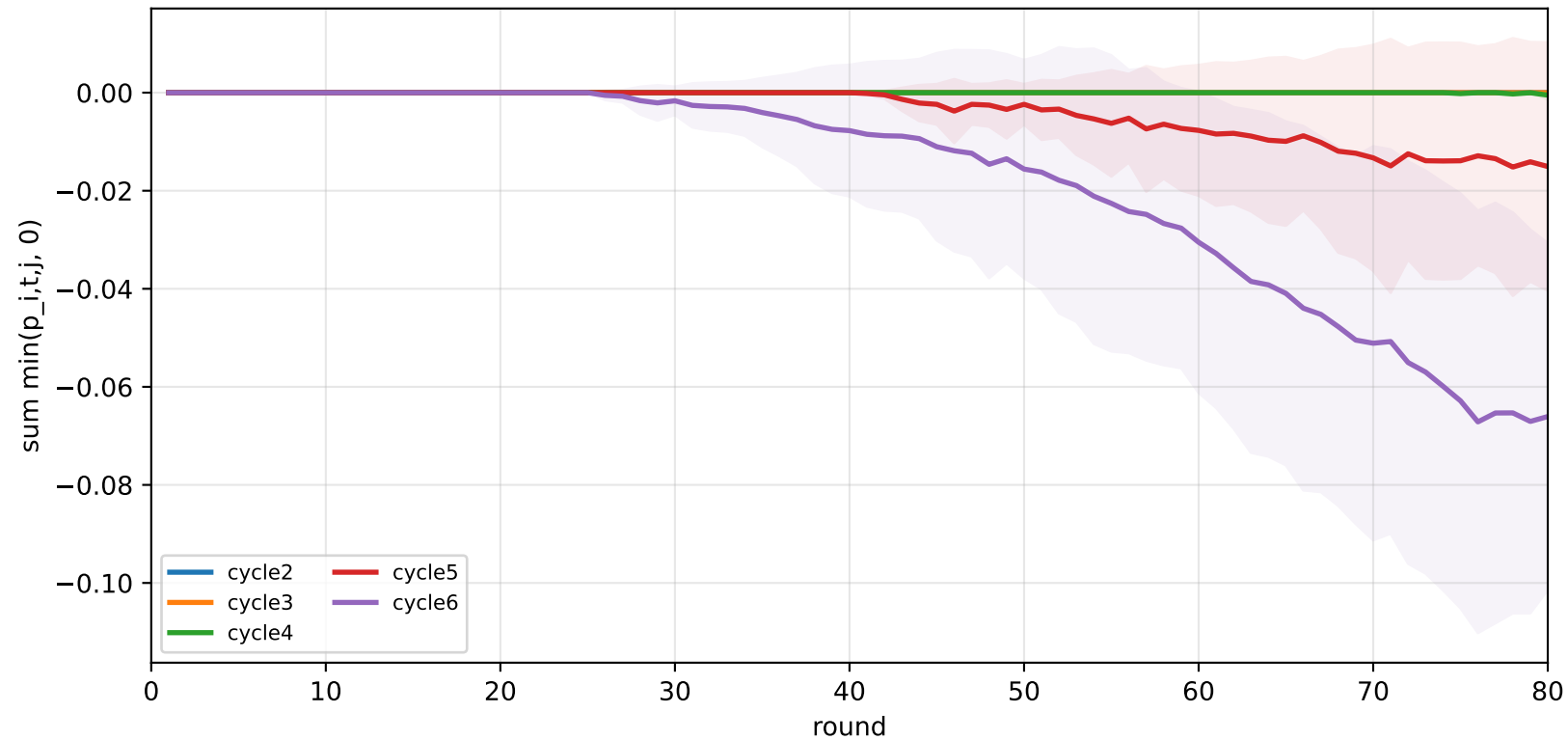
Task-head off-diagonal absolute mass by task



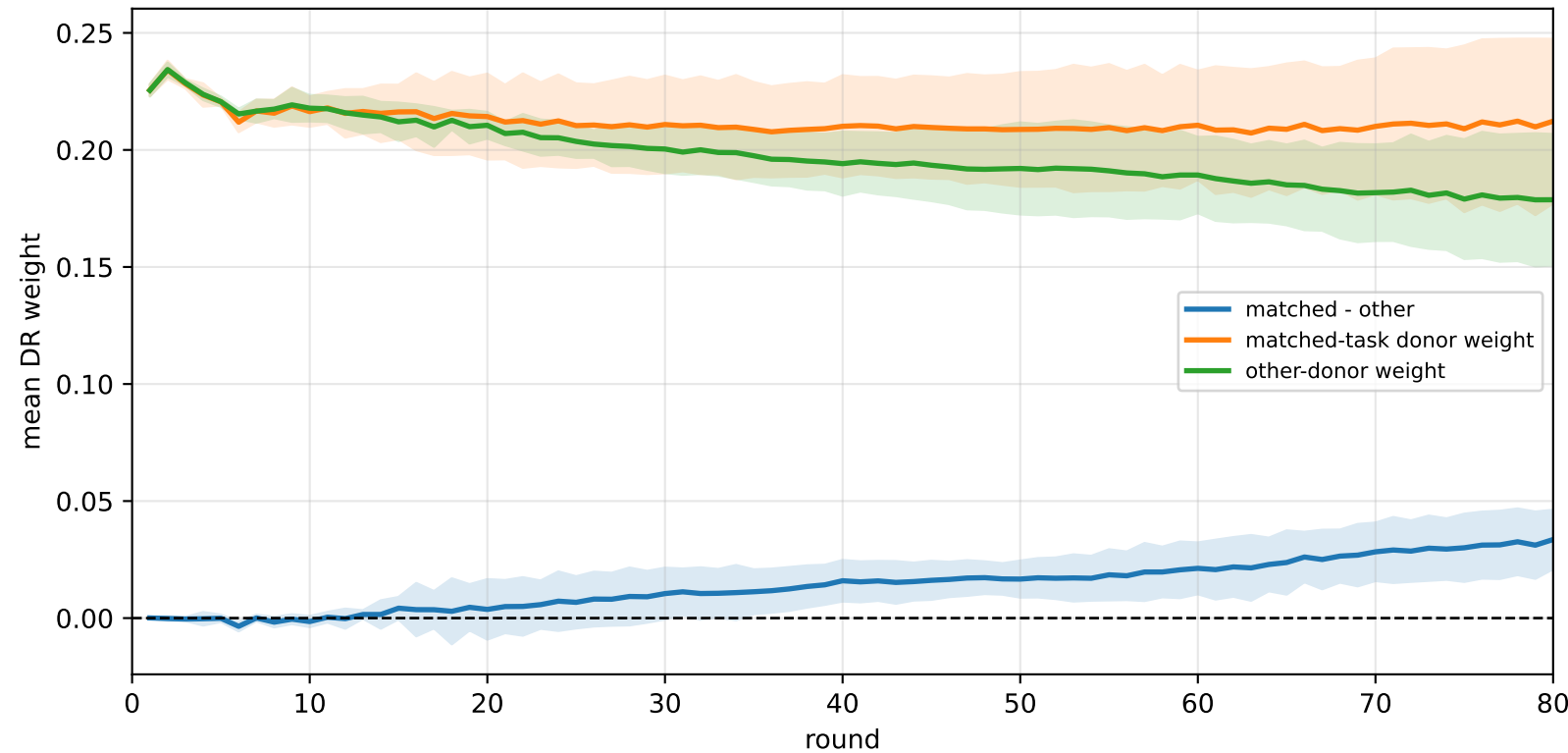
Task-head DR movement away from p0 by task



Task-head off-diagonal negative mass by task



Task-head DR alignment



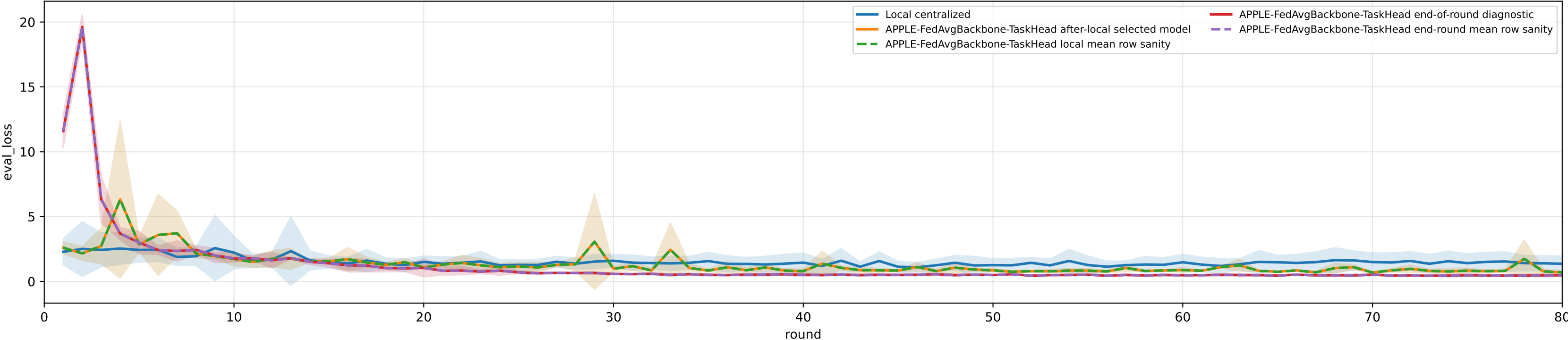
Controlled q-label heterogeneity: Mid | APPLE-FedAvgBackbone-TaskHead diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

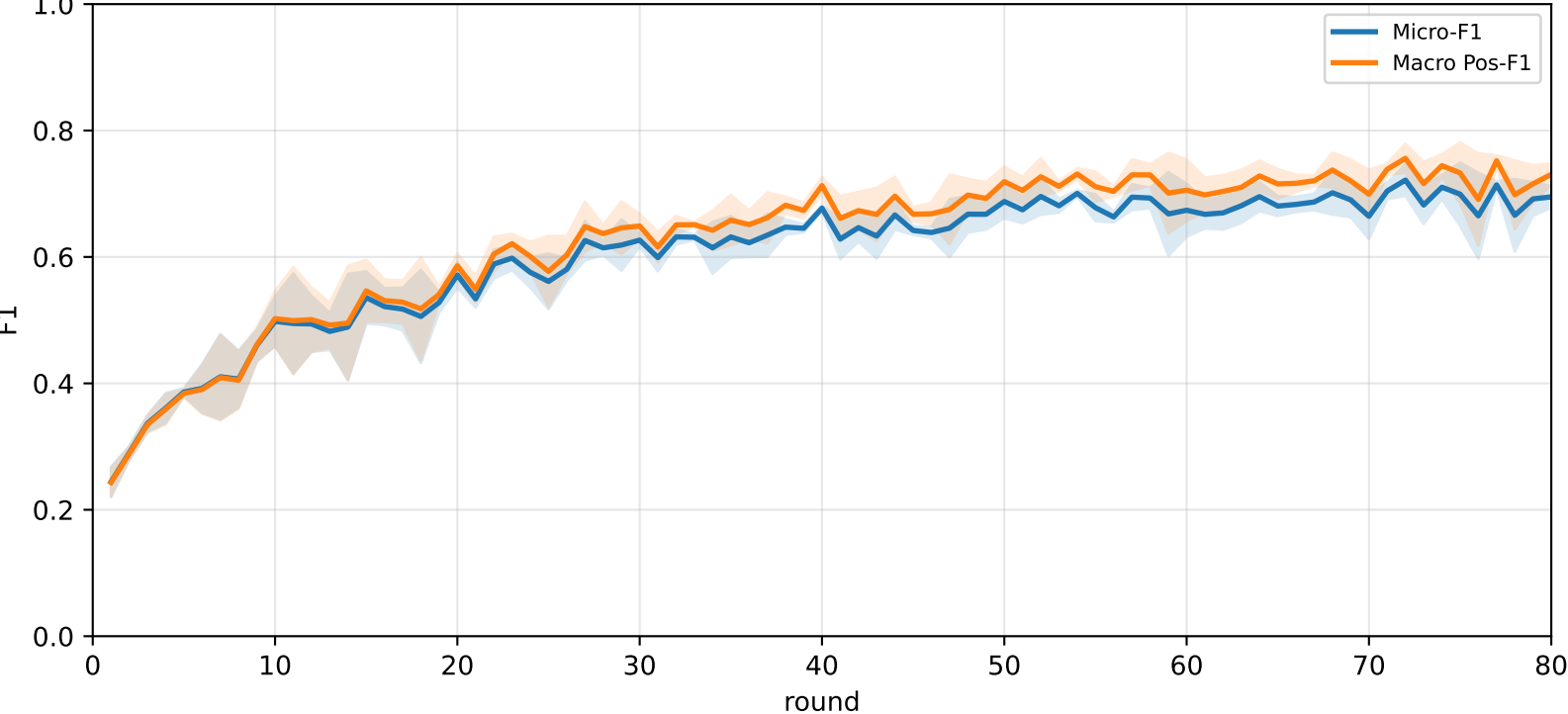
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	60.3% ± 0.4%	63.4% ± 0.4%	74.7% ± 4.2%	70.7% ± 0.1%	60.9% ± 2.0%	57.0% ± 1.1%	47.0% ± 0.4%
APPLE-FedAvgBackbone-TaskHead	69.1% ± 1.6%	71.8% ± 1.0%	84.5% ± 4.7%	81.0% ± 4.3%	67.7% ± 1.0%	67.7% ± 2.5%	56.9% ± 3.8%

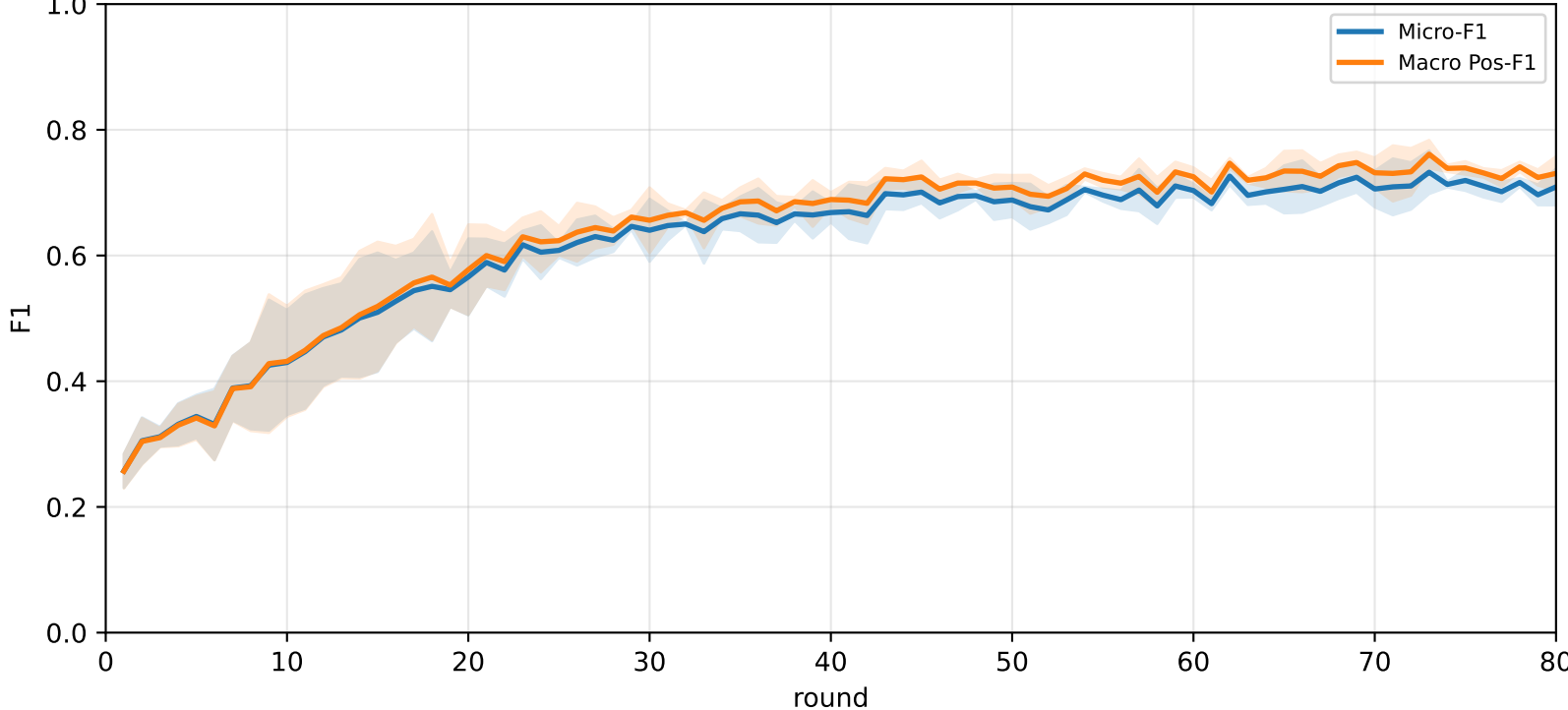
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



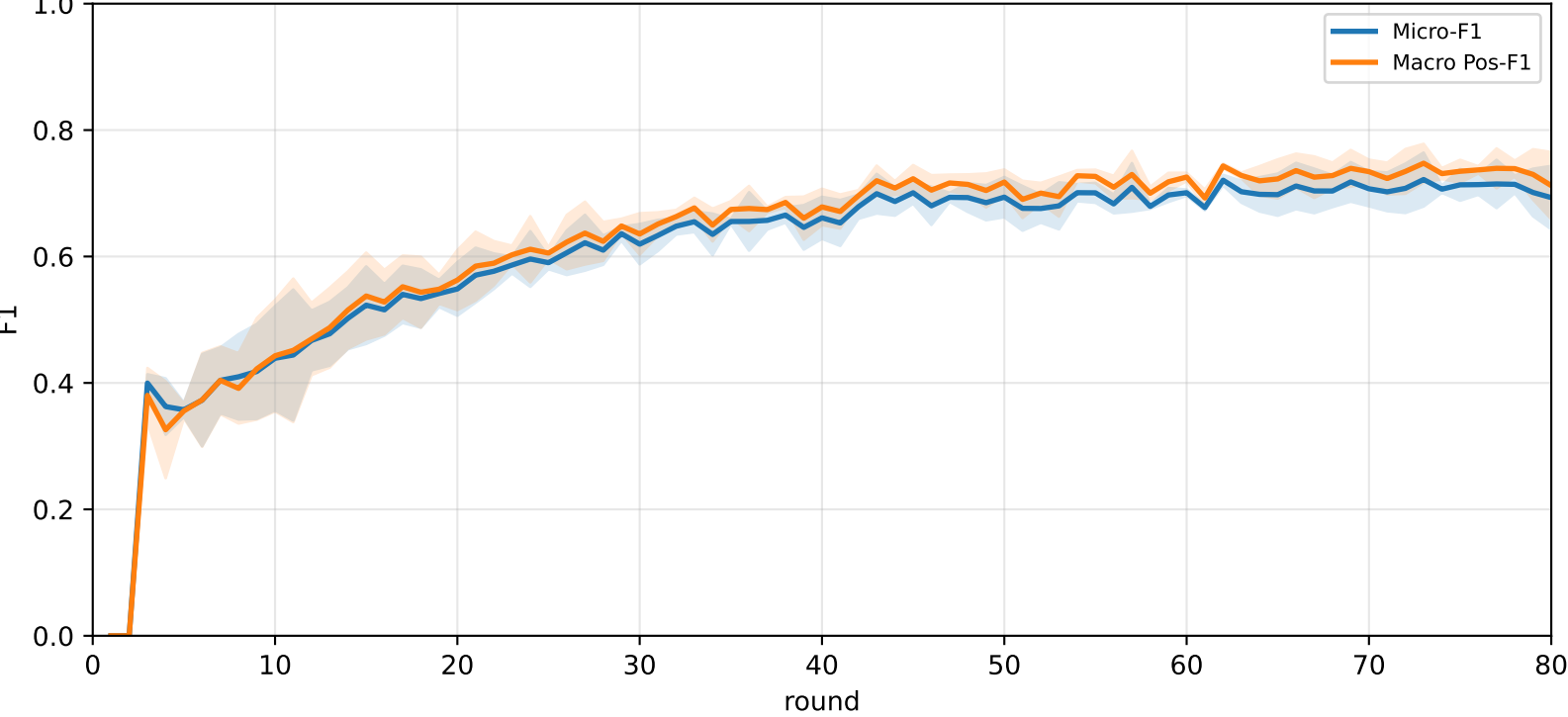
Pooled Local baseline micro / macro F1



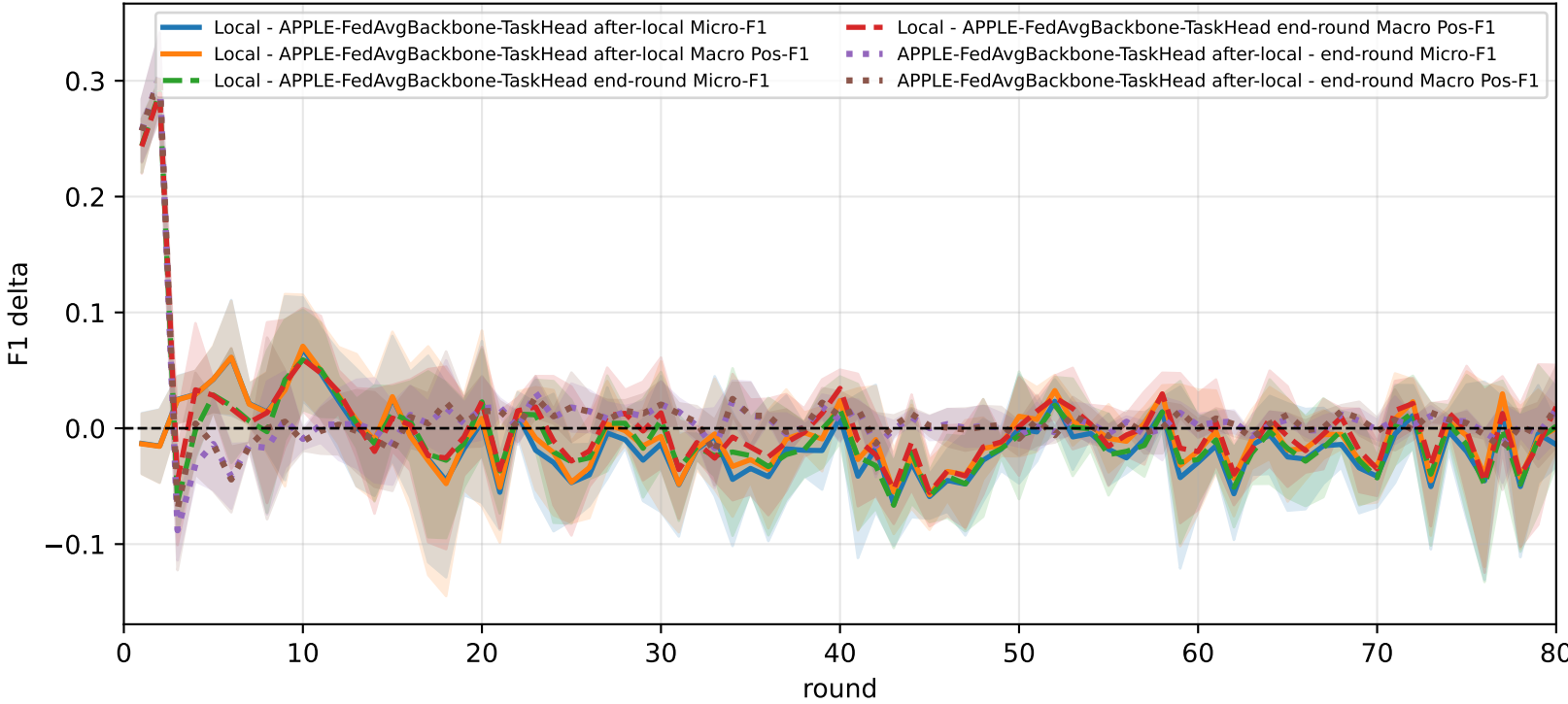
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



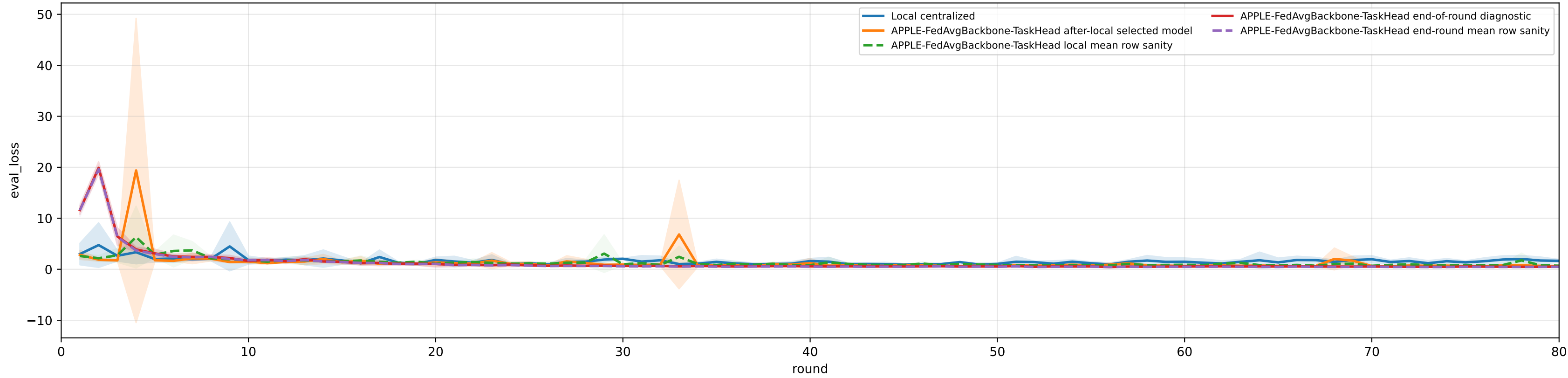
Controlled q-label heterogeneity: Mid | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q_task_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

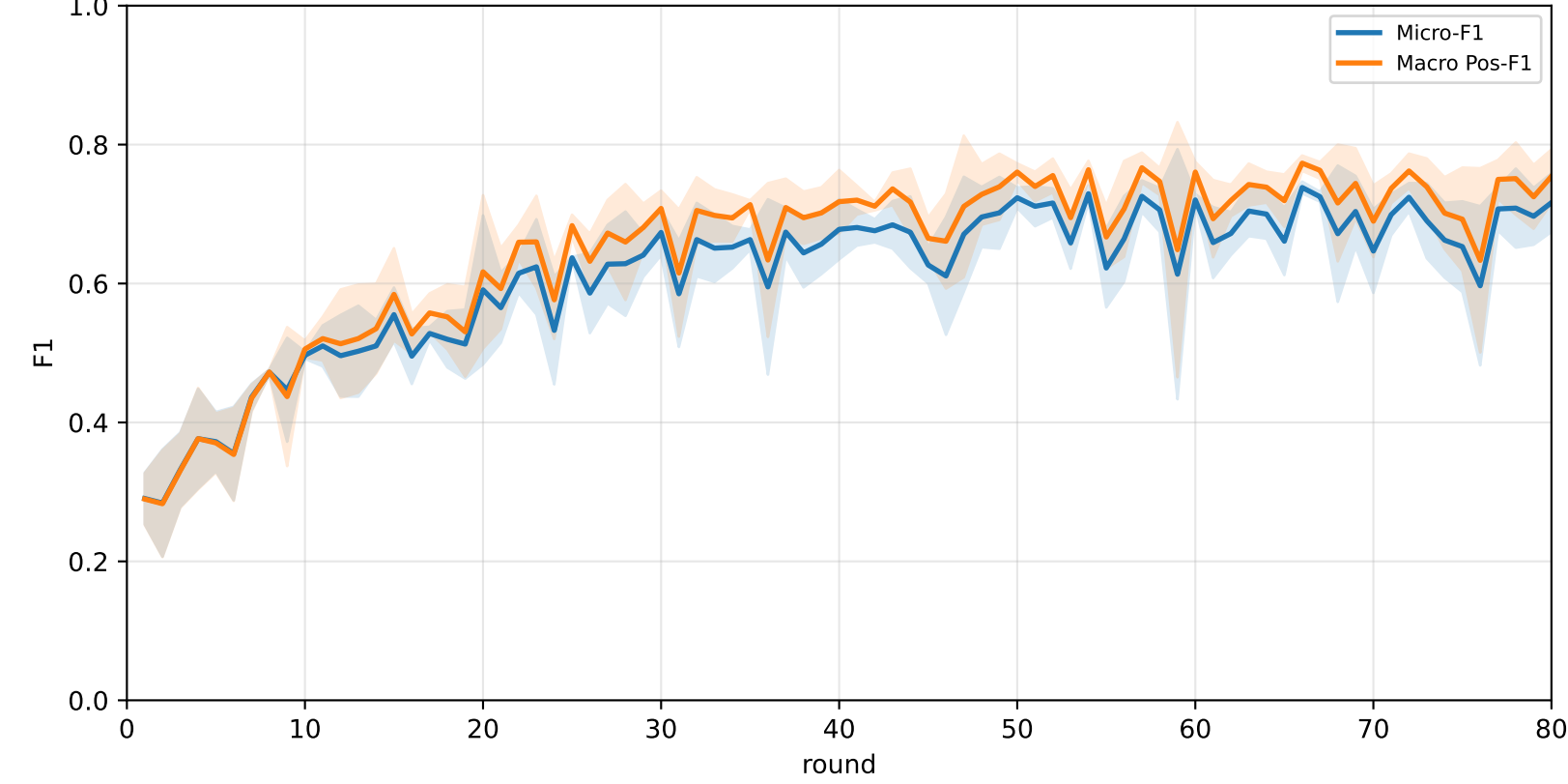
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	62.9% ± 2.7%	67.2% ± 2.6%	84.2% ± 3.3%	78.8% ± 3.0%	67.1% ± 2.5%	59.7% ± 2.9%	46.4% ± 2.7%
APPLE-FedAvgBackbone-TaskHead	70.2% ± 1.0%	72.9% ± 0.9%	83.0% ± 9.1%	85.3% ± 3.3%	71.4% ± 4.3%	66.6% ± 4.0%	58.1% ± 4.4%

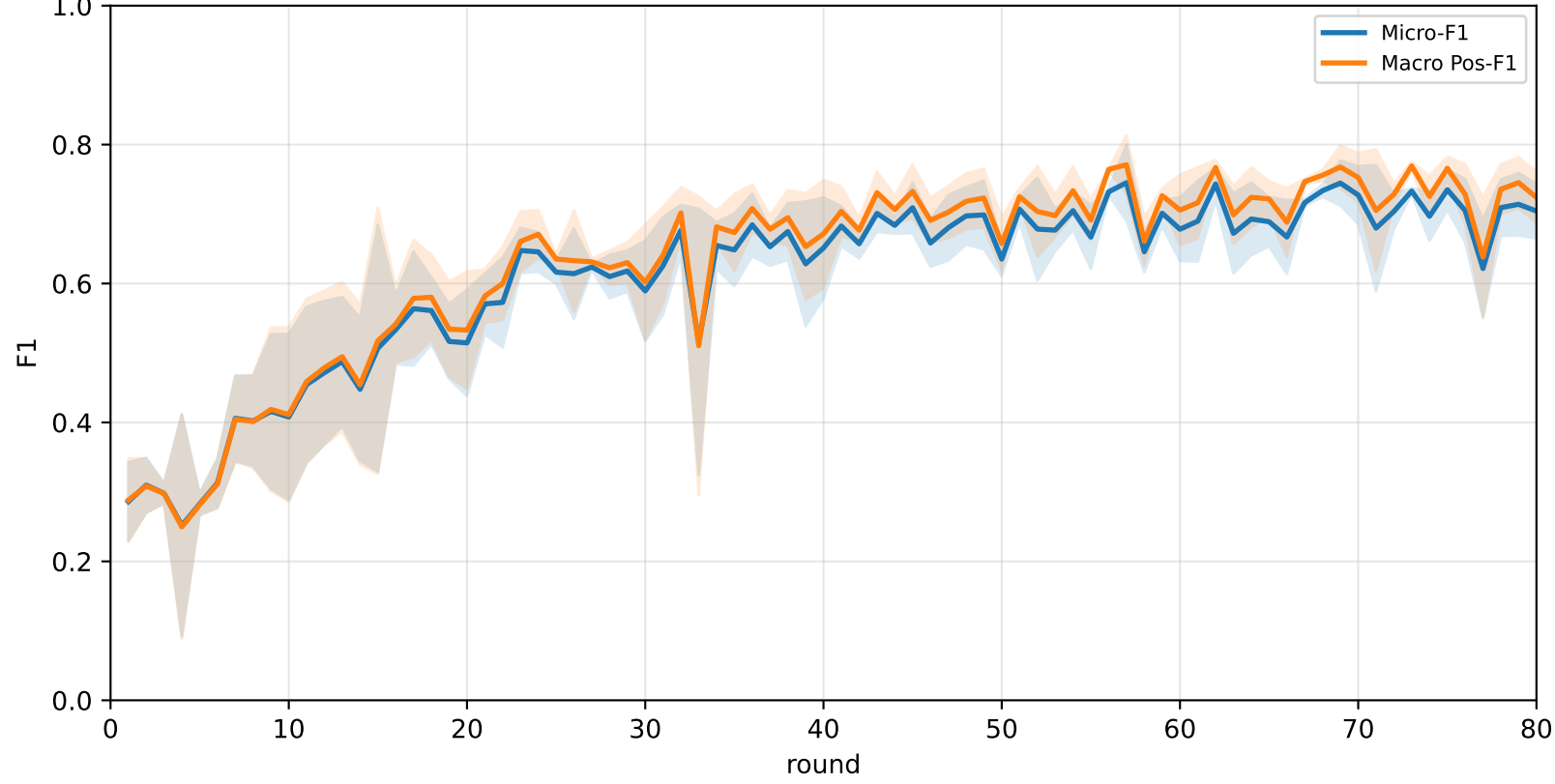
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



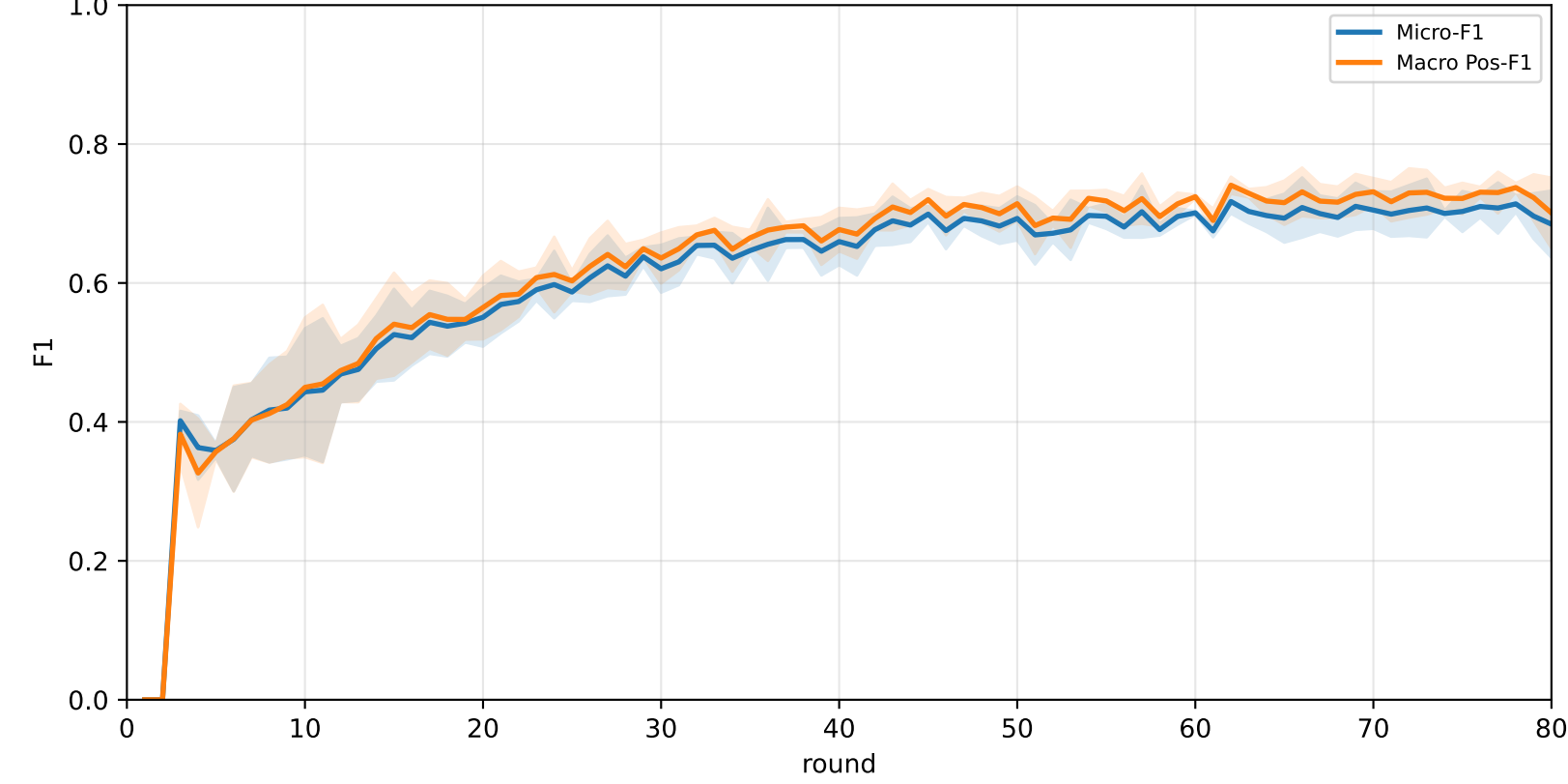
Pooled Local baseline micro / macro F1



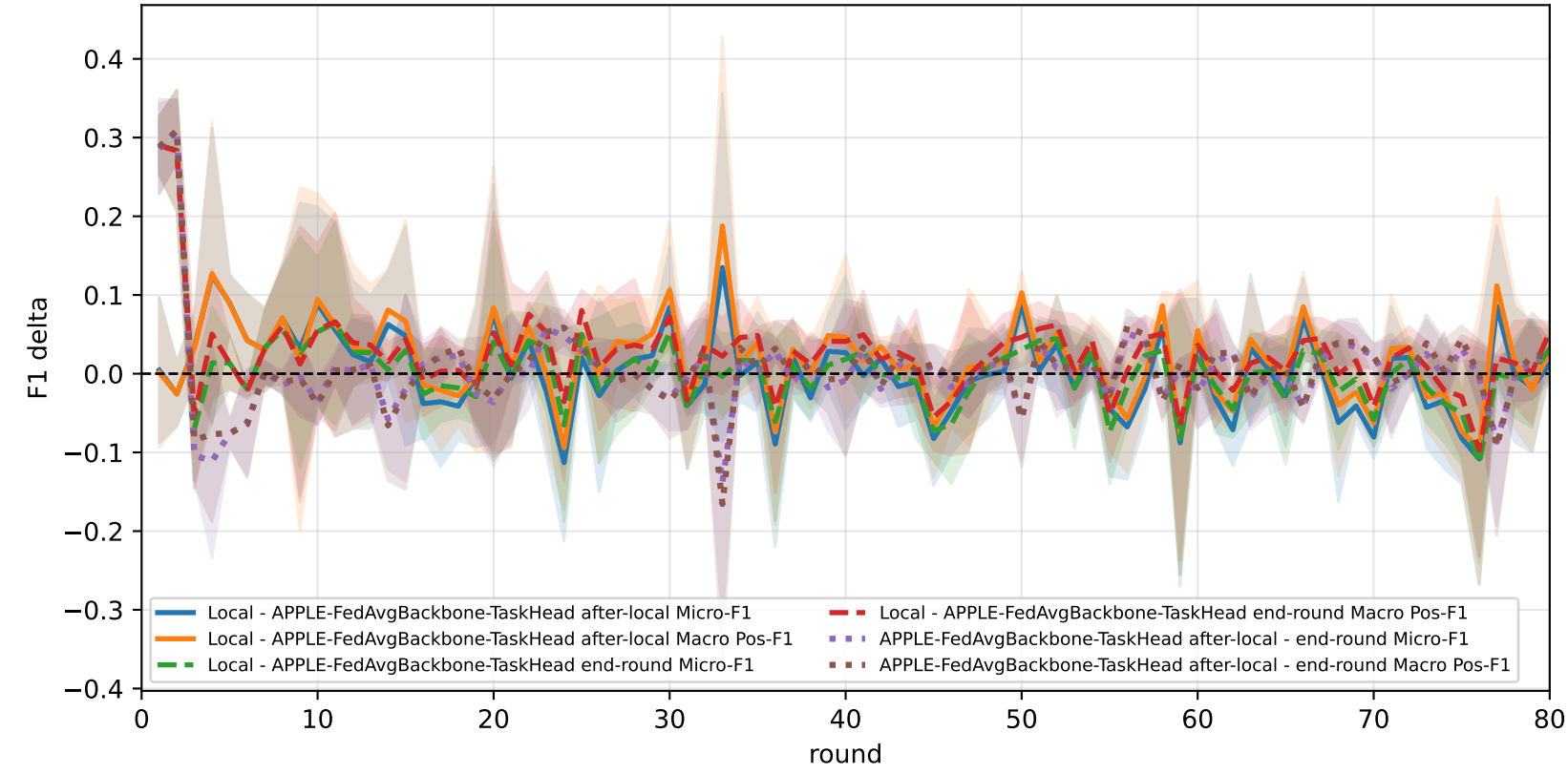
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



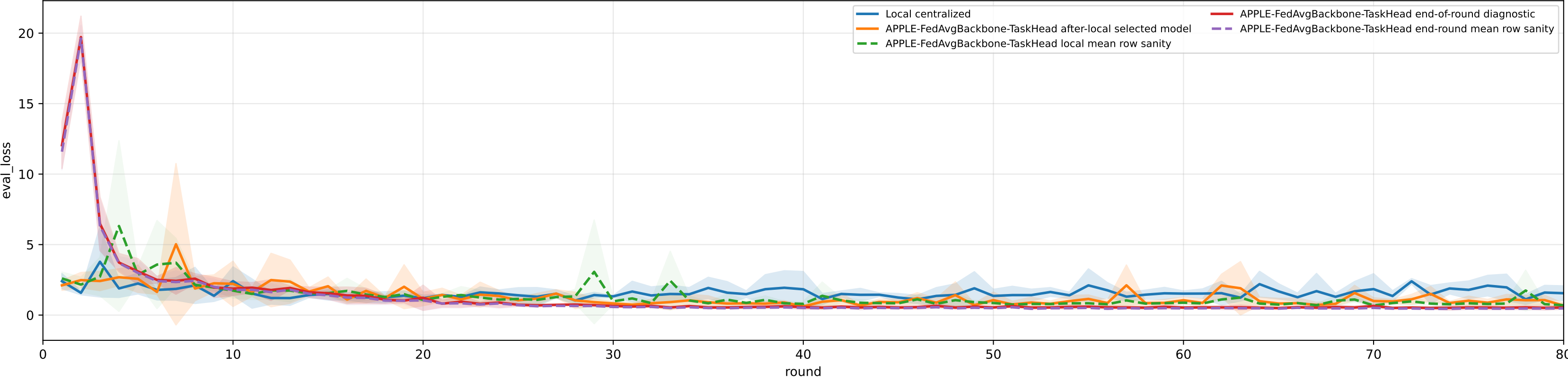
Controlled q-label heterogeneity: Mid | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q task_cycle3 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% |
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

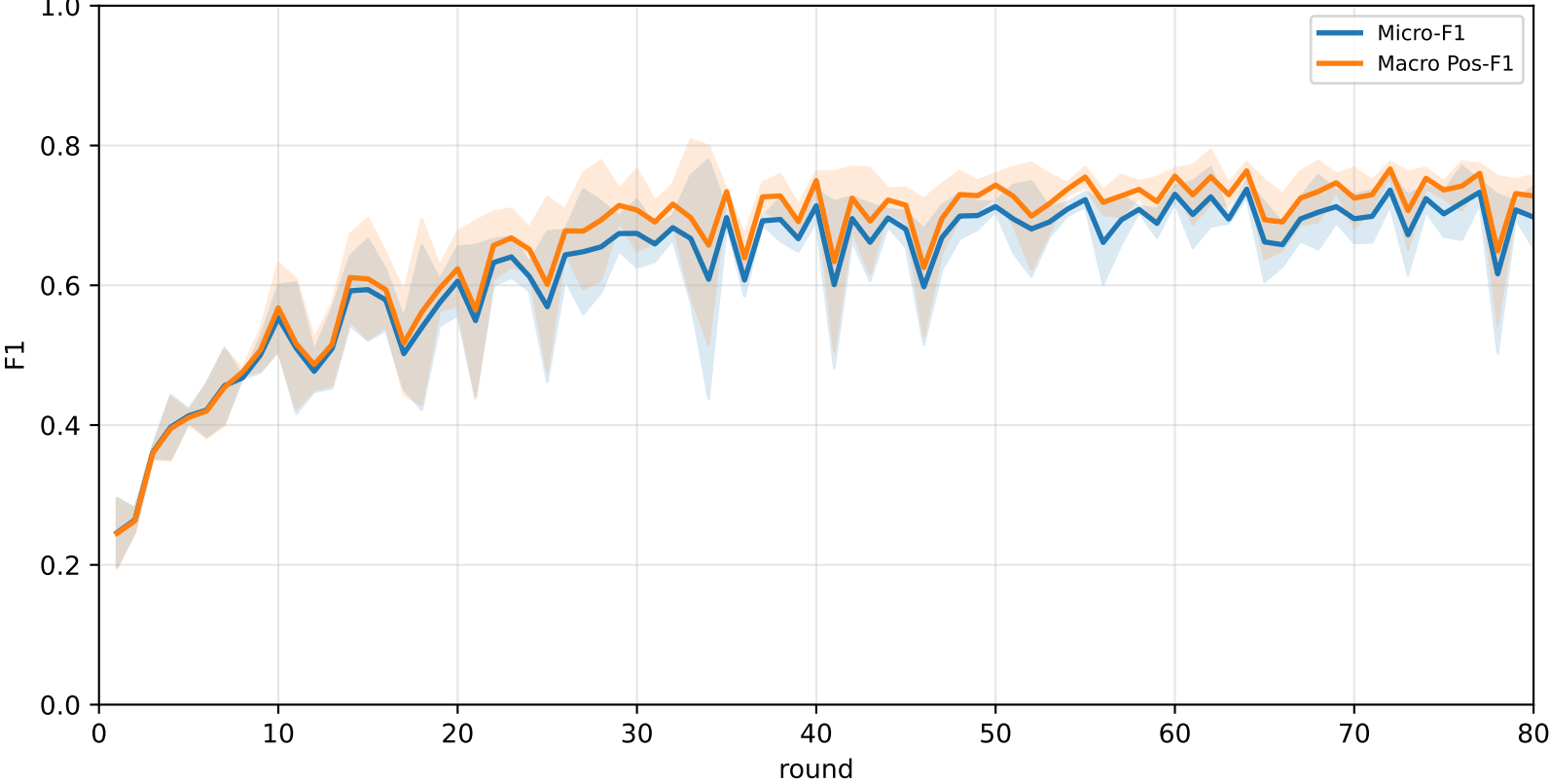
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	55.7% ± 8.8%	57.9% ± 10.2%	66.6% ± 17.9%	62.6% ± 13.2%	58.0% ± 7.4%	56.9% ± 12.7%	45.4% ± 3.0%
APPLE-FedAvgBackbone-TaskHead	69.4% ± 4.0%	71.8% ± 3.2%	85.8% ± 5.4%	78.2% ± 10.8%	67.6% ± 1.9%	68.9% ± 4.4%	58.5% ± 5.4%

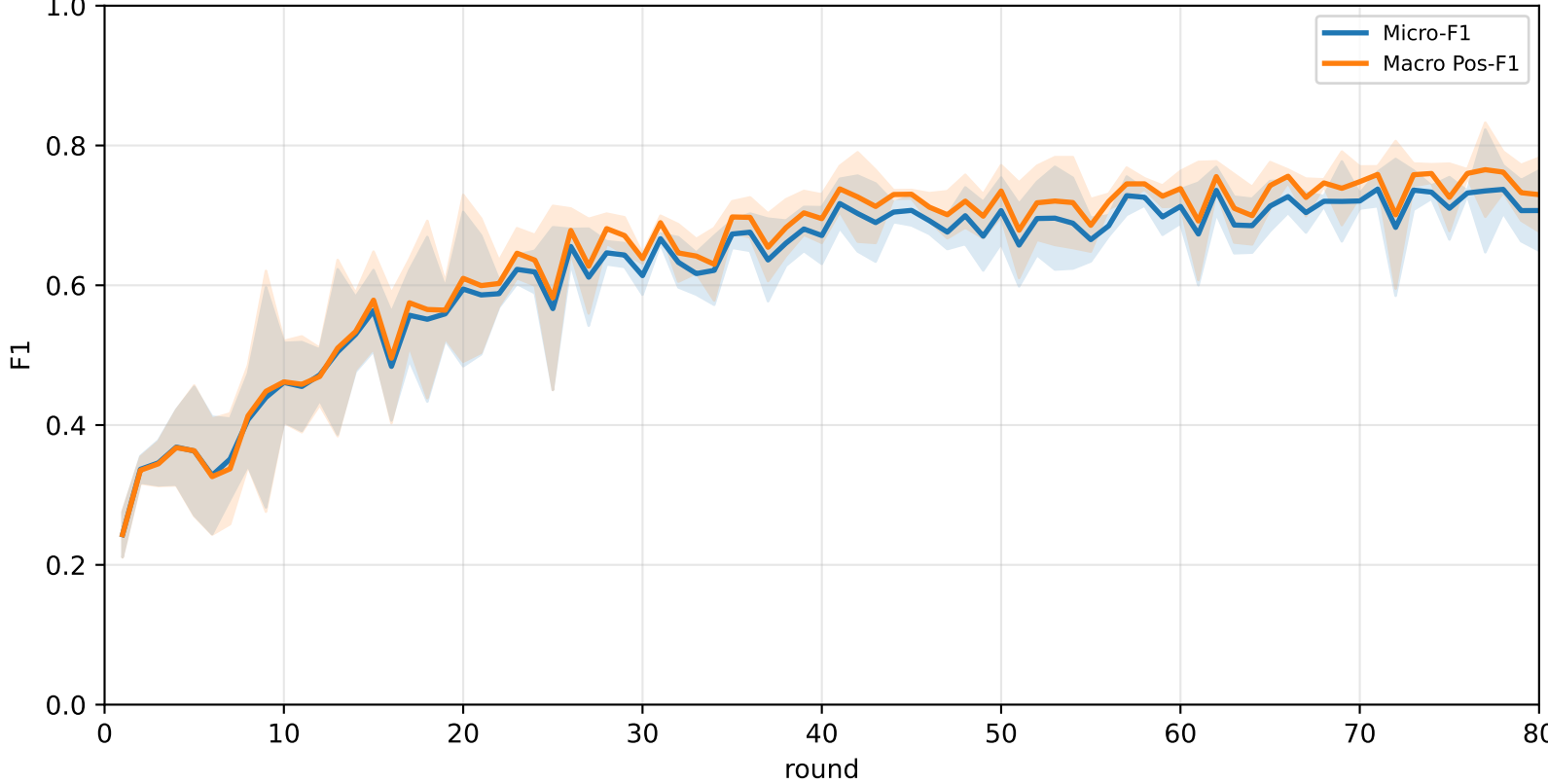
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



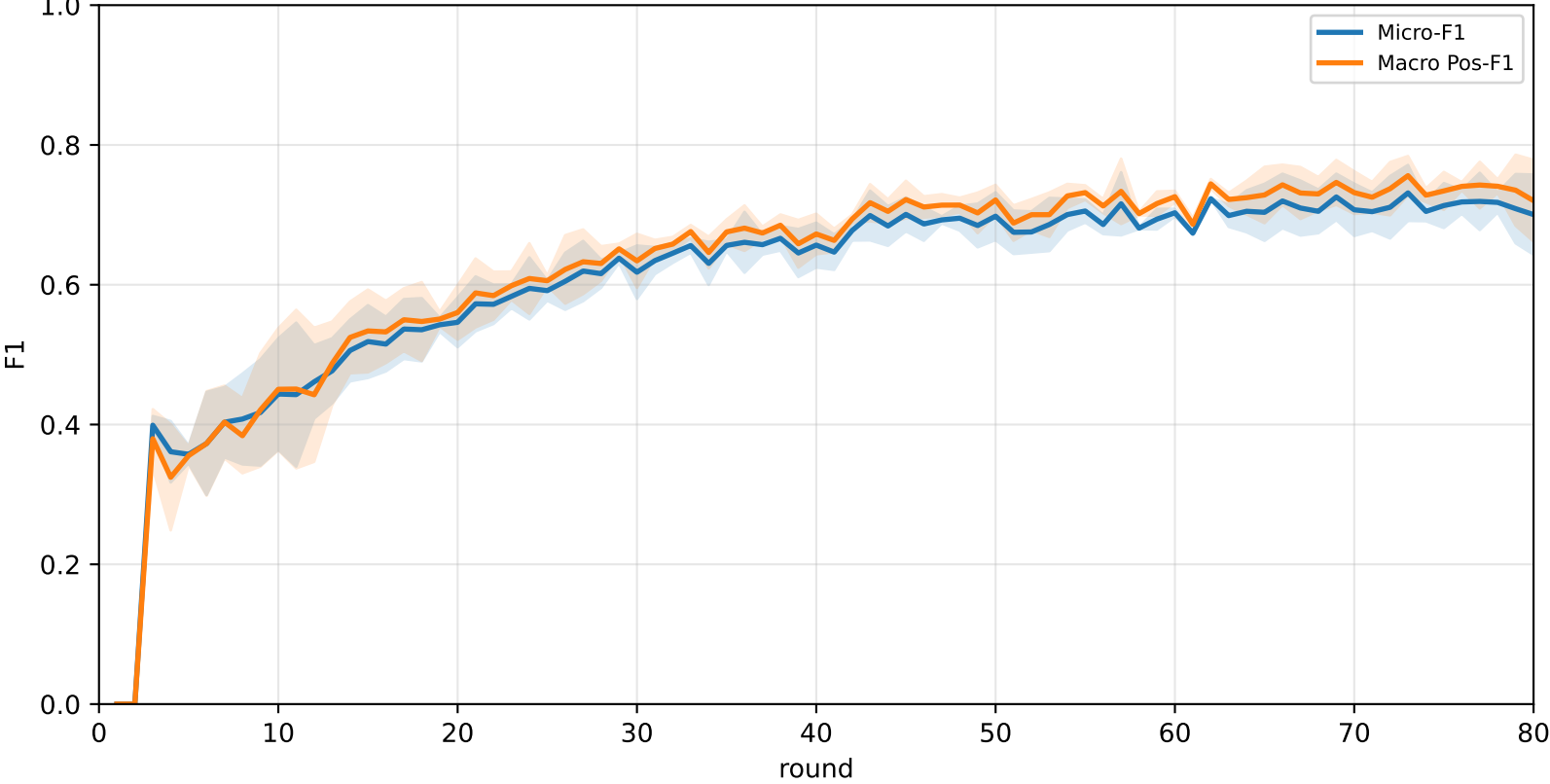
Pooled Local baseline micro / macro F1



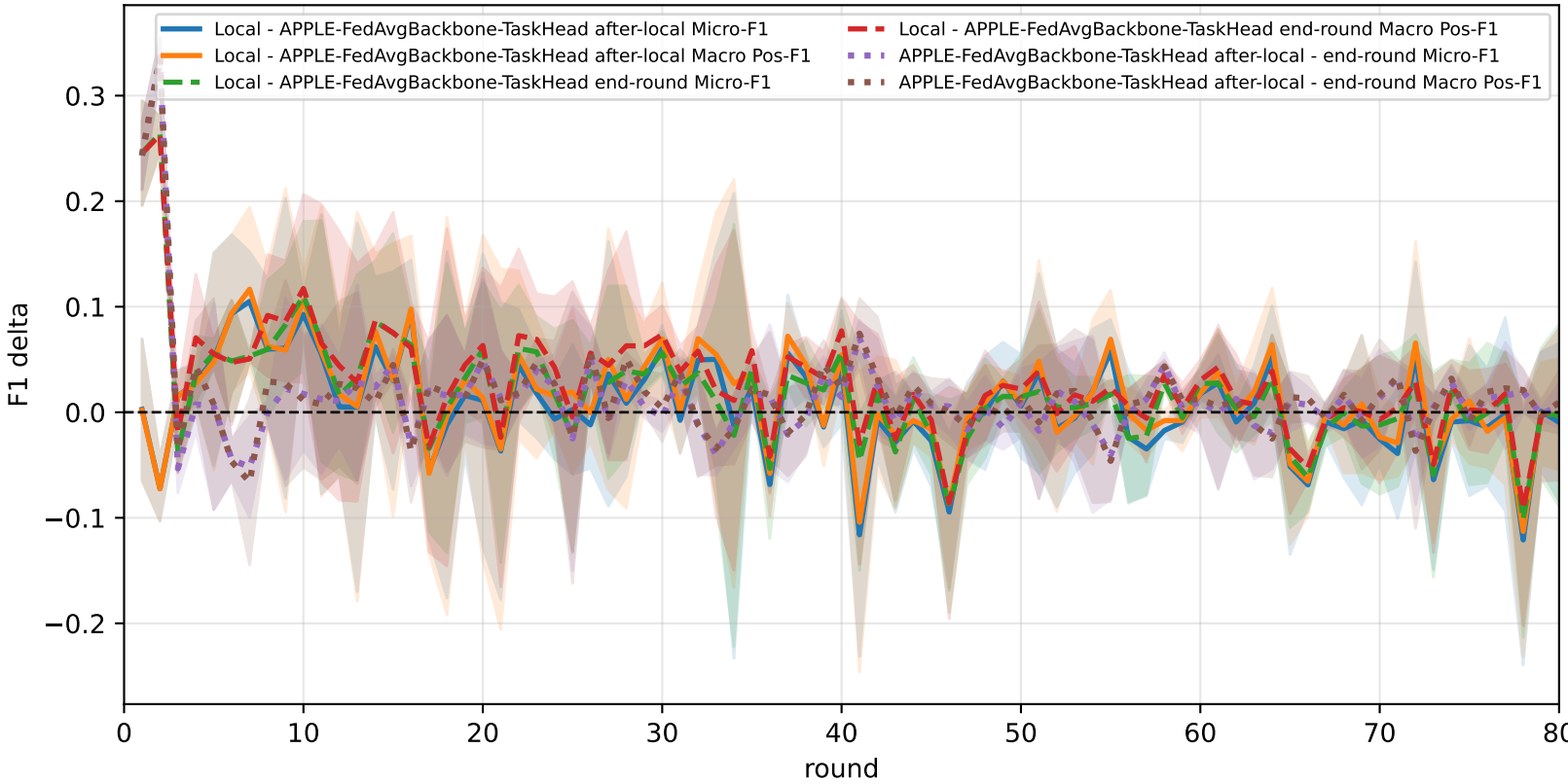
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



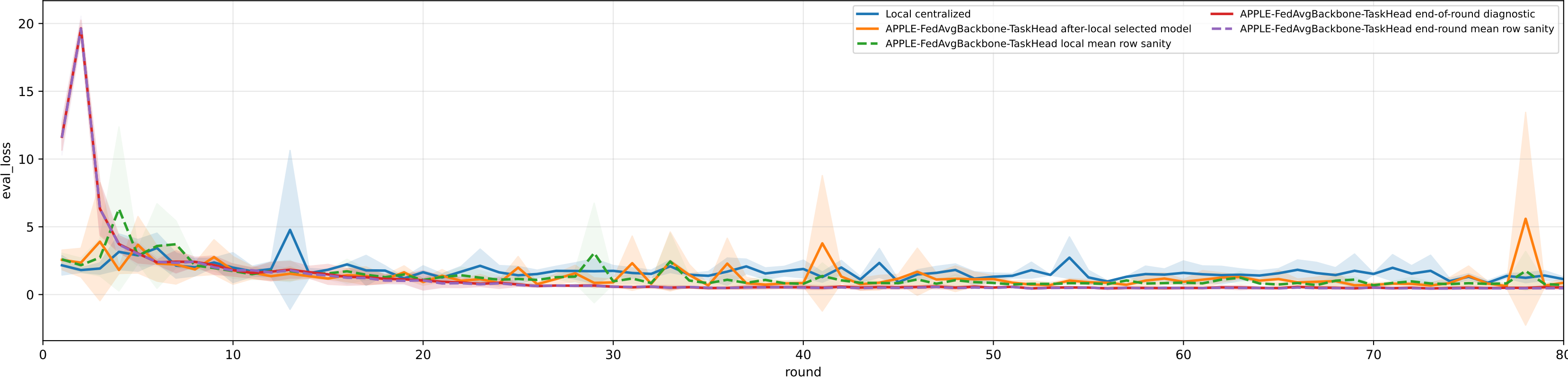
Controlled q-label heterogeneity: Mid | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q task_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% |
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

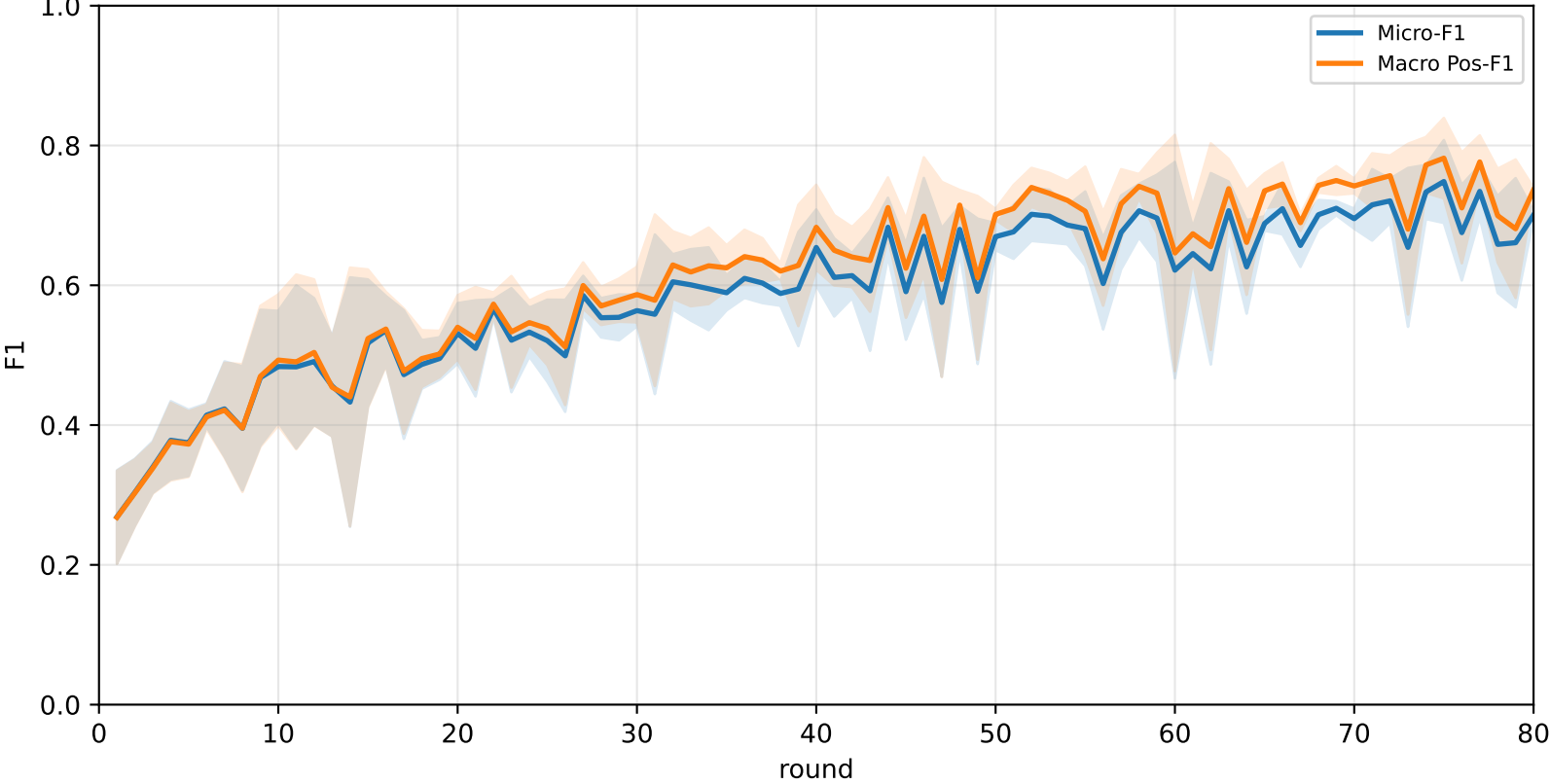
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	57.1% ± 7.9%	59.7% ± 8.6%	71.0% ± 8.9%	68.6% ± 10.4%	60.3% ± 10.1%	53.0% ± 10.3%	45.6% ± 4.2%
APPLE-FedAvgBackbone-TaskHead	68.1% ± 3.0%	71.1% ± 2.8%	84.8% ± 3.3%	80.6% ± 7.1%	65.8% ± 1.0%	69.6% ± 4.7%	54.8% ± 3.9%

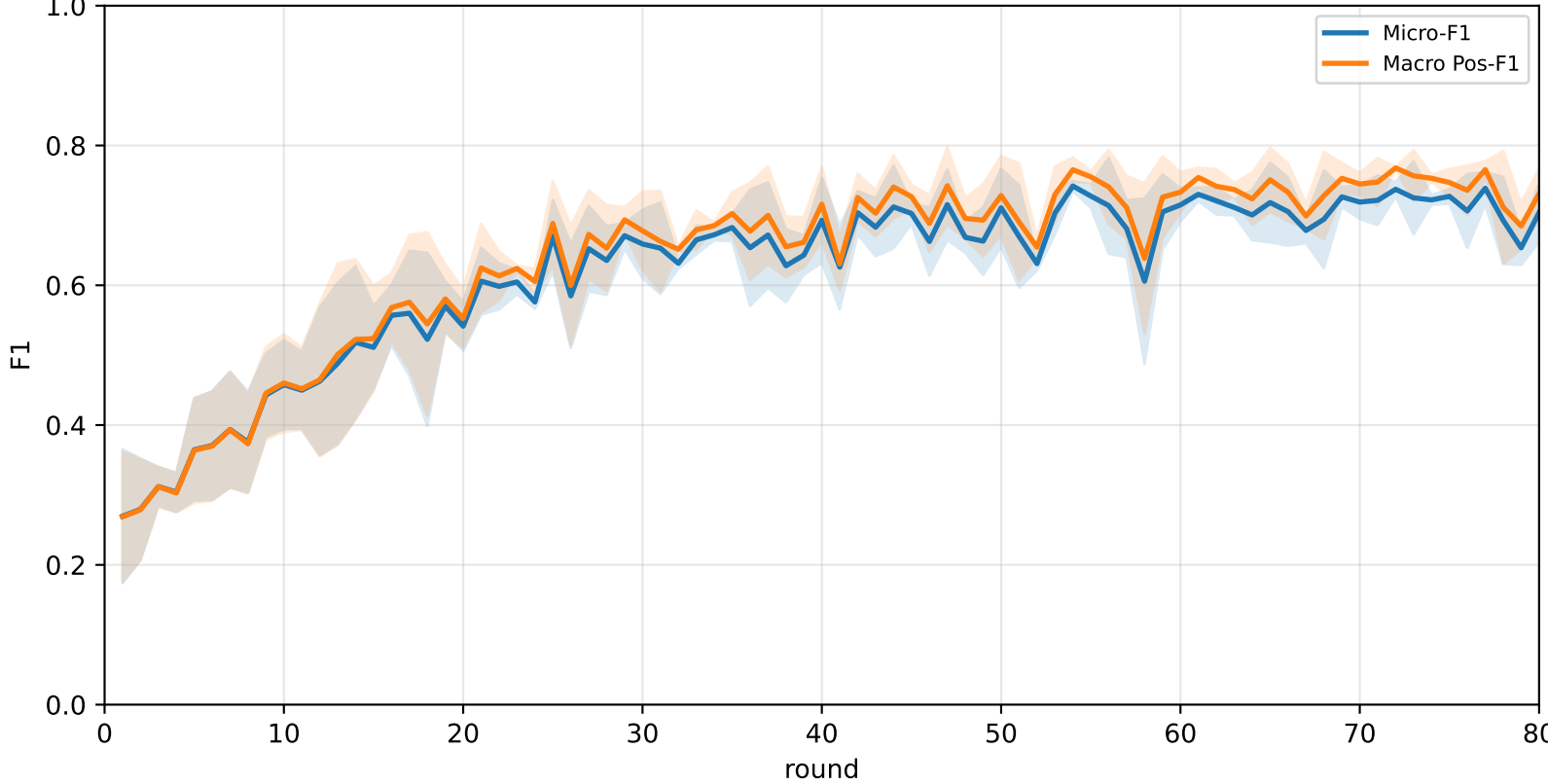
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



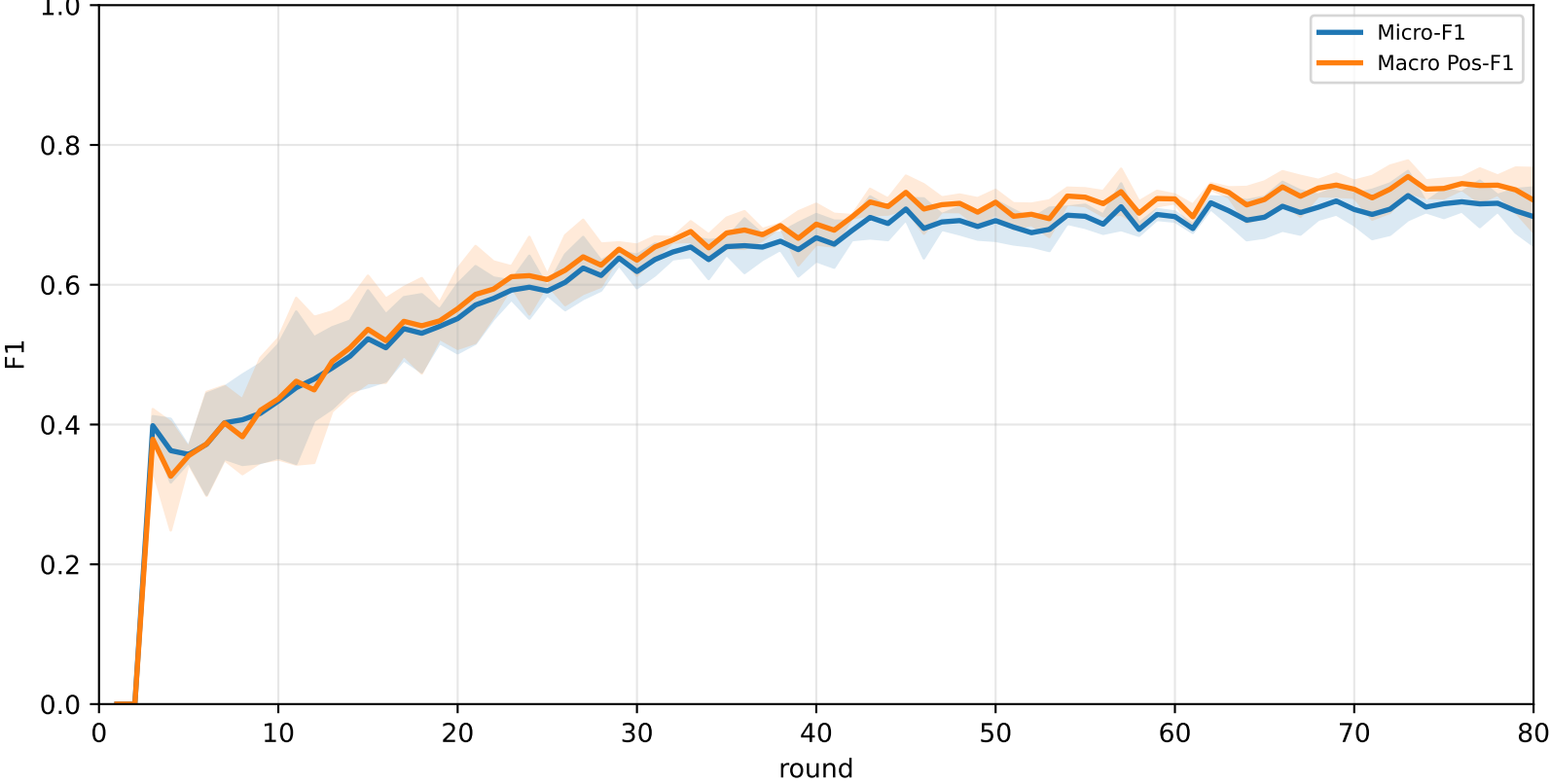
Pooled Local baseline micro / macro F1



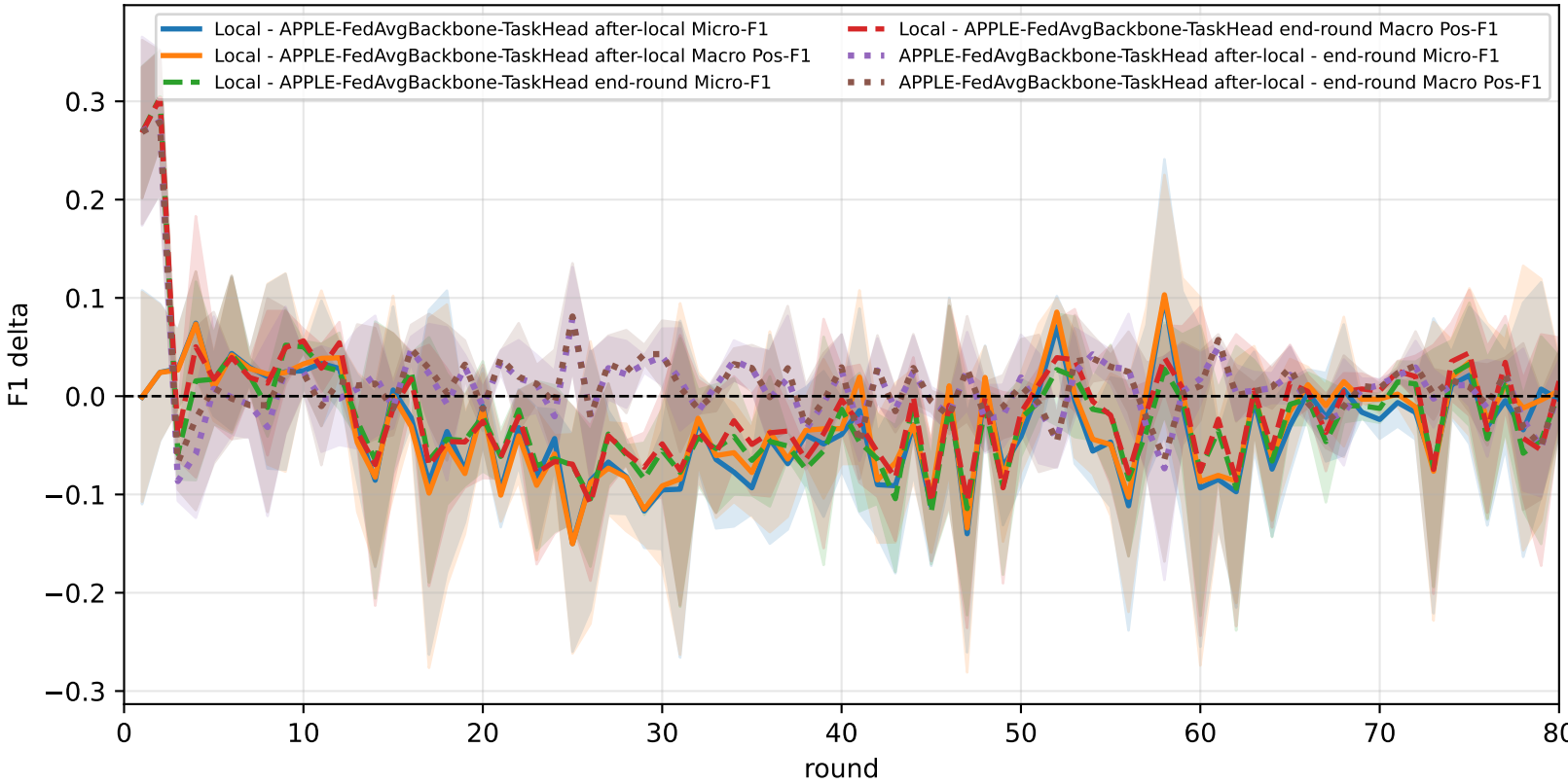
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



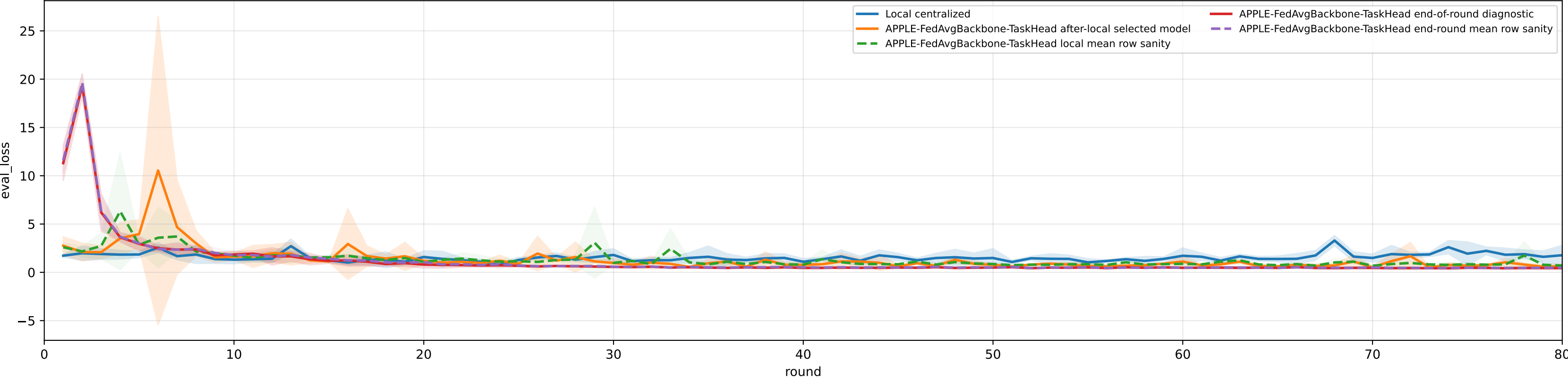
Controlled q-label heterogeneity: Mid | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q task_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% |
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

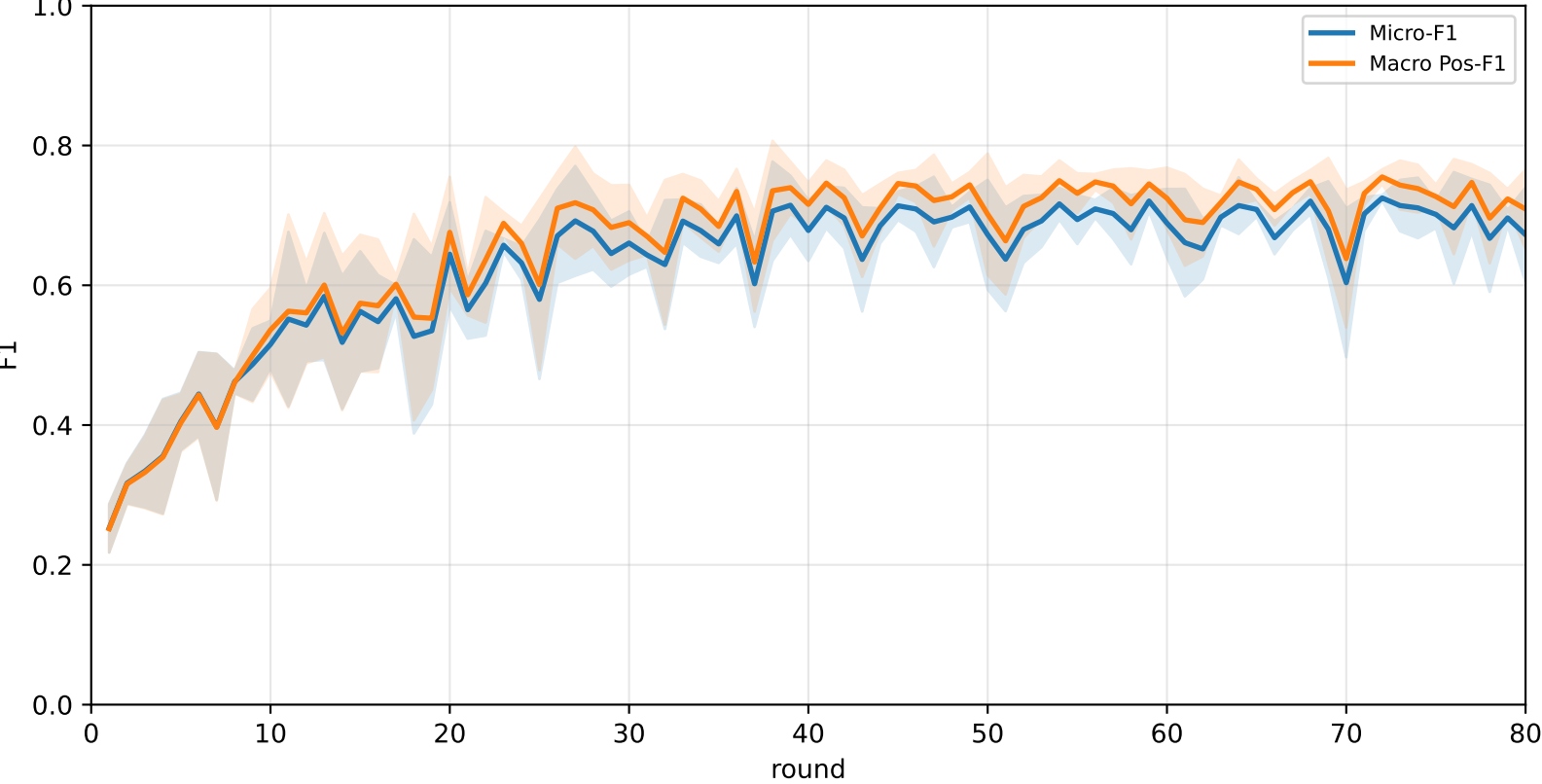
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	62.7% ± 2.4%	65.2% ± 2.6%	76.5% ± 5.8%	75.7% ± 4.0%	59.8% ± 2.9%	64.6% ± 3.0%	49.4% ± 2.2%
APPLE-FedAvgBackbone-TaskHead	72.1% ± 1.2%	74.6% ± 1.3%	87.4% ± 4.8%	85.5% ± 2.1%	72.8% ± 3.6%	66.6% ± 3.7%	60.5% ± 2.0%

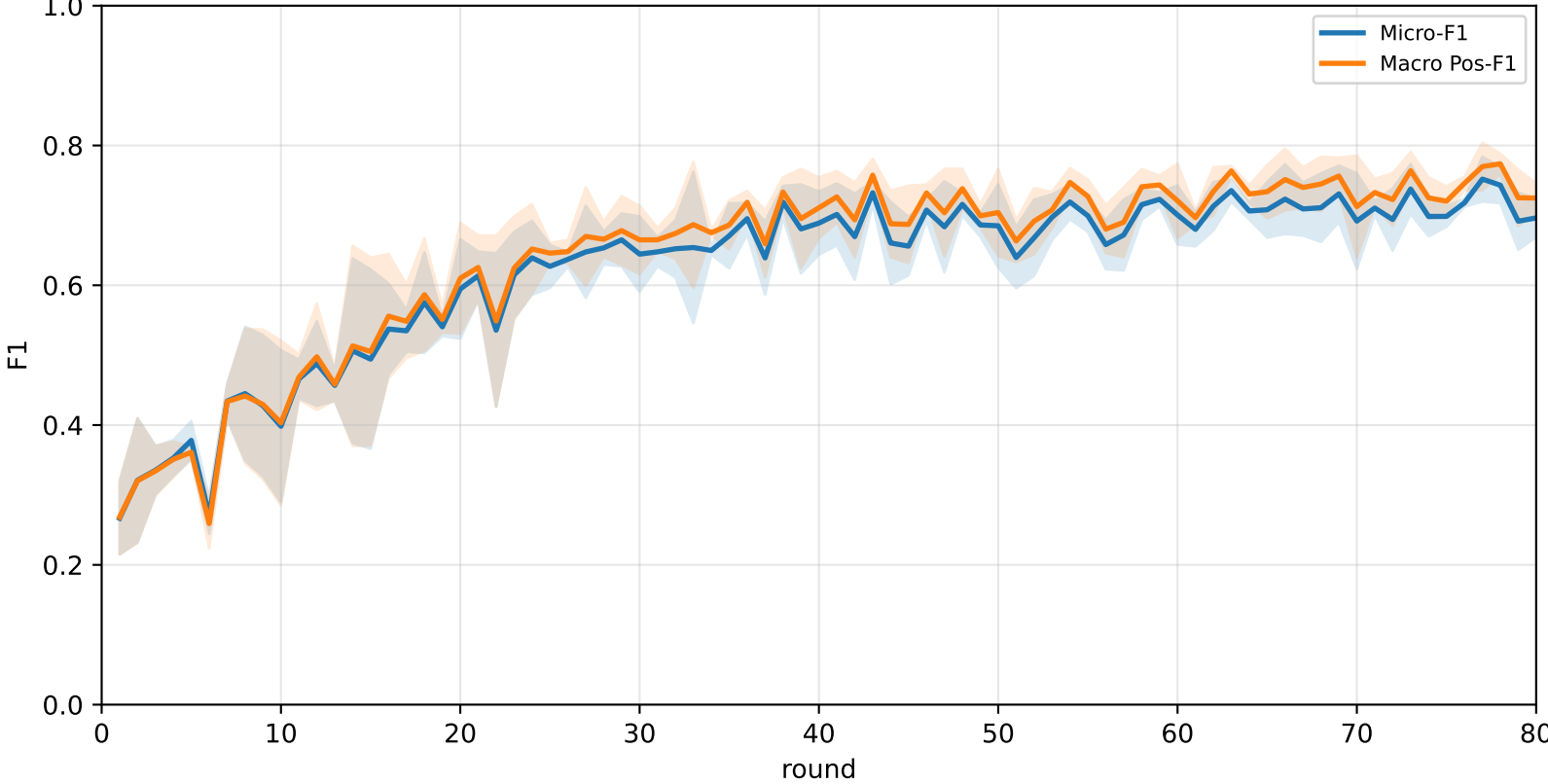
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



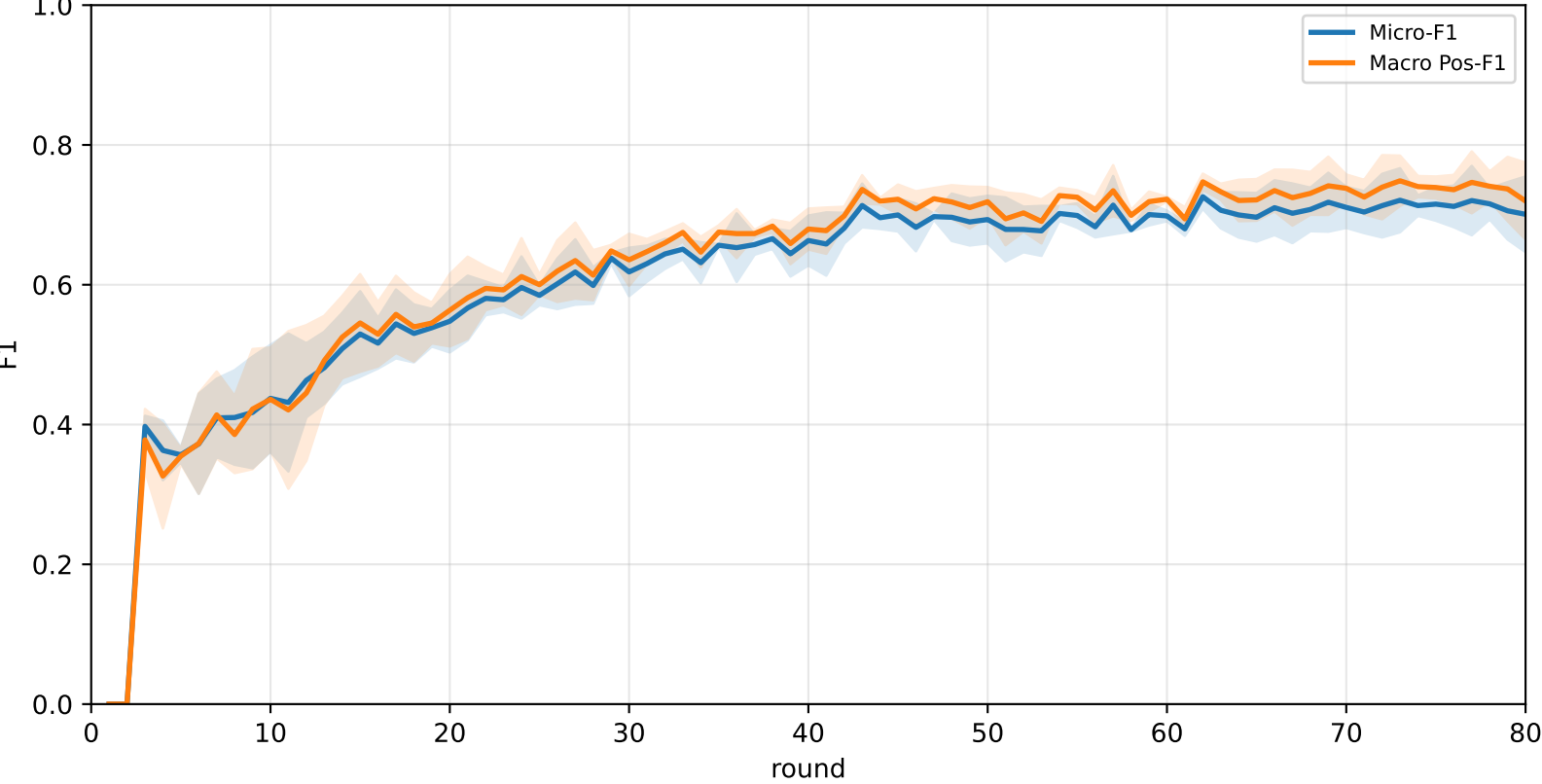
Pooled Local baseline micro / macro F1



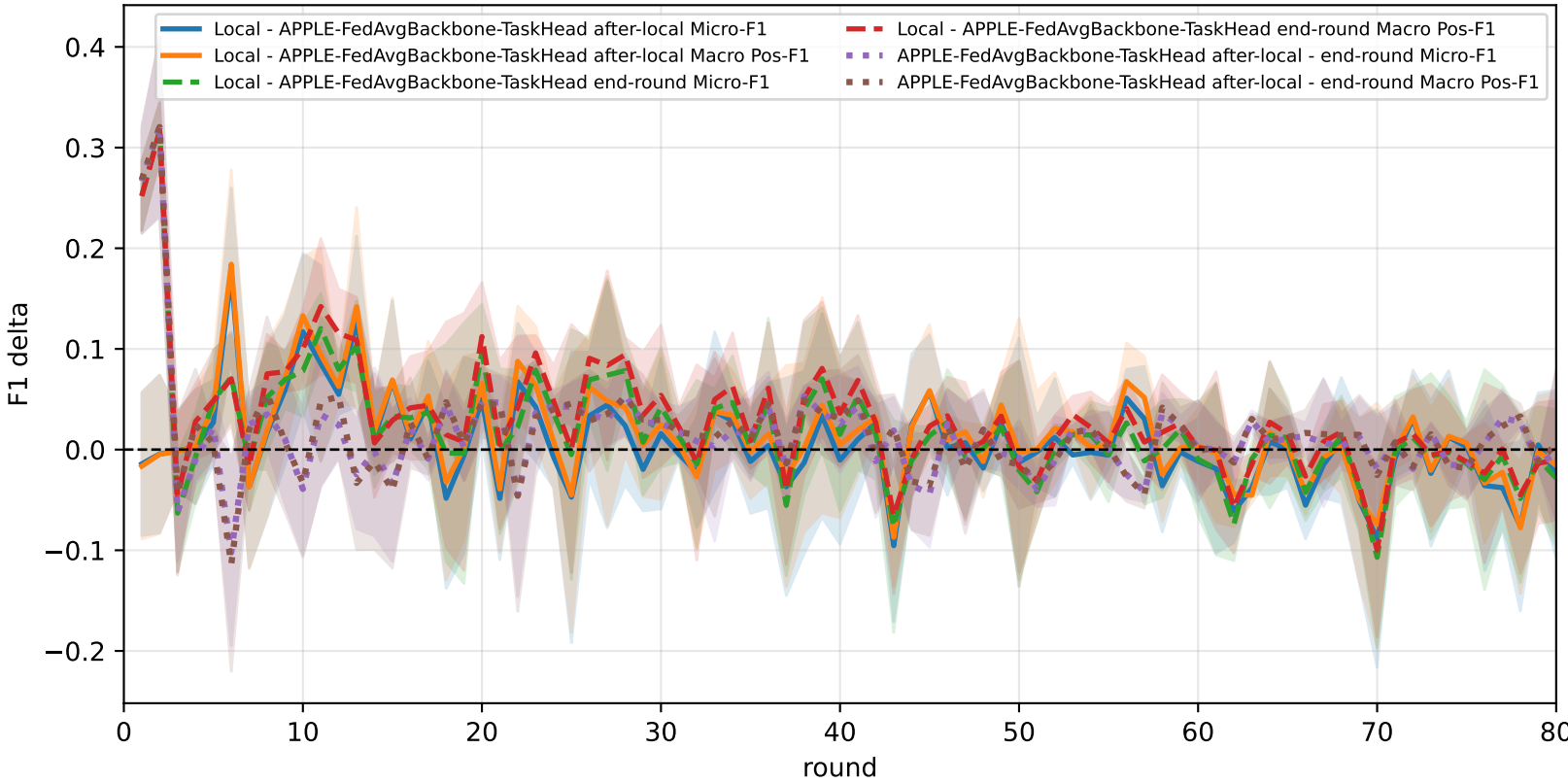
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



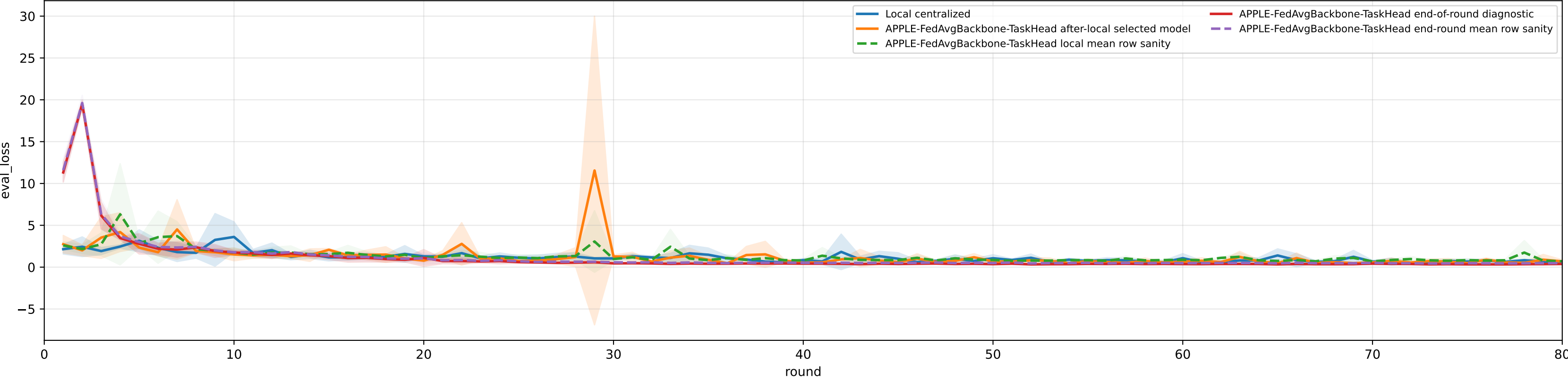
Controlled q-label heterogeneity: Mid | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q task cycle6 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% |
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

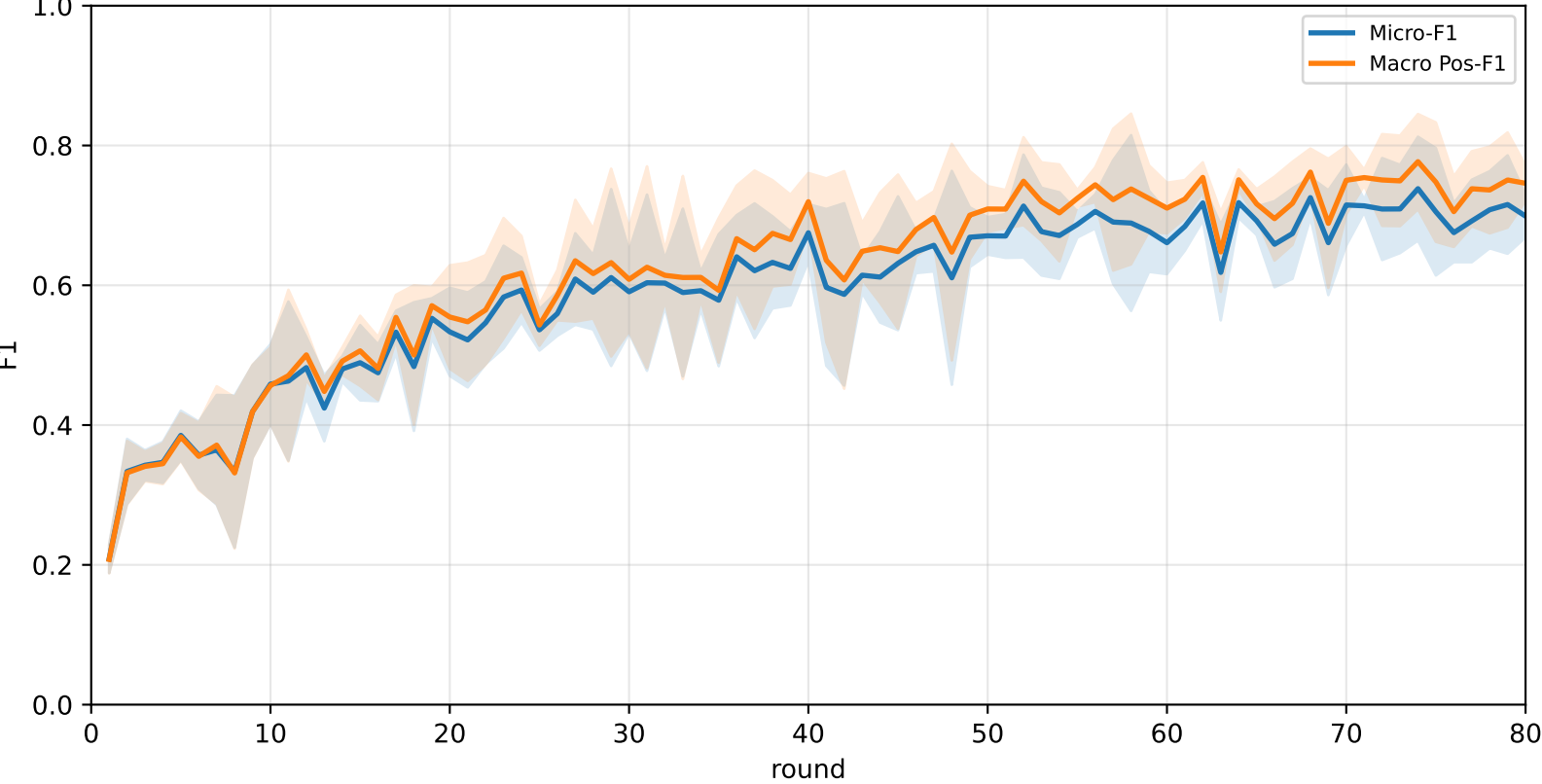
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	63.1% ± 3.5%	67.2% ± 3.7%	86.9% ± 6.6%	77.7% ± 9.8%	63.1% ± 2.3%	58.3% ± 3.3%	49.9% ± 3.9%
APPLE-FedAvgBackbone-TaskHead	65.9% ± 1.8%	68.8% ± 0.7%	82.5% ± 5.9%	77.5% ± 3.0%	62.7% ± 3.9%	67.5% ± 3.4%	53.5% ± 3.8%

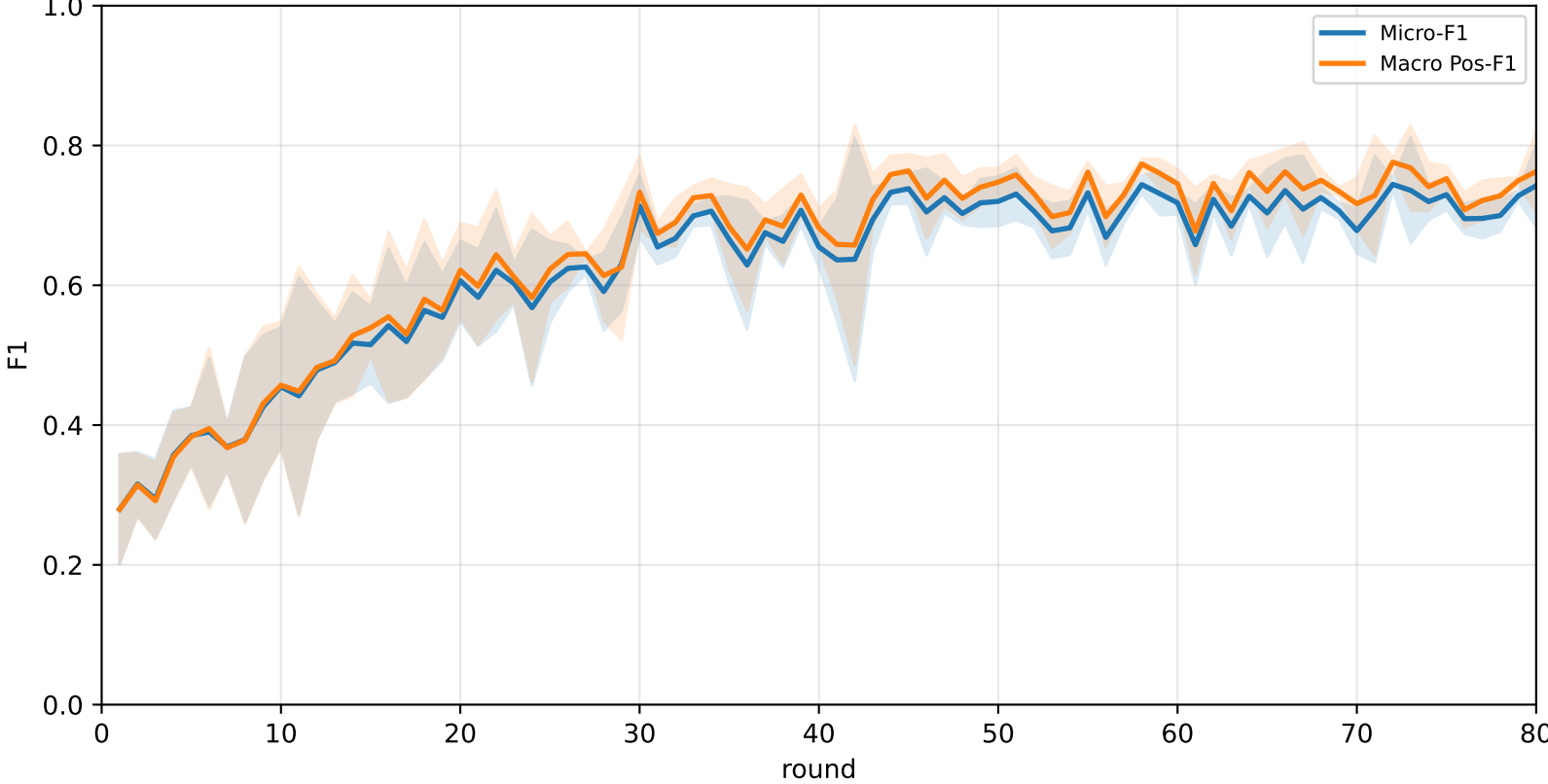
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



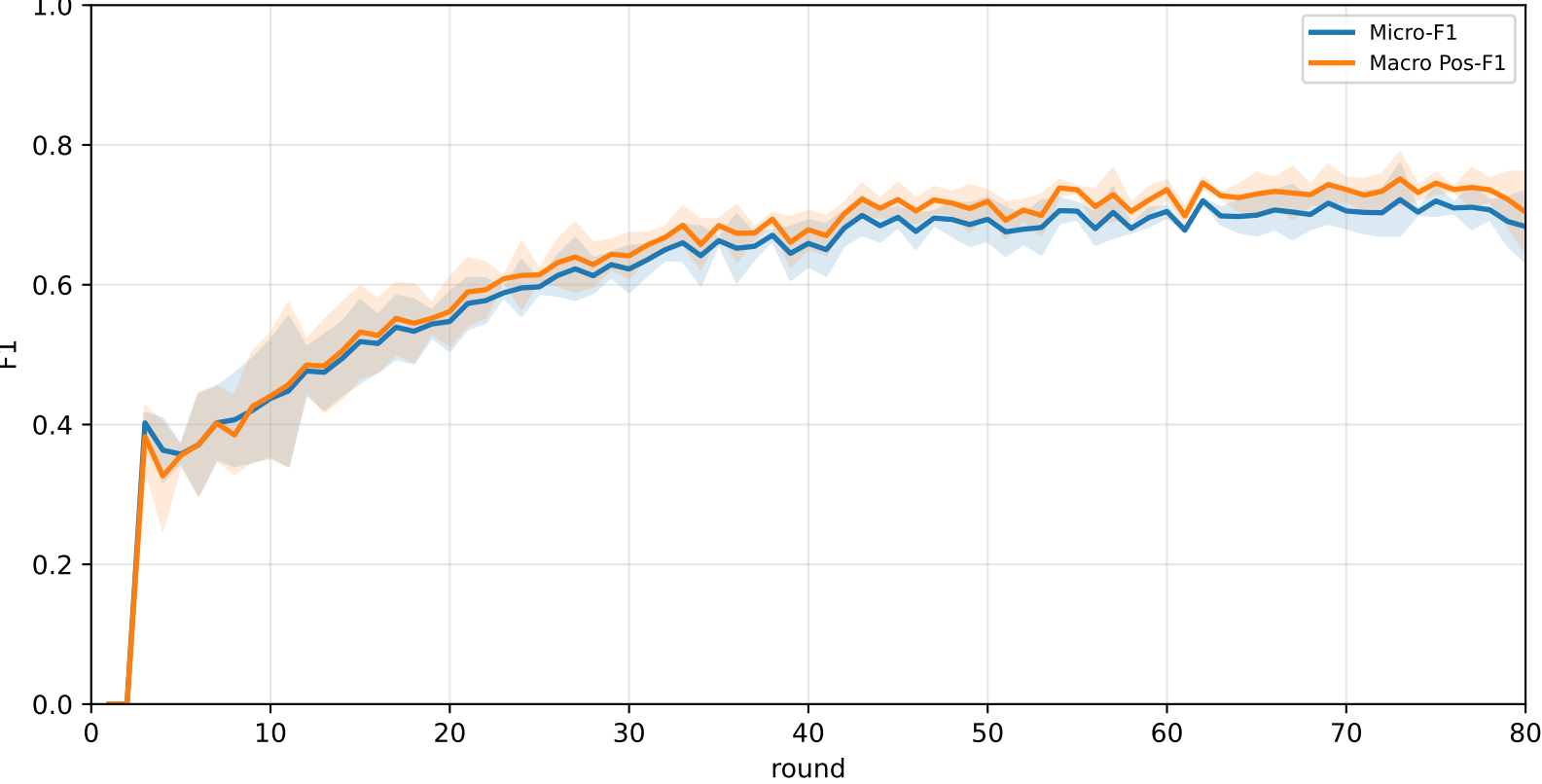
Pooled Local baseline micro / macro F1



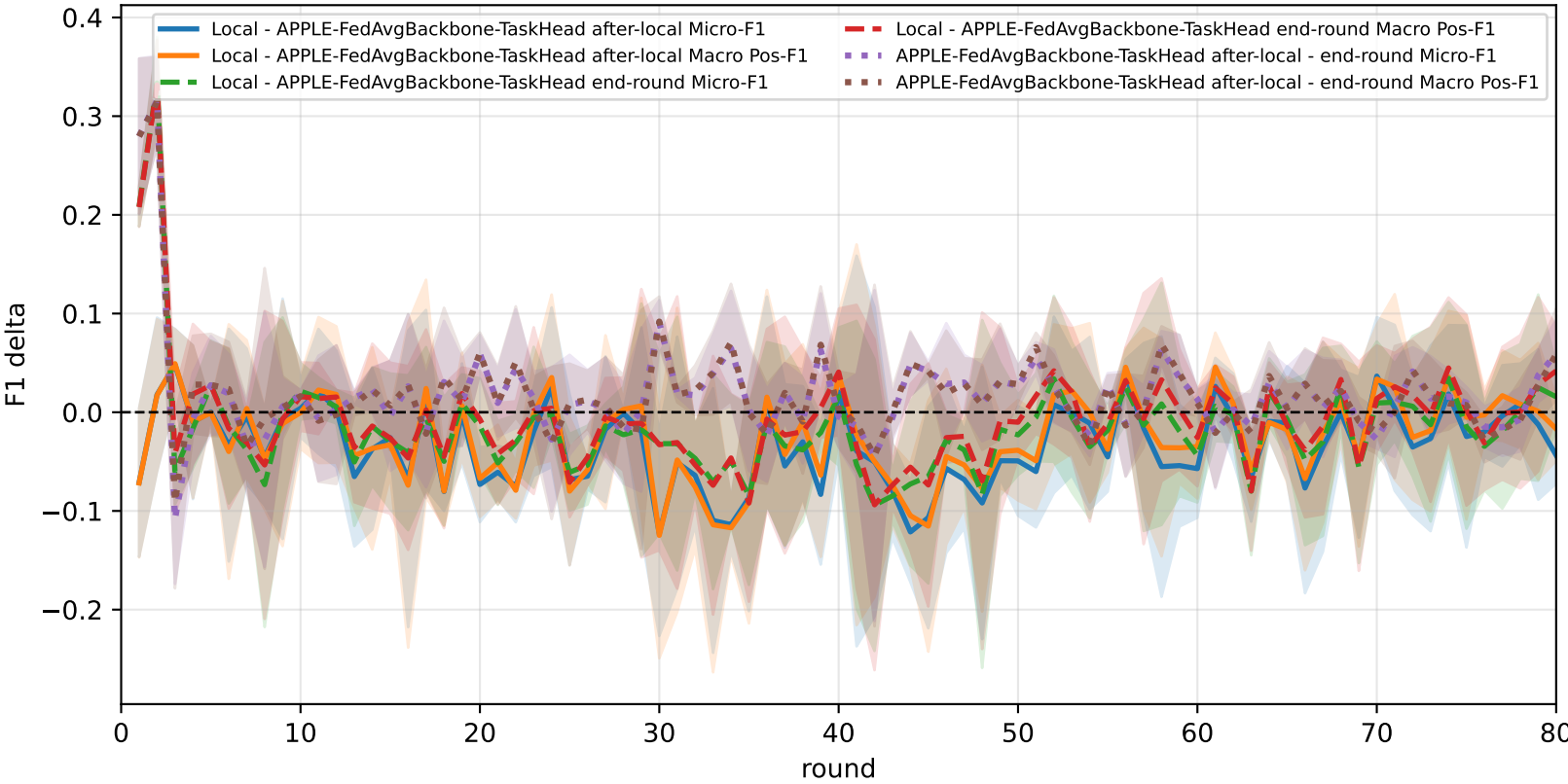
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



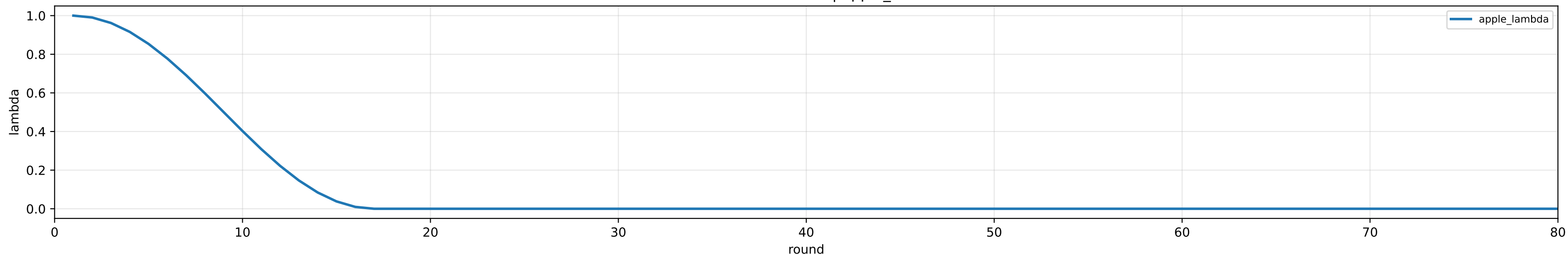
Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



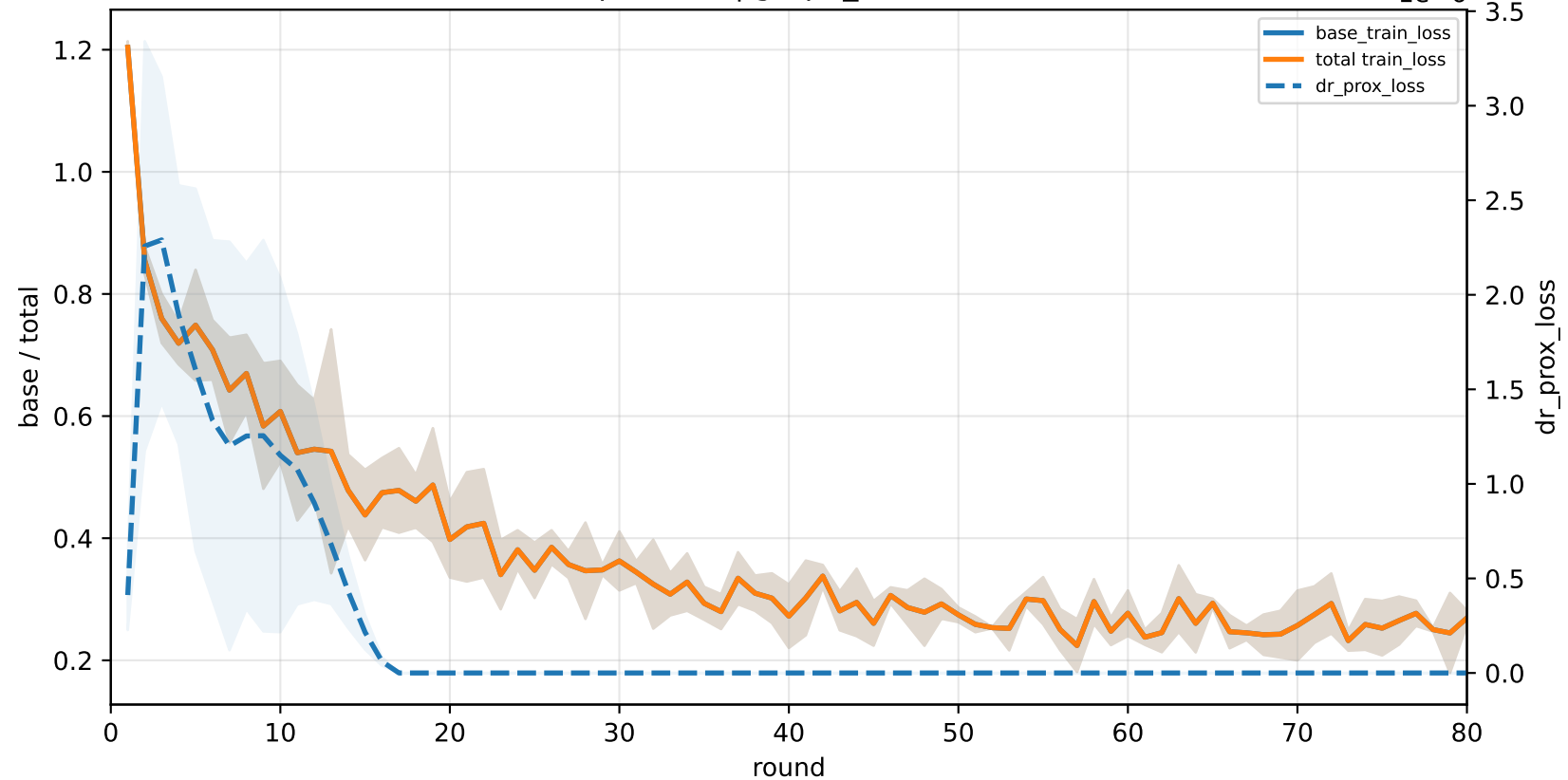
APPLE-FedAvgBackbone-TaskHead training-loss decomposition | Mid heterogeneity

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

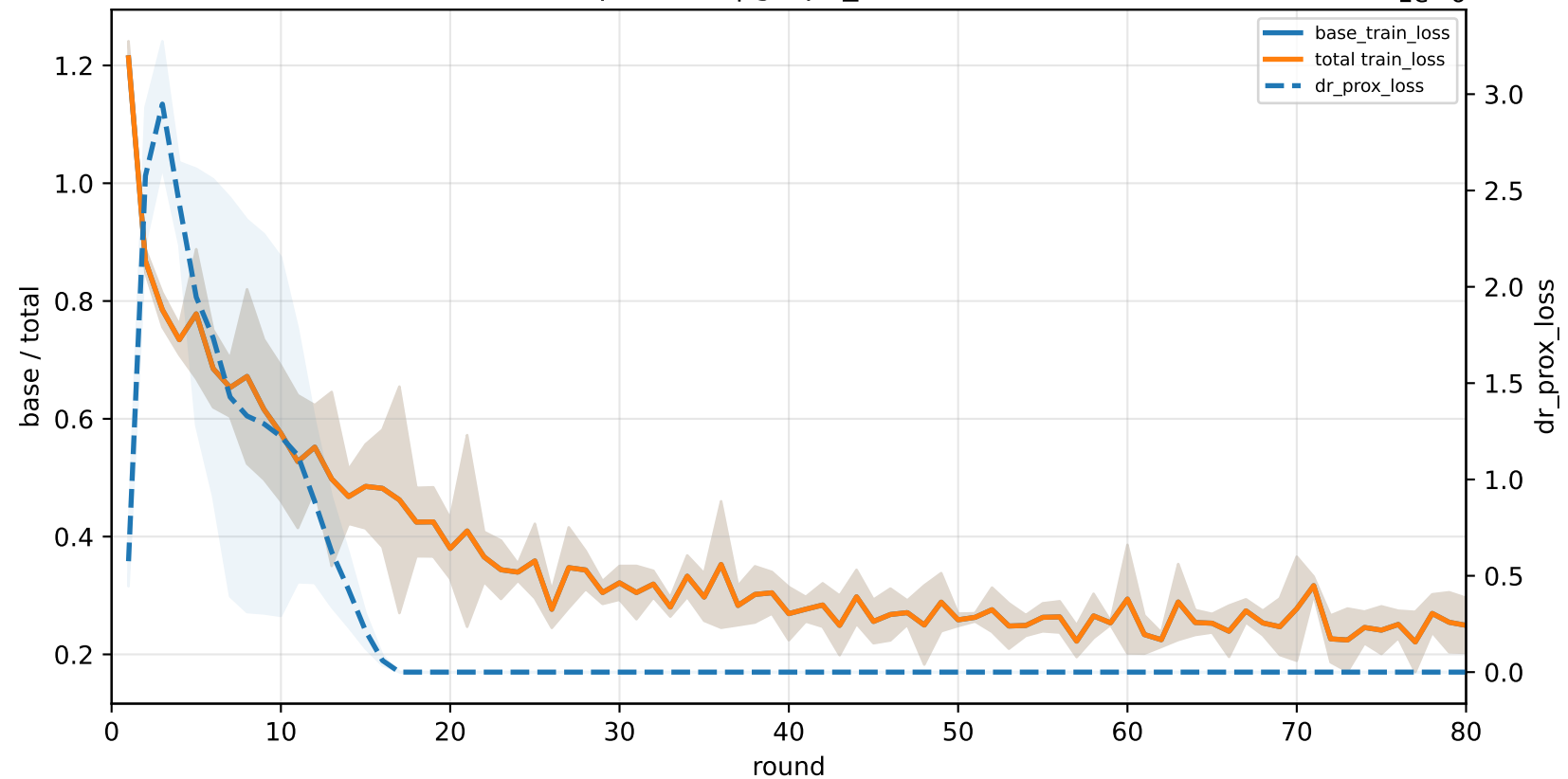
APPLE lambda scheduler | apple_mu=0.001



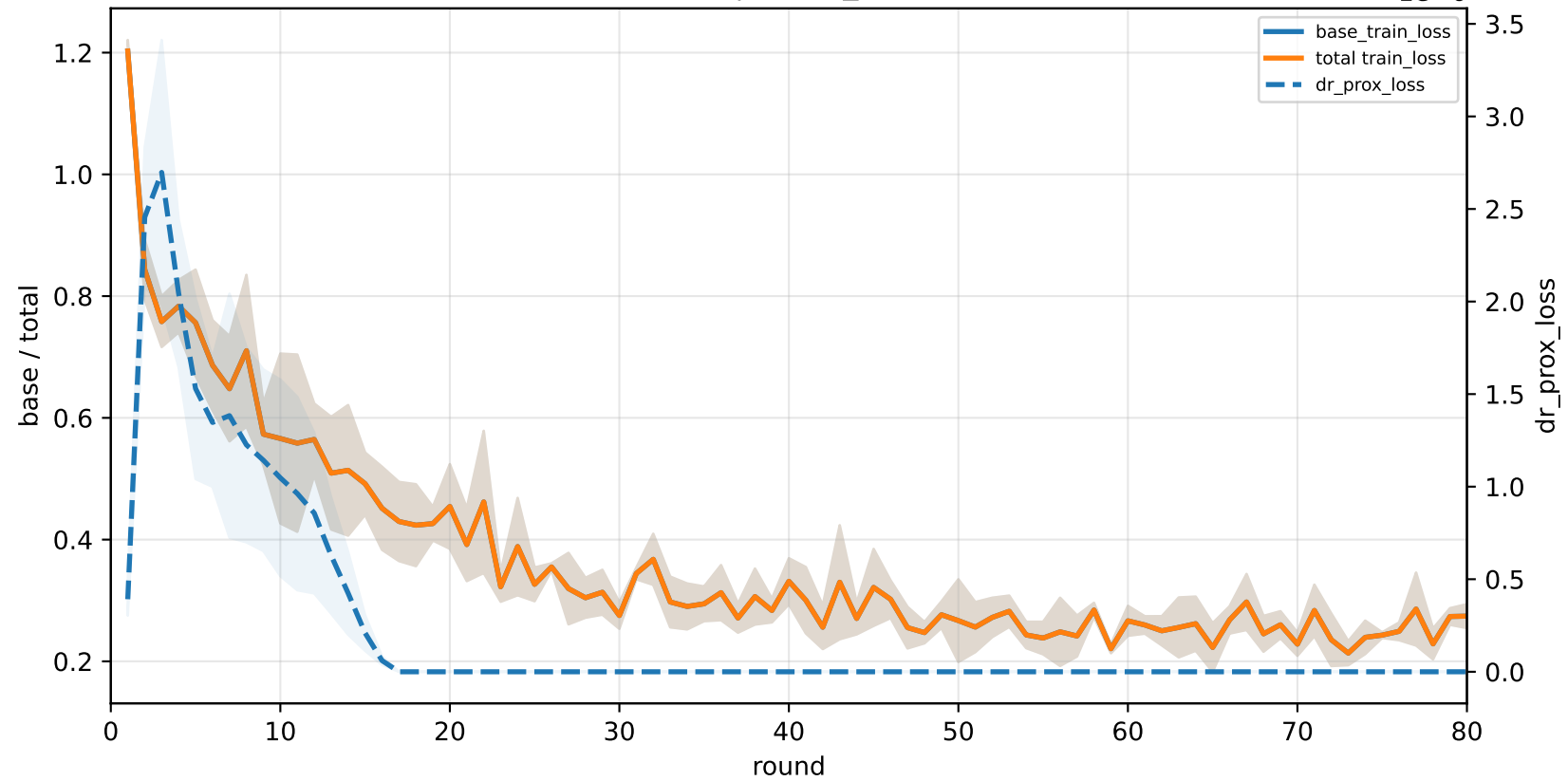
C2-specialist | graph_id=3000005



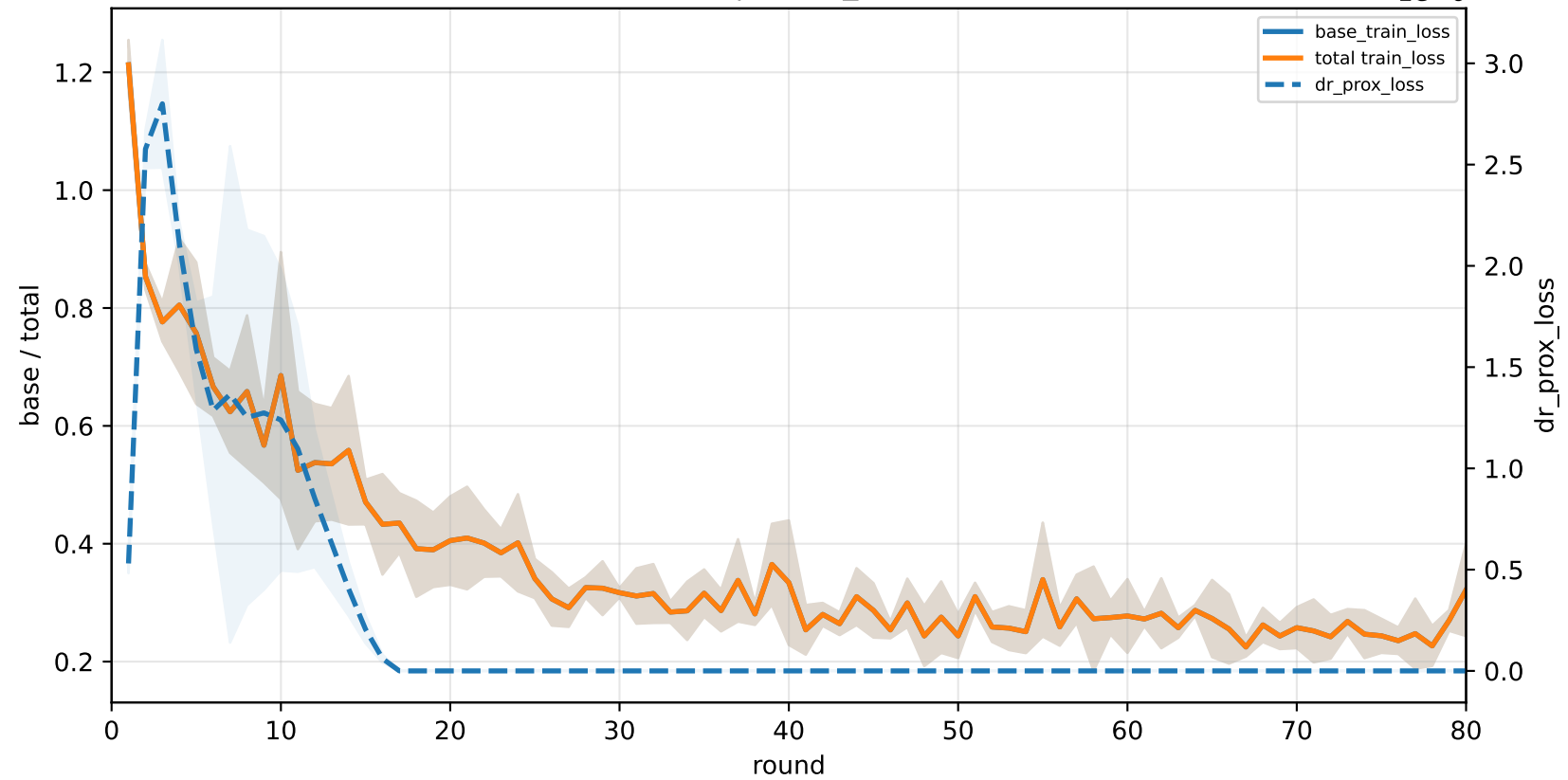
C3-specialist | graph_id=3000006



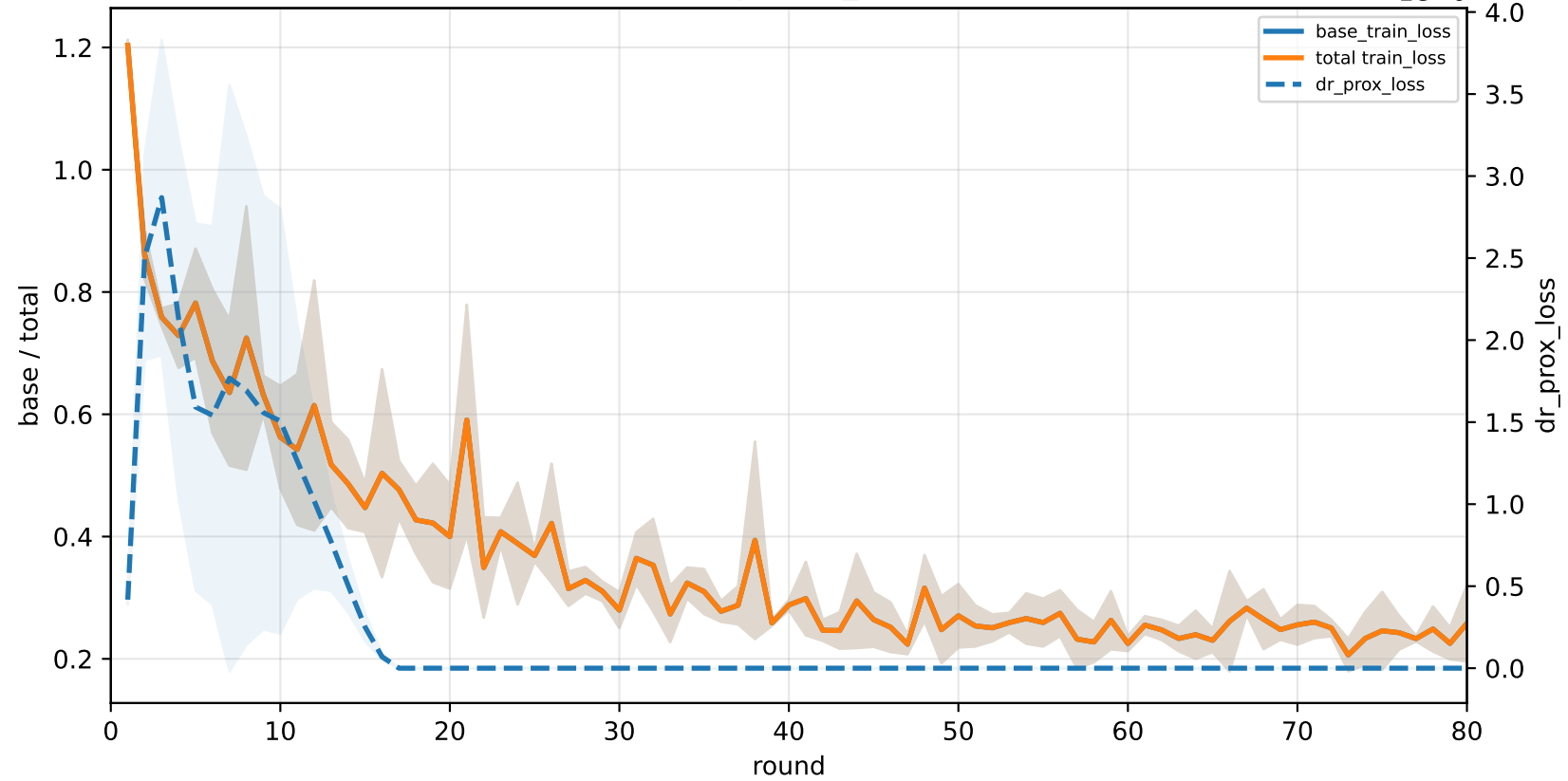
C4-specialist | graph_id=3000007



C5-specialist | graph_id=3000008

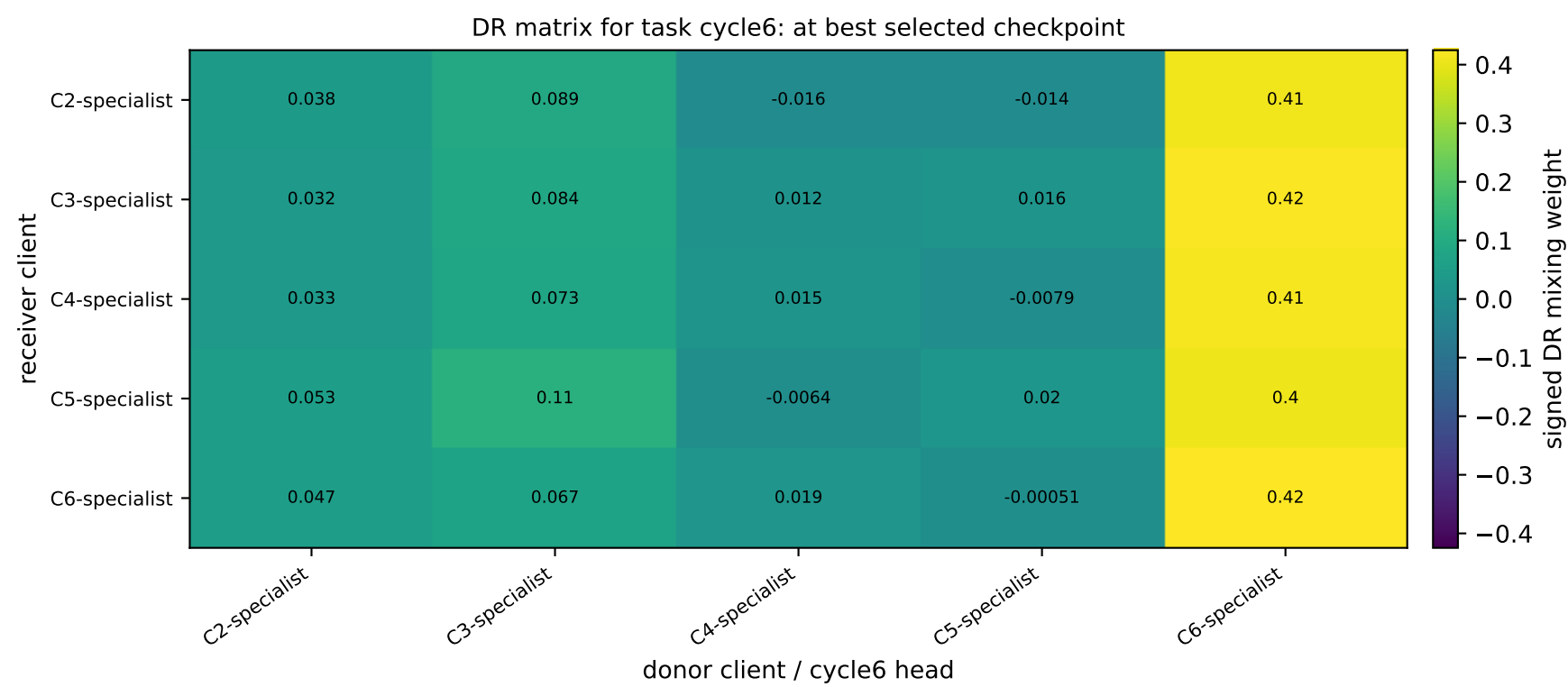
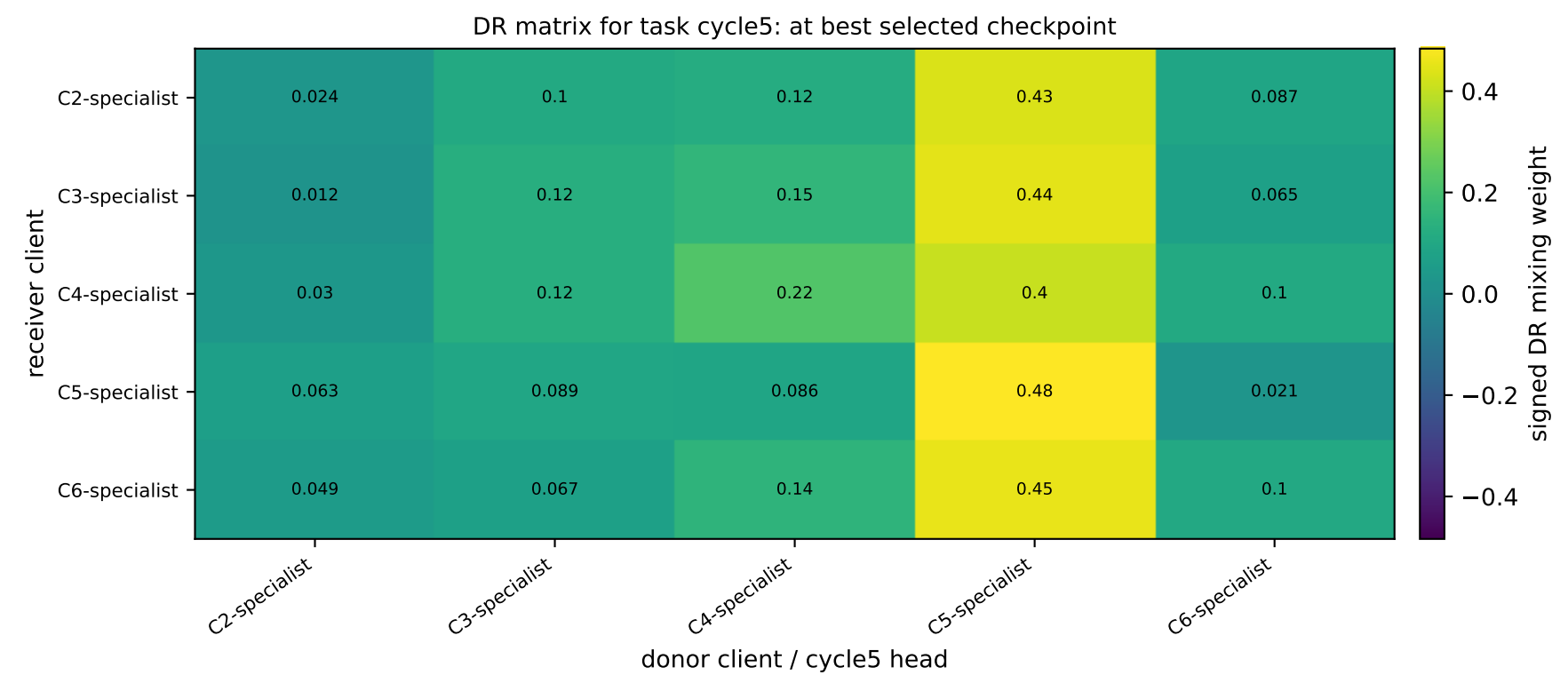
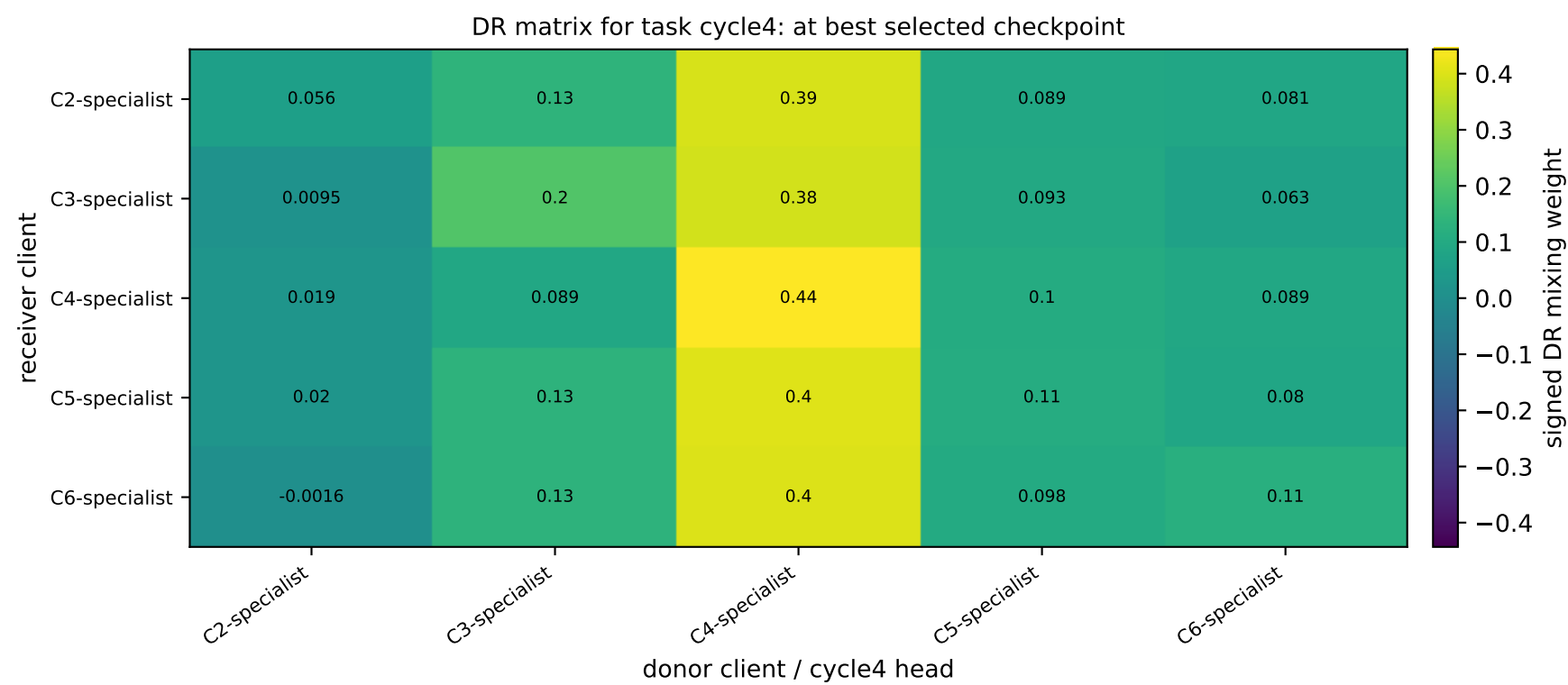
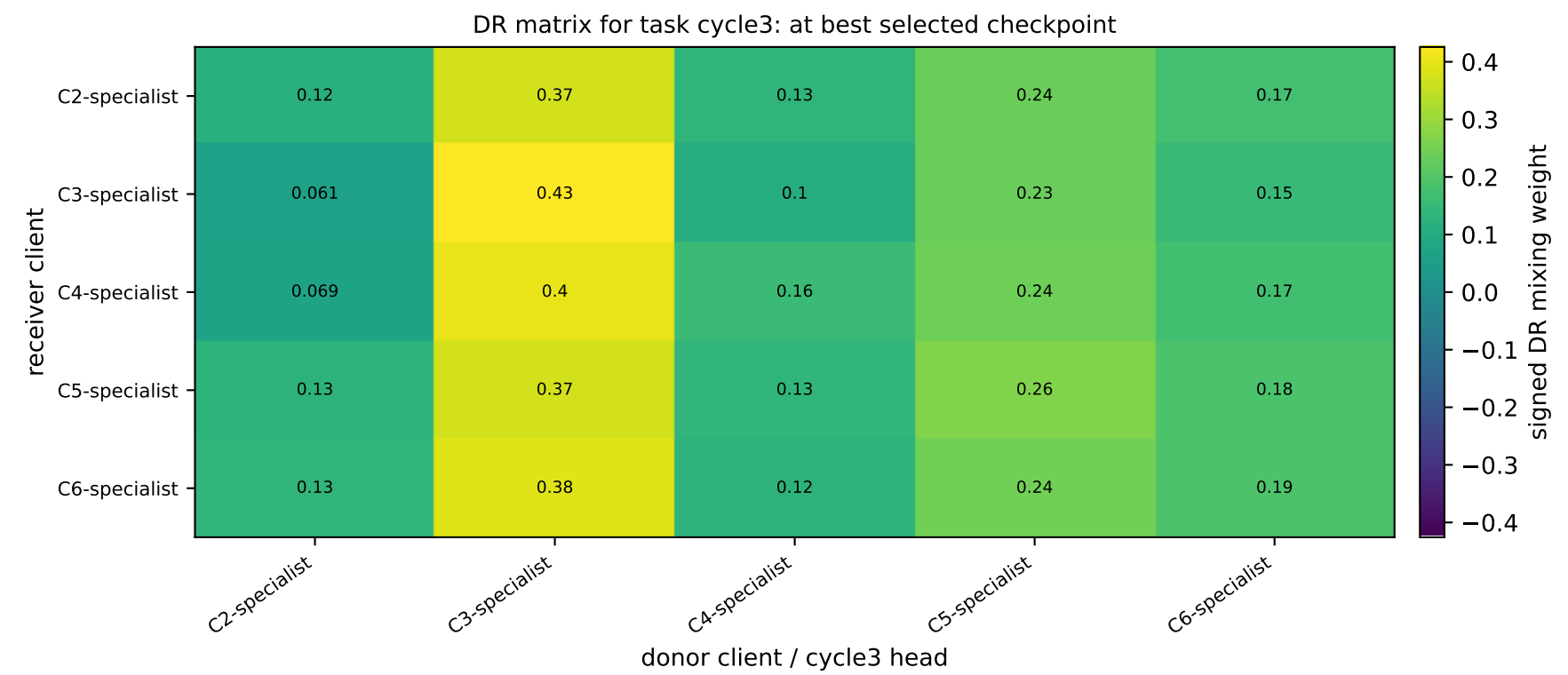
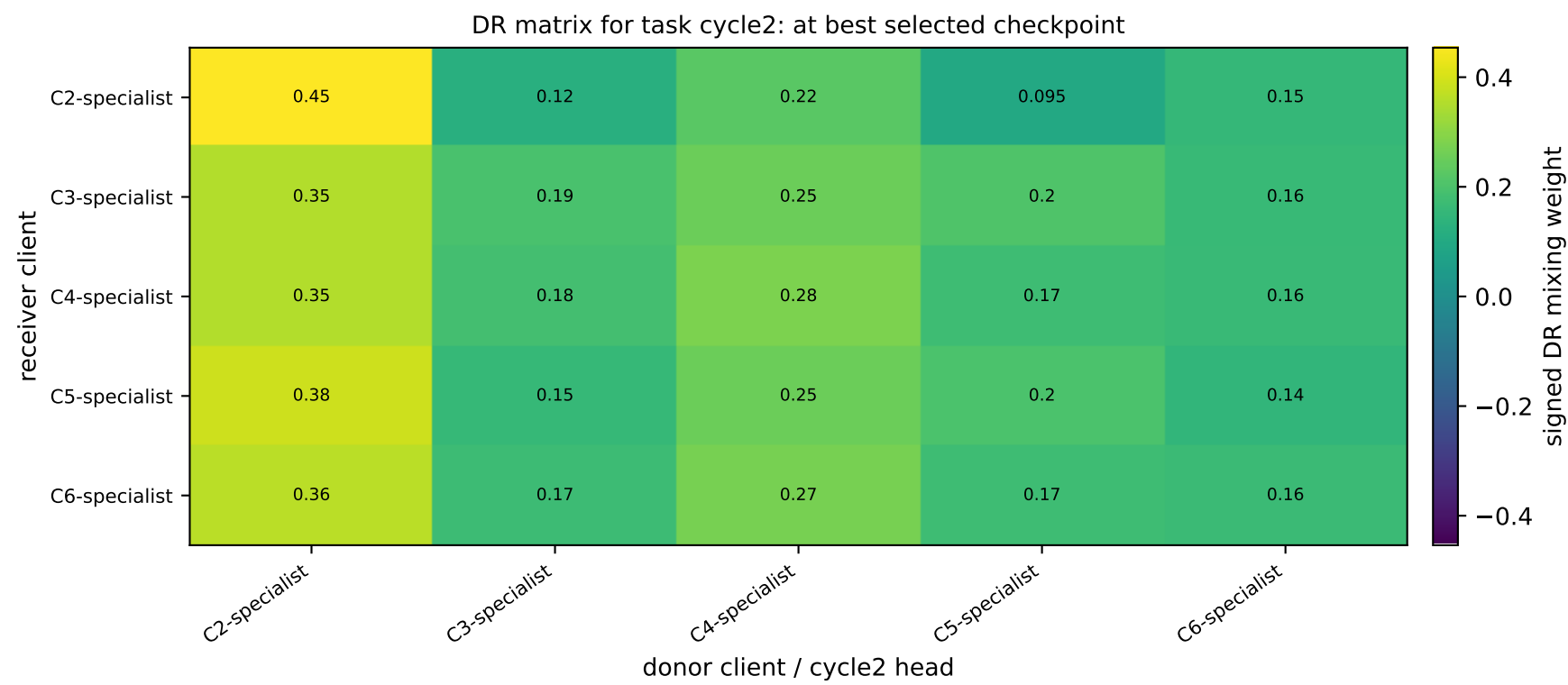


C6-specialist | graph_id=3000009



APPLE-FedAvgBackbone-TaskHead task-wise DR matrices: receiver clients × donor task-heads | Mid

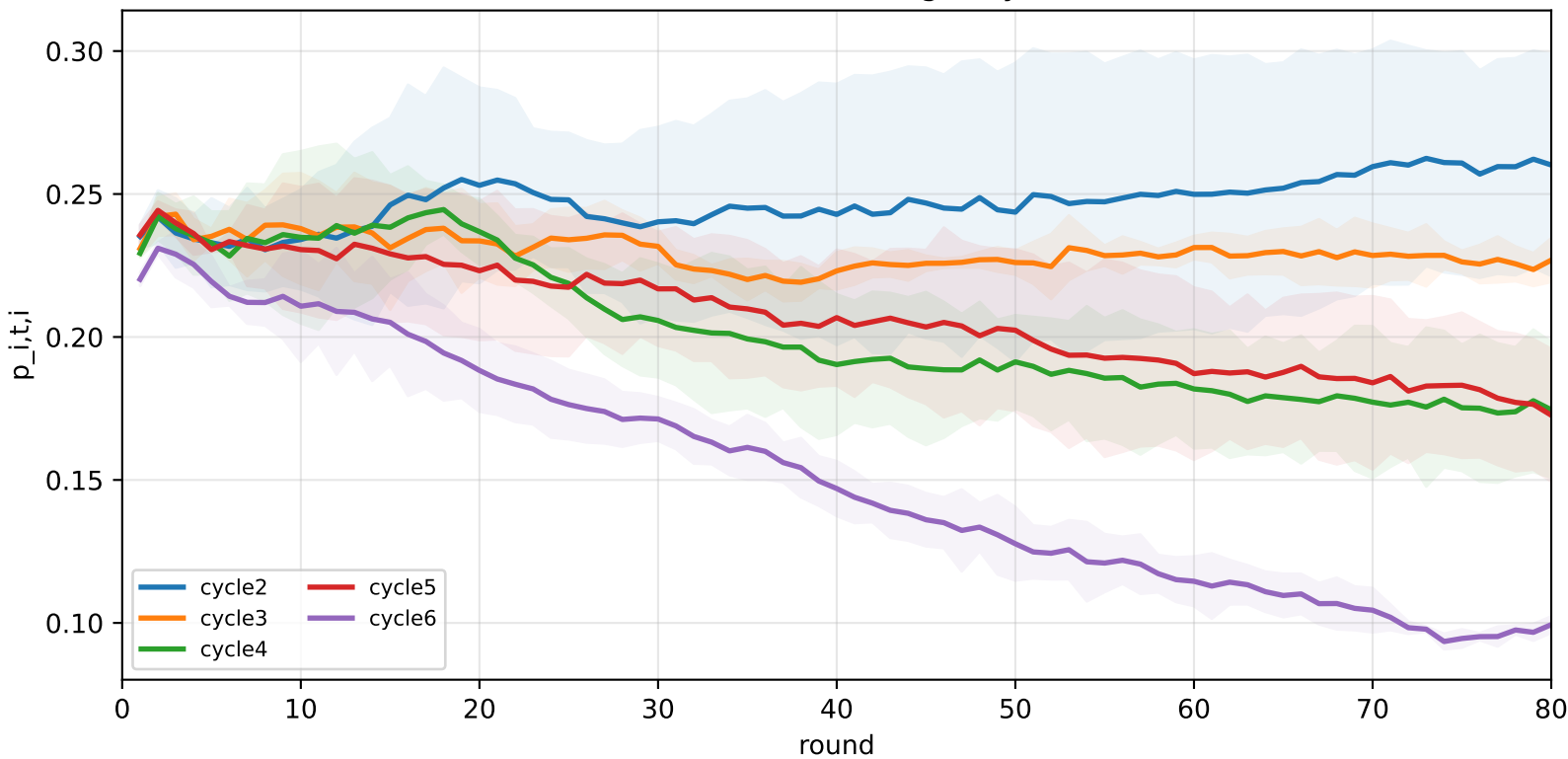
family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120



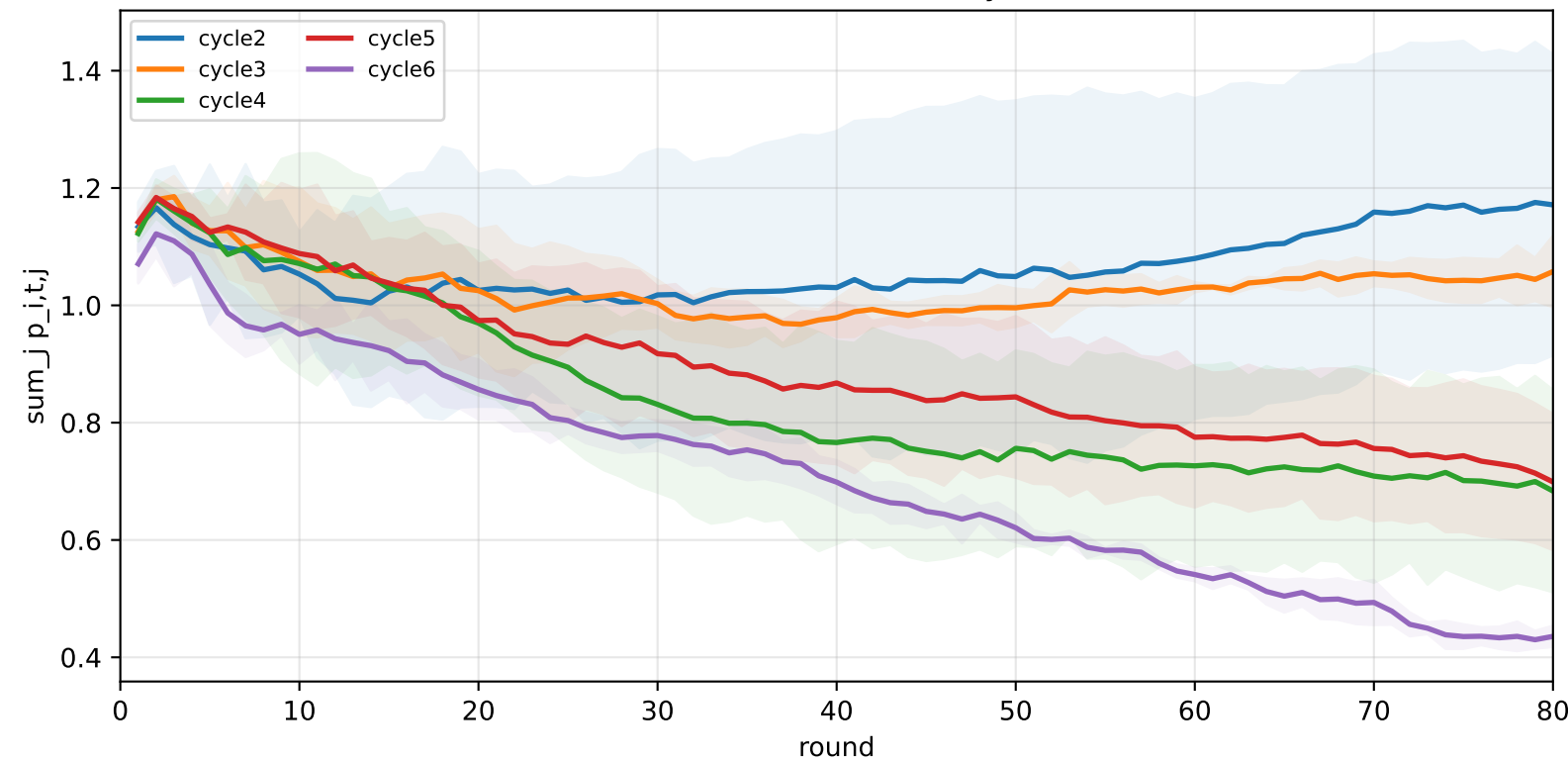
APPLE-FedAvgBackbone-TaskHead task-wise DR dynamics | Mid

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
 family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
 family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
 family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
 family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
 heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

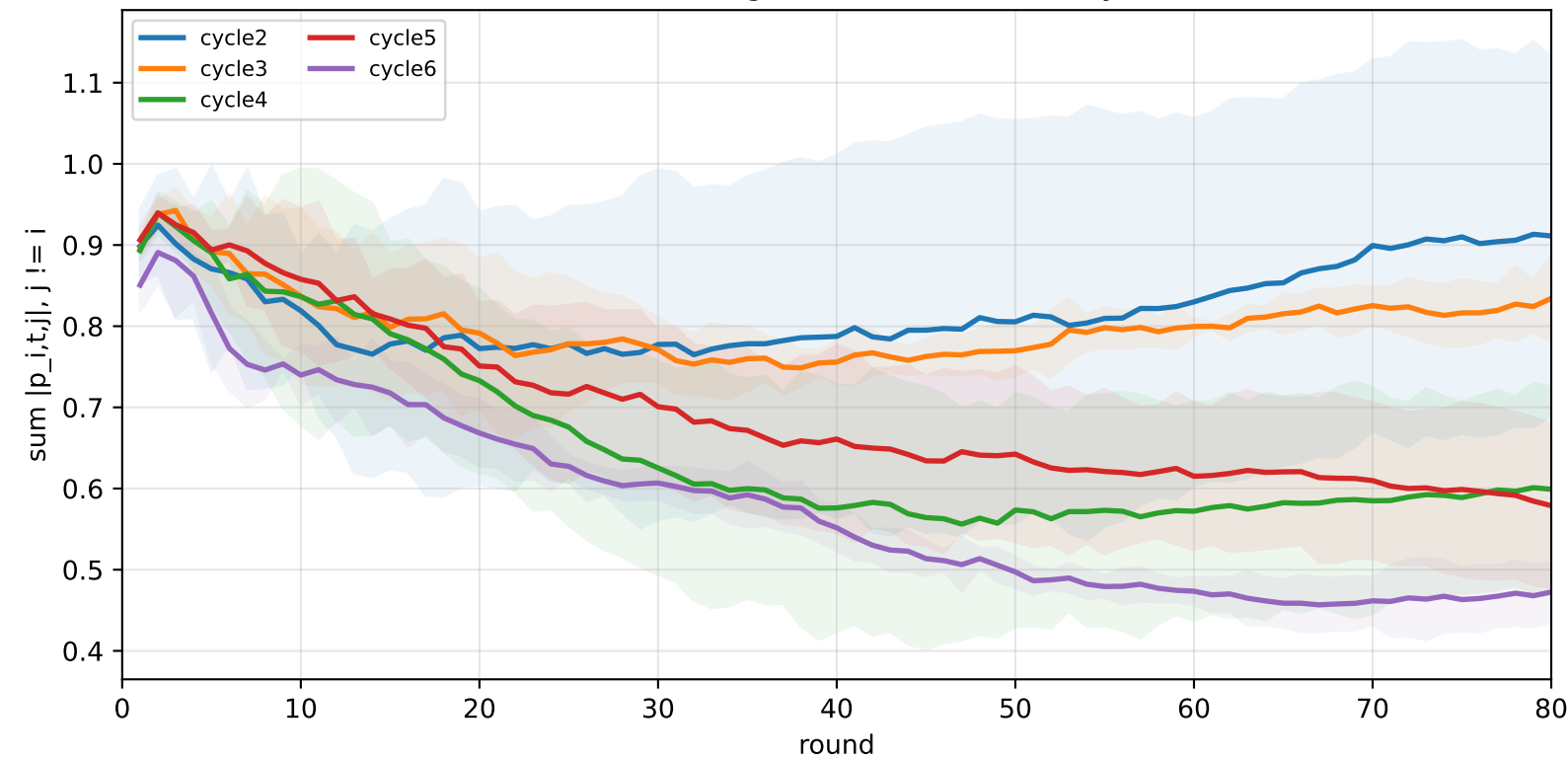
Task-head DR self-weight by task



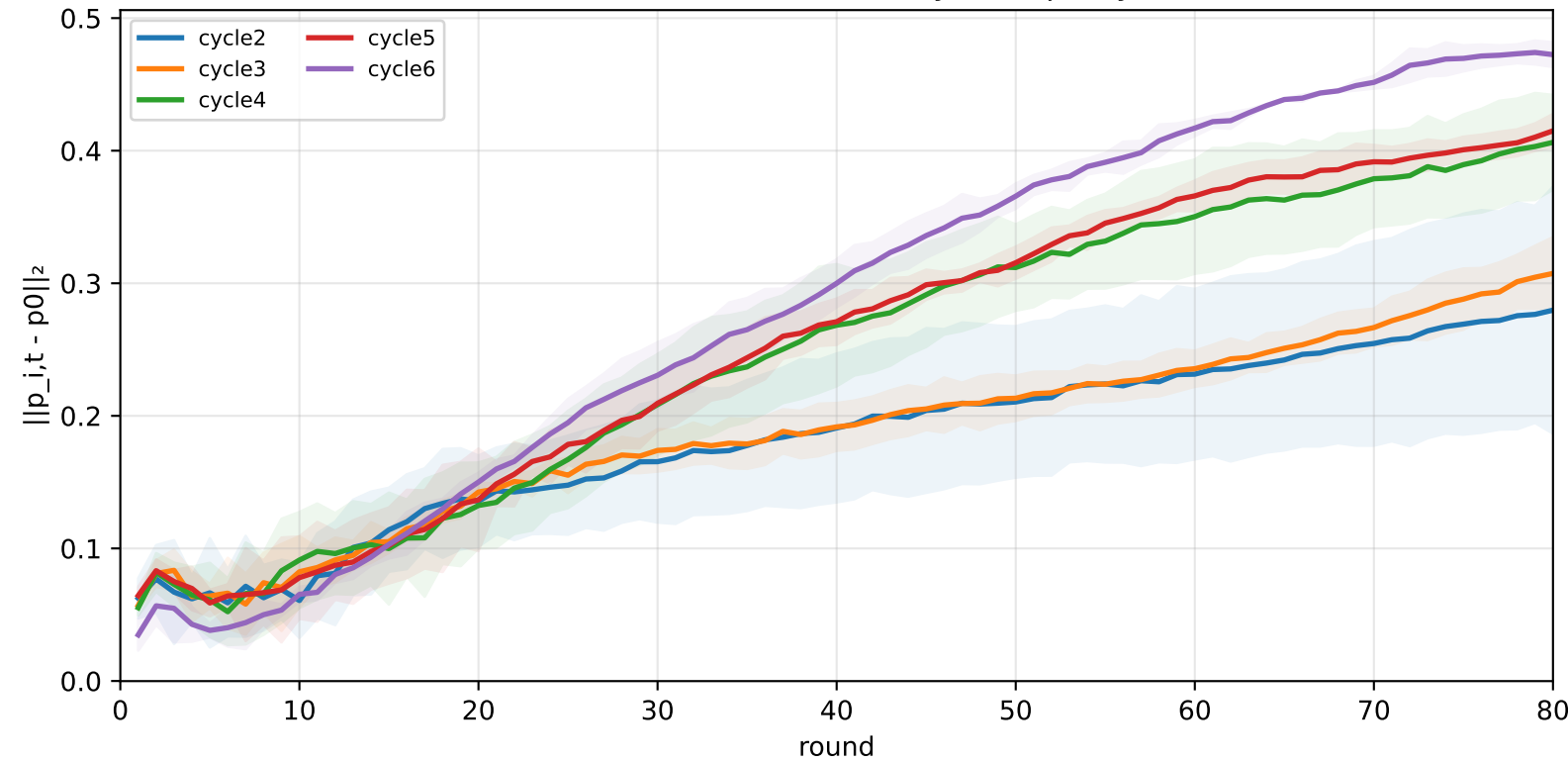
Task-head DR row sum by task



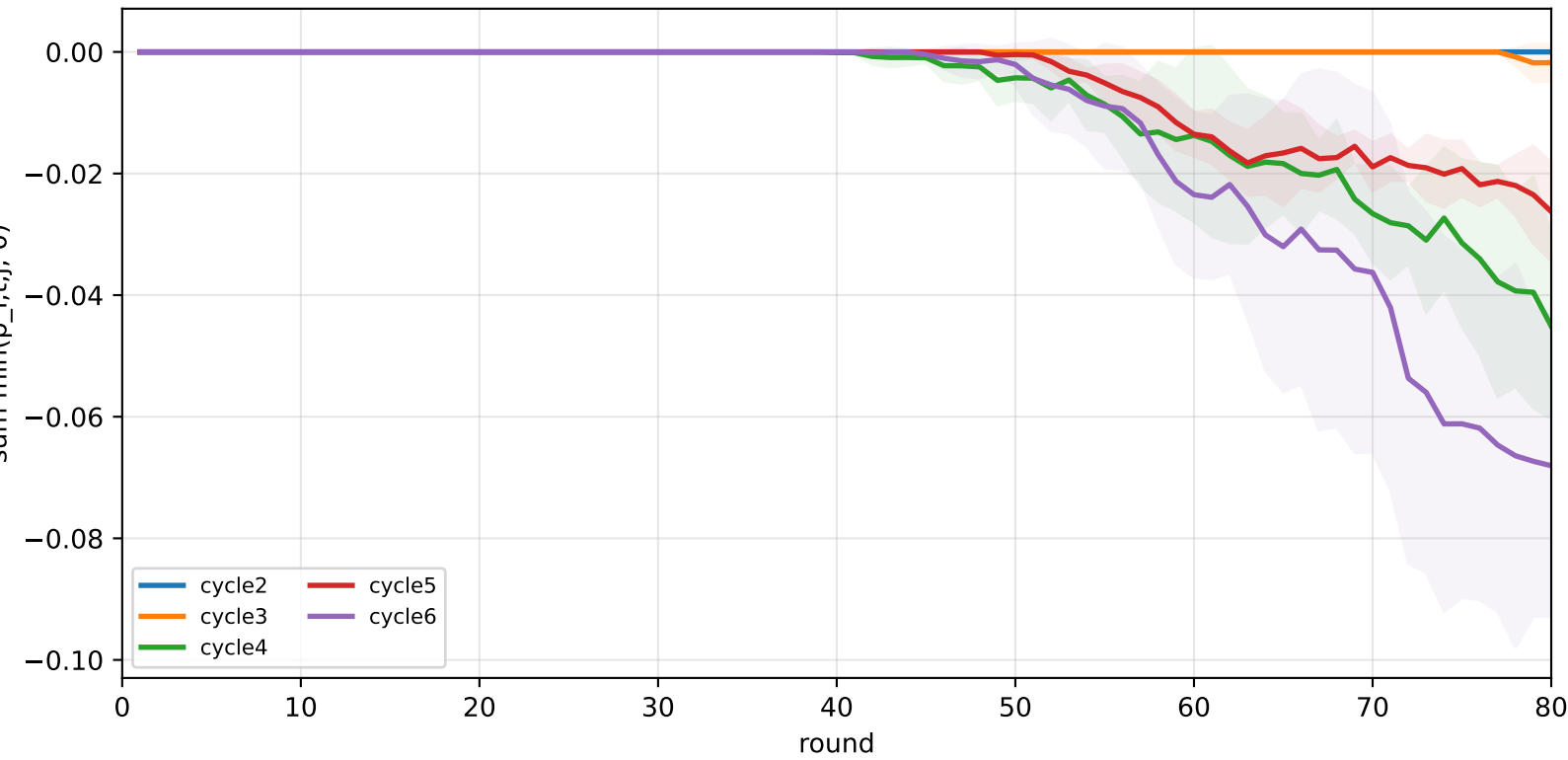
Task-head off-diagonal absolute mass by task



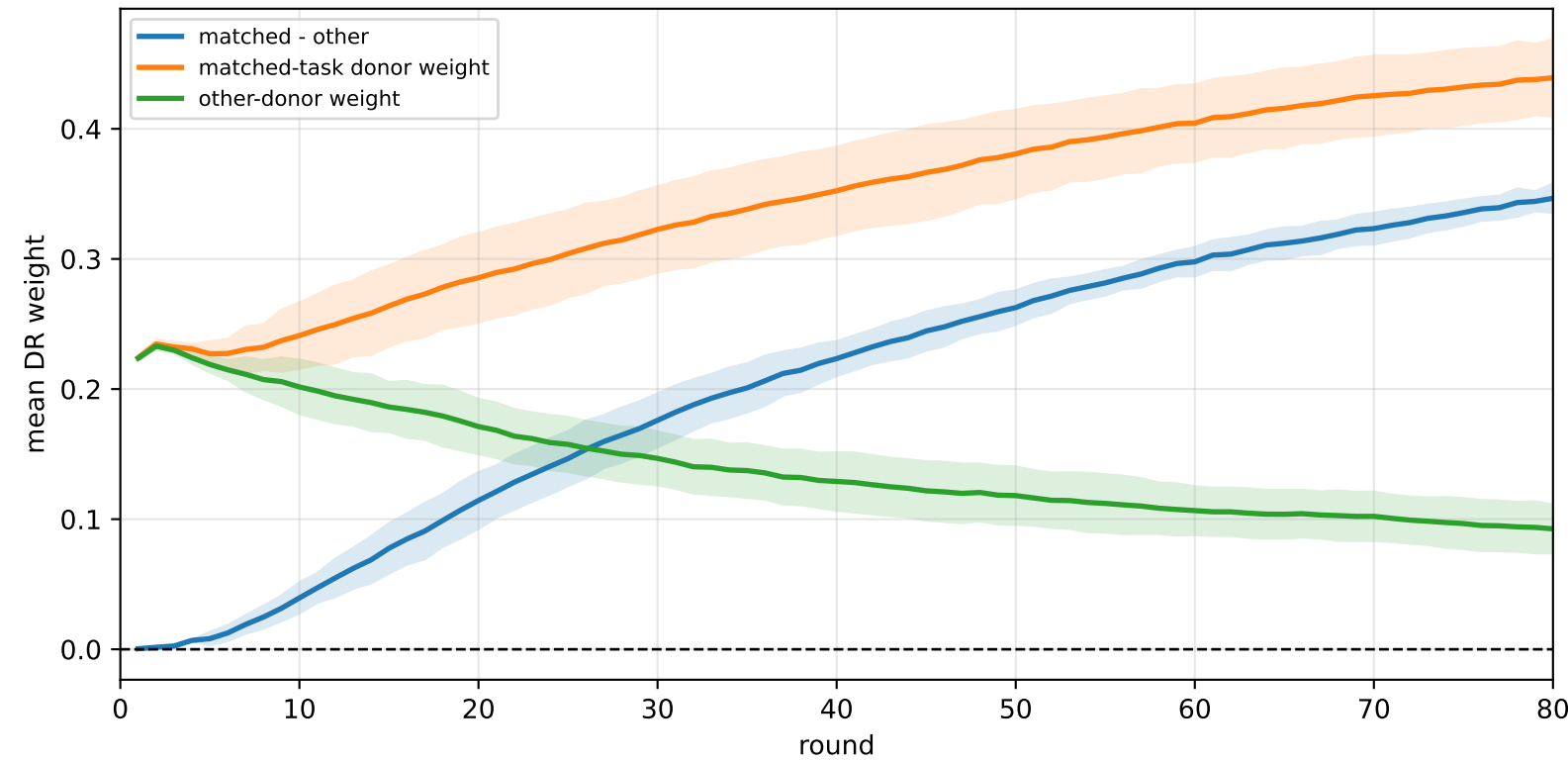
Task-head DR movement away from p0 by task



Task-head off-diagonal negative mass by task



Task-head DR alignment



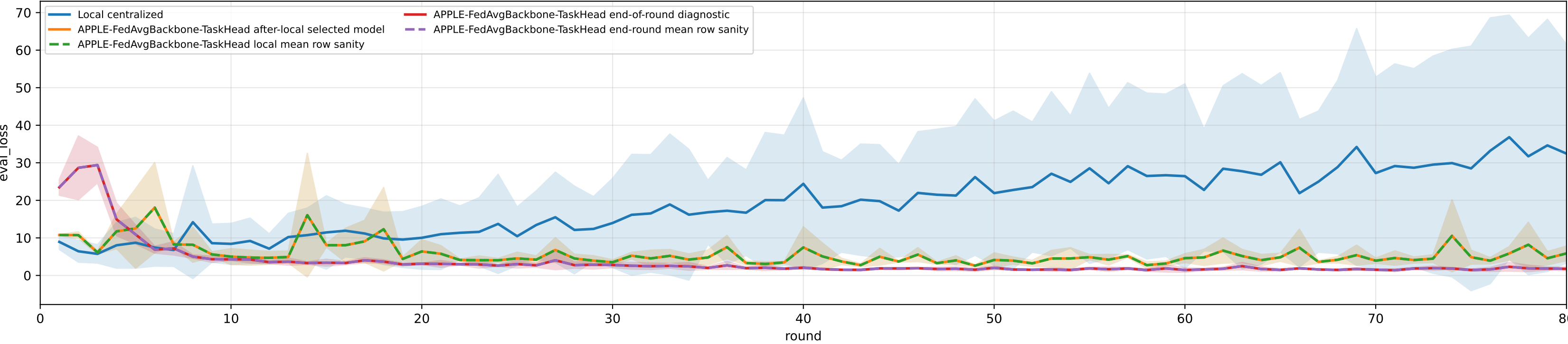
Controlled q-label heterogeneity: High | APPLE-FedAvgBackbone-TaskHead diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

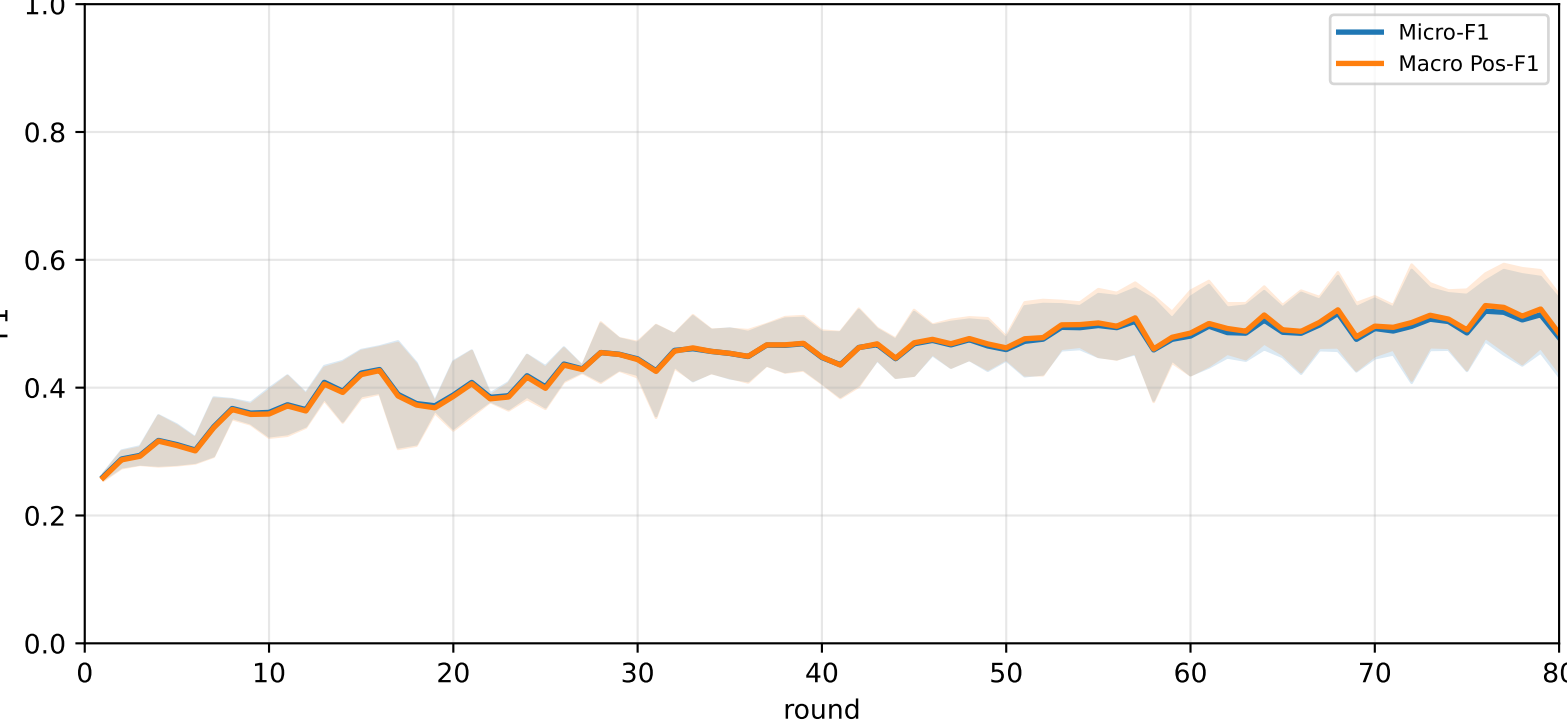
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	29.1% ± 5.1%	29.5% ± 5.3%	22.1% ± 2.8%	26.7% ± 3.4%	26.9% ± 3.1%	26.5% ± 3.3%	30.2% ± 3.0%
APPLE-FedAvgBackbone-TaskHead	58.8% ± 2.9%	60.2% ± 3.1%	61.5% ± 6.2%	62.7% ± 8.3%	61.0% ± 0.9%	62.9% ± 2.0%	50.0% ± 3.6%

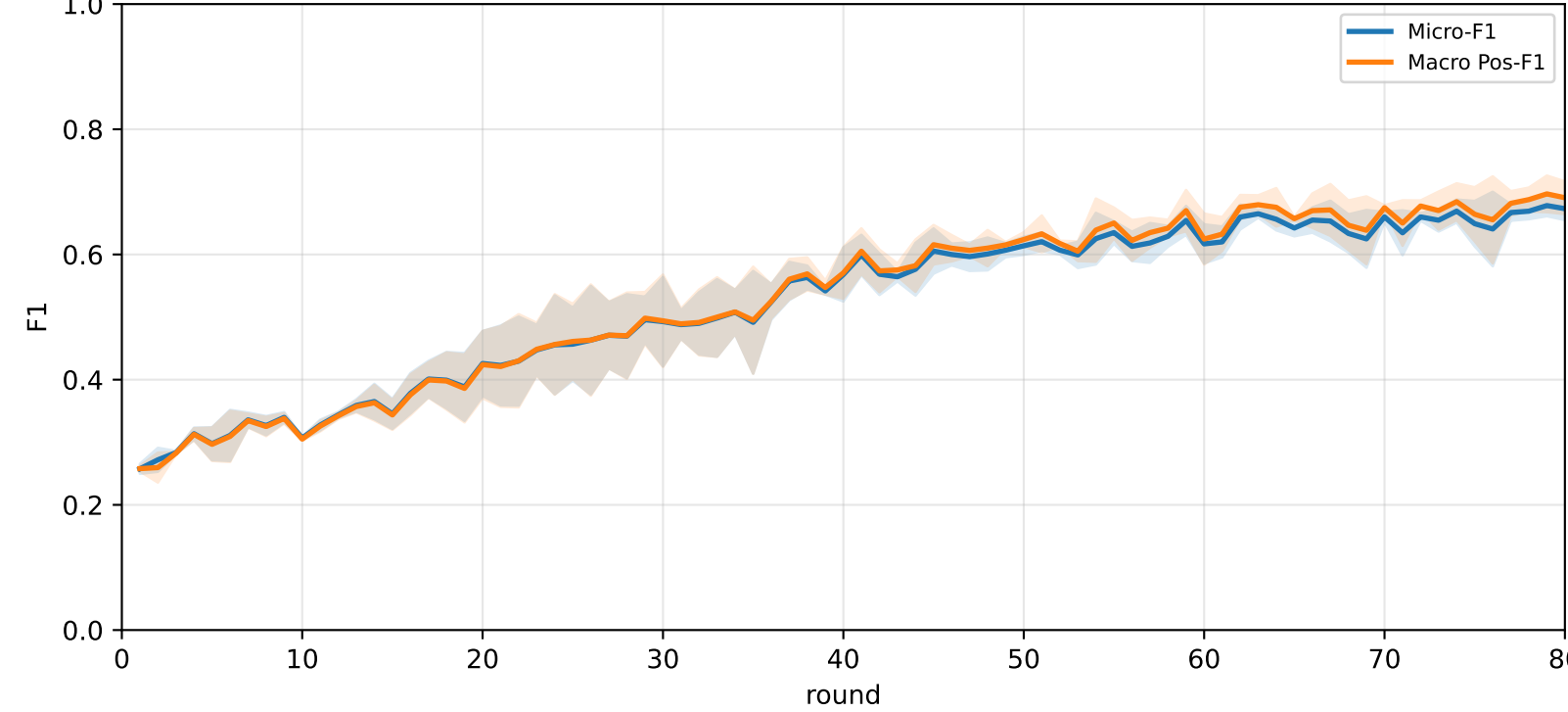
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



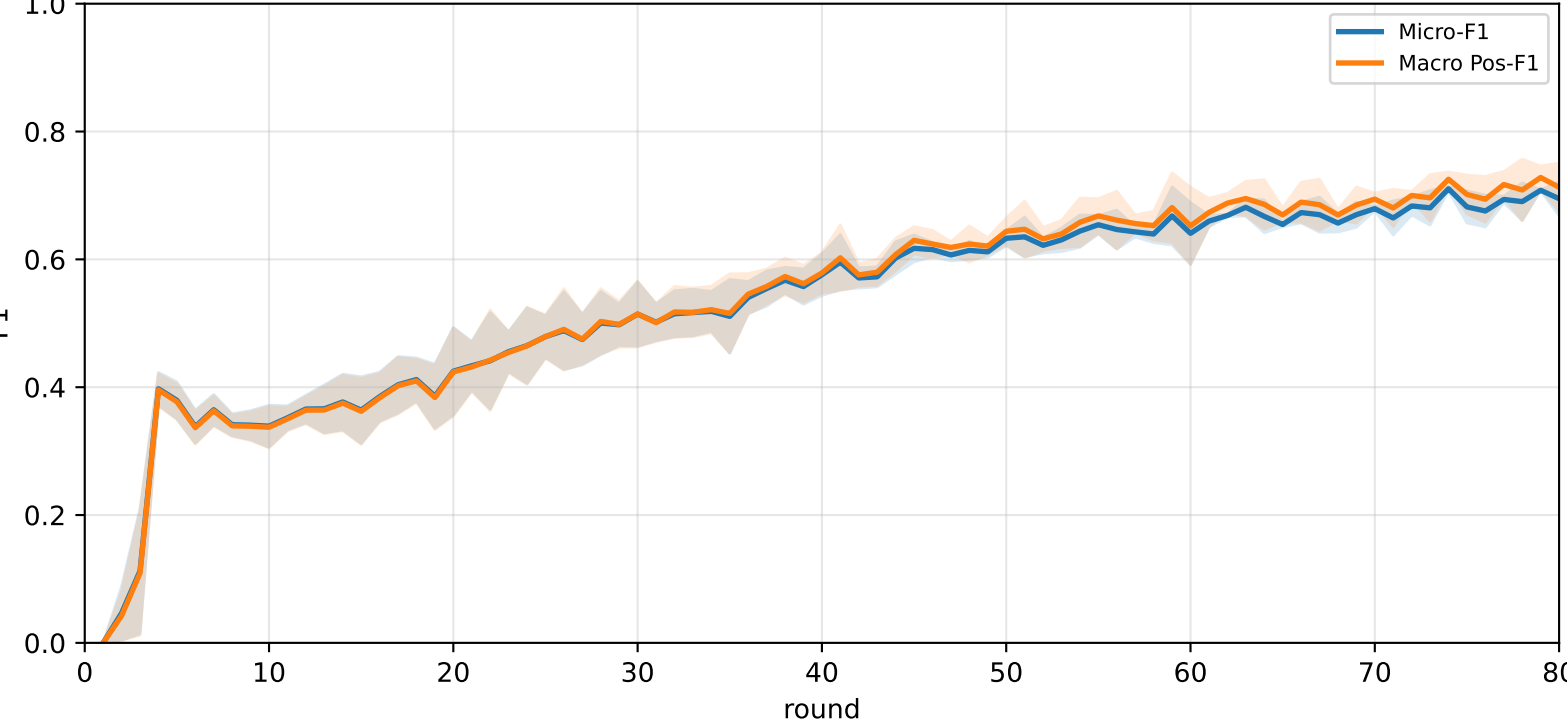
Pooled Local baseline micro / macro F1



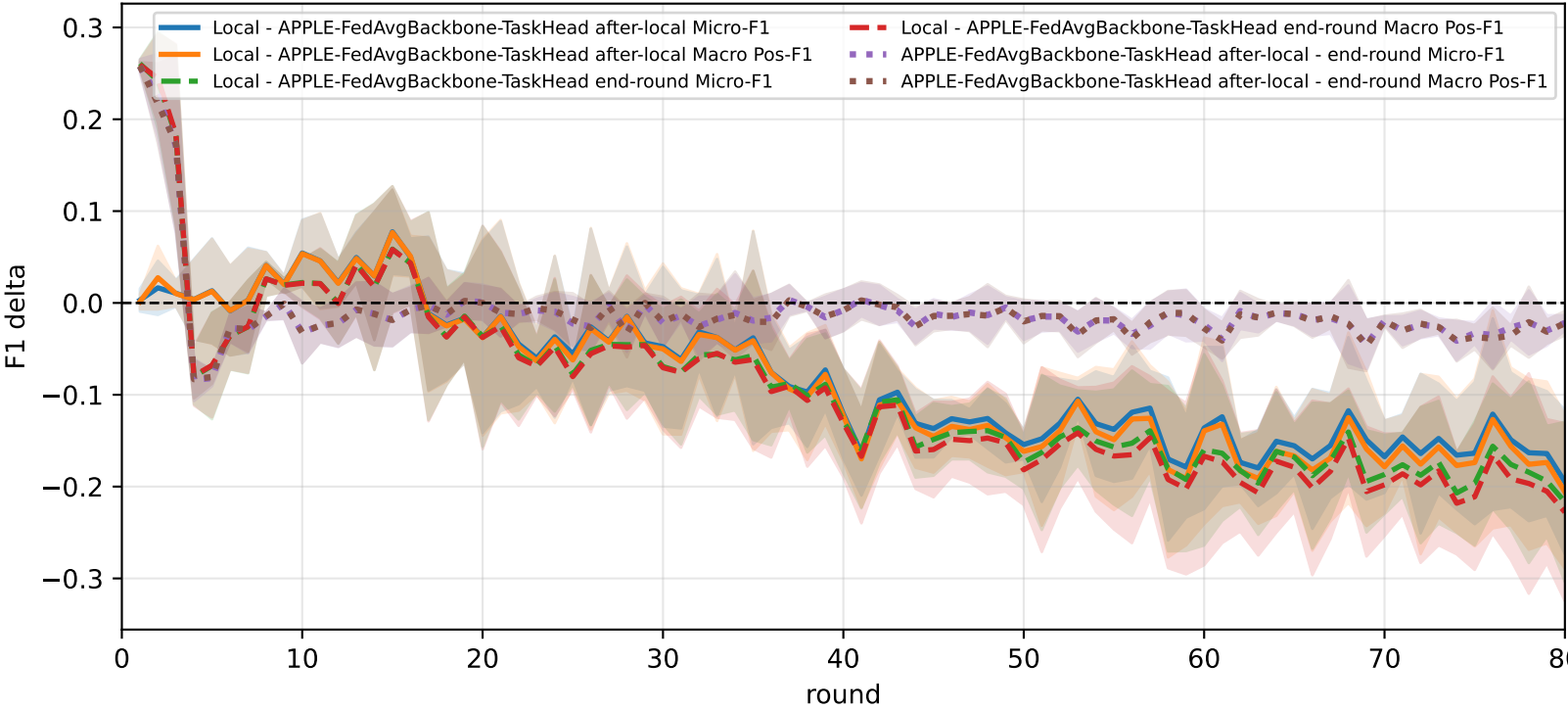
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



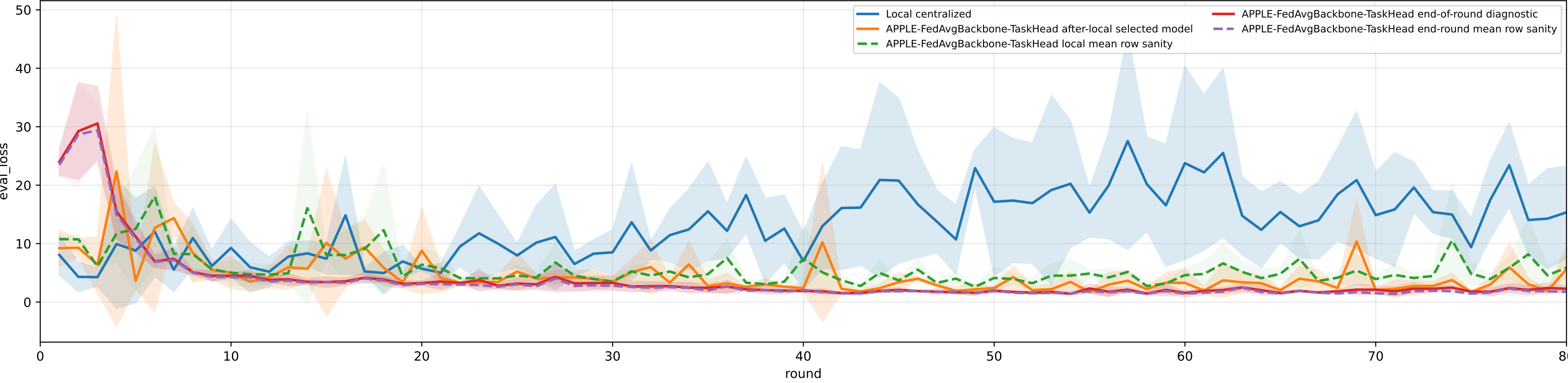
Controlled q-label heterogeneity: High | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q task_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

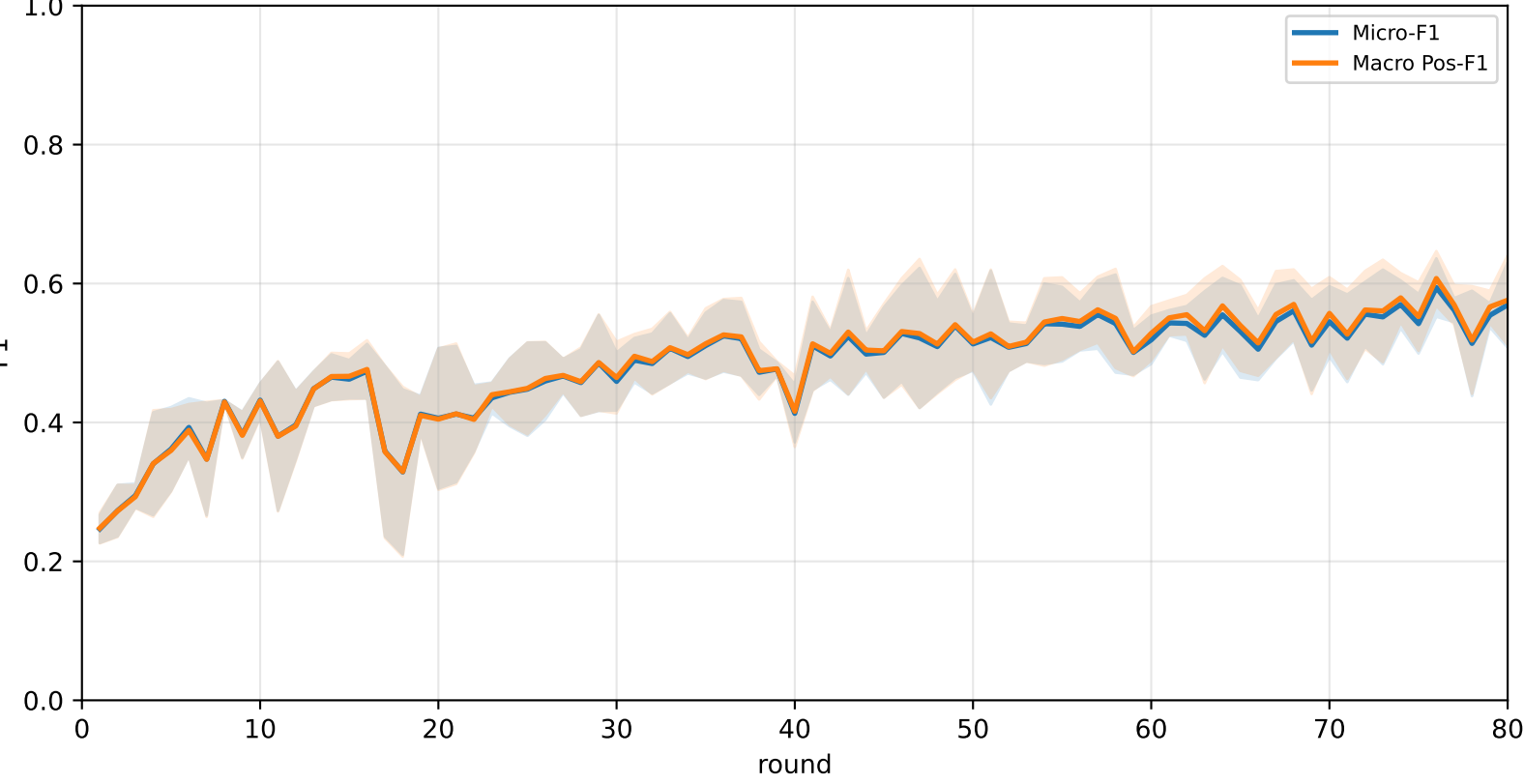
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	32.7% ± 12.1%	33.2% ± 12.8%	31.2% ± 16.3%	35.6% ± 15.5%	35.8% ± 15.4%	31.2% ± 10.3%	32.3% ± 6.7%
APPLE-FedAvgBackbone-TaskHead	56.8% ± 6.3%	57.6% ± 6.0%	53.8% ± 7.4%	62.7% ± 10.2%	60.7% ± 5.6%	59.9% ± 6.9%	50.8% ± 6.9%

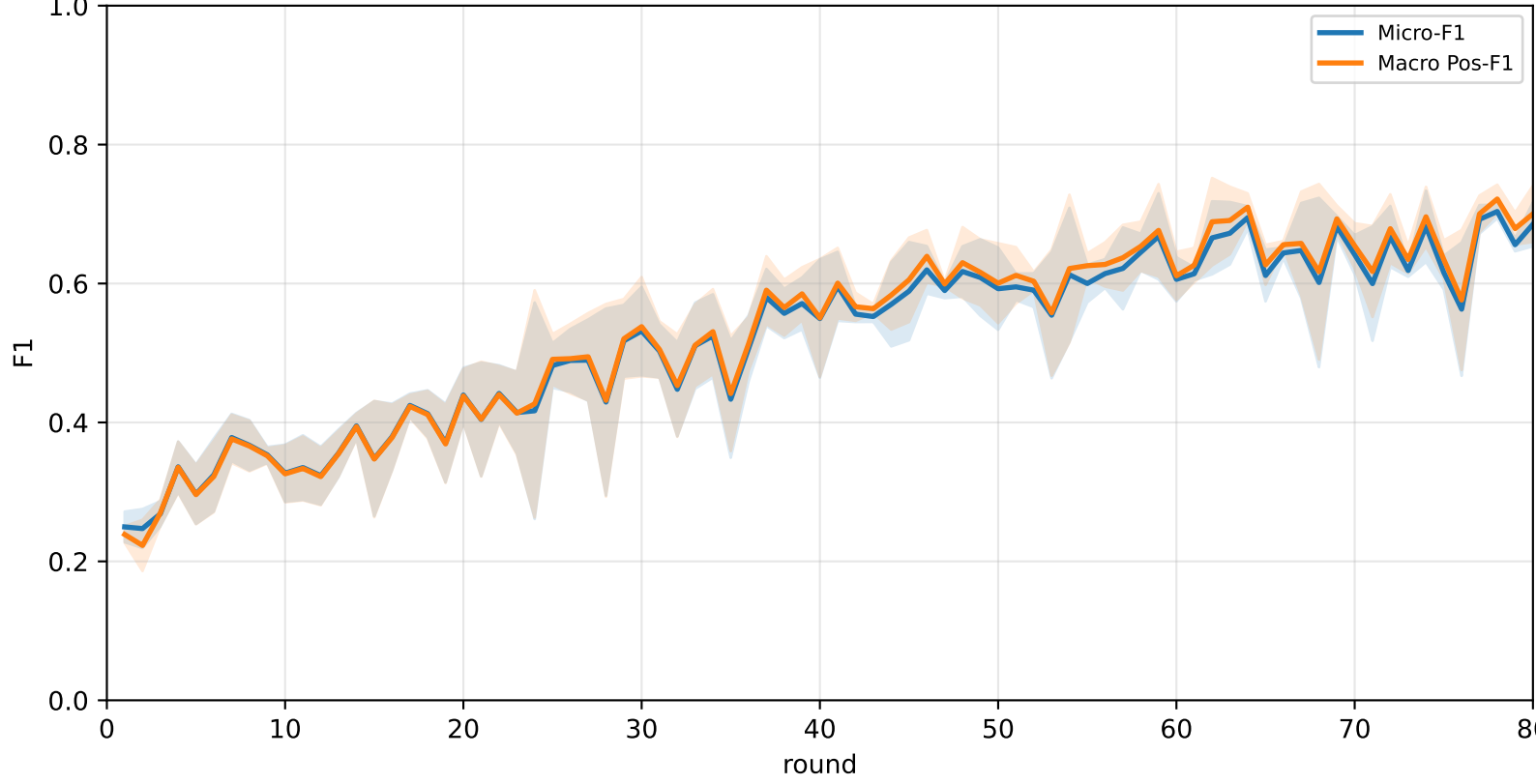
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



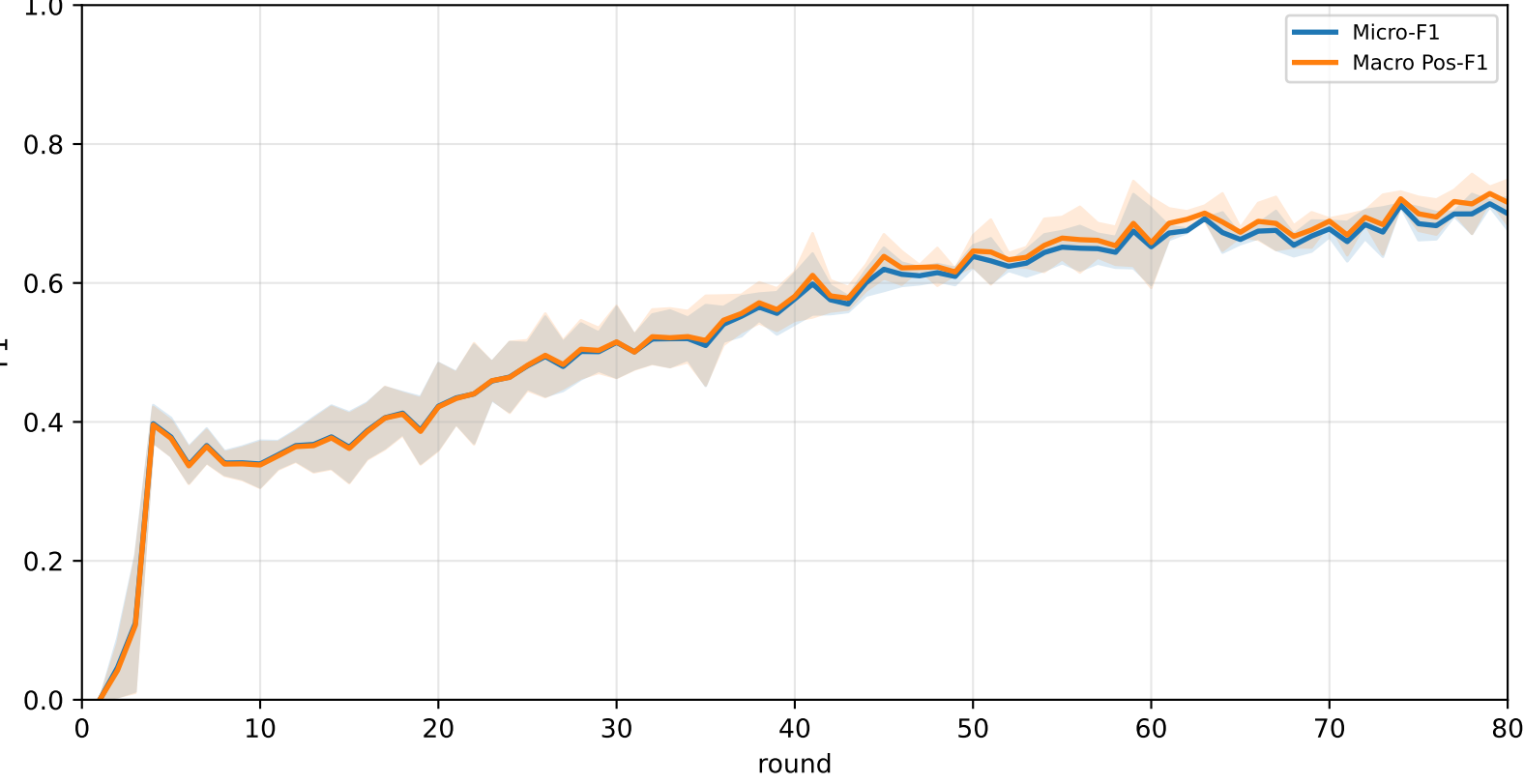
Pooled Local baseline micro / macro F1



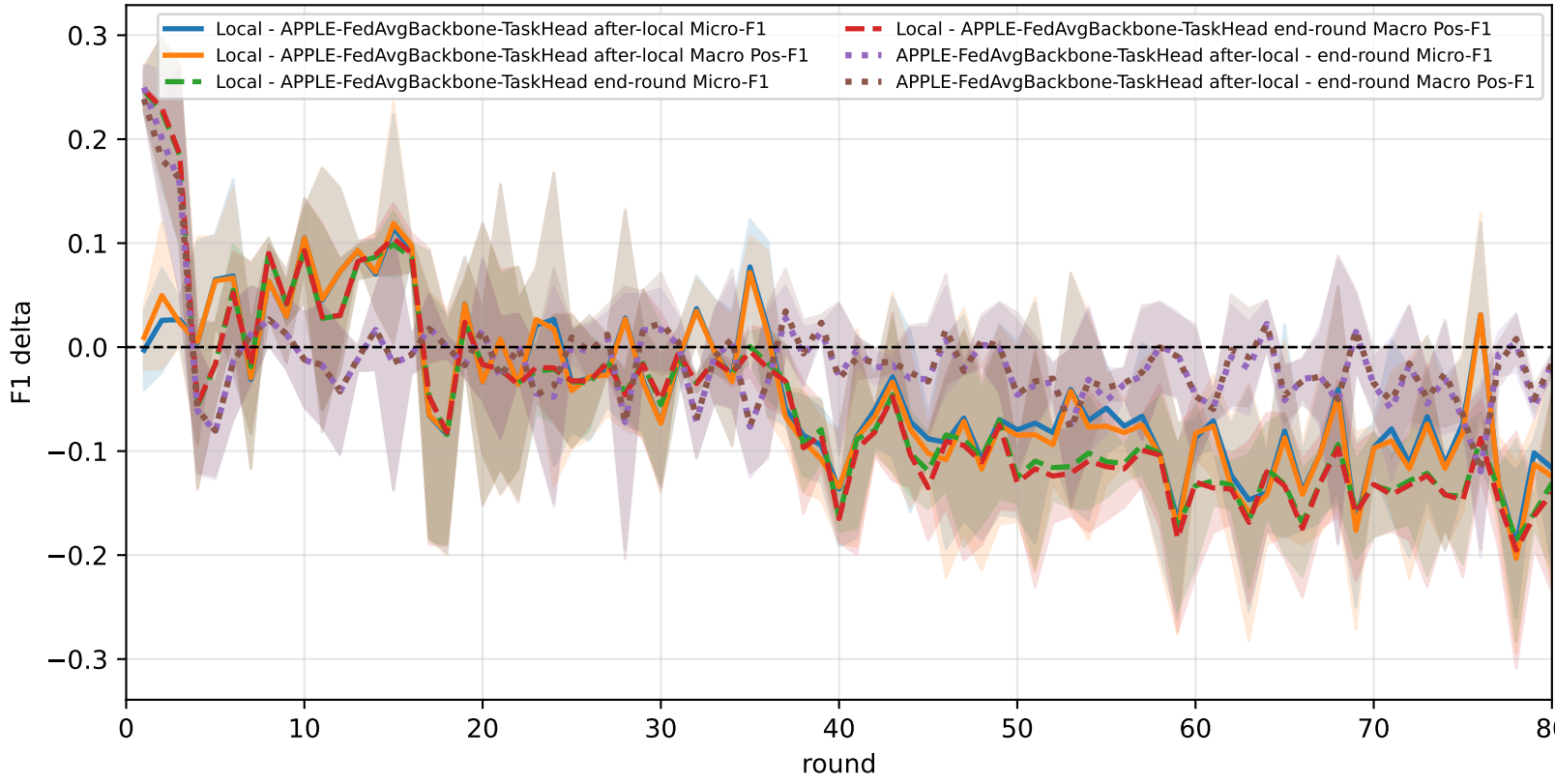
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



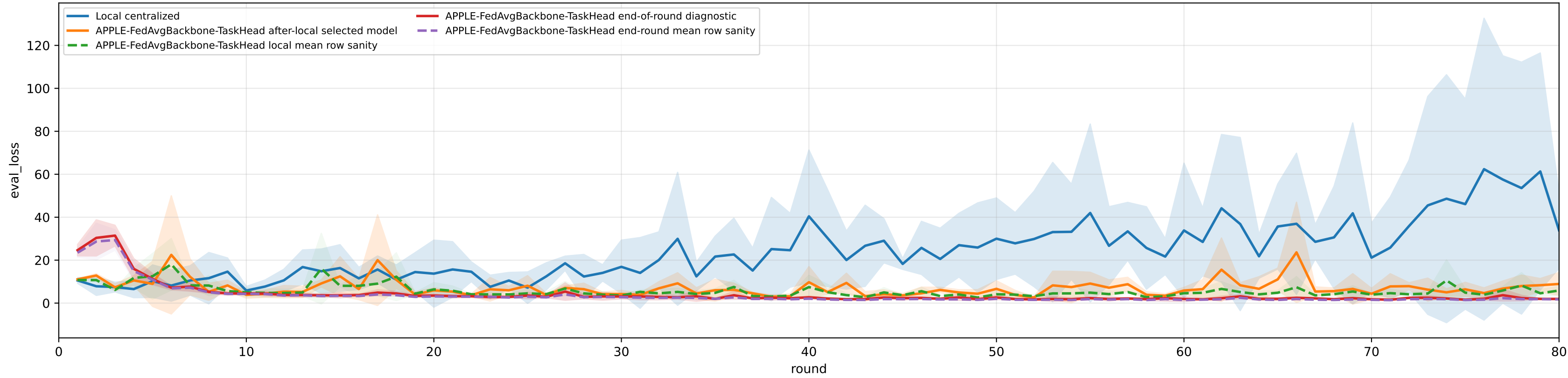
Controlled q-label heterogeneity: High | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q_task_cycle3 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

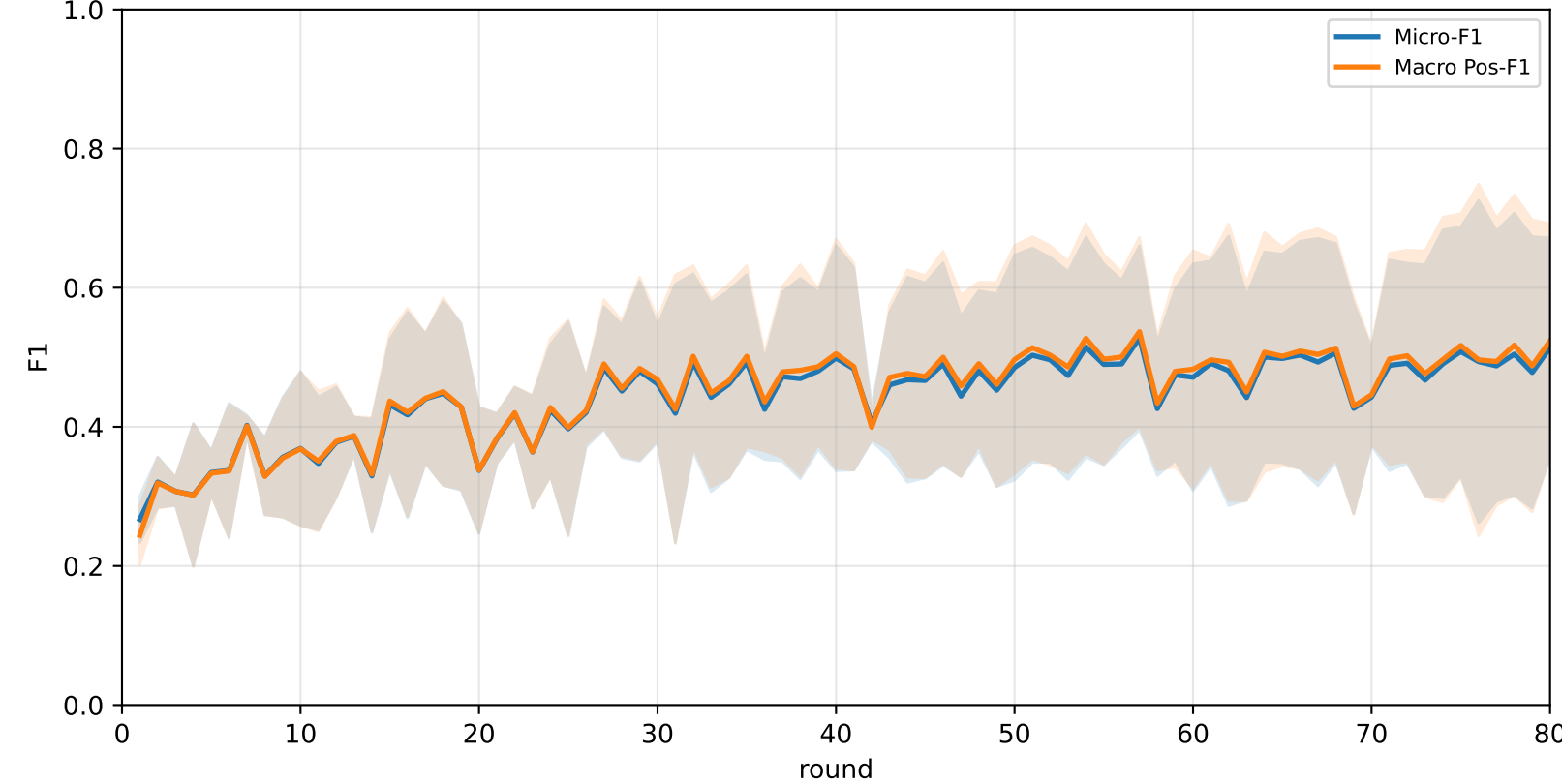
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	24.6% ± 3.5%	24.6% ± 3.5%	20.2% ± 3.0%	23.5% ± 4.2%	24.7% ± 3.6%	25.1% ± 3.4%	29.3% ± 3.3%
APPLE-FedAvgBackbone-TaskHead	59.9% ± 4.4%	61.5% ± 4.8%	68.8% ± 7.2%	58.2% ± 8.1%	64.8% ± 5.7%	66.1% ± 6.9%	49.8% ± 0.5%

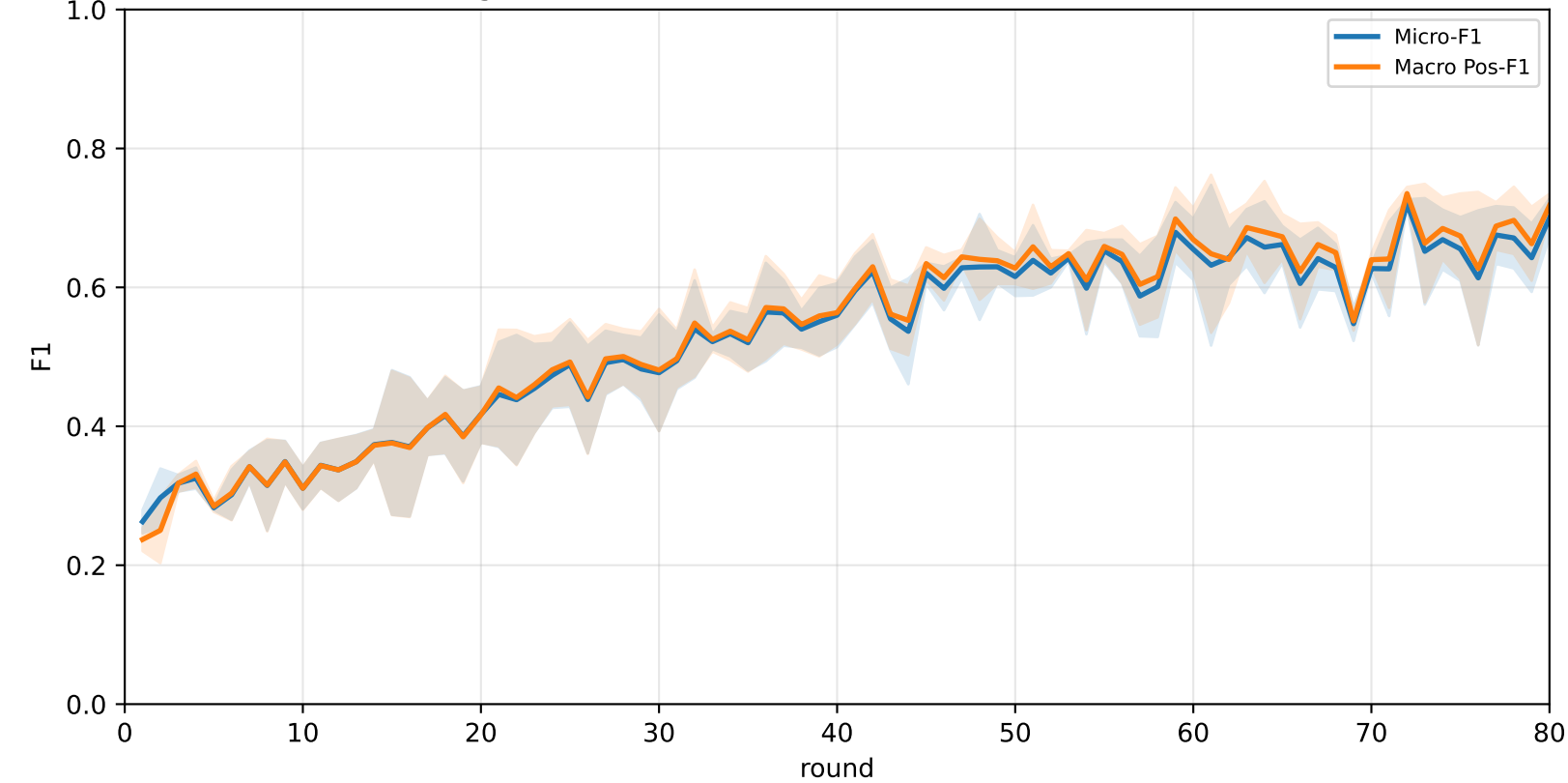
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



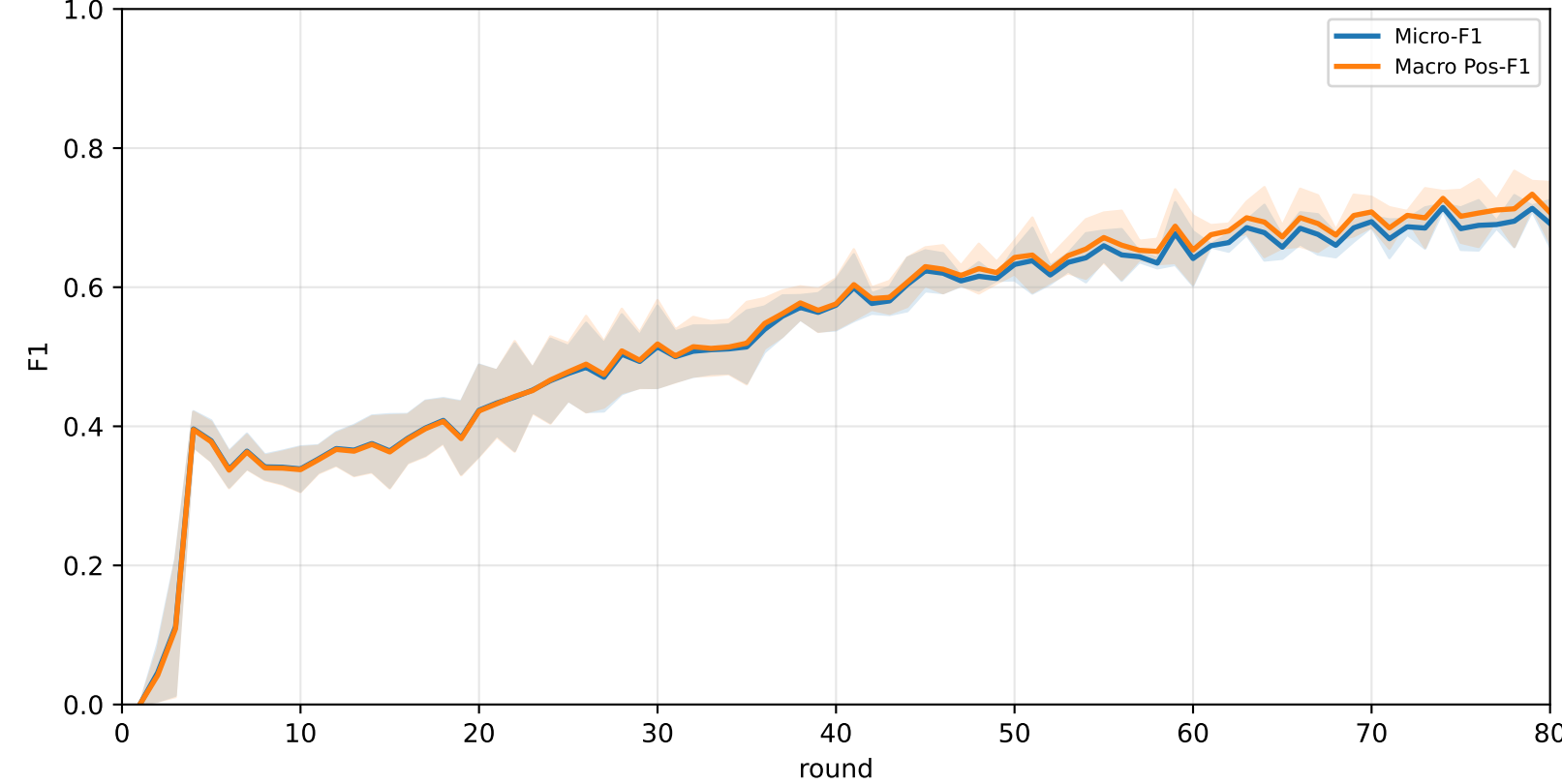
Pooled Local baseline micro / macro F1



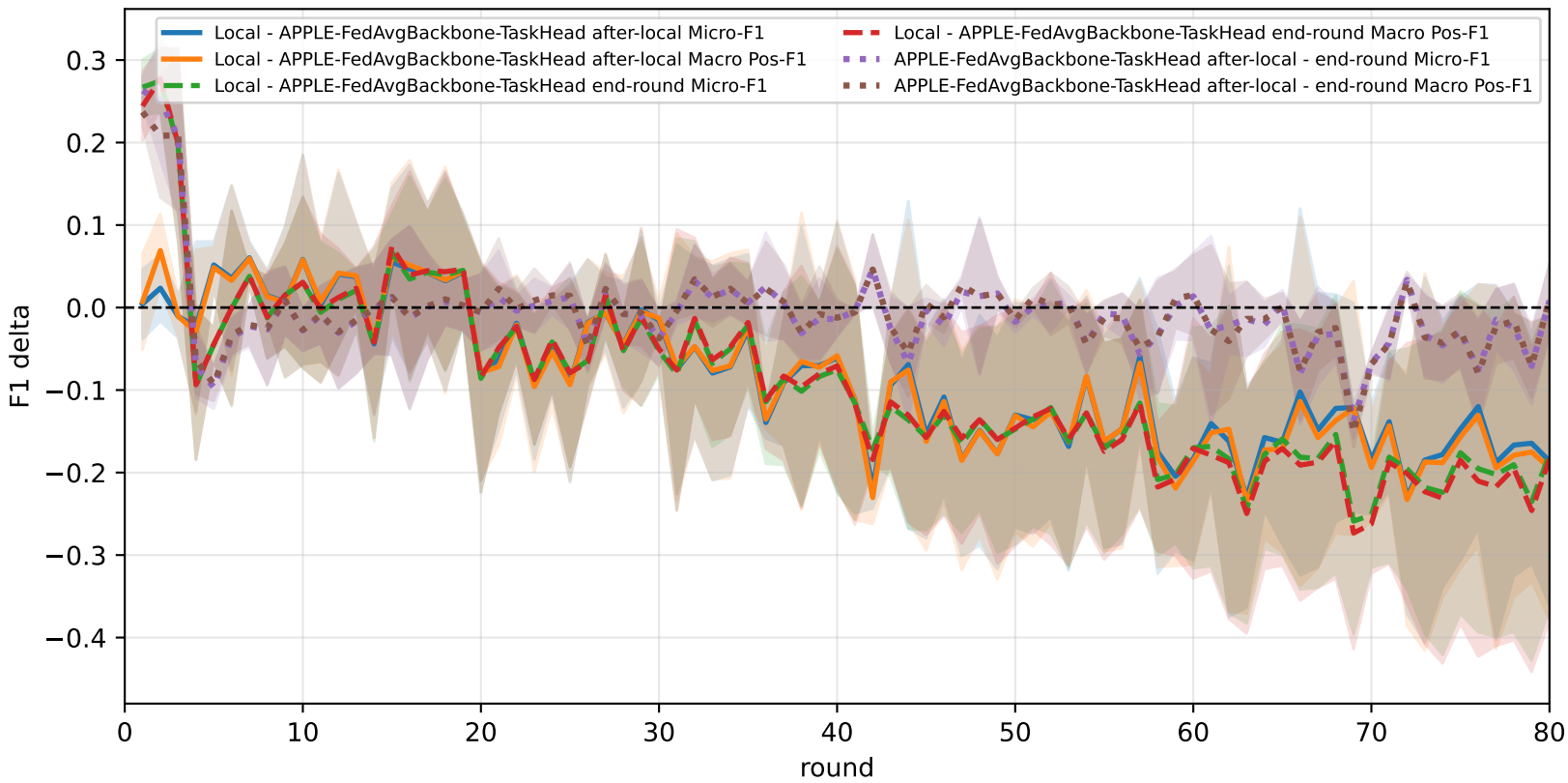
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



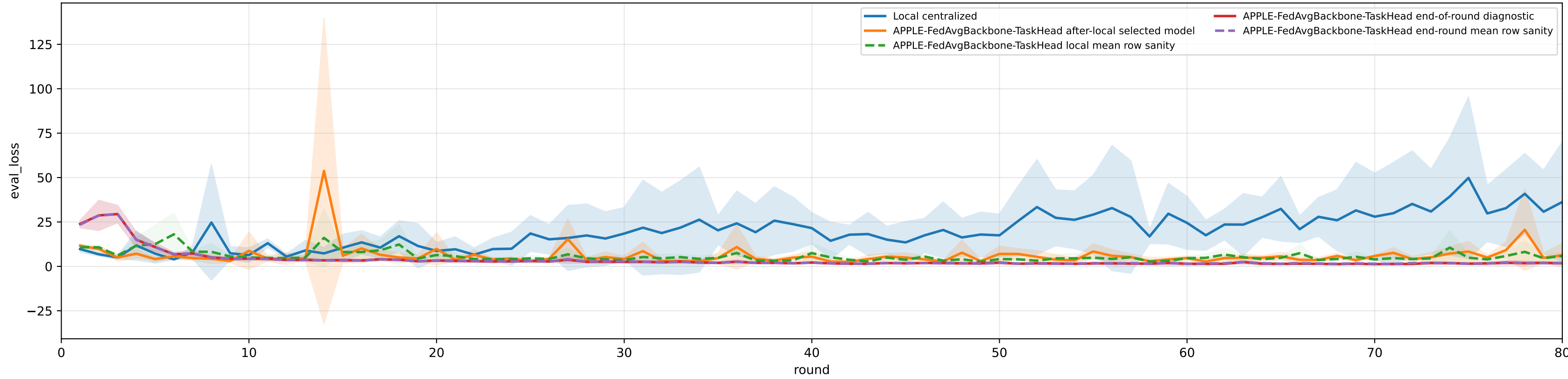
Controlled q-label heterogeneity: High | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q_task_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

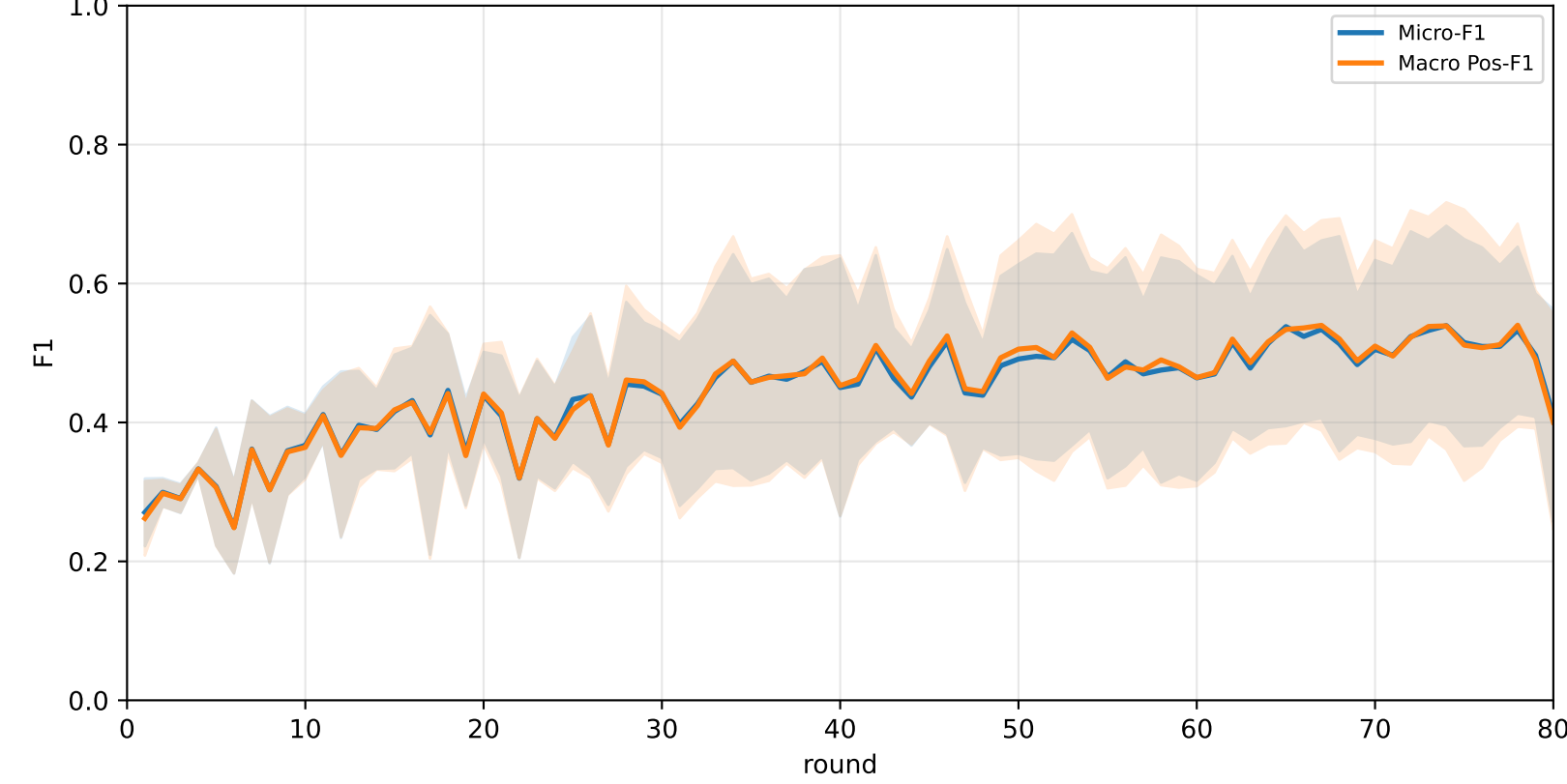
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	20.9% ± 2.3%	21.0% ± 2.5%	16.7% ± 1.9%	21.0% ± 1.6%	20.9% ± 2.8%	20.3% ± 1.5%	26.1% ± 4.9%
APPLE-FedAvgBackbone-TaskHead	60.6% ± 4.8%	62.1% ± 5.8%	65.2% ± 10.2%	66.4% ± 12.9%	56.8% ± 4.0%	70.1% ± 3.5%	52.0% ± 2.8%

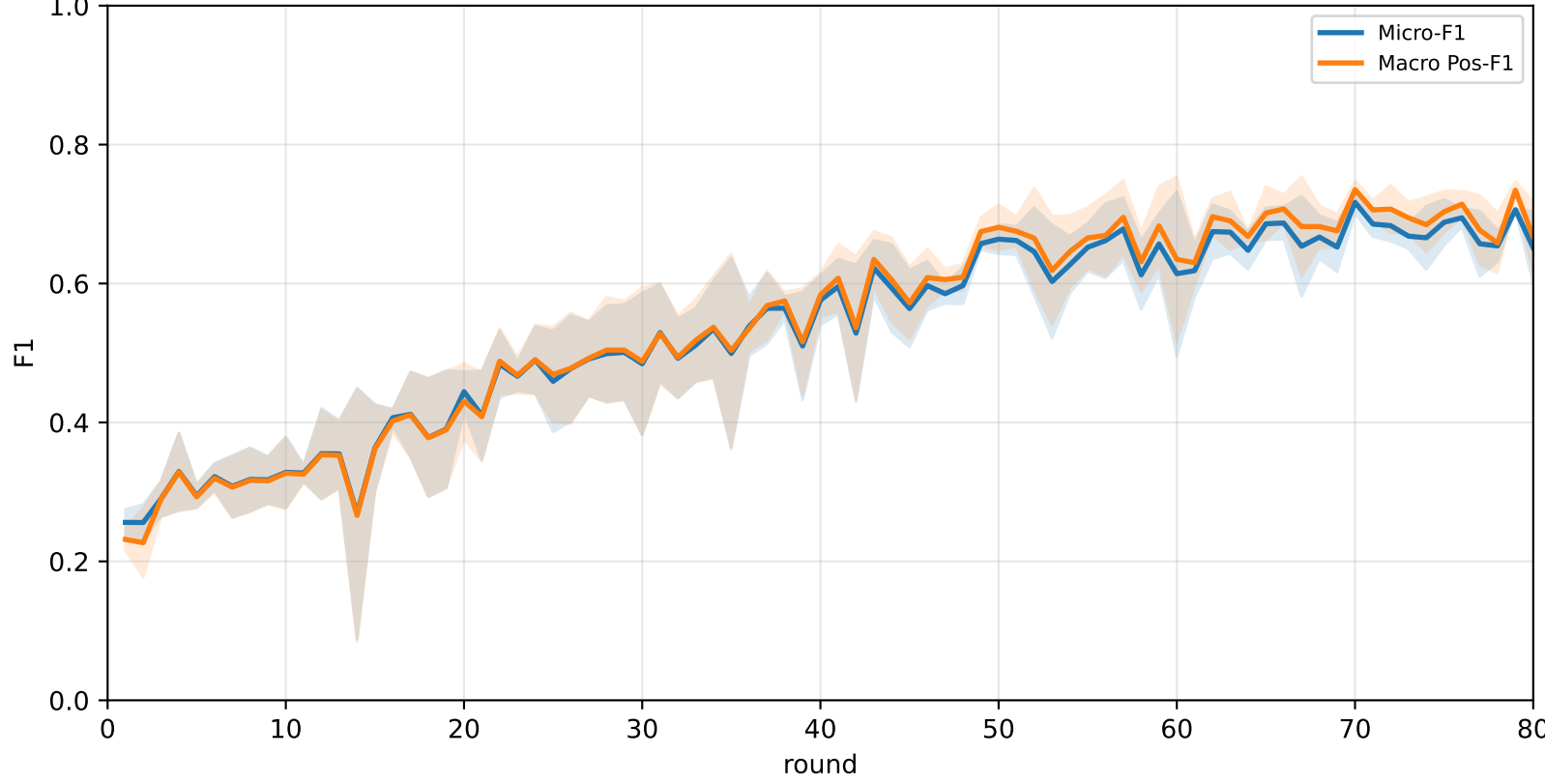
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



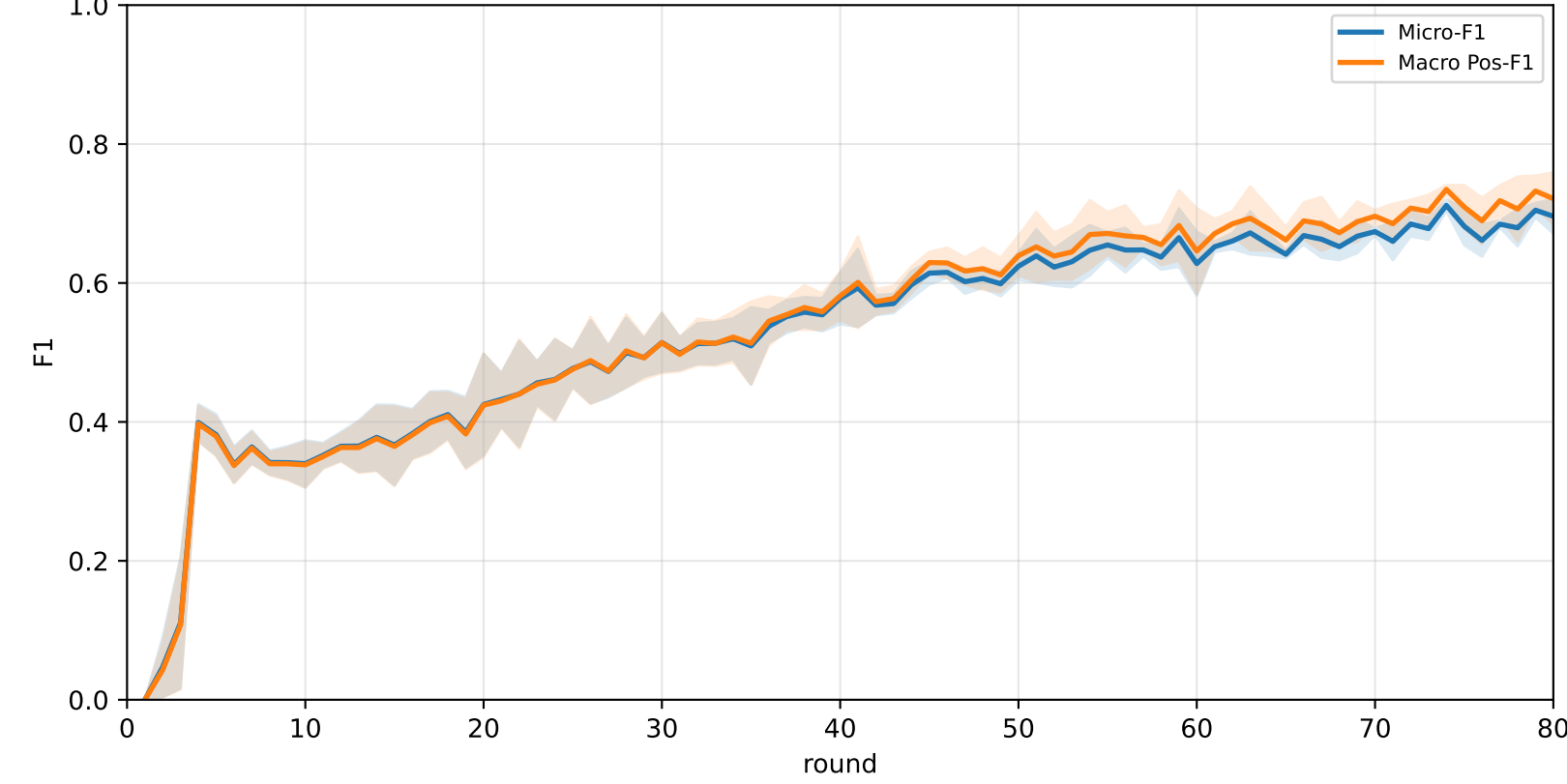
Pooled Local baseline micro / macro F1



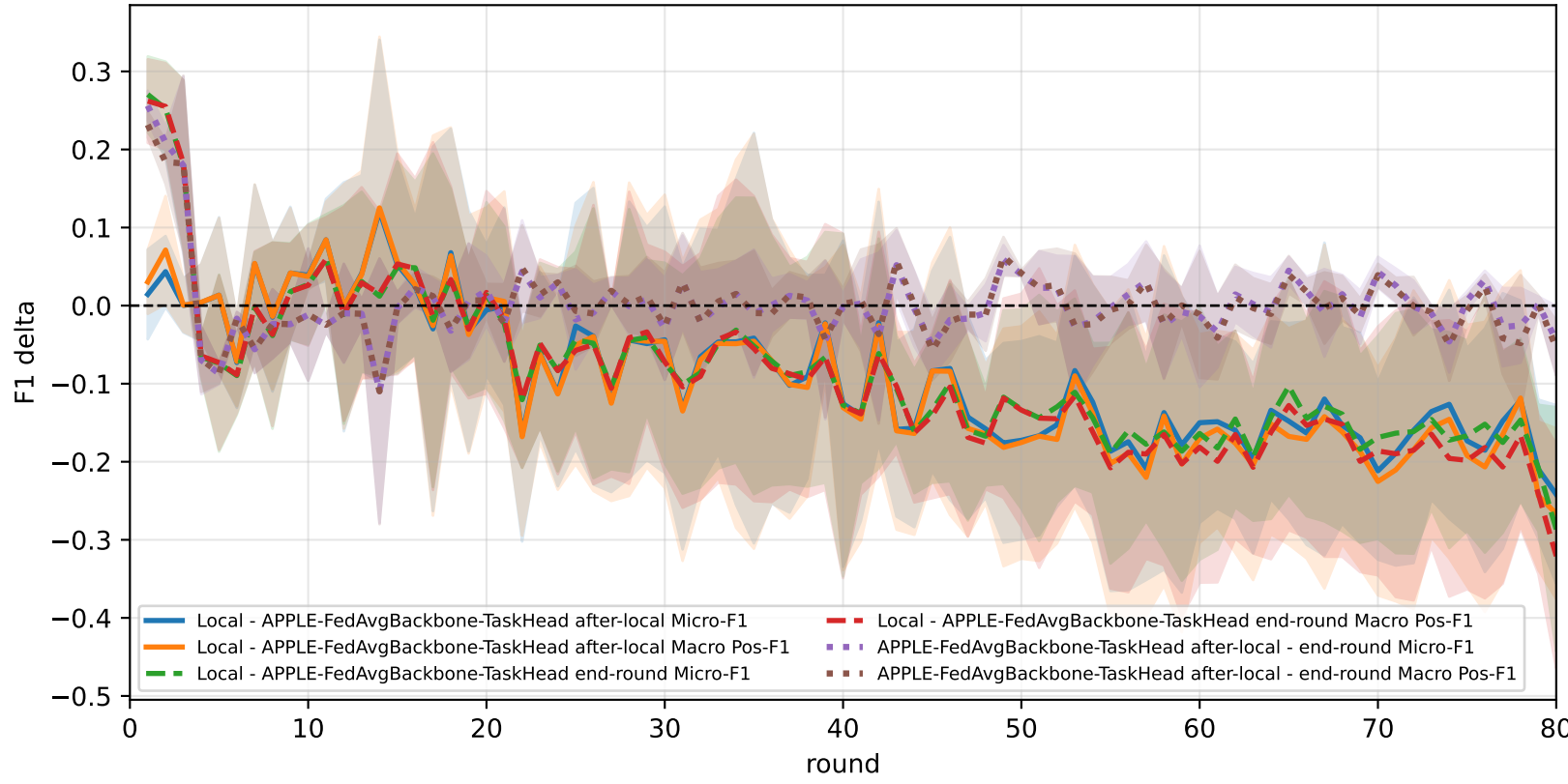
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



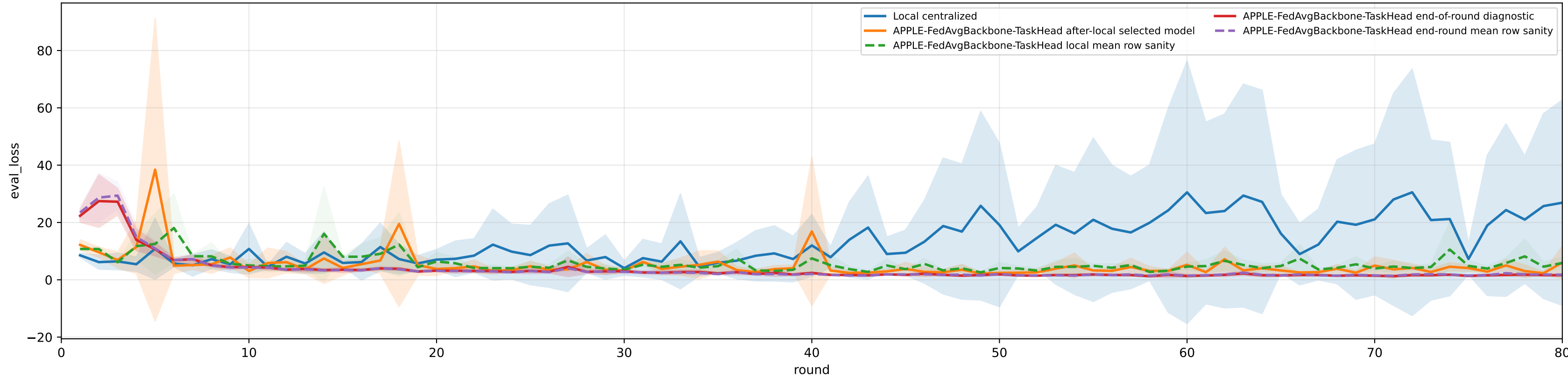
Controlled q-label heterogeneity: High | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q_task_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% |
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

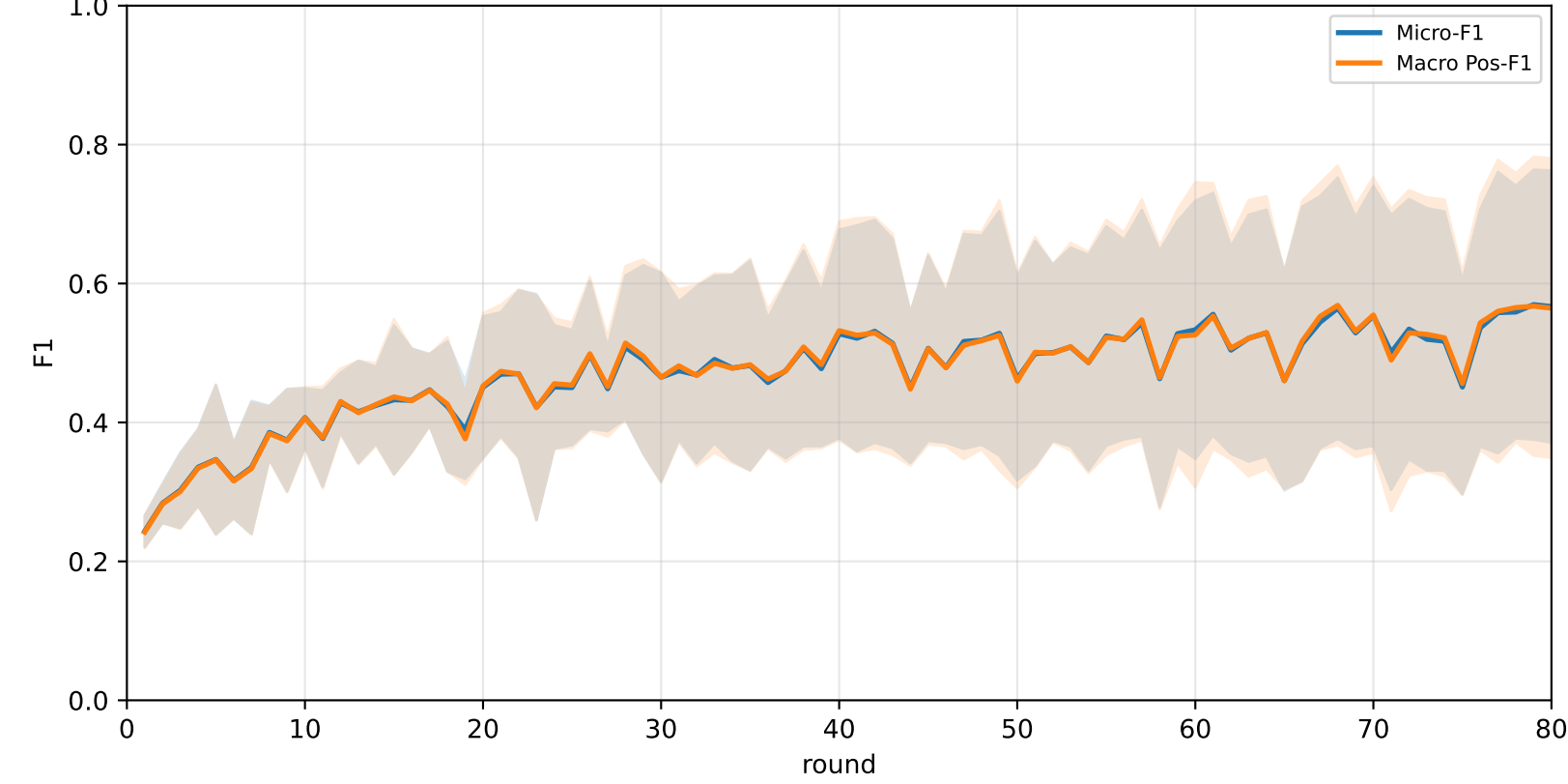
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	34.0% ± 13.0%	34.3% ± 13.3%	29.8% ± 14.7%	32.8% ± 11.9%	34.7% ± 10.6%	37.8% ± 19.0%	36.2% ± 10.7%
APPLE-FedAvgBackbone-TaskHead	56.7% ± 3.8%	57.7% ± 3.9%	59.9% ± 9.0%	61.3% ± 5.3%	58.8% ± 3.7%	59.1% ± 6.8%	49.4% ± 3.6%

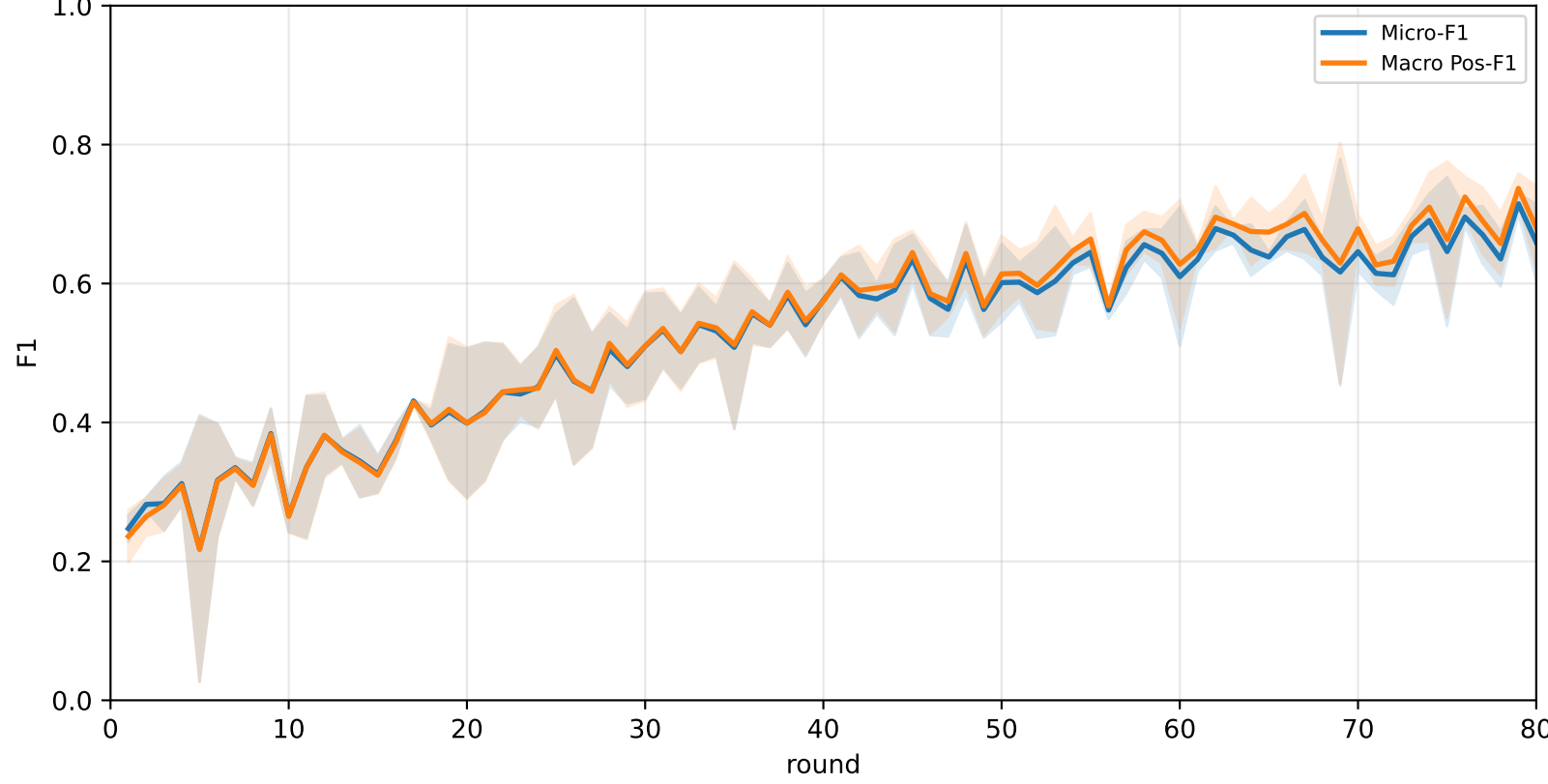
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



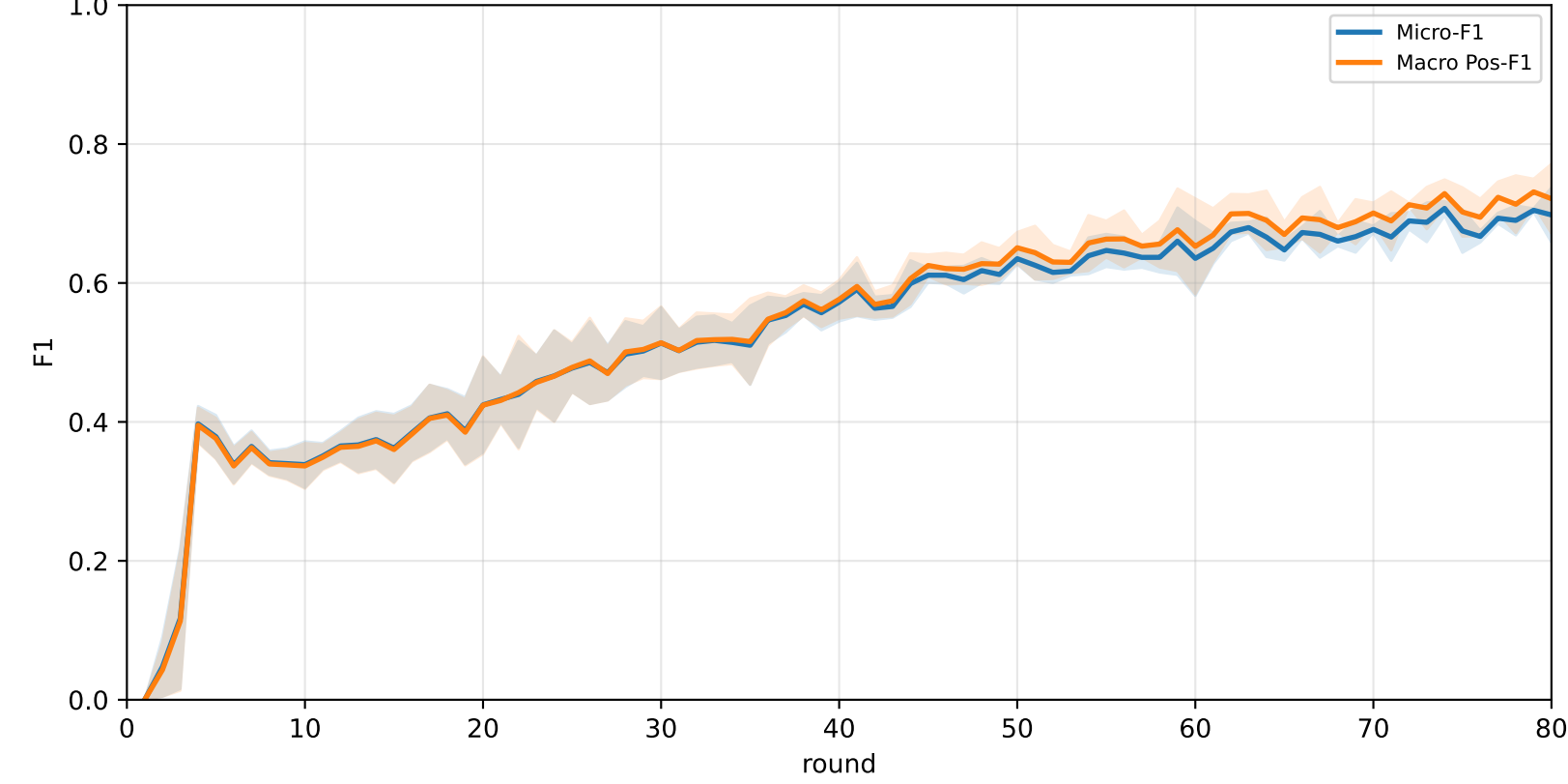
Pooled Local baseline micro / macro F1



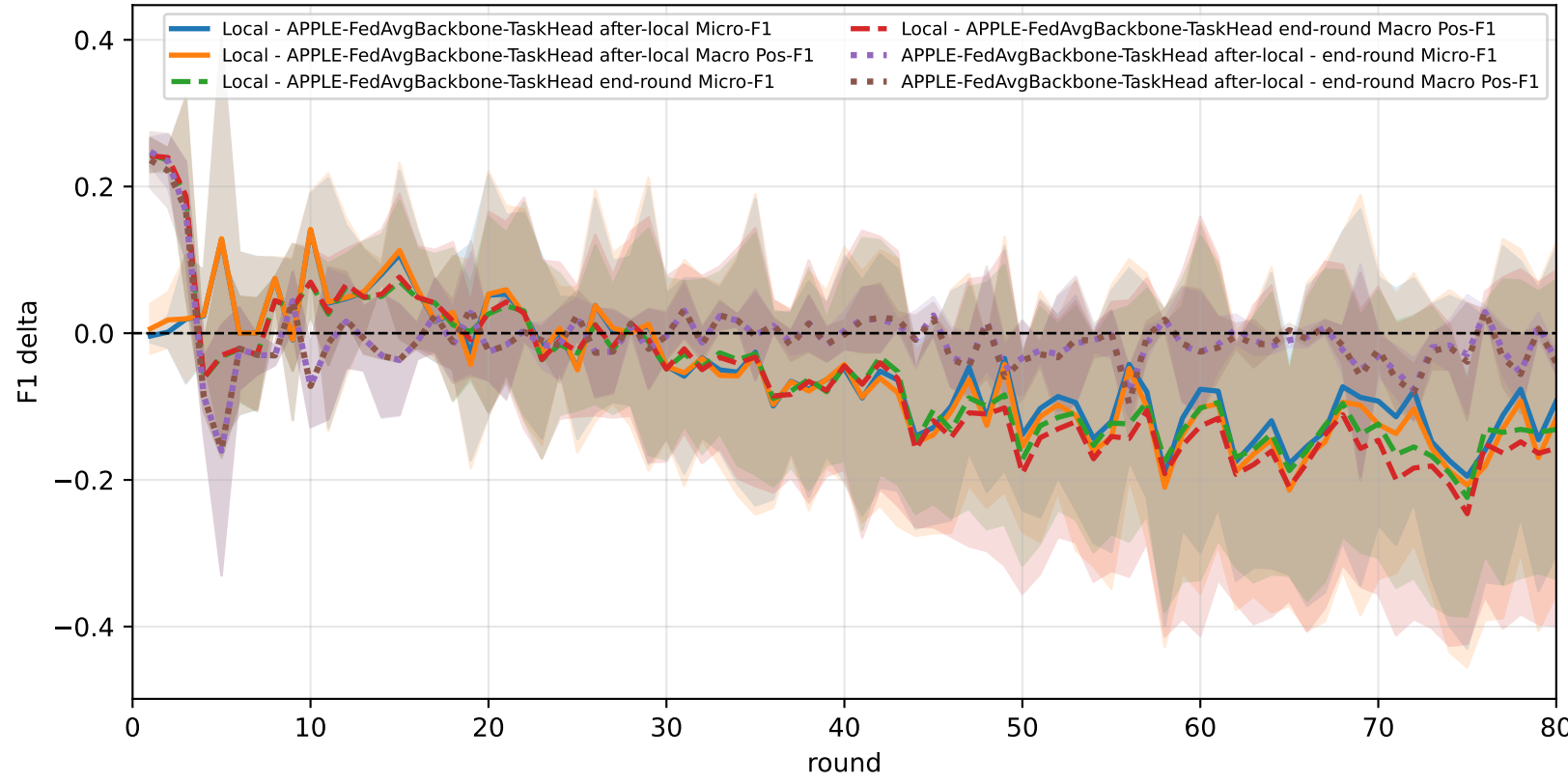
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



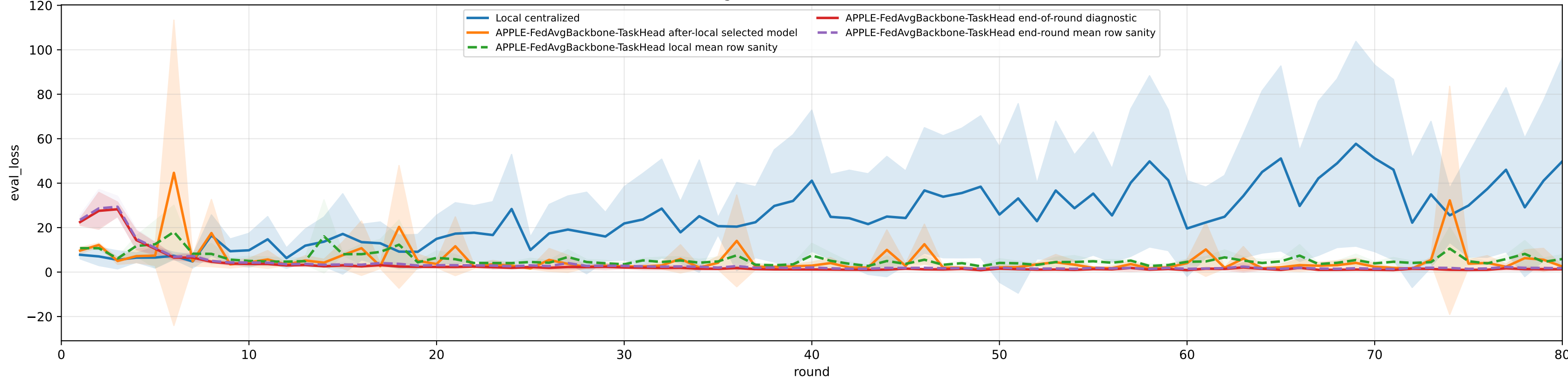
Controlled q-label heterogeneity: High | APPLE-FedAvgBackbone-TaskHead diagnostics |
family=q_task_cycle6 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3% |
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

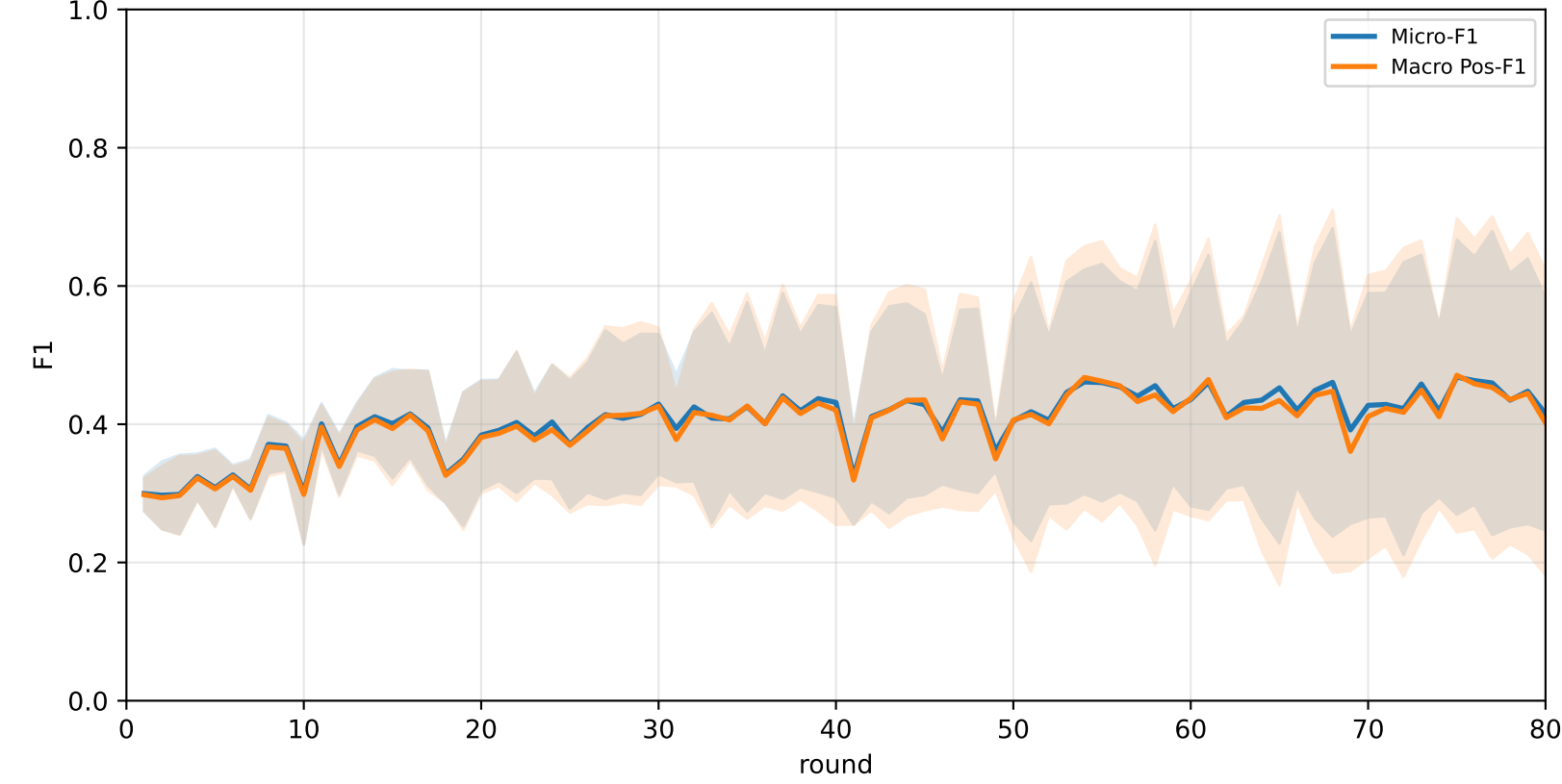
Best test performance (APPLE-FedAvgBackbone-TaskHead selected by after-local validation, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	33.5% ± 12.2%	34.3% ± 13.3%	36.4% ± 23.0%	36.9% ± 15.6%	32.4% ± 10.8%	32.2% ± 9.6%	33.5% ± 7.5%
APPLE-FedAvgBackbone-TaskHead	60.3% ± 2.8%	62.3% ± 3.3%	64.5% ± 8.1%	67.2% ± 8.5%	67.1% ± 3.8%	63.8% ± 2.5%	48.8% ± 4.1%

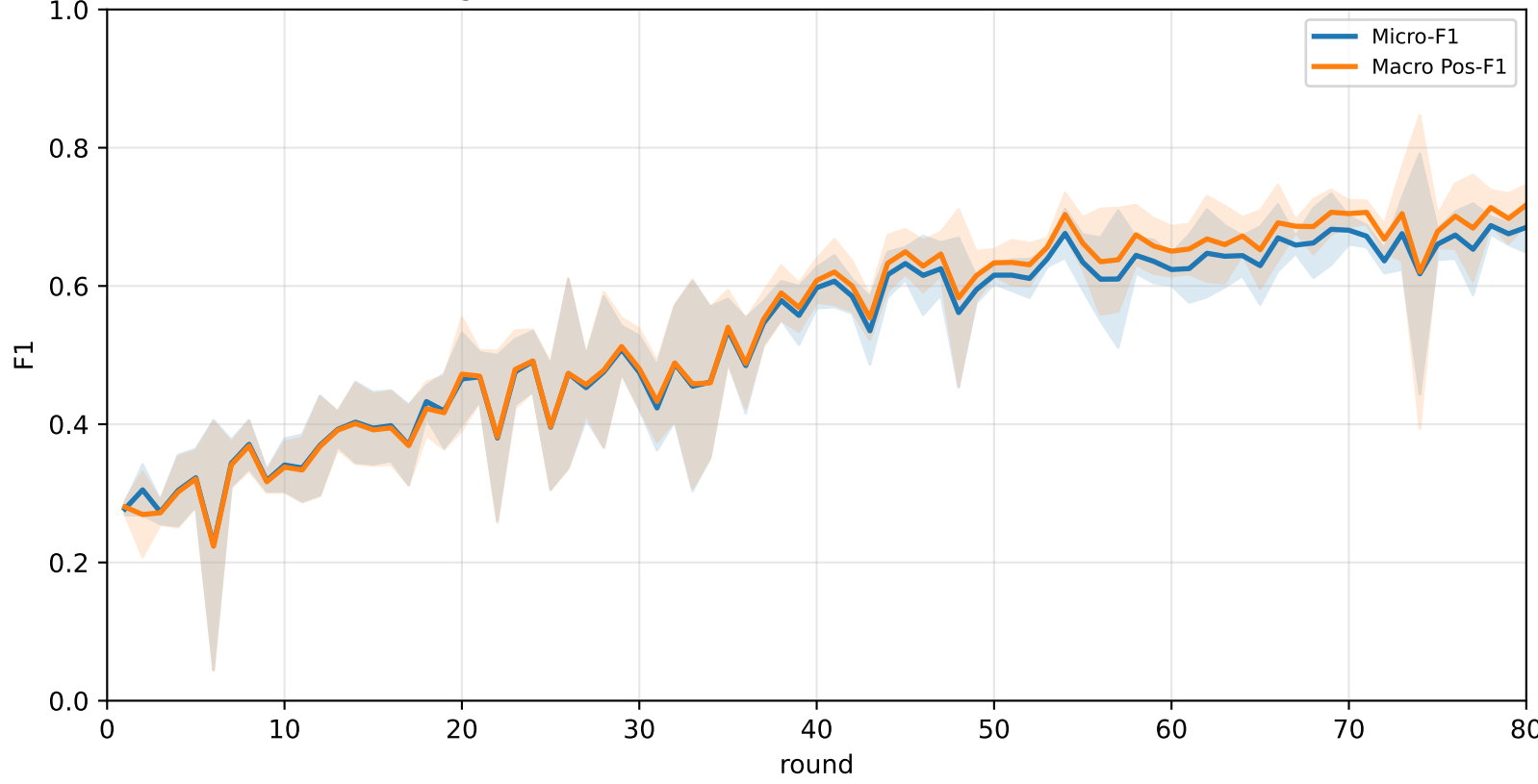
Validation loss: selected APPLE-FedAvgBackbone-TaskHead vs end-of-round mismatch (rounds=80)



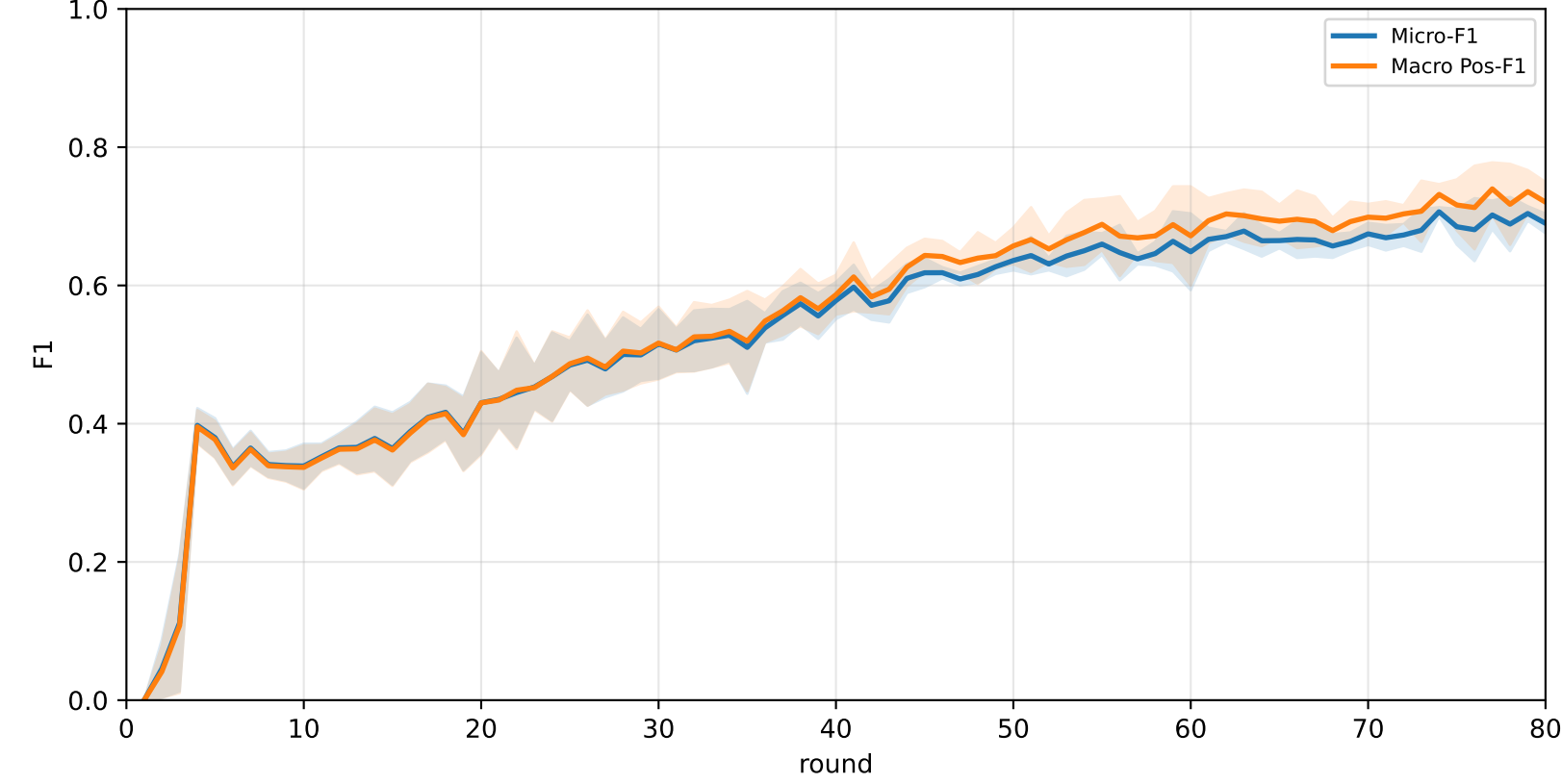
Pooled Local baseline micro / macro F1



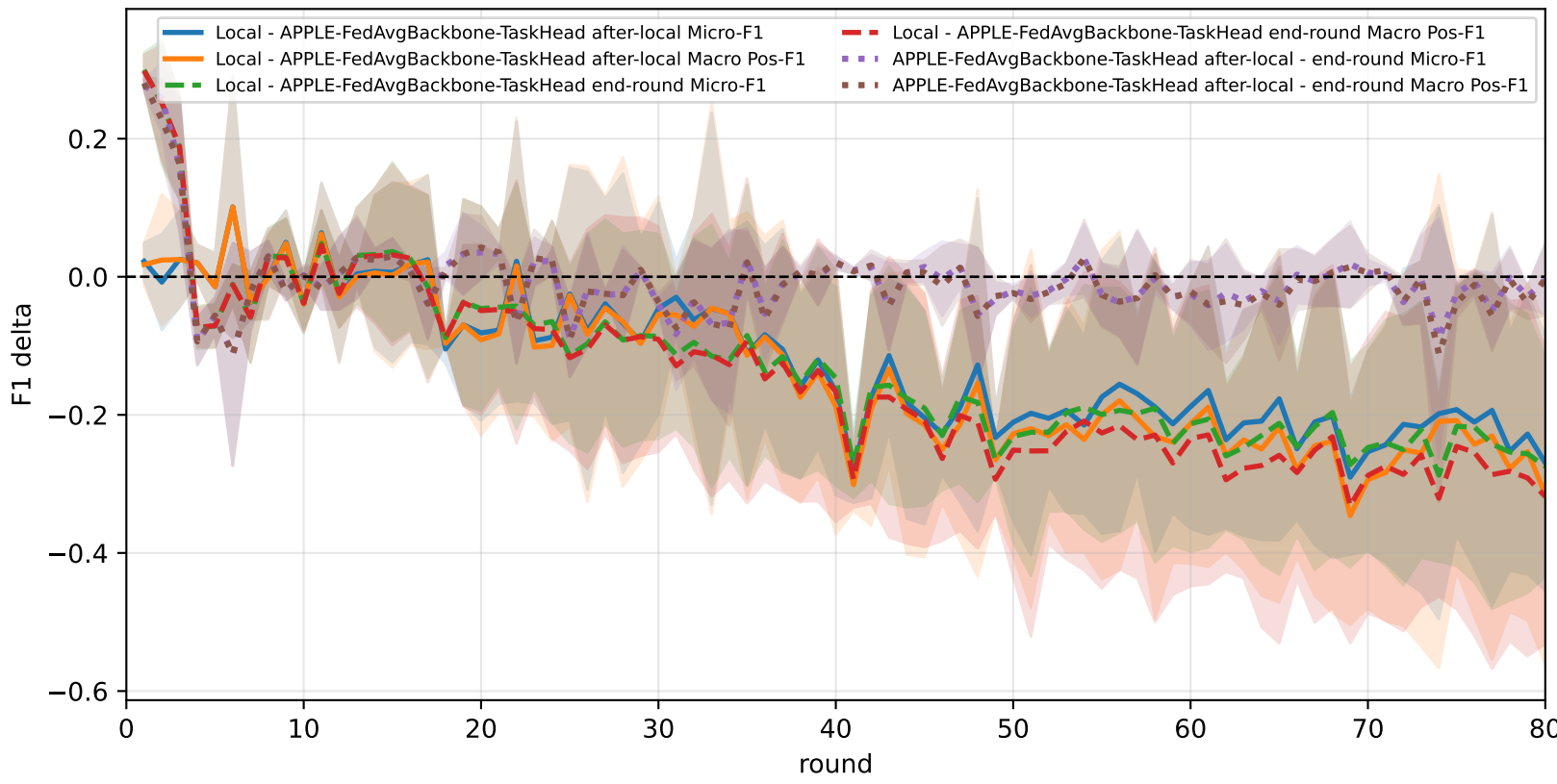
Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-FedAvgBackbone-TaskHead micro / macro F1 (end-of-round diagnostic only)

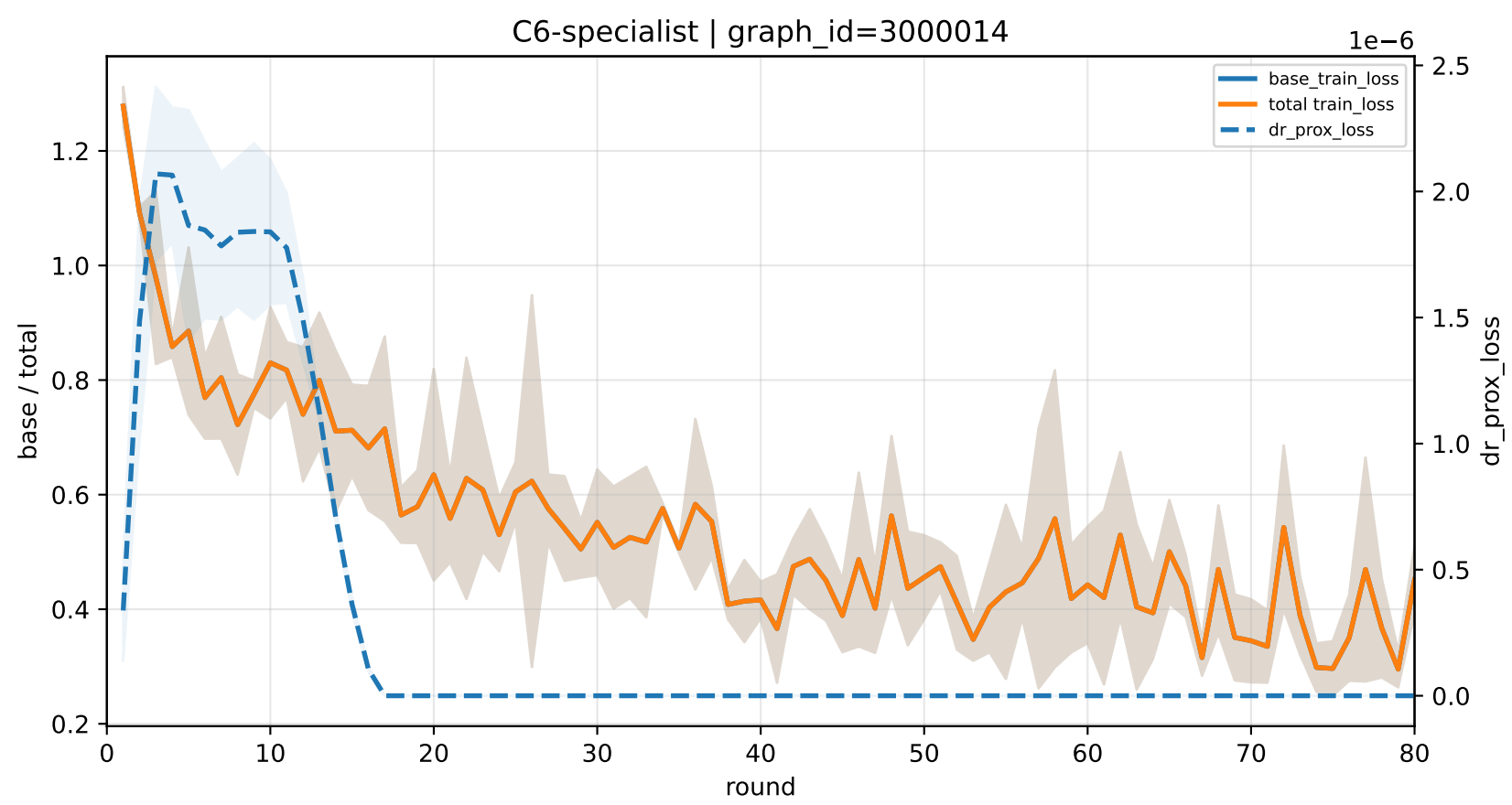
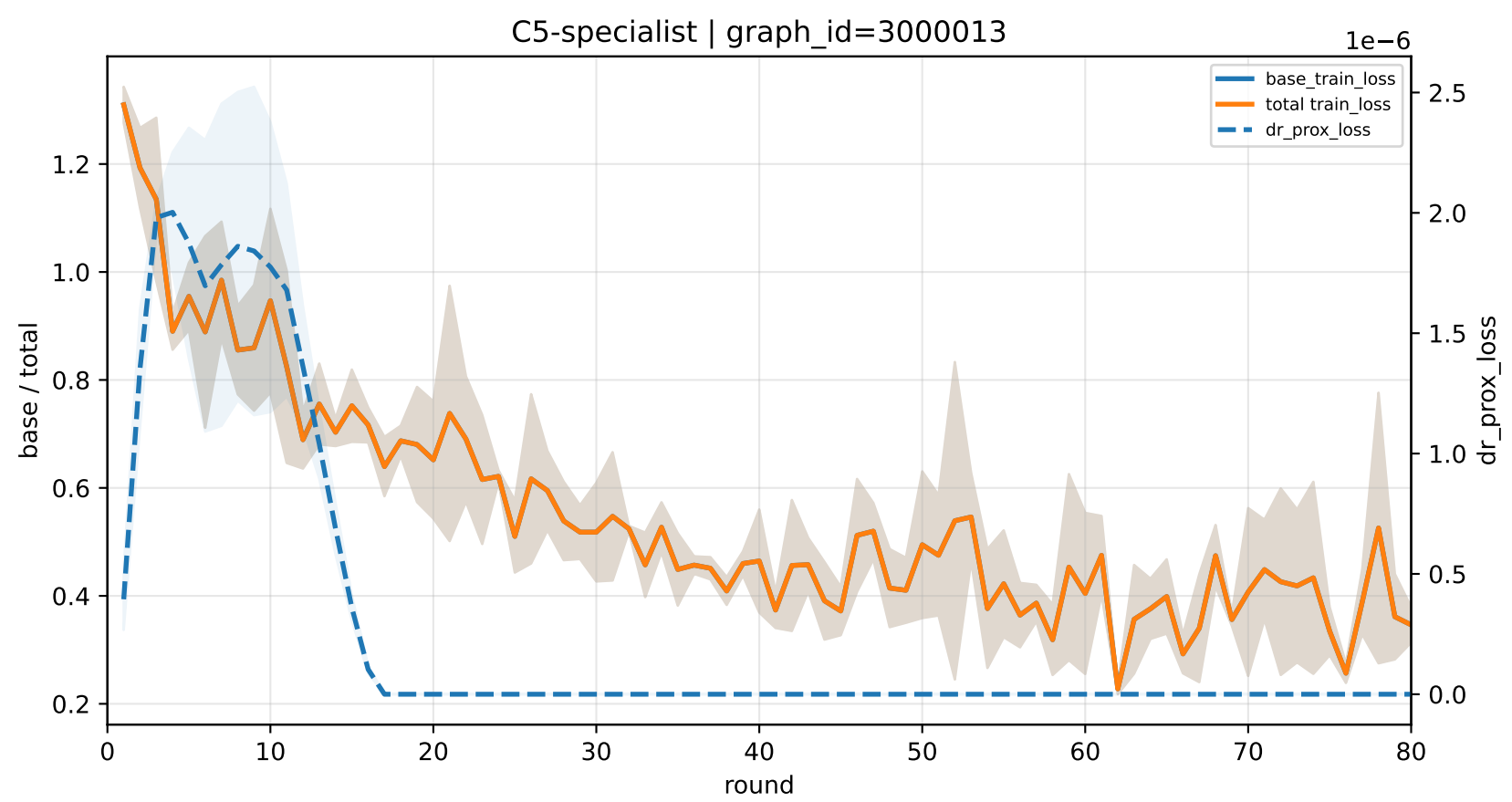
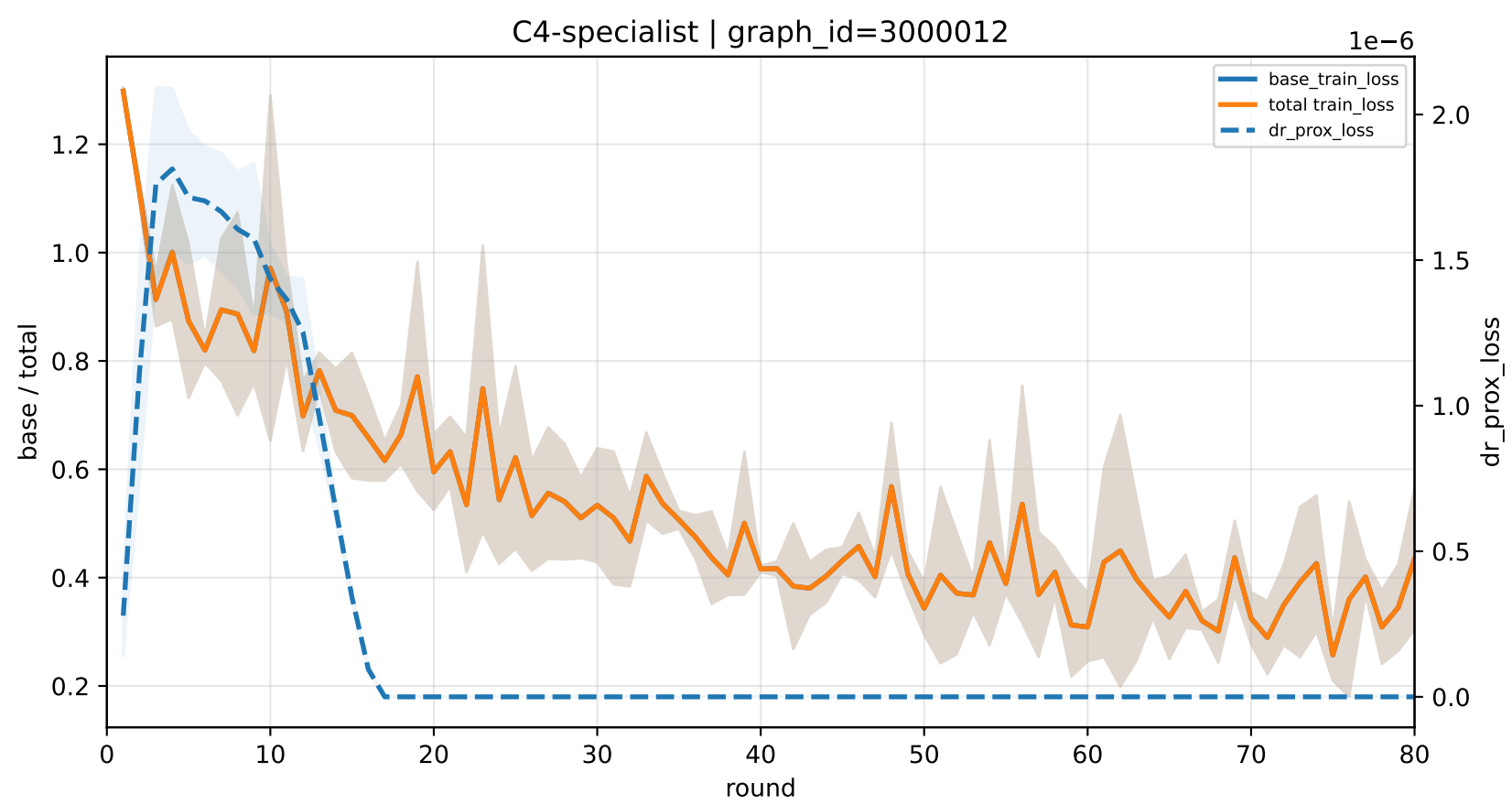
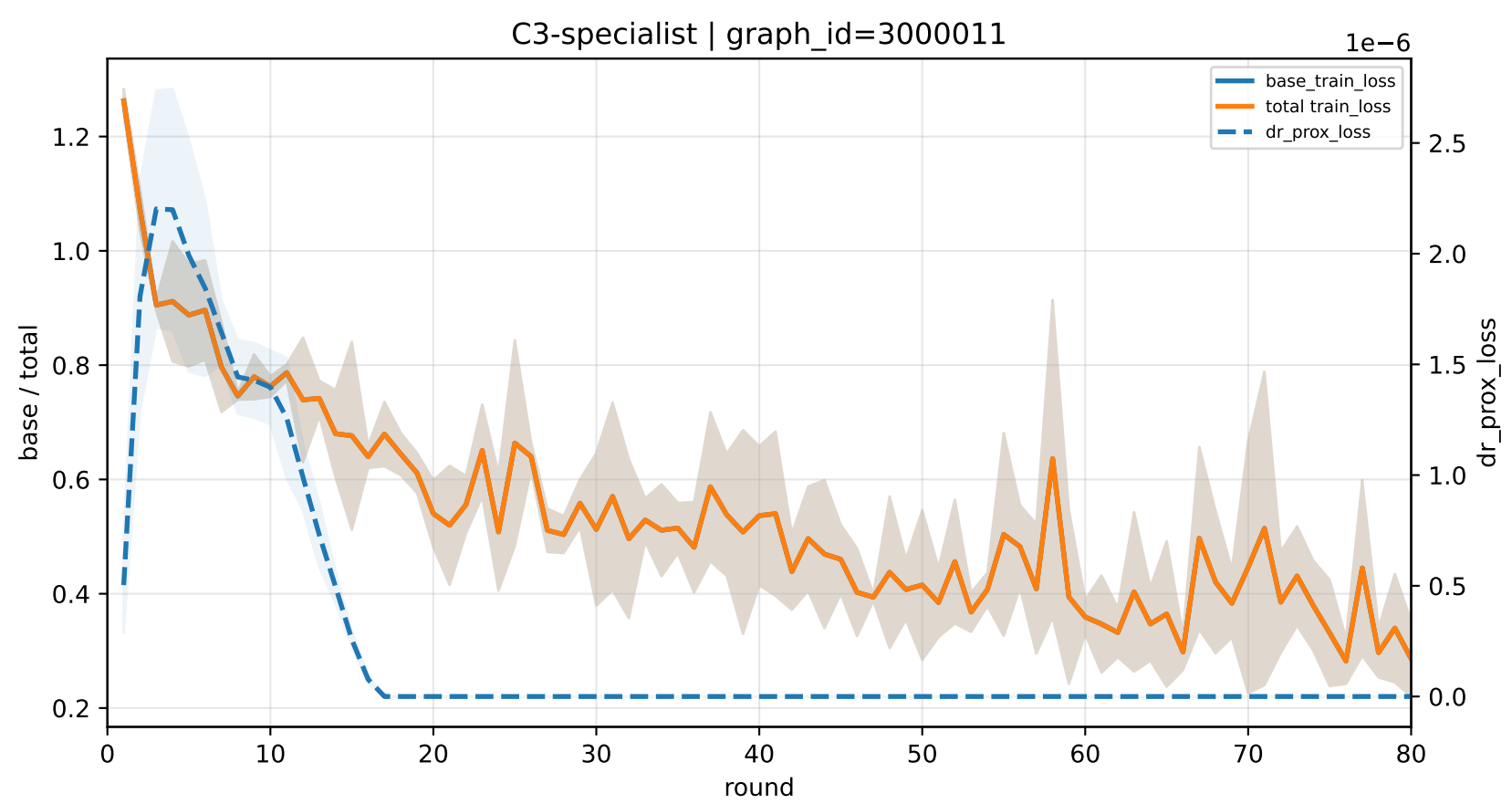
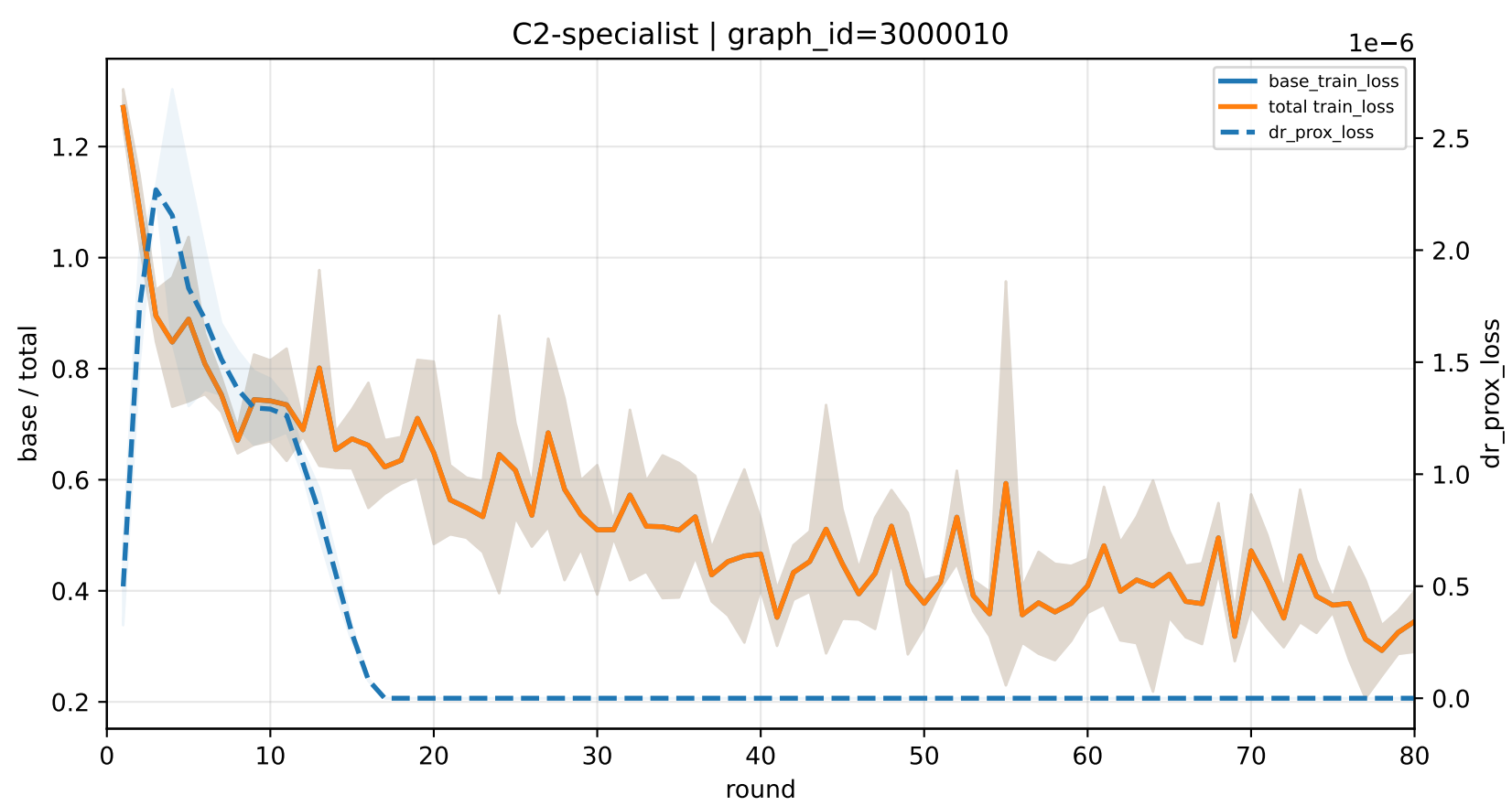
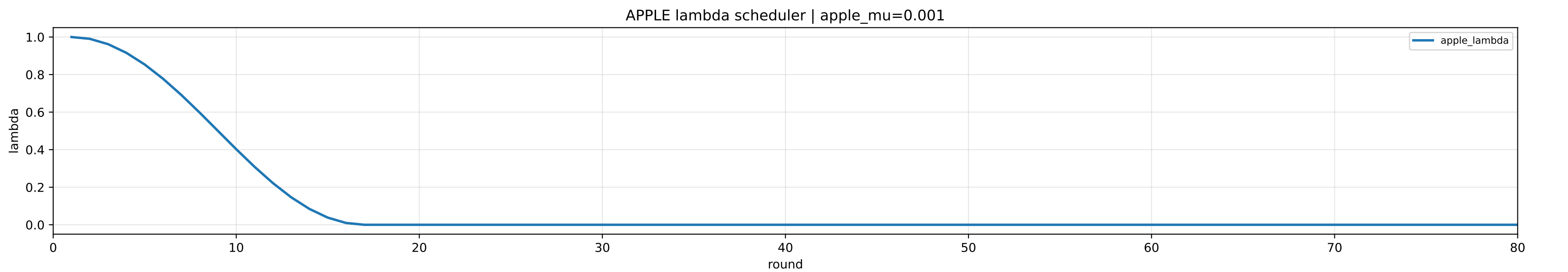


Delta diagnostics: Local gap and APPLE-FedAvgBackbone-TaskHead mismatch gap



APPLE-FedAvgBackbone-TaskHead training-loss decomposition | High heterogeneity

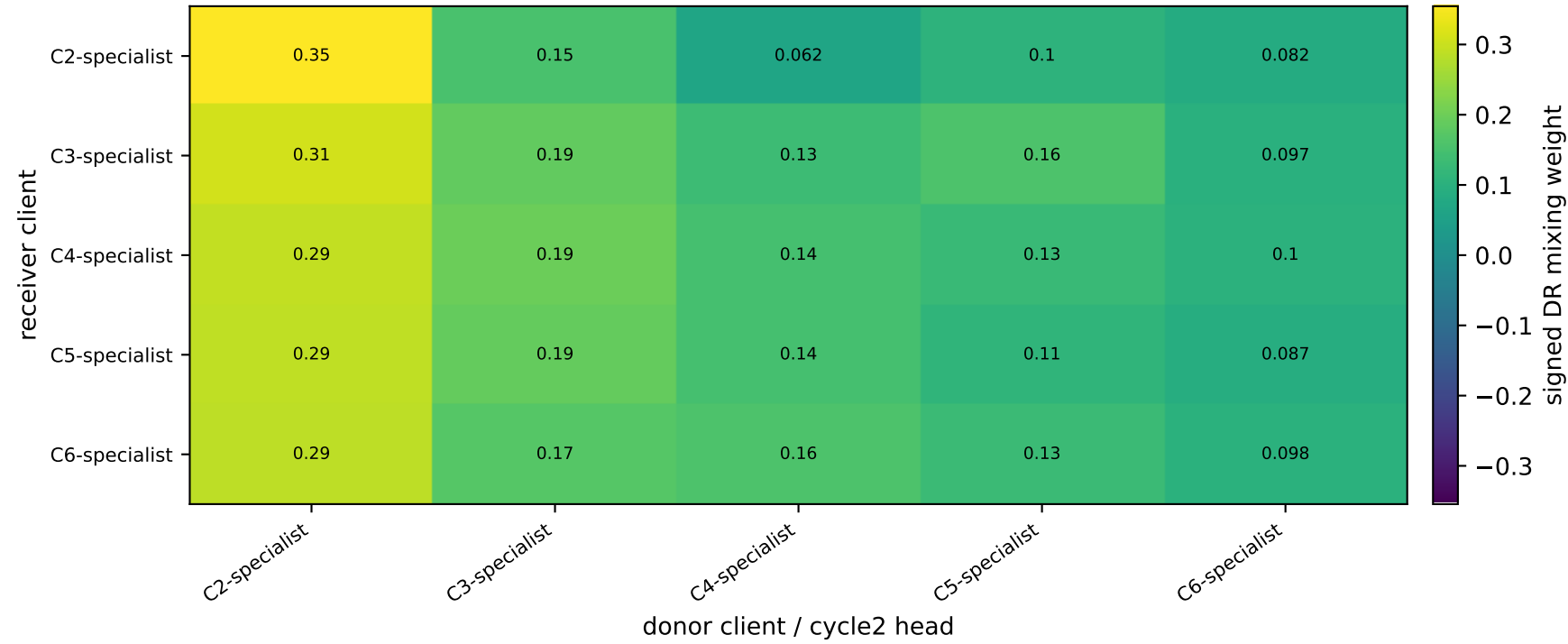
family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399



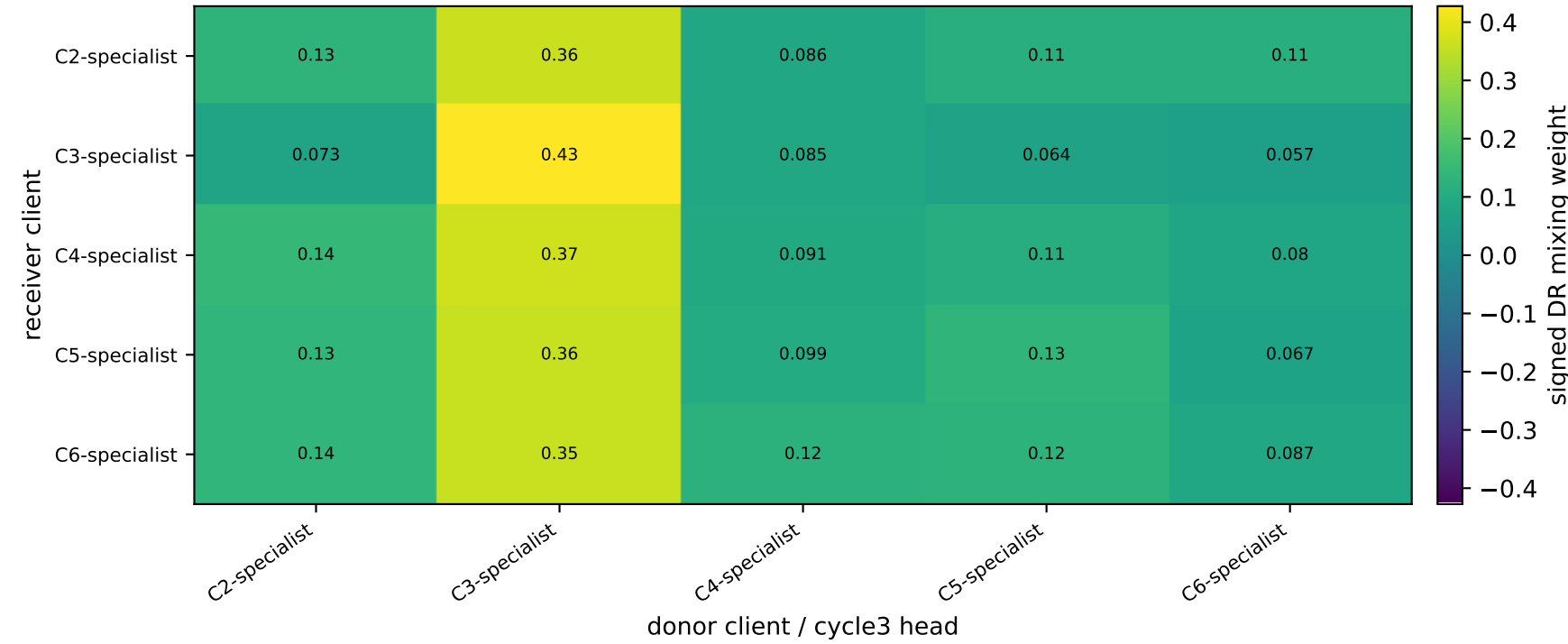
APPLE-FedAvgBackbone-TaskHead task-wise DR matrices: receiver clients × donor task-heads | High

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

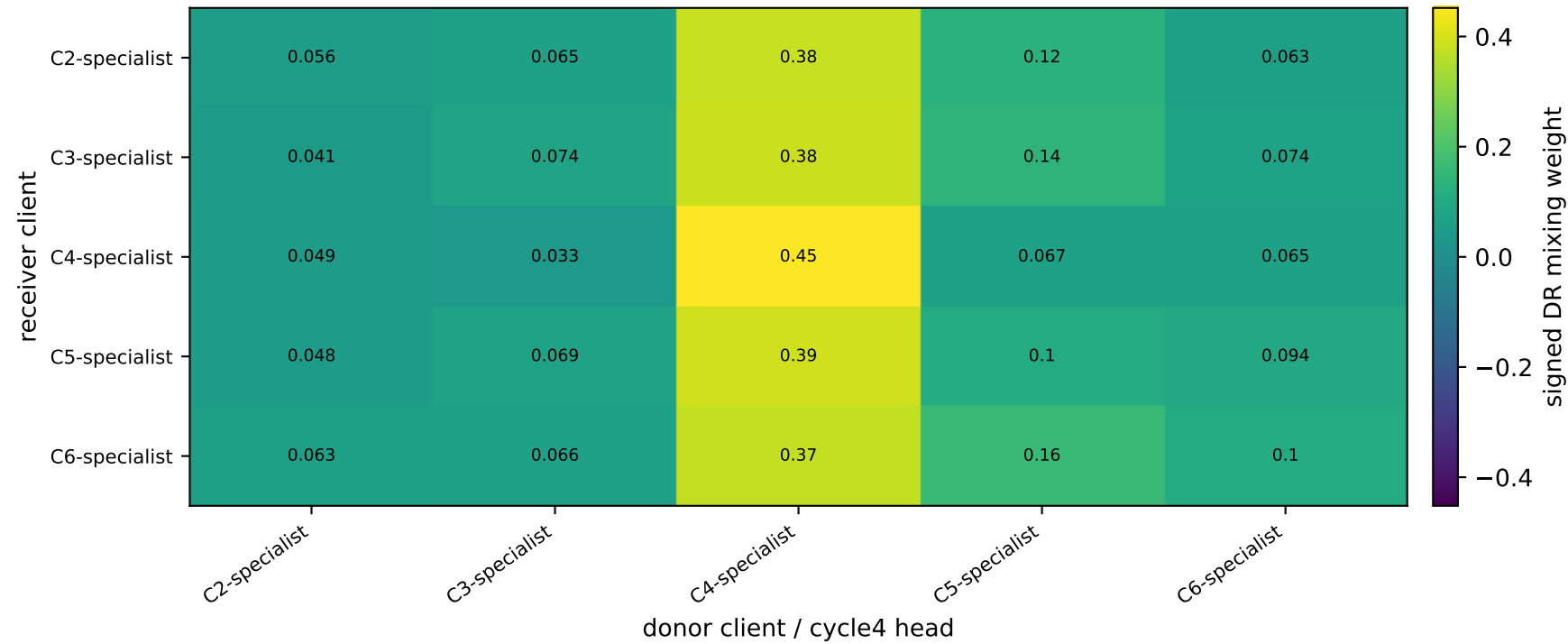
DR matrix for task cycle2: at best selected checkpoint



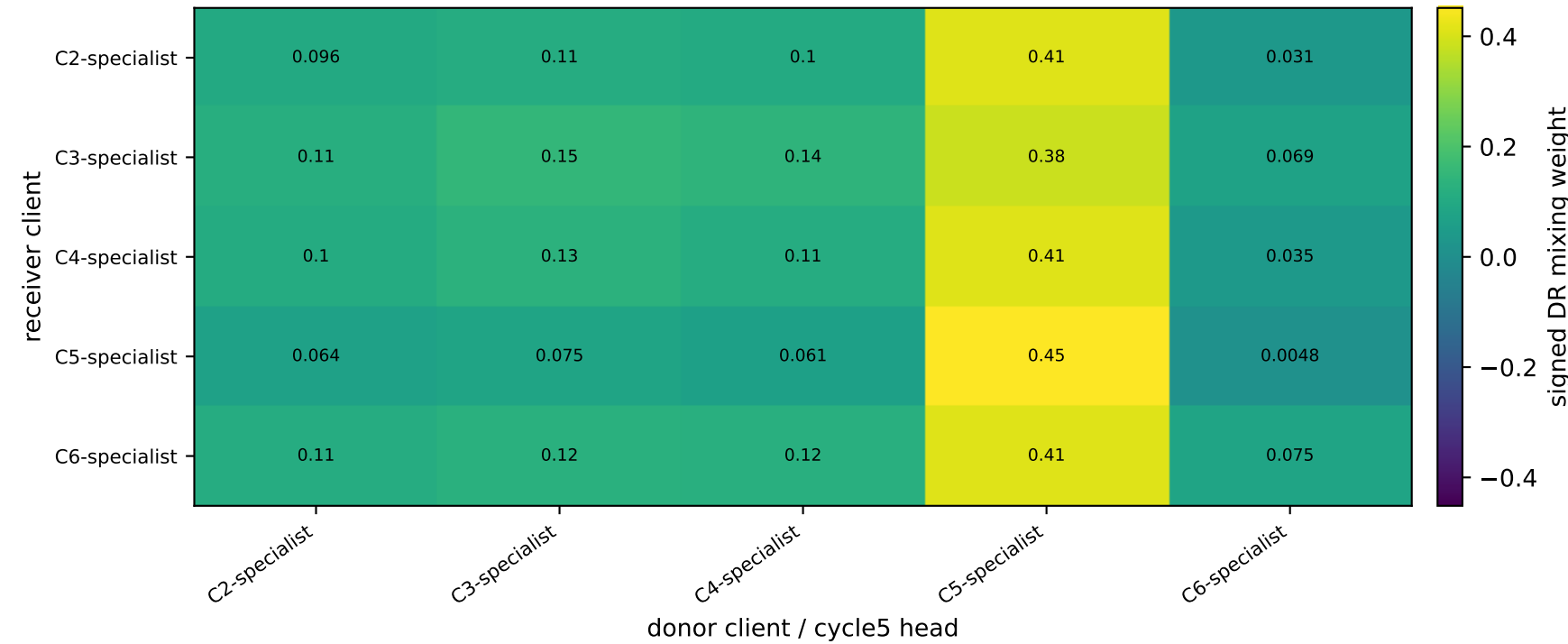
DR matrix for task cycle3: at best selected checkpoint



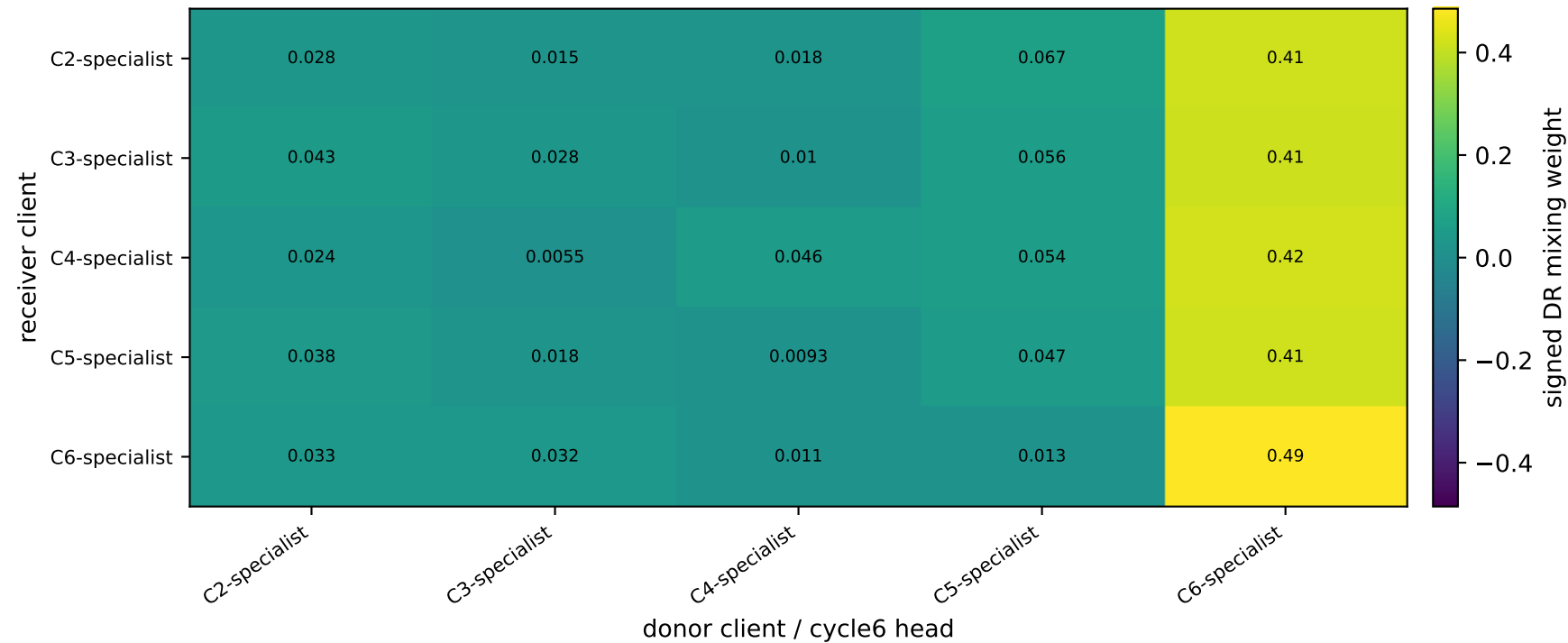
DR matrix for task cycle4: at best selected checkpoint



DR matrix for task cycle5: at best selected checkpoint



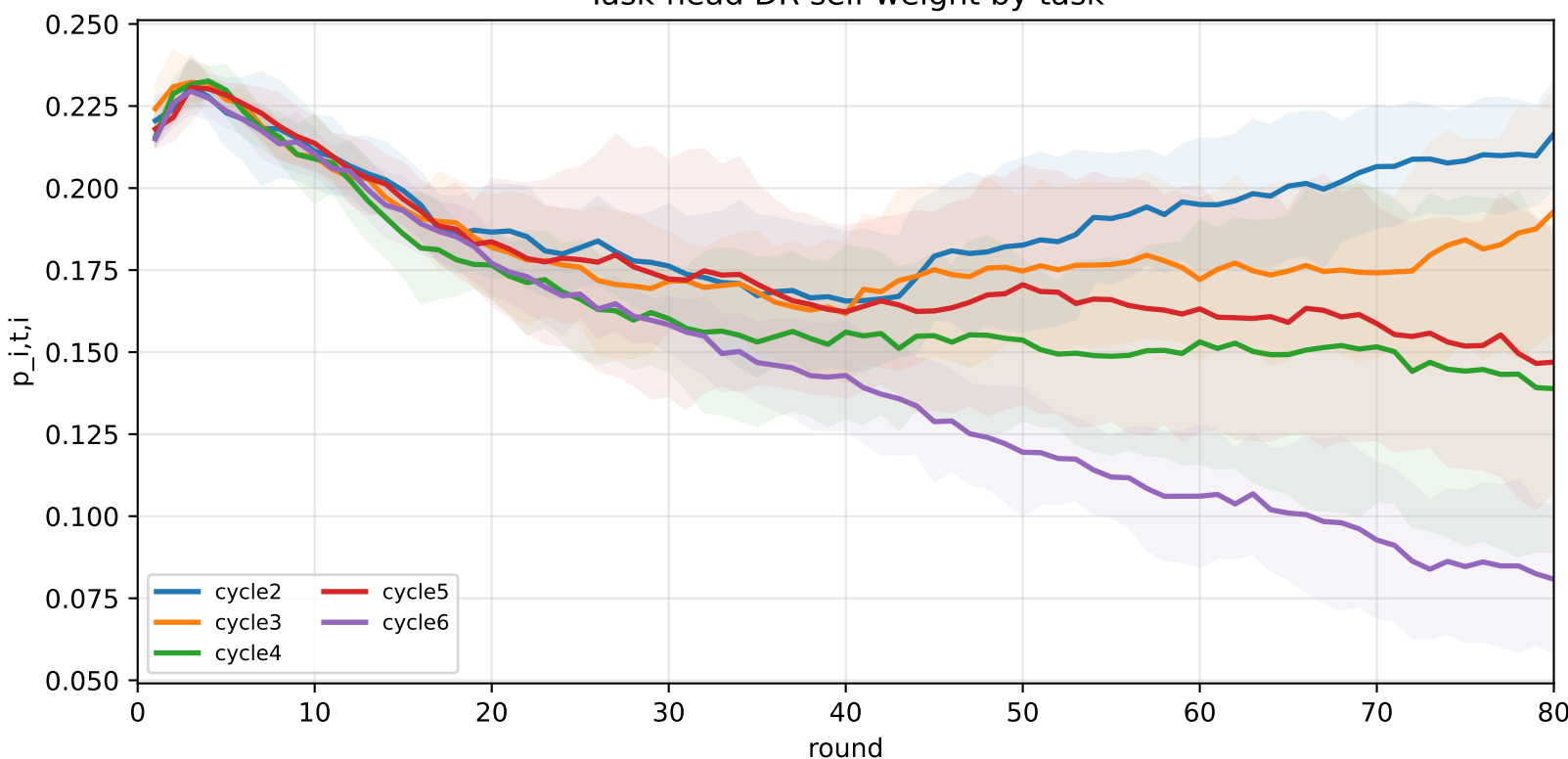
DR matrix for task cycle6: at best selected checkpoint



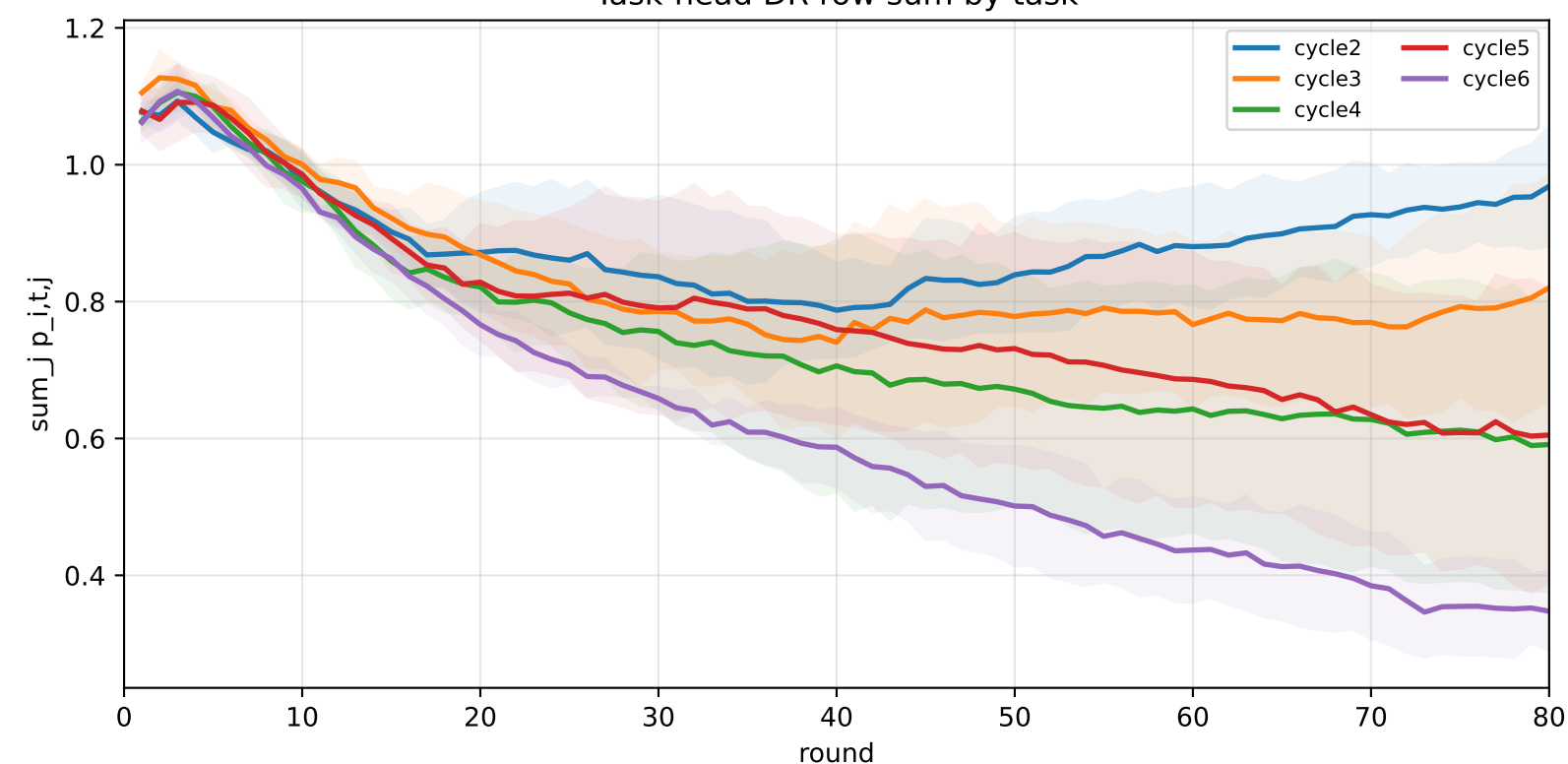
APPLE-FedAvgBackbone-TaskHead task-wise DR dynamics | High

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
 family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
 family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
 family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
 family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
 heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

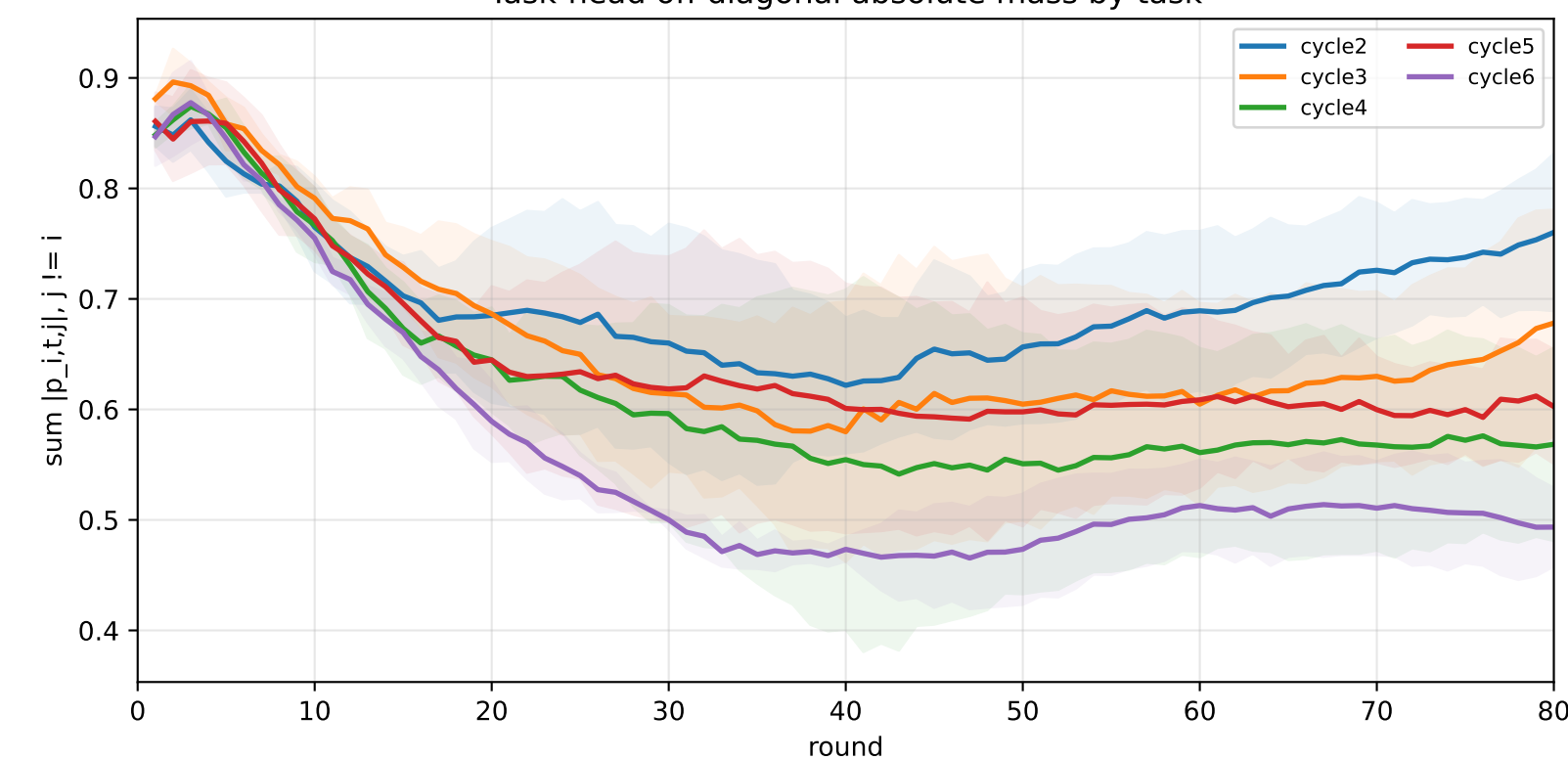
Task-head DR self-weight by task



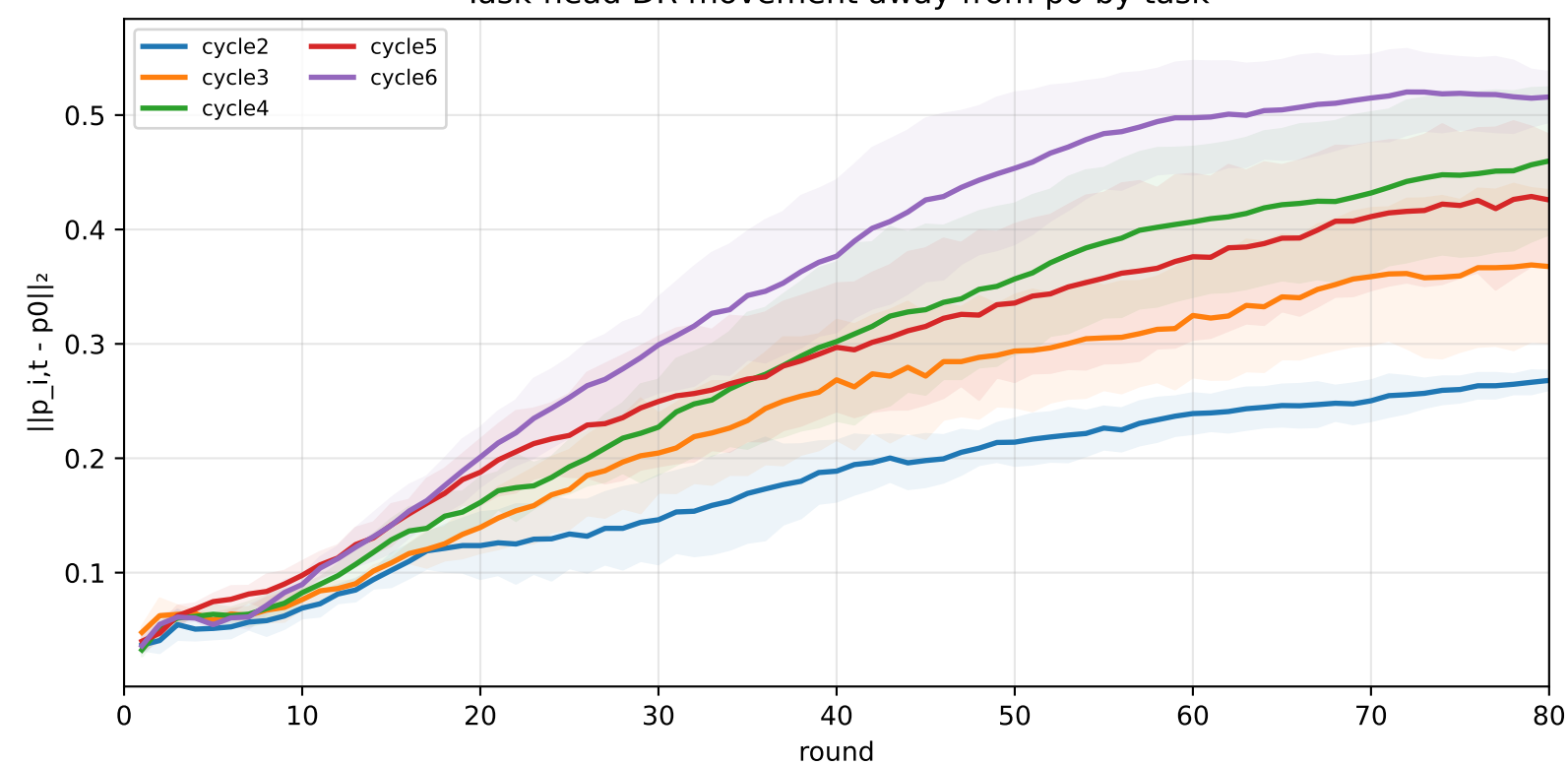
Task-head DR row sum by task



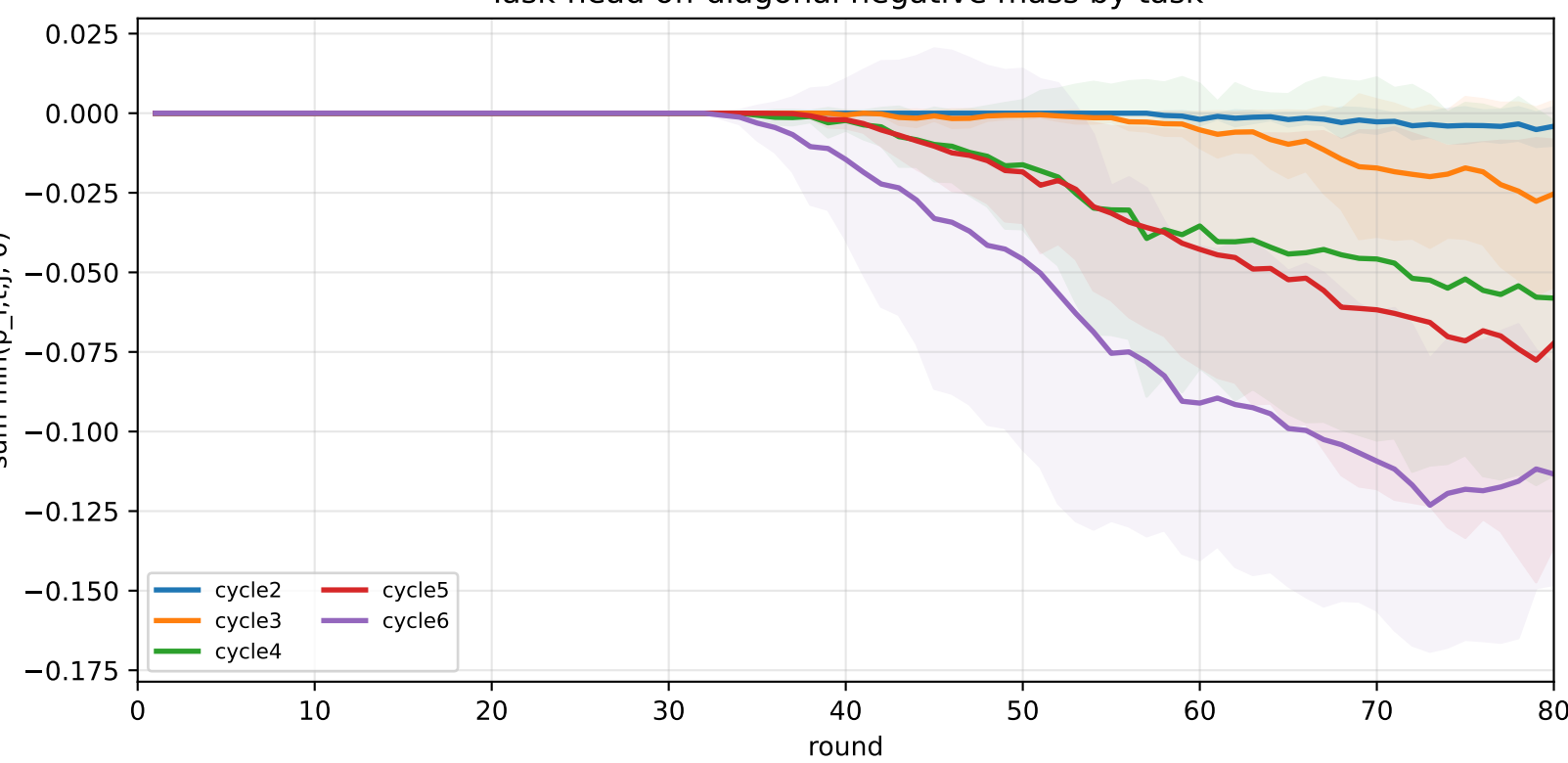
Task-head off-diagonal absolute mass by task



Task-head DR movement away from p0 by task



Task-head off-diagonal negative mass by task



Task-head DR alignment

